

Nonlinear Operator Composition for Robust Reconstruction of Latent Psychometric Representations under Structured Perturbations

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Abstract

Psychometric scoring procedures commonly aggregate or transform observed questionnaire responses into scale- or facet-level scores. Their transparency and reproducibility are valuable; however, deterministic scoring does not, in itself, provide a learned reconstruction mechanism when the observed response vector has been perturbed, an occurrence that is inherently both probable and common in practical questionnaire administration. In real-world questionnaire settings, unintended response deviations may arise from insufficient attention, fatigue, distraction, response uncertainty, or item ambiguity, whereas other deviations may be deliberate; the present study models a controlled subset of these phenomena through structured semi-synthetic transformations. We formulate this problem as a supervised nonlinear functional approximation task in which a corrupted observation is used to reconstruct the facet-level representation defined by the corresponding original response pattern.

In this manuscript, the term “latent psychometric representation” refers specifically to the 30-dimensional facet-level representation operationally derived from the canonical psychometric scoring function applied to the original uncorrupted response vector. The term does not imply that the resulting representation constitutes, in its entirety, an independently observed latent psychological truth, a clinical ground truth, or a directly measured mental state.

We evaluate multilayer perceptron (MLP)-based Gateway configurations on IPIP-NEO-120 and IPIP-NEO-300, selected as complementary 120- and 300-item instantiations within a documented 30-facet psychometric framework. The Gateway formulates reconstruction as a composition of finite-dimensional nonlinear transformations. Configurations were developed through controlled model-selection procedures using performance on corrupted validation data as the primary robustness criterion, whilst the corruption mechanism and its severity remained external evaluation descriptors. We constructed controlled semi-synthetic perturbations using observed human response material for donor-based transformations and deterministic experimental transformations for collapse, randomised modification, and structured alteration regimes. An immutable facet-level Gold representation is computed from the original uncorrupted responses and remains unchanged following perturbation; Gold is never recomputed from the corrupted input. For each corrupted instance, the target definition corresponds to the Gold representation of the original clean participant used as the Traditional reference. Seven structured perturbation families are evaluated, encompassing item-level, facet-level, category-level, collapse, and random response transformations.

The TEST participant and donor populations were held out prior to model training, after which the benchmark was generated and frozen under a predefined, locked perturbation protocol. For donor-based perturbations, donor responses were drawn exclusively from the held-out donor pool assigned to the corresponding split. The perturbation family, severity, donor identity, and specific corruption configuration were retained exclusively as external construction and evaluation metadata and were withheld from the predictive input; the Gateway received only the corrupted response vector. The frozen benchmark remained fully isolated from model development and was not used for training, architecture selection, objective selection, checkpoint selection, or retraining.

The IPIP-NEO-120 validation benchmark contains 61,556 clean observations and 338,444 corrupted observations. The final IPIP-NEO-120 configuration employs the architecture $120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30$, with 69,214 trainable parameters, seed 20260933, and epoch 6. The independent frozen IPIP-NEO-120 TEST contains 61,557 held-out participants, each contributing predefined perturbation conditions, yielding 61,557 clean and 1,723,596 corrupted TEST observations. On the corrupted TEST observations, the final S2 Gateway reduces RMSE from 0.579329 to 0.428449 and MAE from 0.287756 to 0.264785, whilst increasing Pearson correlation from 0.715060 to 0.829916 and Spearman correlation from 0.712385 to 0.798439. These results correspond to relative reductions of 26.044% in RMSE and 7.983% in MAE, together with gains of 0.114856 and 0.086054 in Pearson and Spearman correlation, respectively. At the perturbation-instance level, the Gateway is closer to Gold than the Traditional score in 72.925% of corrupted TEST observations, with no ties. The frozen TEST was not used for training, architecture screening, objective screening, checkpoint selection, or retraining.

The IPIP-NEO-300 validation benchmark contains 21,808 clean observations and 338,444 corrupted observations. The final production Gateway employs the structured architecture $300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 30$, with five six-facet output heads, 220,782 trainable parameters, and the composite objective $0.1675\mathcal{L} * \text{MSE} + 0.8325(1 - \rho * \text{Pearson})$. The model-selection protocol selected epoch 14 on corrupted validation data, with validation RMSE = 0.184836. The independent frozen IPIP-NEO-300 TEST contains 21,808 held-out participants, each contributing predefined perturbation conditions, yielding 21,808 clean and 610,624 corrupted TEST observations. On the corrupted TEST observations, P83.25 E14 reduces RMSE from 0.290377 to 0.223053 and MAE from 0.139890 to 0.137994, whilst increasing Pearson correlation from 0.756625 to 0.833131 and Spearman correlation from 0.747784 to 0.810582. These results correspond to a relative reduction of 23.185% in RMSE, a reduction of 1.355% in MAE, and gains of 0.076506 and 0.062798 in Pearson and Spearman correlation, respectively. At the perturbation-instance level, P83.25 E14 is closer to Gold than Traditional in 69.927% of corrupted TEST observations, with no ties. The model checkpoint was selected exclusively using validation data prior to independent TEST evaluation, and the frozen IPIP-NEO-300 TEST was not used for training, architecture screening, objective screening, checkpoint selection, or retraining.

Across both instrument configurations, the independent corrupted TEST benchmarks therefore exhibit the same directional pattern across all four reported evaluation metrics: lower RMSE, lower MAE, higher Pearson correspondence, and higher Spearman correspondence for the learned Gateway relative to Traditional scoring. All reported indicators improve across both independent TEST configurations, corresponding to 8/8 instrument-metric comparisons showing improvement. This convergence is observed across two instrument configurations within the same 30-facet IPIP-NEO framework and should not be interpreted as evidence of universal external validity across independent psychometric instruments.

Future replications across other instrument families, together with evaluation using architectures independent of the item structure, domains, facets, and scoring functions, are reserved for subsequent versions owing to current resource and time constraints. Likewise, although the Johnson datasets employed in the present study provide a strong and well-documented empirical foundation, there remains scope to increase both sample size and population diversity, including greater ethnic, social, and geographical diversity; such extensions may enable

a broader assessment of result stability under a more heterogeneous population distribution.

The findings are interpreted as evidence of robustness to the specified semi-synthetic perturbation protocol, rather than as evidence of clinical validity, diagnostic utility, autonomous psychological assessment, or replacement of established psychometric measurement. Accordingly, the present study provides a controlled empirical basis for evaluating learned reconstruction under structured perturbations of questionnaire responses, whilst motivating future external replications across psychometric instruments with independent item architectures, domains, facet structures, and scoring functions.

Keywords: psychometric reconstruction; personality assessment; nonlinear function approximation; structured perturbations; response corruption; robust machine learning; latent psychometric representation; psychometric scoring; questionnaire response reconstruction; facet-level reconstruction; semi-synthetic perturbations; structured response transformations; donor-based perturbations; corrupted validation; independent frozen TEST; immutable Gold targets; canonical psychometric scoring; multilayer perceptron; multi-output regression; robust psychometric modelling; reconstruction fidelity; RMSE; MAE; Pearson correlation; Spearman correlation; IPIP-NEO-120; IPIP-NEO-300; 30-facet personality assessment; computational psychometrics; reproducible machine learning; reproducible research.

1. Introduction

Psychometric scoring and reference representation. Personality assessment seeks to translate observed questionnaire responses into interpretable representations of individual differences. Within the Five-Factor Model (FFM), the IPIP-NEO family provides a public-domain operationalisation of five broad domains and thirty narrower facet constructs. The IPIP-NEO-120 was developed as a 120-item short form retaining the thirty-facet structure of the longer 300-item inventory, with four items per facet ([johnson2014ipipneo120](#)). The broader FFM framework and its facet-level organisation have been extensively examined in personality psychology ([mccrae1992fivefactor](#); [kajonius2019ipipneo](#)).

The conventional scoring procedure is deliberately deterministic and transparent. Responses to negatively keyed items are reverse-scored, items designated for each facet are aggregated, and the resulting facet values can be reproduced directly from the observed response vector. This property is valuable for interpretability, auditability, and methodological reproducibility. It also defines a canonical mapping from a complete response vector to a thirty-dimensional psychometric representation. In the present study, we use the term *canonical reference psychometric representation* to denote the representation obtained by applying this established scoring framework to the participant’s original, unperturbed response pattern. This representation constitutes an operational reference target for the reconstruction task and is not interpreted as an independently observed psychological truth, a clinical ground truth, or a directly measured mental state.

Reconstruction problem. The computational problem considered here differs from conventional psychometric scoring in that the model is not merely asked to compute the deterministic score associated with an observed response vector. Instead, the observed vector may have undergone a structured alteration prior to inference, and the task is to reconstruct the canonical reference representation that would have been obtained from the corresponding original response pattern. Accordingly, the objective of the Gateway is neither to learn a new psychometric scoring function, redefine the measured construct, nor replace the canonical scoring function, but rather to approximate a reconstruction function conditioned on a perturbed observation.

Let

$$X_{\text{clean}} \in 1, 2, 3, 4, 5^d, \quad d \in 120, 300, \quad (1)$$

denote an original response vector, and let

$$\mathcal{C}_{\phi, D}$$

denote a structured perturbation operator parameterised by the transformation mechanism ϕ and its severity, which may incorporate donor material D when required by the corresponding perturbation family. For donor-based perturbations, D is selected, where applicable, from the donor registry reserved for the corresponding split and structural transformation unit, whereas transformations that do not require donor material contain no such exogenous input. The corrupted observation is therefore defined as

$$X_{\text{corr}} = \mathcal{C}_{\phi, D}(X_{\text{clean}}). \quad (2)$$

The canonical Gold psychometric representation is defined independently from the original, unperturbed vector,

$$Y_{\text{Gold}} = g(X_{\text{clean}}) \in \mathbb{R}^{30}. \quad (3)$$

where g denotes the established canonical psychometric scoring function. This representation is computed exclusively from the participant's original, uncorrupted response vector and remains immutable throughout perturbation generation, model development, and evaluation; Gold is never recomputed from the corrupted input.

The Traditional procedure applied to the same perturbed observation is defined as

$$Y_{\text{Traditional}} = g(X_{\text{corr}}), \quad (4)$$

whereas the learned Gateway produces

$$Y_{\text{Gateway}} = F_{\theta}(X_{\text{corr}}). \quad (5)$$

Accordingly, the experimental comparison holds both the input observation and the reference target constant: Traditional and Gateway receive exactly the same corrupted response vector X_{corr} , and both are evaluated against the same participant-specific Gold representation,

$$Y_{\text{Gold}} = g(X_{\text{clean}}). \quad (6)$$

The difference between the two methods therefore resides exclusively in the mapping applied to the perturbed input: Traditional reapplies the deterministic scoring function to the observed vector, whereas Gateway learns a nonlinear reconstruction of the original reference representation.

This formulation establishes an explicit distinction between the *observation space*, the *reference representation space*, and the *perturbation process*. The observation space remains the questionnaire item-response vector, with responses recorded on the instrument's five-point response scale. The reference space is the thirty-dimensional facet-level representation defined by the established scoring procedure. The perturbation process modifies only the observed input presented to the reconstruction model and to the Traditional procedure; it does not redefine the Gold target. This separation is central to the experimental design because it permits reconstruction accuracy to be evaluated against a fixed, participant-specific reference rather than

against a target recomputed from the corrupted observation.

For any discrepancy criterion $d(\cdot, \cdot)$ applied to the output representations, the fundamental comparison may be expressed as

$$\mathcal{E}_{\text{Traditional}} = d(Y_{\text{Traditional}}, Y_{\text{Gold}}), \quad (7)$$

versus

$$\mathcal{E}_{\text{Gateway}} = d(Y_{\text{Gateway}}, Y_{\text{Gold}}). \quad (8)$$

RMSE and MAE constitute measures of numerical discrepancy, whereas Pearson and Spearman provide complementary measures of correspondence between the thirty-dimensional profiles. This formulation makes explicit that the two procedures are not evaluated against different targets or different reference representations: both are evaluated against the same Gold representation derived from the same original participant.

Structured semi-synthetic perturbations. The perturbation protocol is deliberately structured across multiple levels of granularity and severity. The evaluated families include item-level substitution, facet-level transplantation, category-level transformations, collapse operations, and random item modifications. Importantly, the protocol is not restricted to arbitrarily generated numerical noise. For selected transformation families, donor material is drawn from observed response material contributed by human participants whose response patterns satisfy predefined eligibility and assignment criteria for donor selection, including facet- and domain-specific criteria where applicable. Donor material is incorporated solely as a component of perturbation construction and not as a source of the Gold target.

For donor-based perturbations, donor responses are drawn exclusively from the donor pool reserved for the corresponding split. The donor material is therefore neither arbitrarily generated nor exchanged across partitions in an uncontrolled manner. Incomplete participants do not provide reference targets: their observed responses contribute solely to the perturbation-construction mechanism, whilst the Gold representation remains associated with the original complete profile of the target participant. The synthetic component of the procedure resides in the experimental transformation mechanism applied to that material: the mechanism controls how the alteration is introduced, whilst donor-based components retain empirically observed human response material. In the remaining regimes, collapse, randomisation, and structured alteration are implemented through deterministic or otherwise controlled experimental transformations. Accordingly, the design constitutes a *semi-synthetic, empirically grounded perturbation protocol*: the perturbation mechanism is experimentally controlled, whilst part of the response content is drawn from observed human material.

The participant and donor populations designated for TEST were held out prior to model training; the TEST benchmark was subsequently materialised, once model development had been fixed, and the resulting dataset was frozen under a locked perturbation protocol. Each held-out participant contributes a predefined set of perturbation conditions; these conditions constitute repeated experimental instances applied to the same held-out participant and should not be interpreted as observations corresponding to independent participants. The independence of the benchmark is therefore established at the level of the held-out participant and donor populations relative to model development, whilst the multiple perturbation conditions represent deliberately defined experimental conditions within those retained populations.

The perturbation family, severity, donor identity, and specific corruption configuration were retained exclusively as external construction and evaluation metadata and were completely

excluded from the predictive input. The Gateway received only the corrupted response vector corresponding to the evaluated instance. No explicit signal identifying the perturbation family, its severity, the provenance of donor material, or the applied configuration was provided to the model as a predictive variable. This separation prevents experimental information used to characterise the corruption from simultaneously functioning as auxiliary predictive information.

The frozen benchmark remained completely isolated from model development. TEST was not used for training, architecture selection, objective selection, checkpoint selection, or retraining. The TEST participant and donor populations were reserved prior to model training, and the benchmark was generated only after that separation had been established, following which it was frozen under the corresponding locked protocol. Thus, final evaluation was conducted on a dataset independent of model development rather than on a subset subsequently reused to optimise architecture, objective function, checkpoint, or any other component of the modelling procedure.

This temporal and functional separation permits three conceptually distinct stages of the experimental procedure to be identified: corrupted validation used for model development and selection; confirmatory evaluation used to verify the behaviour of the selected configuration under criteria defined before confirmation; and the independent frozen TEST reserved for final evaluation. The multi-seed confirmation is not interpreted as confirmatory inference in the sense of a preregistered formal statistical trial, but rather as a predefined procedure for verifying the stability of the selected configuration under explicit robustness criteria.

Nonlinear function approximation. The reconstruction problem may therefore be viewed as a finite-dimensional nonlinear function-approximation task. Classical approximation theory establishes broad approximation capabilities for multilayer feedforward networks over finite-dimensional mappings (**hornik1989universal**). Accordingly, the MLP Gateway is represented as a finite composition of parameterised transformations,

$$F_\theta = \mathcal{T}_L \circ \mathcal{T}_{L-1} \circ \cdots \circ \mathcal{T}_2 \circ \mathcal{T}_1, \quad (9)$$

where each transformation $\mathcal{T}_\ell$ is parameterised by a corresponding subset of model parameters and maps the representation produced by the preceding layer to a new finite-dimensional representation. Let

$$\mathbf{h}_0 = X_{\text{corr}} \quad (10)$$

denote the perturbed response vector supplied to the Gateway. The successive hidden representations are then defined recursively as

$$\mathbf{h}_\ell = \mathcal{T}_\ell(\mathbf{h}_{\ell-1}), \quad \ell = 1, \dots, L-1, \quad (11)$$

with the final transformation producing the reconstructed thirty-dimensional representation,

$$\hat{Y} = F_\theta(X_{\text{corr}}) = \mathcal{T}_L(\mathbf{h}_{L-1}) \in \mathbb{R}^{30}. \quad (12)$$

For a generic hidden transformation, the composition may be expressed as

$$\mathcal{T}_\ell = \mathcal{D}_\ell \circ \mathcal{A}_\ell \circ \mathcal{N}_\ell \circ \mathcal{W}_\ell, \quad (13)$$

where $\mathcal{W}_\ell$ denotes a learned affine transformation, $\mathcal{N}_\ell$ denotes the corresponding normalisation transformation, $\mathcal{A}_\ell$ denotes a nonlinear activation transformation, and $\mathcal{D}_\ell$ denotes the

corresponding regularisation transformation when present. Equivalently, for a hidden layer with input representation $\mathbf{h}_{\ell-1}$,

$$\mathbf{z}_\ell = W_\ell \mathbf{h}_{\ell-1} + \mathbf{b}_\ell, \quad (14)$$

followed by

$$\tilde{\mathbf{z}}_\ell = \mathcal{N}_\ell(\mathbf{z}_\ell), \quad (15)$$

and

$$\mathbf{h}_\ell = \mathcal{D}_\ell(\mathcal{A}_\ell(\tilde{\mathbf{z}}_\ell)). \quad (16)$$

Here, W_ℓ and $\mathbf{b}_\ell$ denote the learned weight matrix and bias vector associated with transformation ℓ , while the remaining operators represent the normalisation, nonlinear activation, and regularisation components specified by the architecture. The complete parameter set is therefore denoted by

$$\theta = \{W_\ell, \mathbf{b}_\ell, \mathcal{N}_\ell, \mathcal{A}_\ell, \mathcal{D}_\ell\}_{\ell=1}^L, \quad (17)$$

with the understanding that only the parameterised components of these transformations contribute trainable parameters.

The resulting Gateway is consequently a finite-dimensional nonlinear mapping from the corrupted questionnaire response space to the canonical reference representation space,

$$F_\theta : \mathcal{X}_d \longrightarrow \mathbb{R}^{30}, \quad (18)$$

such that

$$X_{\text{corr}} \in \mathcal{X}_d \quad \longmapsto \quad \hat{Y} = F_\theta(X_{\text{corr}}). \quad (19)$$

The individual transformations therefore do not operate independently; rather, each transformation receives the representation produced by the preceding transformation, such that the complete Gateway constitutes an ordered composition of learned affine mappings, normalisation operations, nonlinear activation functions, and regularisation transformations. This compositional structure is the basis for treating the reconstruction mechanism as a finite-dimensional nonlinear function approximator.

The terminology *nonlinear operator composition* is used here in this finite-dimensional sense. We do not claim that the model constitutes a *neural operator* in the stricter sense used in the neural-operator literature, where the objective is to learn mappings between function spaces and the formulation is typically infinite-dimensional or discretisation-aware (**kovachki2023neuraloperator**). Rather, the present formulation emphasises the mathematical structure of the Gateway as a composition of nonlinear transformations acting on finite-dimensional questionnaire vectors.

Instrument-specific Gateway configurations. Two complementary instrument configurations are investigated. The IPIP-NEO-120 experiment employs a compact MLP Gateway mapping the 120 item responses to the thirty-dimensional reference representation. The final IPIP-NEO-120 configuration was selected through controlled architecture screening followed by independent multi-seed confirmatory evaluation under robustness criteria defined prior to confirmation, yielding the S2 architecture

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30$$

with 69,214 trainable parameters. The final operating point uses seed 20260933 and epoch 6.

The IPIP-NEO-300 experiment employs a structured-output MLP Gateway mapping the 300 item responses to the same thirty-dimensional representation through five six-facet output heads. The final production configuration was developed through controlled architecture and objective screening on corrupted validation data and subsequently refined through comparative evaluation of metrics, architecture, and hyperparameter tuning. The final selected configuration was P83.25,

$$300 \longrightarrow 344 \longrightarrow 208 \longrightarrow 184 \longrightarrow 5 \times 6. \quad (20)$$

Equivalently, the network may be represented as a sequence of finite-dimensional hidden transformations followed by a structured multi-head output mapping. Let

$$\mathbf{h}_0 = X_{\text{corr}} \in \mathbb{R}^{300} \quad (21)$$

denote the corrupted 300-dimensional questionnaire response vector supplied to the Gateway. The first hidden transformation maps the observation space into a 344-dimensional representation,

$$\mathbf{h}_1 = \mathcal{T}_1(\mathbf{h}_0) \in \mathbb{R}^{344}. \quad (22)$$

The second hidden transformation then produces a 208-dimensional representation,

$$\mathbf{h}_2 = \mathcal{T}_2(\mathbf{h}_1) \in \mathbb{R}^{208}, \quad (23)$$

followed by a third hidden transformation producing the final shared representation,

$$\mathbf{h}_3 = \mathcal{T}_3(\mathbf{h}_2) \in \mathbb{R}^{184}. \quad (24)$$

Thus, the shared nonlinear backbone may be written compactly as

$$\mathbf{h}_3 = \mathcal{T}_3 \circ \mathcal{T}_2 \circ \mathcal{T}_1(X_{\text{corr}}). \quad (25)$$

The 184-dimensional shared representation is subsequently distributed across five parallel output heads. Denoting the k -th head by $\mathcal{H}_k$, each head maps the shared representation into a six-dimensional facet subspace,

$$\mathcal{H}_k : \mathbb{R}^{184} \longrightarrow \mathbb{R}^6, \quad k = 1, \dots, 5. \quad (26)$$

The output of each head is therefore

$$\hat{\mathbf{Y}}^{(k)} = \mathcal{H}_k(\mathbf{h}_3) \in \mathbb{R}^6. \quad (27)$$

The complete thirty-dimensional reconstructed representation is obtained by concatenating the five six-dimensional head outputs,

$$\hat{\mathbf{Y}} = \text{concat}(\hat{\mathbf{Y}}^{(1)}, \hat{\mathbf{Y}}^{(2)}, \hat{\mathbf{Y}}^{(3)}, \hat{\mathbf{Y}}^{(4)}, \hat{\mathbf{Y}}^{(5)}) \in \mathbb{R}^{30}. \quad (28)$$

Consequently, the complete P83.25 Gateway can be written as the structured composition

$$F_{\theta}^{\text{P83.25}} = \text{concat}(\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3, \mathcal{H}_4, \mathcal{H}_5) \circ \mathcal{T}_3 \circ \mathcal{T}_2 \circ \mathcal{T}_1, \quad (29)$$

such that

$$\hat{\mathbf{Y}} = F_{\theta}^{\text{P83.25}}(X_{\text{corr}}) \in \mathbb{R}^{30}. \quad (30)$$

The architecture therefore separates the reconstruction process into two mathematically distinct components: a shared nonlinear representation map,

$$B_{\theta} : \mathbb{R}^{300} \longrightarrow \mathbb{R}^{184}, \quad (31)$$

and a structured output map,

$$H_{\theta} : \mathbb{R}^{184} \longrightarrow \mathbb{R}^{30}, \quad (32)$$

with

$$F_{\theta}^{\text{P83.25}} = H_{\theta} \circ B_{\theta}. \quad (33)$$

This decomposition makes explicit that the five output heads share the same learned 184-dimensional representation before producing their respective six-facet outputs. The structure therefore preserves a common latent computational representation while partitioning the final thirty-dimensional output into five six-dimensional groups.

At the level of individual hidden transformations, each $\mathcal{T}_{\ell}$ may be represented abstractly as a composition of learned affine and nonlinear operations. For an input representation $\mathbf{h}_{\ell-1}$, the affine component is

$$\mathbf{z}_{\ell} = W_{\ell}\mathbf{h}_{\ell-1} + \mathbf{b}_{\ell}, \quad (34)$$

followed by the architecture-specific transformation applied to the resulting representation,

$$\mathbf{h}_{\ell} = \mathcal{T}_{\ell}(\mathbf{h}_{\ell-1}). \quad (35)$$

Accordingly, the dimensional sequence of the shared backbone is

$$\mathbb{R}^{300} \rightarrow \mathbb{R}^{344} \rightarrow \mathbb{R}^{208} \rightarrow \mathbb{R}^{184}, \quad (36)$$

after which the representation branches into

$$\mathbb{R}^{184} \rightarrow \mathbb{R}^6 \quad (37)$$

for each of the five parallel output heads. Since

$$5 \times 6 = 30, \quad (38)$$

the concatenated output recovers the required thirty-dimensional reference representation.

The final production configuration contains 220,782 trainable parameters. The production model was trained on 101,772 TRAIN subjects, with 14,553,396 training examples per epoch and 143 views per subject. The optimisation configuration used a batch size of 8192, learning rate 0.002, weight decay 0.0001, dropout 0.1, BF16 precision, and gradient clipping at 1.0, while the model-selection protocol used a maximum of 120 epochs, a minimum of 5 epochs, an early-stopping patience of 3, and a minimum improvement threshold of 10^{-4} . The independent TEST remained reserved and untouched throughout production training. :contentReference[oaicite:2]index=2

The P83.25 production objective is defined as the weighted combination

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - \rho_{\text{Pearson}}). \quad (39)$$

The objective therefore combines a pointwise squared-error component with a profile-level correlation component. For a thirty-dimensional prediction $\hat{\mathbf{Y}}$ and corresponding canonical reference $\mathbf{Y}_{\text{Gold}}$, the MSE component may be represented as

$$\mathcal{L}_{\text{MSE}} = \frac{1}{30} \sum_{j=1}^{30} (\hat{Y}_j - Y_{\text{Gold},j})^2. \quad (40)$$

The corresponding Pearson profile correlation is represented by

$$\rho_{\text{Pearson}} = \frac{\sum_{j=1}^{30} (\hat{Y}_j - \bar{\hat{Y}}) (Y_{\text{Gold},j} - \bar{Y}_{\text{Gold}})}{\sqrt{\sum_{j=1}^{30} (\hat{Y}_j - \bar{\hat{Y}})^2} \sqrt{\sum_{j=1}^{30} (Y_{\text{Gold},j} - \bar{Y}_{\text{Gold}})^2}}, \quad (41)$$

where

$$\bar{\hat{Y}} = \frac{1}{30} \sum_{j=1}^{30} \hat{Y}_j, \quad \bar{Y}_{\text{Gold}} = \frac{1}{30} \sum_{j=1}^{30} Y_{\text{Gold},j}. \quad (42)$$

The resulting objective can consequently be interpreted as a weighted reconstruction criterion in which numerical fidelity and profile correspondence contribute jointly to the optimisation target. The coefficient 0.1675 determines the contribution of the MSE term, while the coefficient 0.8325 determines the contribution of the Pearson-profile term.

The production checkpoint was selected using corrupted validation RMSE. The best checkpoint occurred at epoch 14 with validation RMSE

$$\text{RMSE}_{\text{val,corr}} = 0.18483594655007943, \quad (43)$$

which was subsequently rounded to 0.184836 for reporting. Training continued beyond this point, but the later observed reduction in raw RMSE did not satisfy the predefined minimum improvement threshold. Epoch 17 therefore triggered early stopping while epoch 14 remained the selected checkpoint. The resulting production checkpoint was subsequently used for the independent frozen TEST evaluation.

Central empirical question. The central empirical question is therefore:

Can a learned nonlinear reconstruction mapping achieve lower discrepancy from the canonical Gold psychometric representation than Traditional deterministic scoring when both procedures are applied to the same perturbed response vector under the specified evaluation protocol?

The foregoing formulation does not presume that the Gateway has access to the cause of the perturbation. The corruption mechanism, its severity, donor identity, and corresponding experimental configuration remain outside the predictive input. The problem is therefore one of inference from the observed response pattern to the reference representation, rather than explicit classification of the corruption type.

Multidimensional evaluation and model confirmation. The evaluation is deliberately multidimensional. Reconstruction is assessed not only through point wise error, but also through profile-level correspondence and stratified behaviour across perturbation families, severity levels, and facets. Pearson and Spearman correlations provide complementary information about correspondence between the predicted and reference thirty-dimensional profiles, whilst RMSE and MAE characterize different aspects of numerical error. This distinction is important because a method may reduce large-magnitude errors without simultaneously minimizing absolute error across all observations.

For the final IPIP-NEO-120 configuration, model selection additionally distinguishes between architecture screening and confirmatory evaluation. The S2 configuration was retained only after independent training across three confirmation seeds, with all three seeds satisfying the predefined robustness criteria and reducing corrupted MAE relative to the V133 control. The final seed-20260933 operating point achieved corrupted-input RMSE = 0.349023, MAE = 0.196198, Pearson = 0.880231, and Spearman = 0.850618. Relative to V133, this corresponds to a small RMSE increase of 0.000194 and an MAE reduction of 0.005228, with Spearman increasing by 0.000157 and Pearson decreasing by 0.000267. Within the corrupted confirmation benchmark used for model selection, Traditional retained a lower MAE than the Gateway. The final configuration is therefore interpreted as a controlled robustness trade-off under predefined tolerance criteria rather than as a uniform improvement over V133 across all metrics.

Confirmatory evaluation and independent final evaluation serve distinct functions. The former verifies that the selected configuration maintains behaviour compatible with the tolerance criteria defined prior to confirmation; the latter constitutes the final comparison against Traditional on the frozen TEST reserved for that purpose. No TEST result is fed back into architecture, objective, checkpoint, or Gateway training.

The Gateway formulation also permits investigation of a hypothesis concerning predictive information across facets and domains. Because the model receives the complete response vector, information associated with facets or domains other than the one directly affected by a perturbation may potentially contain predictive structure useful for reconstructing one or more corrupted facets. In this study, this possibility is treated as an empirically testable property of the joint response representation, termed here *cross-facet predictive information*, rather than as a priori evidence of semantic equivalence, psychological interchangeability, or causal independence among facets. The existence of predictive association between dimensions is therefore not interpreted as evidence that one facet causes another, nor that conceptually differentiated facets represent the same construct.

Scope of robustness and inference. More generally, robustness in machine learning is not a single property but depends upon the class and magnitude of the perturbation under consideration. Recent work has emphasized the need to characterize robustness with respect to specific perturbations and failure regimes rather than treating performance under a single perturbation distribution as universal evidence of robustness (**balendran2025robustness**). Our contribution is accordingly bounded: we investigate robustness to the specified semi-synthetic perturbation distribution and corresponding evaluation protocol within questionnaire response space, and examine whether a learned nonlinear mapping can preserve the canonical reference representation under those conditions.

The study is methodological and does not presume that successful numerical reconstruction establishes clinical validity, diagnostic accuracy, autonomous psychological assessment, or general-purpose psychometric superiority. Nor does it establish universal external validity, universal robustness to untested perturbations, causal relationships among facets, or generalization to in-

struments, populations, response mechanisms, or perturbation regimes not explicitly investigated. Instead, it examines whether cross-item information learned from complete response vectors can provide a reproducible reconstruction mechanism when observations depart from their original configuration through controlled, empirically grounded perturbation processes.

2. Problem Formulation

2.1 Observation Space

Let

$$X_{\text{clean}} = (x_1, \dots, x_d), \quad x_i \in \{1, 2, 3, 4, 5\}, \quad i = 1, \dots, d, \quad (44)$$

denote the complete original item-response vector of a participant, where $d \in \{120, 300\}$ according to the IPIP-NEO instrument configuration considered in the present study. The corresponding observation space is therefore the finite discrete space

$$\mathcal{X}_d = \{1, 2, 3, 4, 5\}^d. \quad (45)$$

Thus,

$$X_{\text{clean}} \in \mathcal{X}_d, \quad d \in \{120, 300\}. \quad (46)$$

The clean response vector represents the participant's original, unperturbed observation and constitutes the source from which the canonical reference psychometric representation is computed. It also serves as the input to the corresponding structured perturbation process, which produces one or more perturbed observations associated with the same participant.

For each resulting perturbation instance, the corrupted response vector remains in the same instrument-specific observation space,

$$X_{\text{corr}} \in \mathcal{X}_d. \quad (47)$$

Accordingly, corruption does not alter the dimensionality or response domain of the questionnaire representation. Instead, it modifies the observed response configuration within the same finite discrete item-response space. The participant-specific canonical reference representation is subsequently associated with the original X_{clean} , rather than being recomputed from any particular X_{corr} .

The present empirical formulation is restricted to the IPIP-NEO-120 and IPIP-NEO-300 configurations, for which $d = 120$ and $d = 300$, respectively. Both configurations share the same thirty-dimensional facet-level reference space while differing in the dimensionality and redundancy of their observed item-response spaces.

2.2 Canonical Reference Psychometric Representation

Let

$$g : \mathcal{X}_d \rightarrow \mathbb{R}^{30} \quad (48)$$

denote the deterministic scoring transformation specified for the IPIP-NEO instrument configuration used in the present study. The resulting representation is defined as

$$Y_{\text{Gold}} = g(X_{\text{clean}}) \in \mathbb{R}^{30}, \quad (49)$$

and is referred to throughout this manuscript as the *canonical reference psychometric representation*.

The reference representation is computed exclusively from the participant's original, unperturbed response vector. In particular, no perturbed observation is used to recompute or redefine the corresponding reference target.

For a negatively keyed item, the conventional reverse-scoring transformation used in the present formulation is

$$x_i^{\text{score}} = 6 - x_i. \quad (50)$$

For a non-reversed item,

$$x_i^{\text{score}} = x_i. \quad (51)$$

Let $\mathcal{I}_j$ denote the set of item indices assigned to facet j , with

$$j \in \{1, \dots, 30\}. \quad (52)$$

The reference value for facet j is then defined as the arithmetic mean of its scored items,

$$y_j = \frac{1}{|\mathcal{I}_j|} \sum_{i \in \mathcal{I}_j} x_i^{\text{score}}. \quad (53)$$

Consequently, the complete thirty-dimensional reference representation is

$$Y_{\text{Gold}} = (y_1, \dots, y_{30}). \quad (54)$$

For IPIP-NEO-120, each facet is represented by four designated items, whereas for IPIP-NEO-300, each facet is represented by ten designated items. Thus, the two instrument configurations differ in item-space dimensionality and within-facet item redundancy, while sharing the same thirty-dimensional facet-level reference space.

The scoring transformation is deterministic and is applied to the original unperturbed response vector X_{clean} . Accordingly, for every perturbation instance associated with a participant, the corresponding reference target remains

$$Y_{\text{Gold}} = g(X_{\text{clean}}), \quad (55)$$

irrespective of the particular corrupted observation subsequently presented to a reconstruction model.

For notational purposes, the same deterministic transformation may equivalently be written as an affine mapping,

$$Y_{\text{Gold}} = AX_{\text{clean}} + \mathbf{b}, \quad (56)$$

where

$$A \in \mathbb{R}^{30 \times d}, \quad \mathbf{b} \in \mathbb{R}^{30}. \quad (57)$$

For facet j , the corresponding row of A contains the item-aggregation weights together with the sign induced by reverse-keying, while $\mathbf{b}$ contains the constants introduced by the reverse-

scoring operation. More explicitly, if r_j denotes the number of negatively keyed items assigned to facet j , then

$$A_{ji} = \begin{cases} -\frac{1}{|\mathcal{I}_j|}, & i \in \mathcal{I}_j \text{ and item } i \text{ is negatively keyed,} \\ \frac{1}{|\mathcal{I}_j|}, & i \in \mathcal{I}_j \text{ and item } i \text{ is not negatively keyed,} \\ 0, & i \notin \mathcal{I}_j, \end{cases} \quad (58)$$

and

$$b_j = \frac{6r_j}{|\mathcal{I}_j|}. \quad (59)$$

This affine form is equivalent to the item-wise scoring and facet-wise aggregation defined above. It therefore makes explicit that the canonical reference representation is a fixed, deterministic transformation of the original response vector rather than an independently observed outcome.

Accordingly, the term *canonical reference psychometric representation* denotes an operational reference target for reconstruction within the experimental framework of this study. It is not interpreted as an independently observed psychological truth, a clinical ground truth, or a direct measurement of latent mental state.

2.3 Structured Semi-Synthetic Perturbation Process

A perturbation realization is represented by a parameter

$$\phi = (c, s, \omega), \quad (60)$$

where c denotes the perturbation family, s denotes its severity or magnitude, and ω contains the realization-specific structural information, including selected items, facets, categories, collapse configurations, and, where applicable, donor assignments.

For donor-based transformations, let D denote the corresponding observed donor response material resolved from the donor assignment specified by ω . The donor material is selected from the donor registry associated with the corresponding data partition and structural transformation unit. For perturbation families that do not require donor material, D is absent and does not constitute an additional input to the reconstruction model.

The corresponding perturbation operator is

$$\mathcal{C}_{\phi,D} : \mathcal{X}_d \rightarrow \mathcal{X}_d, \quad (61)$$

and produces the observed perturbed response vector

$$X_{\text{corr}} = \mathcal{C}_{\phi,D}(X_{\text{clean}}). \quad (62)$$

The perturbation protocol is semi-synthetic and empirically grounded. For donor-based transformations, response values are drawn from observed response patterns of real incomplete participants assigned to the corresponding data partition. These participants do not provide psychometric reference targets; their observed responses serve only as donor material for the construction of controlled perturbations. For perturbation families that do not require donor material, the corresponding transformation is generated according to its specified controlled construction mechanism.

Accordingly, the term *semi-synthetic* refers to the construction mechanism of the perturbation and does not imply that all response values used in the perturbed observation are synthetically generated. Where donor material is required, observed response material is incorporated into the controlled transformation while the participant-specific reference target remains fixed.

The perturbation process therefore modifies the observed response while leaving the participant-specific canonical reference representation unchanged:

$$\boxed{X_{\text{clean}} \rightarrow Y_{\text{Gold}}} \quad \text{and} \quad \boxed{X_{\text{corr}} = \mathcal{C}_{\phi, D}(X_{\text{clean}}) \rightarrow Y_{\text{Gold}}}. \quad (63)$$

The second mapping is the reconstruction problem studied in this work. The reference target is never recomputed from X_{corr} .

2.4 Separation of Predictive Inputs and Perturbation Metadata

The predictive input available to the learned Gateway is exclusively the perturbed response vector X_{corr} . Perturbation metadata and construction variables are not supplied to the model.

Formally,

$$\text{Input}_{\text{Gateway}} = X_{\text{corr}}, \quad (64)$$

whereas

$$c, \quad s, \quad \omega, \quad D \quad (65)$$

are retained externally for dataset construction, auditing, stratified evaluation, and error analysis. Here, D denotes observed donor material only when the corresponding perturbation family requires it.

Thus, the model cannot directly condition its prediction on the known perturbation family, severity label, realization-specific construction metadata, or donor material used to construct the corrupted observation.

The exclusion of these variables from the predictive input is part of the experimental specification: the Gateway receives the same perturbed response vector available for reconstruction rather than an explicit description of how that perturbation was generated.

2.5 Functional Reconstruction

The learned Gateway defines a parametric nonlinear mapping

$$F_{\theta} : \mathcal{X}_d \rightarrow \mathbb{R}^{30}, \quad (66)$$

such that

$$\hat{Y} = F_{\theta}(X_{\text{corr}}). \quad (67)$$

The reconstruction objective is therefore to approximate the canonical reference representation from the perturbed observation:

$$\hat{Y} \approx Y_{\text{Gold}}. \quad (68)$$

More formally, given a distribution over complete response vectors, perturbation realizations, and, where applicable, donor material, the population objective is

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{X_{\text{clean}}, \phi, D} [\mathcal{L}(F_{\theta}(\mathcal{C}_{\phi, D}(X_{\text{clean}})), g(X_{\text{clean}}))], \quad (69)$$

where D is instantiated only for perturbation families whose construction requires donor material.

The learned mapping is therefore optimised with respect to the participant-specific canonical reference associated with the unperturbed response vector rather than with respect to the perturbed observation itself. Consequently, the perturbation mechanism affects the observed input available to the model but does not redefine the target against which reconstruction is evaluated.

2.6 Conventional Deterministic Comparator

The conventional deterministic comparator applies the same established, instrument-specific scoring function used to construct the canonical reference representation directly to the perturbed response vector:

$$Y_{\text{Traditional}} = g(X_{\text{corr}}). \quad (70)$$

The comparator is deterministic and does not involve learned parameters, additional predictive inputs, or perturbation metadata. Its output is determined exclusively by the response vector supplied to the scoring function.

For clean observations, the evaluation input is the original unperturbed response vector, such that

$$X_{\text{corr}} = X_{\text{clean}} \implies Y_{\text{Traditional}} = Y_{\text{Gold}}, \quad (71)$$

by construction, because

$$Y_{\text{Gold}} = g(X_{\text{clean}}). \quad (72)$$

For perturbed observations, the conventional comparator therefore produces

$$Y_{\text{Traditional}} = g(X_{\text{corr}}), \quad (73)$$

whereas the learned Gateway produces

$$\hat{Y} = F_{\theta}(X_{\text{corr}}). \quad (74)$$

Both outputs are evaluated against the same participant-specific canonical reference,

$$Y_{\text{Gold}} = g(X_{\text{clean}}). \quad (75)$$

Thus, the head-to-head comparison is performed under matched observation conditions: both methods receive the same perturbed response vector X_{corr} , while neither method is provided with the perturbation family, severity, realization-specific construction metadata, or donor material as predictive input. The reference representation remains fixed by the participant's original unperturbed response pattern.

Accordingly, the comparison isolates the reconstruction behaviour of the learned Gateway relative to conventional deterministic scoring under the same observed corruption and against the same immutable participant-specific reference target.

3. MLP Gateway Framework

3.1 Composed Nonlinear Mapping

The Gateway is formulated as a finite-dimensional parametric nonlinear mapping from the observed questionnaire response space to the canonical reference psychometric representation space. Let

$$\mathcal{N} : \mathcal{X}_d \rightarrow [-1, 1]^d \quad (76)$$

denote the element-wise input normalization

$$\mathcal{N}(x_i) = \frac{x_i - 3}{2}, \quad (77)$$

which maps the original five-point response scale $\{1, 2, 3, 4, 5\}$ to the interval $[-1, 1]$.

The Gateway can then be written as the composition

$$F_\theta = \mathcal{G}_\theta \circ \mathcal{N}, \quad (78)$$

where $\mathcal{G}_\theta$ denotes the learned multilayer transformation acting on the normalized response space. More generally, the learned transformation may be expressed as the finite composition

$$\mathcal{G}_\theta = \mathcal{T}_L \circ \mathcal{T}_{L-1} \circ \cdots \circ \mathcal{T}_2 \circ \mathcal{T}_1. \quad (79)$$

For a hidden layer ℓ , the pre-normalization affine state is

$$\mathbf{z}_\ell = W_\ell \mathbf{h}_{\ell-1} + \mathbf{b}_\ell, \quad (80)$$

followed in the final IPIP-NEO-120 S2 configuration by Layer Normalization, the GELU nonlinearity, and dropout during training:

$$\mathbf{h}_\ell = \text{Dropout}_\ell (\text{GELU} (\text{LayerNorm}_\ell (\mathbf{z}_\ell))). \quad (81)$$

The dropout operation is active during training with rate 0.10 and is disabled in evaluation mode. Consequently, the reported inference mapping is deterministic for a fixed input and a fixed frozen parameter state.

For the final IPIP-NEO-120 S2 configuration, the learned transformation is instantiated as

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (82)$$

with three hidden transformations. The normalized input is

$$\mathbf{h}_0 = \mathcal{N}(X_{\text{corr}}) \in [-1, 1]^{120}. \quad (83)$$

The first hidden transformation is

$$\mathbf{h}_{192} = \text{Dropout} (\text{GELU} (\text{LayerNorm}_{192} (W_1 \mathbf{h}_0 + \mathbf{b}_1))), \quad (84)$$

where

$$W_1 \in \mathbb{R}^{192 \times 120}, \quad \mathbf{b}_1 \in \mathbb{R}^{192}. \quad (85)$$

The second hidden transformation is

$$\mathbf{h}_{128}^{\text{pre}} = \text{Dropout}(\text{GELU}(\text{LayerNorm}_{128}(W_2 \mathbf{h}_{192} + \mathbf{b}_2))), \quad (86)$$

where

$$W_2 \in \mathbb{R}^{128 \times 192}, \quad \mathbf{b}_2 \in \mathbb{R}^{128}. \quad (87)$$

The third hidden transformation is

$$\mathbf{h}_{128}^{\text{wide}} = \text{Dropout}(\text{GELU}(\text{LayerNorm}_{128}(W_3 \mathbf{h}_{128}^{\text{pre}} + \mathbf{b}_3))), \quad (88)$$

where

$$W_3 \in \mathbb{R}^{128 \times 128}, \quad \mathbf{b}_3 \in \mathbb{R}^{128}. \quad (89)$$

The final output layer is purely affine:

$$\hat{Y} = W_{\text{out}} \mathbf{h}_{128}^{\text{wide}} + \mathbf{b}_{\text{out}}, \quad (90)$$

with

$$W_{\text{out}} \in \mathbb{R}^{30 \times 128}, \quad \mathbf{b}_{\text{out}} \in \mathbb{R}^{30}. \quad (91)$$

No output activation, clipping, or additional deterministic post-processing is imposed in the reported inference pipeline.

The dimensional structure of the complete forward mapping is therefore

$$\mathbb{R}^{120} \xrightarrow{\mathcal{N}} \mathbb{R}^{120} \xrightarrow{\mathcal{T}_1} \mathbb{R}^{192} \xrightarrow{\mathcal{T}_2} \mathbb{R}^{128} \xrightarrow{\mathcal{T}_3} \mathbb{R}^{128} \xrightarrow{W_{\text{out}}, \mathbf{b}_{\text{out}}} \mathbb{R}^{30}. \quad (92)$$

The 69,214 trainable parameters of the final S2 configuration can be decomposed explicitly into affine-layer parameters, Layer Normalization parameters, and output-layer parameters:

$$N_{\theta} = (120 \times 192 + 192) + (2 \times 192) + (192 \times 128 + 128) + (2 \times 128) + (128 \times 128 + 128) + (2 \times 128) + (128 \times 30 + 30), \quad (93)$$

giving

$$N_{\theta} = 69,214. \quad (94)$$

Thus, the reported parameter count includes the trainable scale and bias parameters associated with each of the three Layer Normalization operations, in addition to the weights and biases of the affine transformations.

For the final IPIP-NEO-120 S2 configuration, the selected operating point corresponds to seed 20260933 and epoch 6. The architecture was retained following controlled architecture screening and subsequent three-seed confirmation under predefined robustness guardrails. The confirmation procedure evaluated seeds 20260931, 20260932, and 20260933, with all three seeds satisfying the recorded guardrails and reducing corrupted-validation MAE relative to the V133 control configuration.

The preceding V133 configuration was

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30, \quad (95)$$

whereas the final S2 configuration was

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30. \quad (96)$$

The principal architectural change was therefore the replacement of the 64-dimensional third hidden representation by a 128-dimensional third hidden representation. The final hidden representation consequently remains 128-dimensional immediately before projection into the thirty-dimensional canonical reference space, rather than passing through the 64-dimensional bottleneck present in V133.

This construction represents the Gateway as a finite composition of parameterized affine maps, normalization operations, nonlinear activations, and regularization applied during training. The resulting learned mapping is explicitly nonlinear because of the GELU transformations, while remaining a finite-dimensional mapping from the instrument-specific item-response space to the thirty-dimensional canonical reference representation space.

3.2 IPIP-NEO-300 Gateway Development and P83.25 Selection

The IPIP-NEO-300 Gateway was developed through a staged experimental programme in which representational architecture and optimisation objective were treated as distinct experimental factors. The development process therefore did not consist of a single undifferentiated sequence of architecture changes. Instead, architecture screening, objective screening, joint production evaluation, P83.25 architectural refinement, and full production training were treated as separate methodological stages.

The IPIP-NEO-300 observation space is

$$\mathcal{X}_{300} = \{1, 2, 3, 4, 5\}^{300}, \quad (97)$$

and the Gateway maps the complete 300-dimensional response vector to the thirty-dimensional canonical reference representation.

The initial compact Gateway used the mapping

$$300 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30. \quad (98)$$

This compact architecture was subsequently retained as the fixed reference architecture for the independent objective-screening experiment. Nine candidate training objectives were evaluated under this fixed representational geometry. The objective selected at this stage was

$$\mathcal{L}_H = 0.90 \mathcal{L}_{\text{MSE}} + 0.10 (1 - \rho_{\text{Pearson}}), \quad (99)$$

with corrupted-validation performance reported as RMSE = 0.278249, MAE = 0.213398, Pearson correlation = 0.767012, and Spearman correlation = 0.739026. The objective-screening stage therefore isolated the optimisation criterion independently of the later structured-output architectural development.

In parallel, a separate architecture-screening programme evaluated seven candidate architectures on a 10,000-subject screening subset using corrupted validation RMSE as the principal screening criterion. The independent TEST population remained reserved and was not used for either architecture or objective screening. The architecture-screening procedure identified a structured-output configuration, denoted G , with a shared representation followed by five parallel output heads of six facets each.

The initial structured-output architecture was

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}, \quad (100)$$

where

$$5 \times 6 = 30. \quad (101)$$

Thus, the output structure preserves the thirty-dimensional target space while introducing an explicit partition of the output coordinates into five six-facet groups. The shared trunk produces a common latent representation, after which each output head generates six facet values.

The architecture and objective screening stages were deliberately kept separate. Architecture screening therefore investigated representational geometry, whereas objective screening investigated the mathematical form of the optimisation criterion under a fixed compact architecture. This separation was designed to prevent the two interventions from being treated as a single undifferentiated model change.

The two screening factors were subsequently combined in the intermediate $G + H$ production configuration,

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}, \quad (102)$$

with 121,054 trainable parameters and objective

$$\mathcal{L}_{G+H} = 0.90 \mathcal{L}_{\text{MSE}} + 0.10 (1 - \rho_{\text{Pearson}}). \quad (103)$$

The best corrupted-validation RMSE of this intermediate production stage was 0.188995 at epoch 9. The corresponding validation profile exhibited a metric-specific trade-off, with lower RMSE but higher MAE than the conventional comparator in that intermediate benchmark, alongside improved correlation measures. The $G + H$ configuration was consequently retained as an intermediate development stage rather than as the final production model.

This result motivated a separate optimisation branch in which substantially greater weight was assigned to profile-level Pearson correspondence. The resulting P83.25 objective was defined as

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - \rho_{\text{Pearson}}). \quad (104)$$

For each observation n , the Pearson component is computed across its thirty predicted and reference facet values. Let

$$\hat{\bar{y}}_n = \frac{1}{30} \sum_{j=1}^{30} \hat{y}_{nj}, \quad \bar{y}_n = \frac{1}{30} \sum_{j=1}^{30} y_{nj}. \quad (105)$$

The profile-level correlation is then

$$\rho_n = \frac{\sum_{j=1}^{30} (\hat{y}_{nj} - \hat{\bar{y}}_n) (y_{nj} - \bar{y}_n)}{\sqrt{\sum_{j=1}^{30} (\hat{y}_{nj} - \hat{\bar{y}}_n)^2 + \varepsilon} \sqrt{\sum_{j=1}^{30} (y_{nj} - \bar{y}_n)^2 + \varepsilon}}, \quad \varepsilon = 10^{-8}. \quad (106)$$

The profile-correlation loss is therefore

$$\mathcal{L}_{\text{Pearson}} = 1 - \frac{1}{N} \sum_{n=1}^N \rho_n, \quad (107)$$

and the complete P83.25 optimisation objective is

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 \mathcal{L}_{\text{Pearson}}. \quad (108)$$

Thus, the Pearson component acts on the internal thirty-dimensional profile of each observation rather than representing a single correlation computed across the pooled dataset.

The P83.25 branch was not evaluated merely by replacing the objective in the previous production checkpoint. Instead, a new controlled architecture screening stage was performed on a 10,000-subject subset while TEST remained untouched. The search evaluated multiple hidden-dimensional configurations under the fixed P83.25 objective, including the structured control,

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}, \quad (109)$$

and the principal wider candidate,

$$300 \rightarrow 320 \rightarrow 192 \rightarrow 128 \rightarrow \{6, 6, 6, 6, 6\}. \quad (110)$$

The subsequent refinement produced the final screened geometry

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (111)$$

with 220,782 trainable parameters. On the internal refinement benchmark, this configuration yielded $\text{RMSE} = 0.220199$, $\text{MAE} = 0.147551$, $\text{Pearson correlation} = 0.861929$, and $\text{Spearman correlation} = 0.842994$. Relative to the principal WIDE configuration, the corresponding changes were $\Delta\text{RMSE} = -0.003109$, $\Delta\text{MAE} = -0.004251$, $\Delta\text{Pearson} = +0.002119$, and $\Delta\text{Spearman} = +0.002648$. This result was used for architectural refinement and was not treated as the final independent generalisation result.

The final P83.25 production architecture consists of a shared nonlinear trunk followed by five parallel linear output heads. For an input $X_{\text{corr}} \in \mathcal{X}_{300}$, the model first applies the same fixed response normalization,

$$\mathbf{h}_0 = \mathcal{N}(X_{\text{corr}}) = \frac{X_{\text{corr}} - 3}{2}, \quad (112)$$

so that

$$\mathbf{h}_0 \in [-1, 1]^{300}. \quad (113)$$

The first hidden transformation is

$$\mathbf{h}_{344} = \text{Dropout}(\text{GELU}(\text{LayerNorm}_{344}(W_1 \mathbf{h}_0 + \mathbf{b}_1))), \quad (114)$$

where

$$W_1 \in \mathbb{R}^{344 \times 300}, \quad \mathbf{b}_1 \in \mathbb{R}^{344}. \quad (115)$$

The second hidden transformation is

$$\mathbf{h}_{208} = \text{Dropout}(\text{GELU}(\text{LayerNorm}_{208}(W_2 \mathbf{h}_{344} + \mathbf{b}_2))), \quad (116)$$

where

$$W_2 \in \mathbb{R}^{208 \times 344}, \quad \mathbf{b}_2 \in \mathbb{R}^{208}. \quad (117)$$

The third hidden transformation is

$$\mathbf{h}_{184} = \text{Dropout}(\text{GELU}(\text{LayerNorm}_{184}(W_3 \mathbf{h}_{208} + \mathbf{b}_3))), \quad (118)$$

where

$$W_3 \in \mathbb{R}^{184 \times 208}, \quad \mathbf{b}_3 \in \mathbb{R}^{184}. \quad (119)$$

The final shared representation is therefore

$$\mathbf{h}_{184} \in \mathbb{R}^{184}. \quad (120)$$

Five independent linear output heads then map this representation into five six-dimensional facet groups:

$$\hat{Y}^{(k)} = H_k(\mathbf{h}_{184}) = W_k^{\text{head}} \mathbf{h}_{184} + \mathbf{b}_k^{\text{head}} \in \mathbb{R}^6, \quad k = 1, \dots, 5, \quad (121)$$

with

$$W_k^{\text{head}} \in \mathbb{R}^{6 \times 184}, \quad \mathbf{b}_k^{\text{head}} \in \mathbb{R}^6. \quad (122)$$

The complete thirty-dimensional prediction is obtained by concatenation:

$$\hat{Y} = \text{Concat}(\hat{Y}^{(1)}, \hat{Y}^{(2)}, \hat{Y}^{(3)}, \hat{Y}^{(4)}, \hat{Y}^{(5)}) \in \mathbb{R}^{30}. \quad (123)$$

No output activation or output clipping is applied. The model therefore produces the thirty reconstructed facet values directly from the five linear output heads.

The complete dimensional sequence is consequently

$$\mathbb{R}^{300} \rightarrow \mathbb{R}^{344} \rightarrow \mathbb{R}^{208} \rightarrow \mathbb{R}^{184} \rightarrow \mathbb{R}^6 \times 5 \rightarrow \mathbb{R}^{30}. \quad (124)$$

The 220,782 trainable parameters can be accounted for exactly as

$$\begin{aligned} N_\theta &= (300 \times 344 + 344) + (2 \times 344) \\ &\quad + (344 \times 208 + 208) + (2 \times 208) \\ &\quad + (208 \times 184 + 184) + (2 \times 184) \\ &\quad + 5(184 \times 6 + 6), \end{aligned} \quad (125)$$

which gives

$$N_\theta = 220,782. \quad (126)$$

The count therefore includes the weights and biases of the three affine transformations, the trainable scale and bias parameters of the three Layer Normalization operations, and the parameters of the five independent output heads.

The final production model was retrained from the full TRAIN population rather than using the 10,000-subject architecture-screening fit as the final model. The production training set contained 101,772 subjects and 14,553,396 examples per epoch, corresponding to exactly 143 views per subject:

$$101,772 \times 143 = 14,553,396. \quad (127)$$

The 143-view exposure schedule consists of three clean views and twenty corrupted views for each of seven perturbation families,

$$3 + 7 \times 20 = 143. \quad (128)$$

Training used a subject-isolated SIES schedule. For each epoch, all TRAIN subjects were processed in 143 rounds. Within each round, the subjects were randomly permuted, while each subject was assigned one view according to a fixed subject-specific rank and a cyclic offset indexed by the round. The view assignment therefore traverses the 143 available views over the course of an epoch. Duplicate subject identities within a batch were explicitly prohibited.

The production optimisation used AdamW with batch size 8,192, learning rate 0.002, weight decay 0.0001, dropout rate 0.10, gradient clipping norm 1.0, and BF16 mixed precision. The production run was initialised with global seed 20260920, while the SIES scheduling process used seed 20261697. The configured training horizon was a minimum of five and a maximum of 120 epochs, with early-stopping patience of three epochs and a minimum corrupted-validation RMSE improvement of 10^{-4} .

For checkpoint selection, the sole improvement criterion was corrupted validation RMSE. Clean validation RMSE, MAE, Pearson correlation, and Spearman correlation were recorded but were not used as the checkpoint selection criterion. After each epoch, the current checkpoint was saved as the last checkpoint, while the best checkpoint was updated only when the corrupted-validation RMSE improved by more than the predefined 10^{-4} threshold.

The production validation benchmark contained 21,808 clean observations and 338,444 corrupted observations. The selected checkpoint was epoch 14, with

$$\text{RMSE}_{\text{val,corr}} = 0.18483594655007943. \quad (129)$$

Training continued through epoch 17. At epoch 17, the raw corrupted-validation RMSE was approximately 0.184769, but the improvement relative to the epoch-14 checkpoint was only approximately

$$0.18483594655007943 - 0.184769 \approx 6.7 \times 10^{-5}, \quad (130)$$

which was below the predefined 10^{-4} minimum improvement threshold. Consequently, epoch 17 did not replace epoch 14 as the best checkpoint, and early stopping was triggered after the third consecutive epoch without a valid improvement.

The final P83.25 production checkpoint is therefore the epoch-14 model rather than the last epoch executed. The corresponding architecture is

$$\boxed{300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}} \quad (131)$$

with 220,782 trainable parameters and objective

$$\boxed{\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - \rho_{\text{Pearson}})}. \quad (132)$$

The epoch-14 artifact was fixed before the independent TEST evaluation. The TEST population was explicitly recorded as reserved and untouched during this production training procedure. No TEST observation was used for training, architecture selection, objective selection, or checkpoint selection.

Unlike the IPIP-NEO-120 S2 procedure, the present P83.25 production block documents a single final production seed rather than a subsequent multi-seed confirmation stage. Accordingly, the reported P83.25 production result is associated with the specified training seed and selection protocol, without attributing additional multi-seed confirmation to this stage.

3.3 Instrument-Specific Gateway Configurations

The study uses instrument-specific Gateway configurations. The IPIP-NEO-120 and IPIP-NEO-300 experiments map their respective complete item-response vectors to the same thirty-dimensional canonical reference representation, while preserving the different observational dimensionalities of the questionnaires.

The IPIP-NEO-120 configuration operates on 120 item responses, whereas the IPIP-NEO-300 configuration operates on 300 item responses. The shared 30-dimensional output space makes the two experiments directly comparable at the level of the reconstructed operational psychometric representation, while the instrument-specific architectures preserve the dimensional structure of each observation space.

The final IPIP-NEO-120 Gateway is the S2 configuration

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (133)$$

with 69,214 trainable parameters, seed 20260933, and epoch 6 as the frozen operating point following architecture screening and independent multi-seed confirmation.

The IPIP-NEO-300 production configuration is the P83.25 architecture

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (134)$$

where the five six-dimensional heads together produce the same 30-dimensional output space. This configuration contains 220,782 trainable parameters and was fixed at epoch 14 under the production validation checkpoint protocol. The IPIP-NEO-300 production run used a single global training seed, 20260920; unlike the IPIP-NEO-120 S2 pathway, it was not subjected to the same three-seed confirmatory stage.

In both instruments, the TEST population was kept outside architecture screening, objective screening, checkpoint selection, and retraining. The shared output dimensionality therefore defines a common reconstruction space, whereas the input dimensionality and internal architecture remain instrument-specific.

3.4 IPIP-NEO-120 Gateway

The initial IPIP-NEO-120 Gateway adopted the compact mapping

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30. \quad (135)$$

The three hidden transformations used Layer Normalization, GELU activations, and dropout with probability $p = 0.10$. The final layer mapped the 64-dimensional hidden representation directly to the 30-dimensional canonical facet representation. The initial configuration contained 58,910 trainable parameters.

For this architecture, the trainable parameter count is obtained from four affine transformations and three Layer Normalization stages:

$$\begin{aligned}
P_{V_{133}} &= (120 \times 192 + 192) + 2(192) \\
&\quad + (192 \times 128 + 128) + 2(128) \\
&\quad + (128 \times 64 + 64) + 2(64) \\
&\quad + (64 \times 30 + 30) \\
&= 58,910.
\end{aligned} \tag{136}$$

Here, each $2d$ Layer Normalization contribution corresponds to one trainable scale parameter and one trainable bias parameter for a d -dimensional normalized representation. Dropout and GELU introduce no additional trainable parameters.

This configuration constituted the starting point of the IPIP-120 robustness program. Model development subsequently proceeded through an explicit sequence of representation auditing, constrained architecture exploration, confirmatory retraining, and checkpoint freezing. The architecture was therefore not modified solely in response to the aggregate validation score of a single training run.

3.4.1 Dimensional and Representation Audit

The initial Gateway was subjected to a dedicated dimensional audit to determine whether the internal representation exhibited a restrictive compression point. Particular attention was given to the transition

$$128 \rightarrow 64, \tag{137}$$

because the 64-dimensional representation constituted the narrowest hidden state preceding the final 30-dimensional output.

The audit examined effective representational dimensionality, spectral structure, corruption-family discriminability, and downstream gradient sensitivity. In the initial Gateway, the H64 representation exhibited an entropy-based effective rank of approximately 20.18, while approximately 24 dimensions were sufficient to account for 95% of the observed spectral energy. These quantities characterize the empirical numerical structure of the learned representation and should not be interpreted as direct estimates of psychological dimensionality.

For a representation matrix with non-negative spectral contributions λ_i , the entropy effective rank is defined as

$$R_{\text{eff}} = \exp \left(- \sum_{i:p_i>0} p_i \log p_i \right), \tag{138}$$

with

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \tag{139}$$

The corresponding empirical spectral-energy dimension at threshold $\tau = 0.95$ is the smallest integer k satisfying

$$k_{0.95} = \min \left\{ k : \frac{\sum_{i=1}^k \lambda_i^2}{\sum_i \lambda_i^2} \geq 0.95 \right\}. \tag{140}$$

These quantities measure different aspects of representation concentration: the effective rank summarizes the entropy of the normalized spectral mass, whereas $k_{0.95}$ describes the minimum number of ordered components required to account for 95% of the measured squared spectral energy. They are therefore not interchangeable measures.

A complementary linear-probe analysis measured the ability of the hidden representation to discriminate corruption families. The balanced accuracy of a probe operating on the H64 representation of the initial Gateway was approximately 0.227. The corresponding wider representation examined during the S2 analysis reached approximately 0.266, providing an additional indication that the wider state retained more information relevant to the corruption structure being evaluated.

The dimensional audit was interpreted as a diagnostic rather than as proof of a causal bottleneck. Effective rank, spectral concentration, probe accuracy, and gradient sensitivity characterize different properties of the learned numerical representation and were therefore considered jointly rather than treated as interchangeable evidence. In particular, the observed differences do not establish that the 64-dimensional state is itself the causal source of the downstream error difference.

3.4.2 Alternative Architecture Tests

To distinguish the effect of preserving information from the effect of merely increasing the nominal number of neurons after a compression step, an explicit expansion architecture was evaluated:

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 128 \rightarrow 30. \quad (141)$$

This configuration retained the original 128-to-64 compression and expanded the representation only after the bottleneck. Its performance did not recover the behavior obtained by maintaining the 128-dimensional representation throughout the third hidden transformation.

The result was treated as evidence consistent with an information-preservation hypothesis: once the intermediate representation had been compressed from 128 to 64 dimensions, a subsequent expansion layer did not reliably recover the information discarded by the earlier transformation. This interpretation is deliberately bounded to the evaluated numerical representation and does not imply a general theorem concerning information loss in all neural architectures.

The architecture investigation therefore shifted from post-bottleneck expansion toward preservation of the wider intermediate representation.

3.4.3 Internal Architecture Tuning

A constrained architecture search was subsequently performed around the initial Gateway while preserving the 120-dimensional input, 30-dimensional output, normalization scheme, nonlinearities, and general training protocol. The principal alternatives included wider third hidden representations and a post-bottleneck expansion configuration.

The selected configuration was

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30. \quad (142)$$

This configuration was designated S2 and contains 69,214 trainable parameters. The three hidden transformations use Layer Normalization, GELU activations, and dropout with probability $p = 0.10$, while the final 128-dimensional representation is mapped directly to the 30-dimensional canonical facet outputs.

The exact S2 trainable parameter count is

$$\begin{aligned}
P_{S2} &= (120 \times 192 + 192) + 2(192) \\
&\quad + (192 \times 128 + 128) + 2(128) \\
&\quad + (128 \times 128 + 128) + 2(128) \\
&\quad + (128 \times 30 + 30) \\
&= 69,214.
\end{aligned} \tag{143}$$

During constrained screening, S2 achieved corrupted-validation $\text{RMSE} = 0.348362$ and $\text{MAE} = 0.196240$ at its selected screening epoch. These results were evaluated jointly because RMSE and MAE encode different aspects of the error distribution. RMSE gives greater weight to large deviations through the squared-error term, whereas MAE weights deviations linearly. Consequently, a candidate that reduces RMSE while substantially increasing MAE may improve the upper tail of the error distribution while worsening the typical absolute deviation.

For this reason, architecture selection was not defined as a search for the numerically smallest RMSE irrespective of all other criteria. RMSE remained the primary robustness metric, while candidate configurations were evaluated under predefined guardrails that also considered MAE and profile-level correspondence. The objective was therefore to identify a configuration that preserved the primary robustness behaviour without accepting an uncontrolled deterioration in the absolute-error distribution.

This distinction is important for interpreting the final architecture. The V133 control achieved corrupted-validation $\text{RMSE} = 0.348829$ and $\text{MAE} = 0.201426$, whereas the final S2 checkpoint achieved $\text{RMSE} = 0.349023$ and $\text{MAE} = 0.196198$. Thus, the final production checkpoint does not claim the lowest RMSE observed among all evaluated checkpoints. Instead, it preserves RMSE within the predefined confirmation regime while reducing MAE and maintaining closely matched profile-level correspondence. The architecture was consequently retained on the basis of a controlled metric trade-off rather than single-metric minimization.

The distinction between the screening operating point and the final operating point is maintained throughout this section. The screening result of $\text{RMSE} = 0.348362$ and the later seed-specific confirmation result of $\text{RMSE} = 0.349023$ are not treated as interchangeable measurements. The former characterizes the constrained architecture-screening stage, whereas the latter characterizes the frozen seed-20260933 confirmation checkpoint.

The architecture-screening stage was additionally examined through three descriptive diagnostics. These diagnostics characterize the geometry of the candidate set, the metric-specific changes relative to the designated reference, and the rank association between objective components and observed validation outcomes. They are reported as screening diagnostics only and were not introduced as additional post-hoc selection criteria.

Architectural Topology and Parameter Geometry

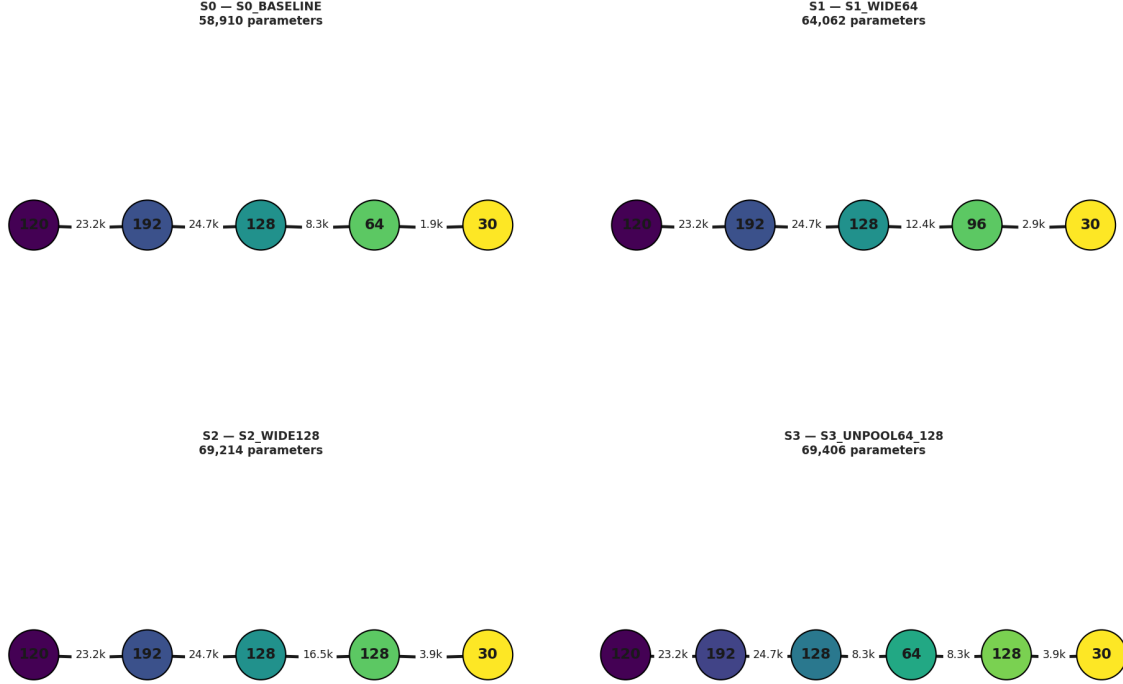


Figure 1: Architectural topology and parameter geometry of the principal IPIP-NEO-120 candidates evaluated during constrained architecture screening. S0 corresponds to the V133 control architecture $120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30$; S1 widens the terminal hidden representation to 96 units; S2 preserves a 128-dimensional terminal hidden representation; and S3 introduces a post-bottleneck expansion $120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 128 \rightarrow 30$. Parameter counts are shown for each configuration.

Let an architecture candidate c be represented by its ordered layer-width vector

$$\mathcal{A}_c = [d_0, d_1, \dots, d_L], \quad (144)$$

where $d_0 = 120$ is the input dimension and $d_L = 30$ is the canonical output dimension.

For a feed-forward candidate containing affine transformations and explicitly parameterized normalization layers, the total number of trainable parameters is

$$P_c = \sum_{\ell=1}^L (d_{\ell-1}d_\ell + d_\ell) + P_{\text{norm},c} + P_{\text{other},c}, \quad (145)$$

where $P_{\text{norm},c}$ denotes the trainable parameters introduced by normalization layers and $P_{\text{other},c}$ denotes any additional explicitly parameterized components of the evaluated implementation.

For the architectures considered here, GELU activations and dropout do not contribute trainable parameters. Layer Normalization over a d -dimensional state contributes $2d$ trainable parameters.

The candidate set used for constrained screening is therefore defined over architectures

satisfying the fixed input, output, normalization, activation, and training-protocol constraints.

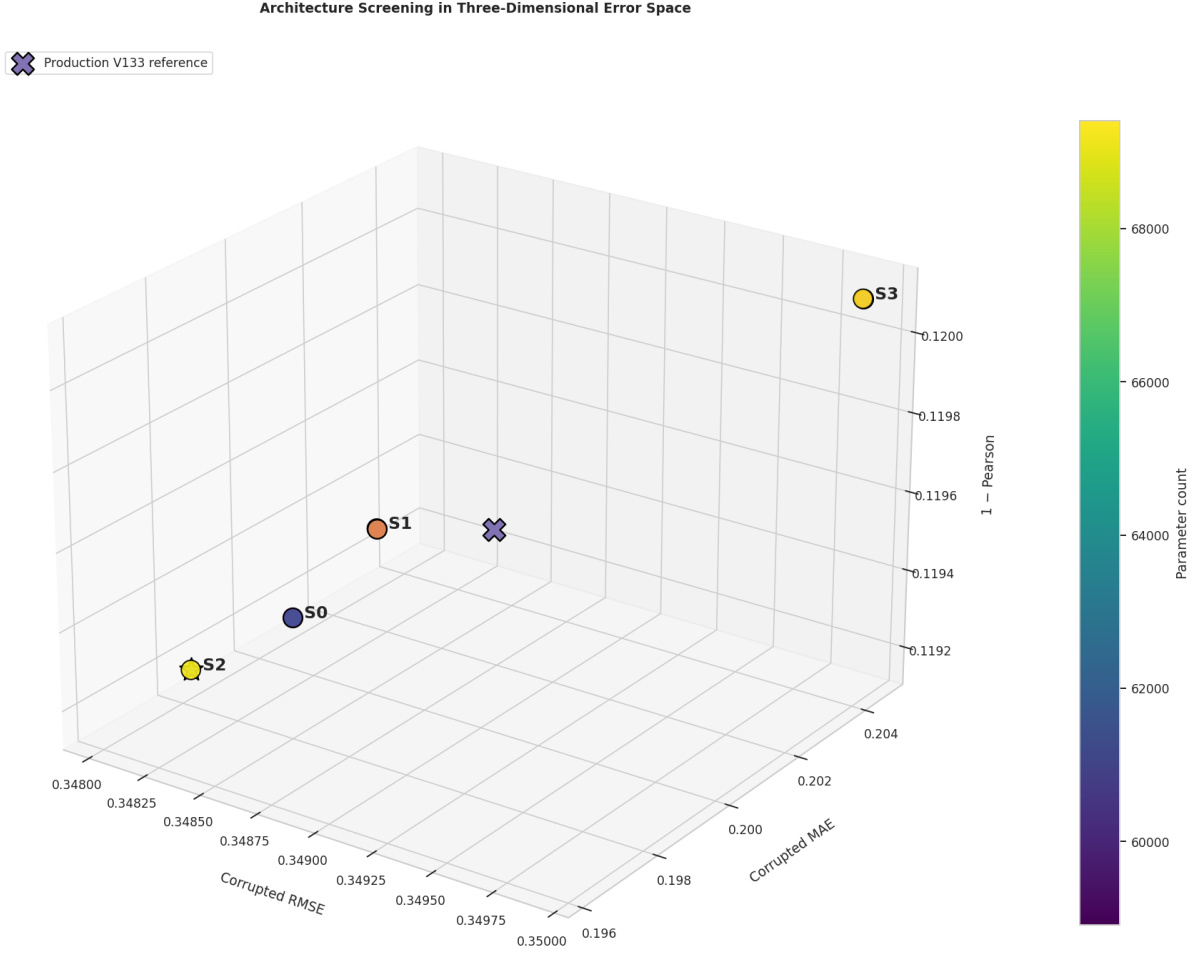


Figure 2: Architecture screening in three-dimensional corrupted-validation error space for the principal IPIP-NEO-120 candidates. Candidate architectures are represented using corrupted RMSE, corrupted MAE, and $1 - \text{Pearson}$, with marker color indicating parameter count. S0 represents the V133 production reference. The visualization is descriptive of the principal candidate architectures and does not itself define the final selection rule.

For candidate architecture c , the validation-error vector is defined as

$$\mathbf{e}_c = [RMSE_c, MAE_c, 1 - r_c], \quad (146)$$

where r_c denotes the corresponding profile-level Pearson correlation.

The V133 control is denoted by

$$\mathbf{e}_{V133} = [RMSE_{V133}, MAE_{V133}, 1 - r_{V133}]. \quad (147)$$

For an architecture candidate c , its displacement relative to V133 is

$$\Delta \mathbf{e}_c = \mathbf{e}_c - \mathbf{e}_{V133}. \quad (148)$$

This representation preserves the three metric dimensions independently and does not collapse them into a single composite performance score.

Performance Geometry and Non-Dominated Set

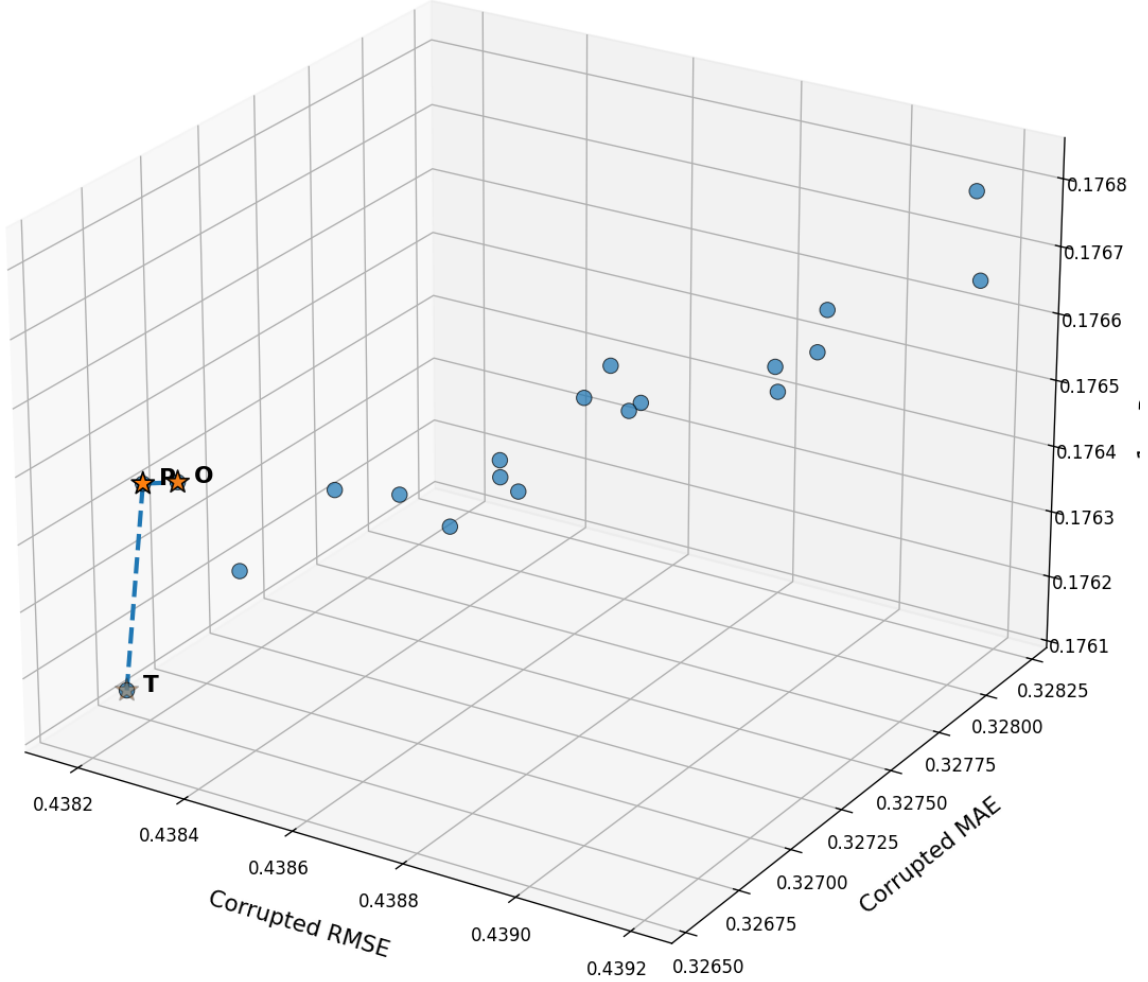


Figure 3: Performance geometry of the evaluated IPIP-NEO-120 Gateway candidates during constrained architecture screening. Candidate configurations are represented in corrupted-validation metric space using RMSE, MAE, and the corresponding profile-correlation dimension. The displayed non-dominated structure illustrates the multi-objective nature of the screening landscape. The figure is descriptive of the evaluated candidate set and does not by itself define the final operating point.

For candidate c evaluated on the screening set, let

$$\mathbf{m}_c = [RMSE_c, MAE_c, r_c, \rho_c] \quad (149)$$

denote its validation-performance vector. Here, r_c denotes Pearson correspondence and ρ_c denotes Spearman correspondence.

The screening representation in Fig. 3 therefore treats architecture evaluation as a multi-dimensional performance problem rather than as an optimization of RMSE alone. No claim of universal dominance is inferred from the displayed geometry.

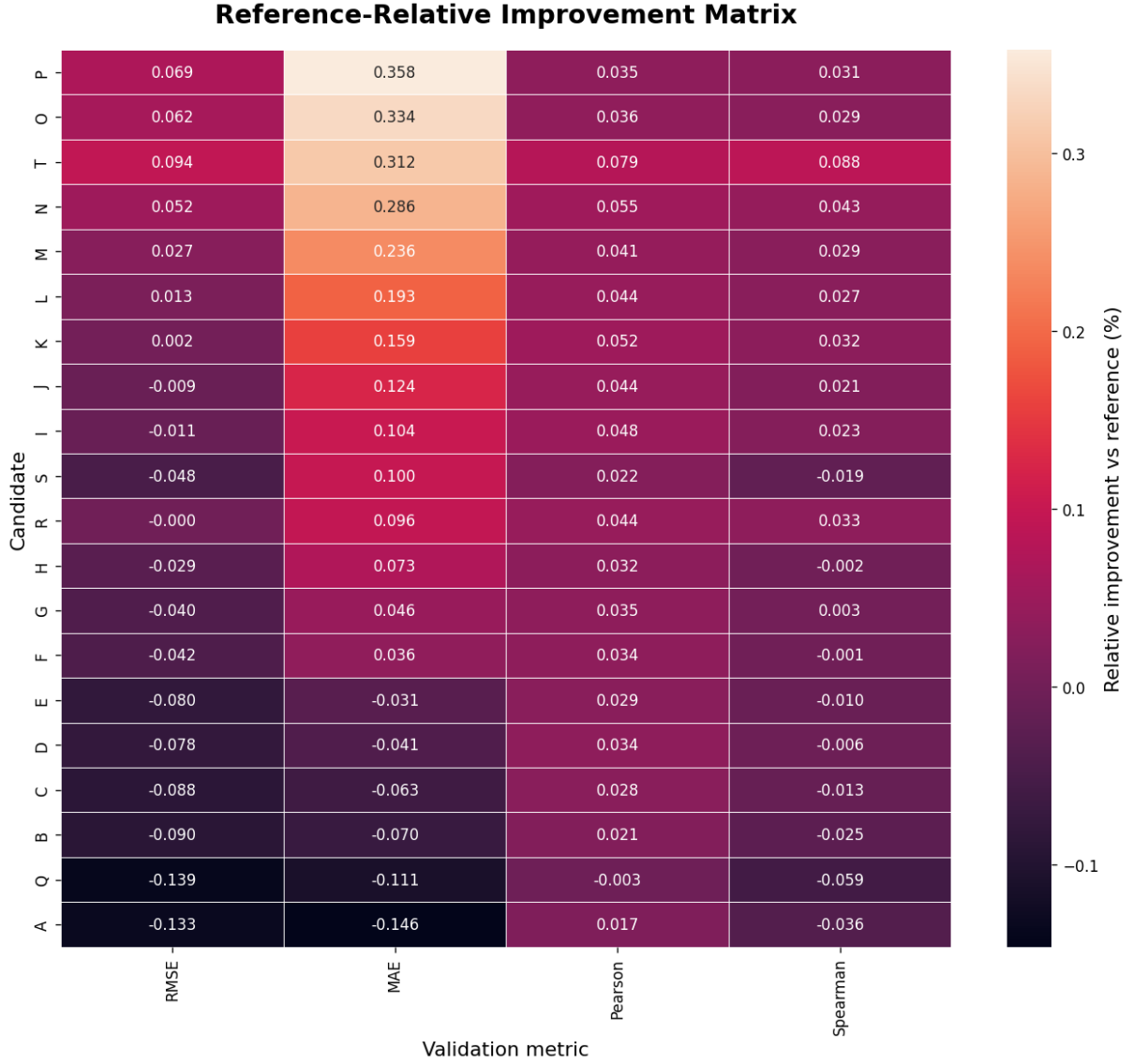


Figure 4: Reference-relative improvement matrix for the evaluated IPIP-NEO-120 Gateway candidates. Each cell reports the relative change of a candidate with respect to the designated screening reference for corrupted RMSE, MAE, Pearson correlation, and Spearman correlation. The matrix preserves metric-specific directionality and is used as a descriptive screening diagnostic rather than as a composite ranking criterion.

For candidate c and metric M , the relative change with respect to the designated reference configuration r is

$$\Delta_{\text{rel}}^{(c,r)}(M) = \frac{M_c - M_r}{|M_r|}. \quad (150)$$

For RMSE and MAE, smaller values represent smaller numerical error; for Pearson and Spearman correlation, larger values represent greater agreement with the canonical reference. Consequently, the signs in Fig. 4 are interpreted separately for each metric and are not collapsed into a single scalar score.

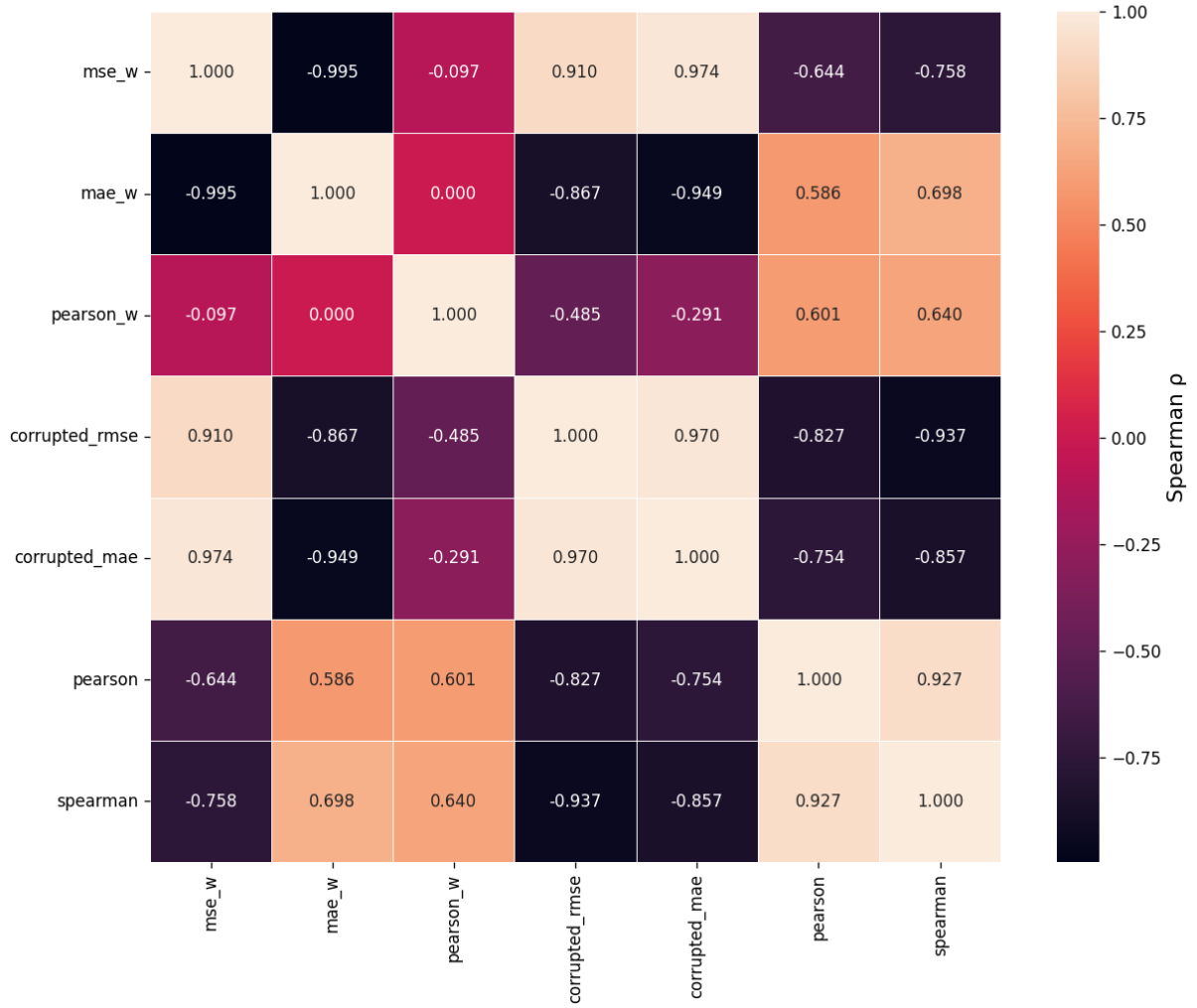
Rank-Correlation Structure Between Objective Weights and Outcomes

Figure 5: Spearman rank-correlation structure between objective-weight variables and observed corrupted-validation outcomes across the architecture-screening candidates. The matrix characterizes monotonic co-variation within the screening set. These associations are descriptive and are not interpreted as causal effects of objective weighting on model performance.

For two screening variables X and Y , the Spearman rank correlation is

$$\rho_s(X, Y) = \text{corr}(\text{rank}(X), \text{rank}(Y)). \quad (151)$$

Accordingly, Fig. 5 is used only to characterize the structure of the constrained screening set. In particular, strong correlation between an objective component and a validation metric does not establish that changing that objective component would causally produce the observed metric change.

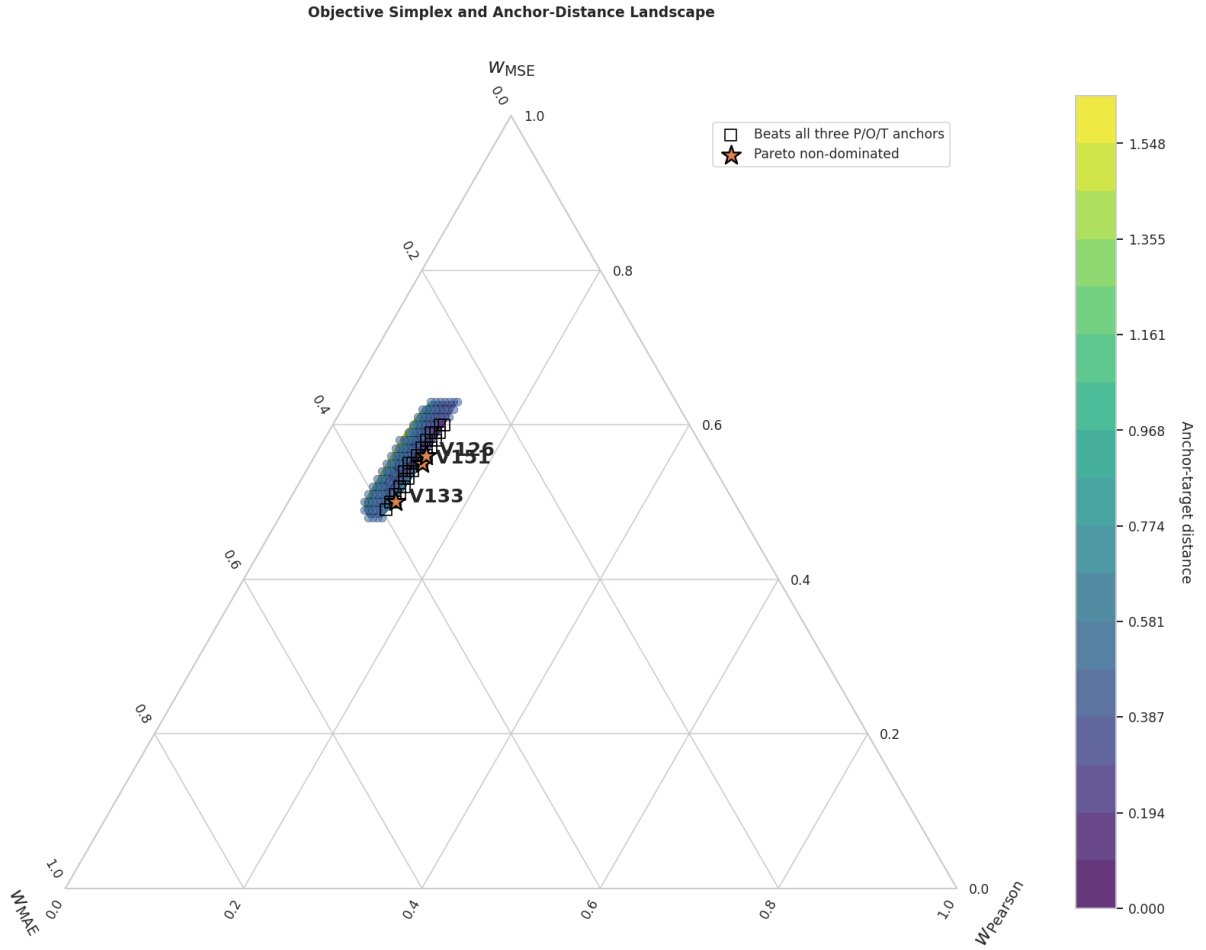


Figure 6: Objective-simplex and anchor-distance landscape for the constrained architecture-screening search. Each candidate is represented by its objective weights $(w_{\text{MSE}}, w_{\text{MAE}}, w_{\text{Pearson}})$ satisfying $w_{\text{MSE}} + w_{\text{MAE}} + w_{\text{Pearson}} = 1$. Color encodes the corresponding distance to the designated anchor-target configuration. The displayed geometry characterizes the search space and is not interpreted as an independent optimization criterion.

The objective-weight vectors belong to the simplex

$$\mathcal{S}_2 = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^3 : \sum_{k=1}^3 w_k = 1 \right\}, \quad (152)$$

with

$$\mathbf{w} = [w_{\text{MSE}}, w_{\text{MAE}}, w_{\text{Pearson}}]. \quad (153)$$

The simplex representation describes the admissible objective-weight combinations examined in the internal screening stage. It should not be interpreted as evidence that the objective weights themselves form a psychological parameter space.

The anchor-distance quantity shown in Fig. 6 is treated as a screening-space diagnostic and does not replace the predefined validation guardrails used for final architecture confirmation.

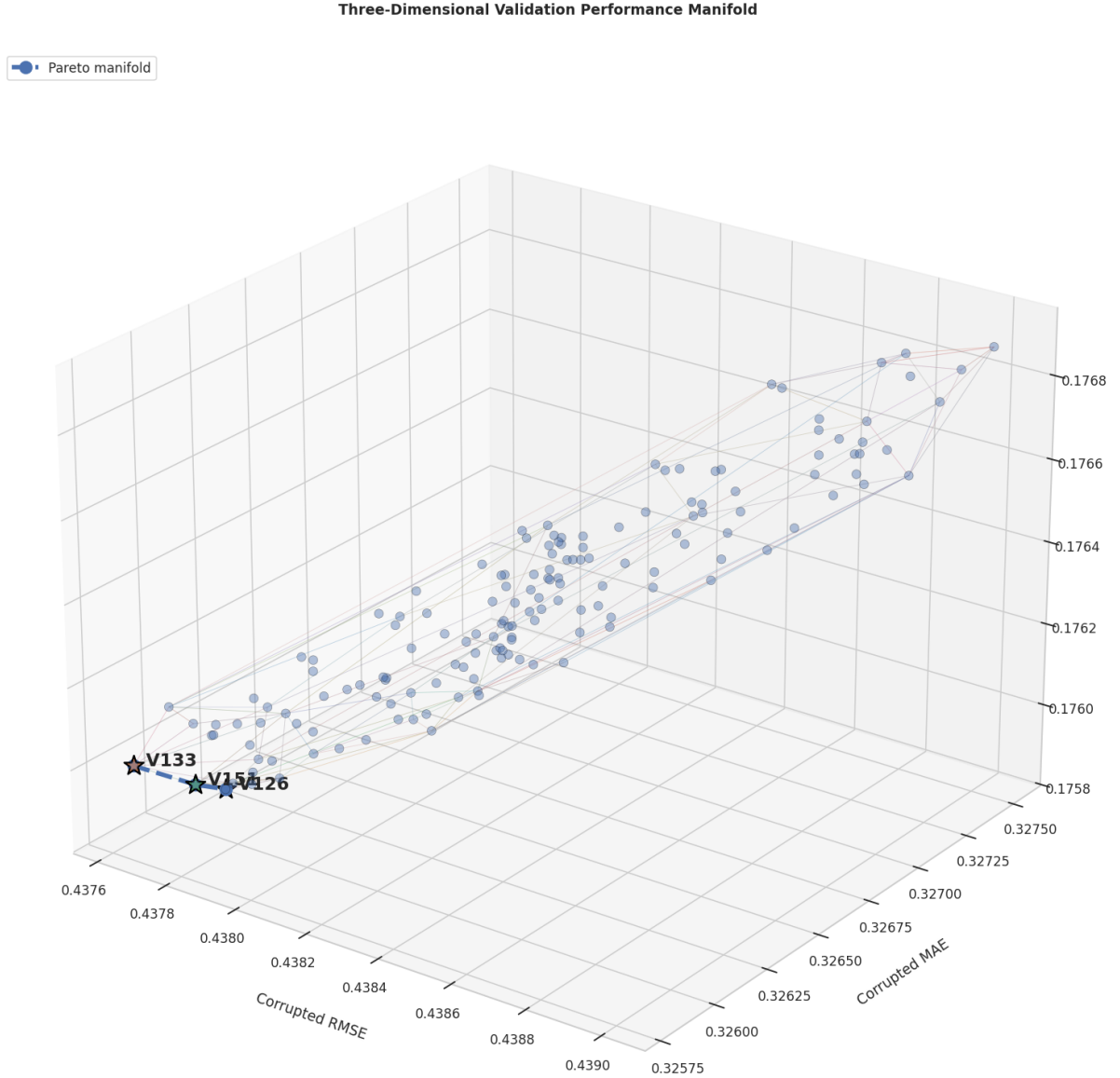


Figure 7: Three-dimensional validation-performance manifold of the evaluated architecture-screening candidates. The axes represent corrupted RMSE, corrupted MAE, and the corresponding profile-correlation dimension. The displayed candidate cloud and non-dominated structure provide a geometric description of the validation search landscape. The visualization is descriptive and does not constitute a separate model-selection rule.

The validation-performance manifold provides a geometric representation of the screening vector

$$\mathbf{v}_c = [RMSE_c, MAE_c, r_c] \quad (154)$$

for candidate c . Candidate separation in this space is used only to describe the structure of the evaluated search landscape. It does not imply that Euclidean distance in this metric space is itself a psychometric or statistical objective.

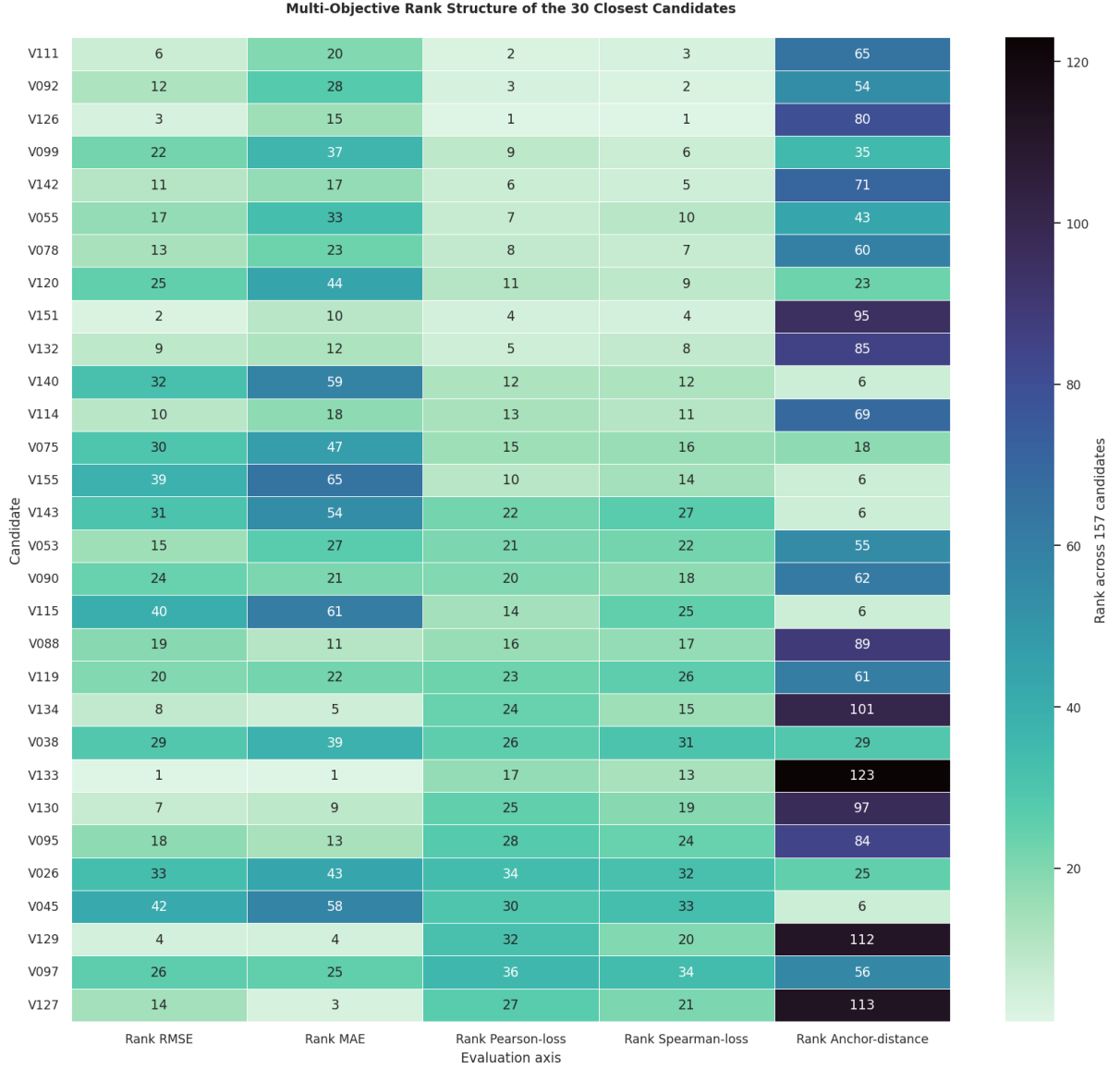


Figure 8: Multi-objective rank structure of the 30 closest architecture-search candidates. Candidate positions are reported across corrupted RMSE, MAE, Pearson-loss, Spearman-loss, and anchor-distance evaluation axes. The rank-based representation is descriptive of the screening set and does not define an additional post-hoc ranking rule for the final operating point.

For candidate c and evaluation axis k , let $\text{rank}_k(c)$ denote its rank within the screening population. The displayed rank vector is therefore

$$\mathbf{q}_c = [\text{rank}_{RMSE}(c), \text{rank}_{MAE}(c), \text{rank}_P(c), \text{rank}_S(c), \text{rank}_A(c)] . \quad (155)$$

These ranks are retained as descriptive diagnostics and are not aggregated into an undisclosed composite candidate score.

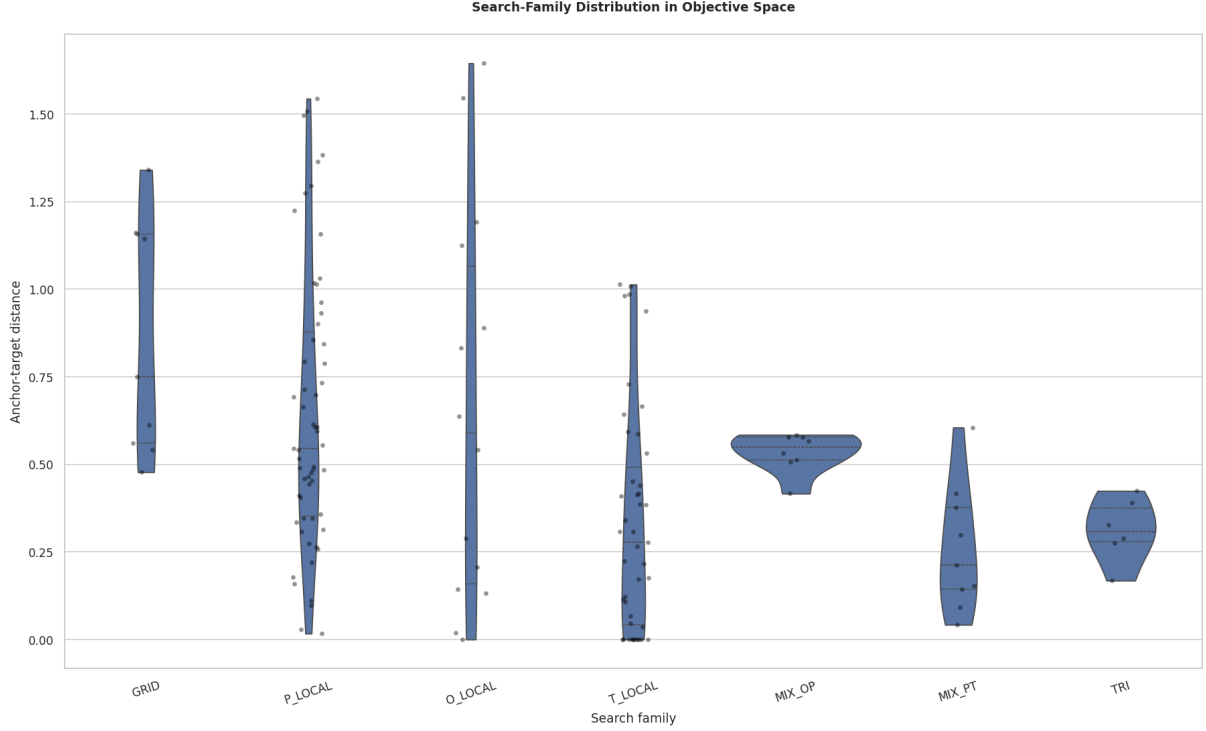


Figure 9: Search-family distribution in the objective-space exploration. The distribution of anchor-target distances is shown for the predefined search families used during constrained architecture screening, including grid, local, mixed, and trial configurations. The figure characterizes how the search procedure populated the evaluated objective space and is not interpreted as evidence that any search family is intrinsically superior.

Let $\mathcal{F}$ denote the set of predefined search families and d_c the anchor-target distance of candidate c . For a family $f \in \mathcal{F}$, its empirical distribution is

$$\hat{G}_f(t) = \frac{1}{N_f} \sum_{c \in \mathcal{C}_f} \mathbb{I}(d_c \leq t), \quad (156)$$

where $\mathcal{C}_f$ denotes the candidate subset generated by search family f and $N_f = |\mathcal{C}_f|$. This distribution is used solely to describe the coverage and concentration of the search procedure.

3.4.4 Confirmatory Multi-Seed Evaluation

Following architecture screening, the S2 configuration was subjected to an independent confirmatory retraining procedure using three seeds:

$$\{20260931, 20260932, 20260933\}. \quad (157)$$

Each seed was evaluated under the same predefined confirmation protocol on the full corrupted-validation benchmark. The confirmation criterion required the candidate to remain within the allowed RMSE, Pearson, and Spearman tolerances relative to V133 while reducing corrupted MAE.

All three confirmation seeds satisfied the predefined guardrails and all three reduced corrupted MAE relative to V133. The resulting confirmation metrics were:

Table 1: Final S2 multi-seed confirmation on corrupted validation.

Seed	RMSE	MAE	Pearson	Spearman	Guardrails
20260931	0.349045	0.196393	0.880235	0.849576	Pass
20260932	0.349288	0.196121	0.880109	0.849640	Pass
20260933	0.349023	0.196198	0.880231	0.850618	Pass

The confirmation therefore produced 3/3 guardrail passes and 3/3 MAE improvements relative to V133. The final operating point was fixed to seed 20260933 at epoch 6.

Its corrupted-validation performance was

$$RMSE = 0.349023, \quad MAE = 0.196198, \quad (158)$$

with profile-level correspondence

$$r_{\text{Pearson}} = 0.880231, \quad \rho_{\text{Spearman}} = 0.850618. \quad (159)$$

Relative to V133, the final S2 checkpoint therefore yielded

$$\Delta RMSE = 0.349023 - 0.348829 = +0.000194, \quad (160)$$

$$\Delta MAE = 0.196198 - 0.201426 = -0.005228, \quad (161)$$

$$\Delta r_{\text{Pearson}} = 0.880231 - 0.880498 = -0.000267, \quad (162)$$

and

$$\Delta \rho_{\text{Spearman}} = 0.850618 - 0.850461 = +0.000157. \quad (163)$$

These values are reported to preserve the direction and magnitude of the metric-specific trade-off. The final S2 operating point should therefore not be interpreted as uniformly superior to V133 across all metrics. Rather, it represents a constrained operating point in which the predefined RMSE and correlation tolerances were maintained while corrupted MAE improved consistently across the three confirmation seeds.

The three-seed stage was used as a stability verification procedure under the predefined guardrails. It is not treated as three independent participants, nor as a preregistered statistical trial, and it does not constitute an inferential estimate of population-level sampling variability.

The independent IPIP-NEO-120 TEST population remained outside architecture selection, objective selection, hyperparameter tuning, checkpoint selection, confirmatory retraining, and post-confirmation adjustment. The TEST benchmark was not loaded or used to determine the final operating point.

3.4.5 Final Checkpoint Selection and Freezing

The final production checkpoint was frozen from seed 20260933 at epoch 6. The resulting Gateway is

$$F_{\theta, \text{S2}}^{(120)} : \mathcal{X}_{120} \longrightarrow \mathbb{R}^{30}. \quad (164)$$

Its frozen operating configuration is therefore defined by

$$\boxed{120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30} \quad (165)$$

with 69,214 trainable parameters.

The sealed checkpoint is identified by the SHA-256 digest

$$\text{SHA256}_{\text{S2}} = 679\text{c}88\text{d}15552\text{ee}6\text{efaf}78\text{cb}0\text{b}3\text{e}288\text{c}09\text{b}3\text{f}00\text{af}53\text{cf}6\text{d}4100807830323\text{e}6992. \quad (166)$$

The cryptographic digest is used as an artifact-integrity control and does not replace independent reproduction of the training or evaluation procedure.

After confirmatory selection, the checkpoint was reloaded under strict architectural compatibility checks and frozen as the final IPIP-NEO-120 production benchmark. The independent TEST population remained outside this freezing procedure.

The resulting methodological sequence is therefore explicitly separated into three stages:

$$\text{Architecture screening} \rightarrow \text{Multi-seed confirmation} \rightarrow \text{Checkpoint freezing}. \quad (167)$$

The independent TEST evaluation follows these stages as a separate benchmark and is not part of the architecture-selection loop.

3.4.6 Local Objective and Adaptation Perturbation Analysis

To further characterize the local screening behaviour around the V133 control, a separate perturbation analysis was performed over objective-weight configurations and adaptation regimes. The purpose of this analysis was to describe local numerical sensitivity to changes in objective composition and adaptation mode without incorporating these diagnostics into the subsequent selection of the final S2 operating point.

The diagnostic distinguishes two adaptation regimes, denoted *head-only* and *full-network*, and evaluates nine predefined objective-weight configurations under each regime. The resulting local diagnostic therefore comprises

$$N_{\text{obj}} = 9, \quad N_{\text{adapt}} = 2, \quad N_{\text{local}} = N_{\text{obj}}N_{\text{adapt}} = 18 \quad (168)$$

evaluated objective-adaptation configurations.

The diagnostic record reports one completed training epoch for each evaluated configuration, with 286/286 optimisation steps recorded for the corresponding run. This stage was therefore a deliberately local perturbation diagnostic rather than a substitute for the subsequent full confirmatory training procedure. The final S2 checkpoint was not selected from these local runs.

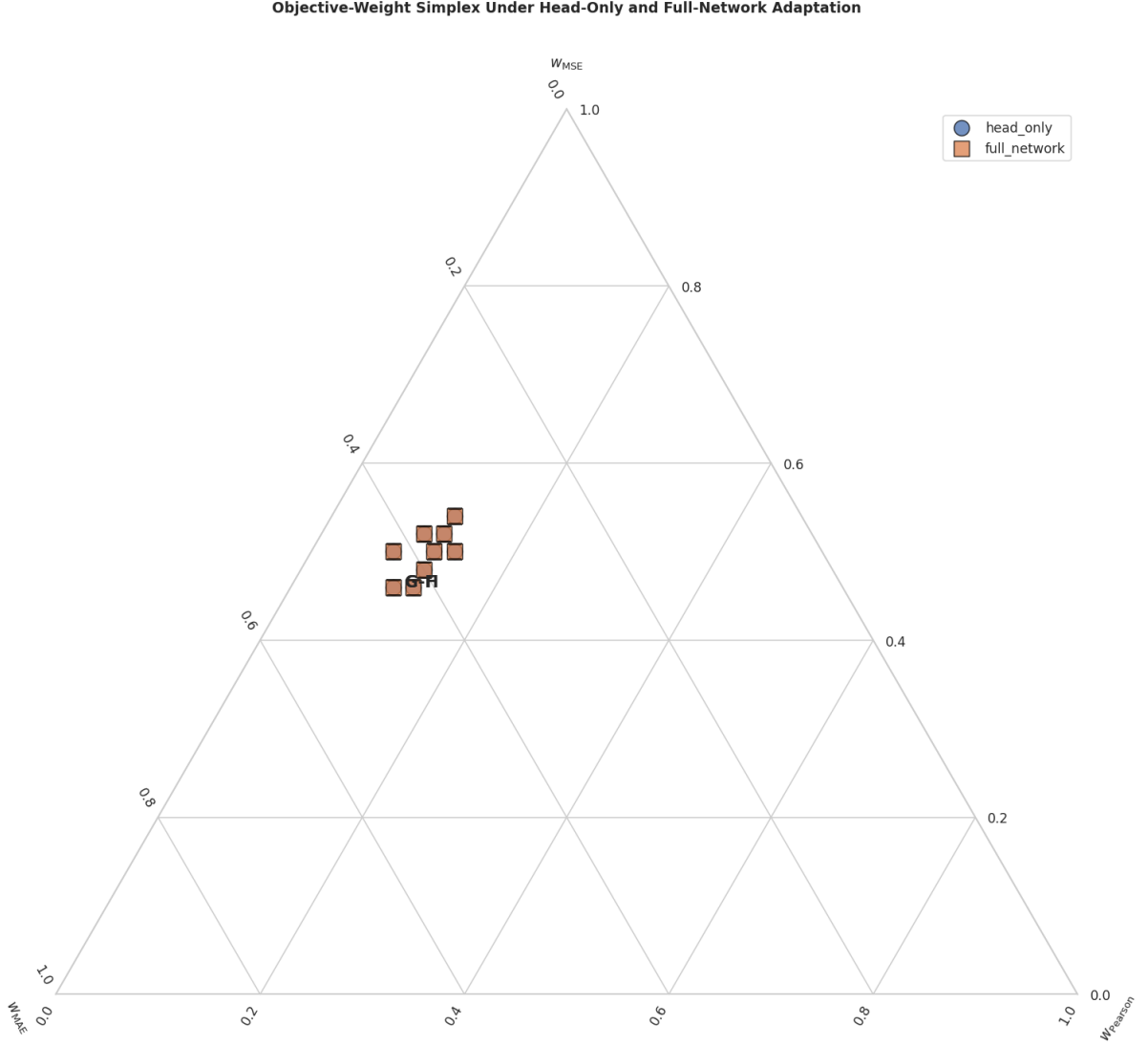


Figure 10: Objective-weight simplex under head-only and full-network adaptation. Each point represents an evaluated objective-weight configuration satisfying $w_{\text{MSE}} + w_{\text{MAE}} + w_{\text{Pearson}} = 1$. Marker shape distinguishes the adaptation regime. The figure characterizes the local objective-space perturbations explored around the reference configuration and does not define an independent selection rule.

Let the objective-weight vector be

$$\mathbf{w} = (w_{\text{MSE}}, w_{\text{MAE}}, w_{\text{Pearson}}), \quad (169)$$

subject to

$$w_{\text{MSE}} + w_{\text{MAE}} + w_{\text{Pearson}} = 1, \quad w_{\text{MSE}}, w_{\text{MAE}}, w_{\text{Pearson}} \geq 0. \quad (170)$$

The admissible objective-weight space is therefore the simplex

$$\mathcal{S}_{\text{obj}} = \left\{ \mathbf{w} \in \mathbb{R}_{\geq 0}^3 : \sum_{k=1}^3 w_k = 1 \right\}. \quad (171)$$

The notation $\mathcal{S}_{\text{obj}}$ is used here specifically to distinguish the objective-weight simplex from the S2 architecture designation used elsewhere in the manuscript.

For a given objective-weight vector, the diagnostic objective is represented as

$$\mathcal{L}_{\mathbf{w}} = w_{\text{MSE}}\mathcal{L}_{\text{MSE}} + w_{\text{MAE}}\mathcal{L}_{\text{MAE}} + w_{\text{Pearson}}(1 - r_{\text{Pearson}}). \quad (172)$$

The diagnostic log reports the Pearson component in abbreviated form as “Pearson”; operationally, the correlation contribution is represented as the loss term $1 - r_{\text{Pearson}}$. Thus, increasing the Pearson weight increases the contribution of profile correspondence to the optimization objective rather than directly rewarding a larger scalar loss.

The nine evaluated objective configurations were

Table 2: Objective-weight configurations evaluated in the local IPIP-NEO-120 perturbation diagnostic.

Configuration	w_{MSE}	w_{MAE}	w_{Pearson}
V133_BASE	0.50	0.38	0.12
LOCAL_A	0.46	0.42	0.12
LOCAL_B	0.48	0.40	0.12
LOCAL_C	0.52	0.36	0.12
LOCAL_D	0.50	0.42	0.08
LOCAL_E	0.50	0.36	0.14
LOCAL_F	0.54	0.34	0.12
LOCAL_G	0.46	0.44	0.10
LOCAL_H	0.52	0.38	0.10

Each row satisfies the simplex constraint exactly. The objective grid therefore constitutes a finite, explicitly enumerated local perturbation set rather than a continuous optimisation over the simplex.

Let

$$a \in \{\text{head-only}, \text{full-network}\} \quad (173)$$

denote the adaptation regime used by the diagnostic. The local evaluation set can then be represented as

$$\mathcal{D}_{\text{local}} = \{(\mathbf{w}_c, a_c, \mathbf{m}_c)\}_{c=1}^{N_{\text{local}}}, \quad (174)$$

where $\mathbf{m}_c$ denotes the validation-performance vector observed for configuration c .

The adaptation labels identify the two protocol-defined training regimes evaluated in the diagnostic. They are used here to distinguish the numerical training configurations and are not interpreted as general intervention classes or as causal treatments.

The corresponding validation-performance vector is

$$\mathbf{m}_c = [\text{RMSE}_c, \text{MAE}_c, r_c, \rho_c], \quad (175)$$

where r_c denotes Pearson correspondence and ρ_c denotes Spearman rank correspondence with the canonical reference representation.

The purpose of this representation is therefore to characterize the local behaviour of the evaluated system under controlled changes in objective composition and adaptation regime, not to construct an additional composite performance score.

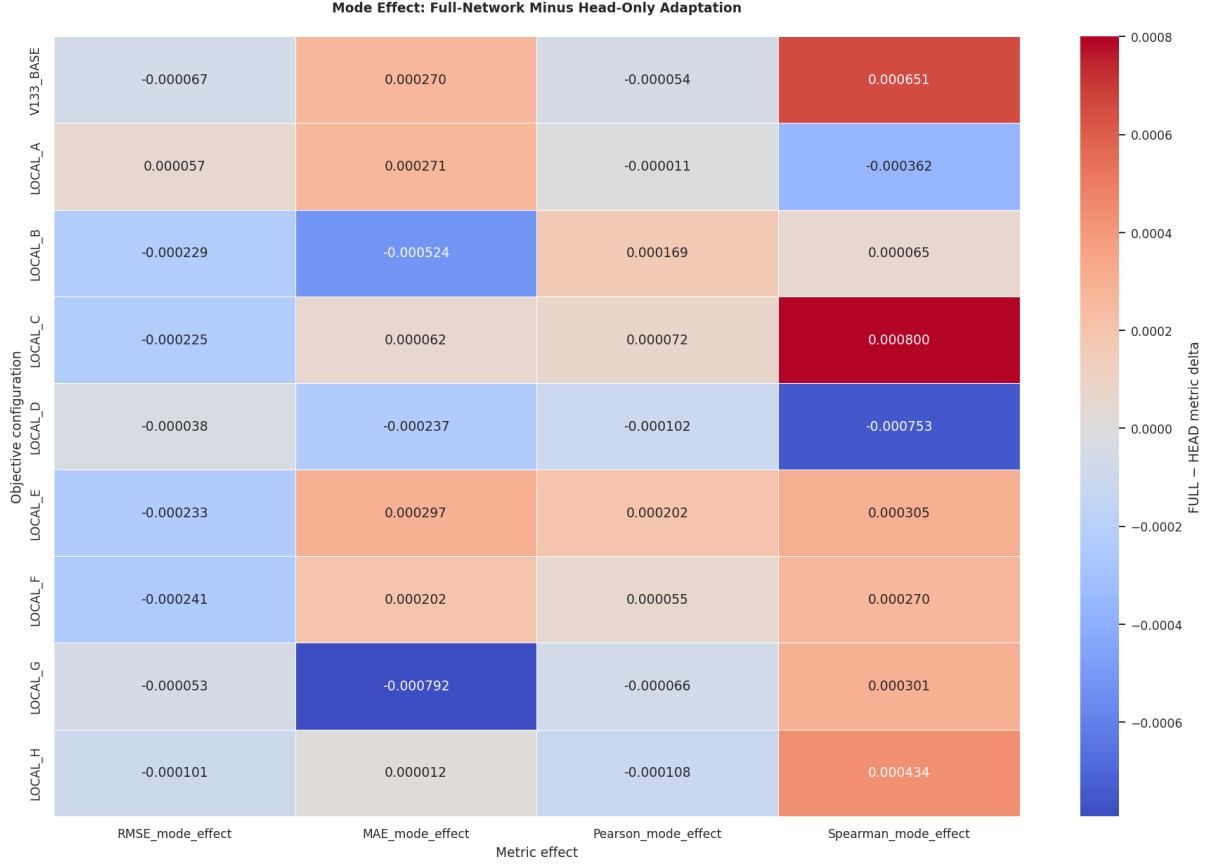


Figure 11: Mode effect of full-network relative to head-only adaptation across the evaluated local objective configurations. Each cell reports the metric difference $\text{FULL} - \text{HEAD}$ for corrupted RMSE, MAE, Pearson, and Spearman correspondence. The matrix provides a descriptive decomposition of adaptation mode contrasts and is not interpreted as a causal effect estimate.

For objective configuration g and validation metric M , the corresponding adaptation-mode contrast is defined as

$$\Delta M_g^{\text{mode}} = M_g^{(\text{full})} - M_g^{(\text{head})}. \quad (176)$$

Accordingly,

$$\Delta \mathbf{M}_g^{\text{mode}} = [\Delta \text{RMSE}_g^{\text{mode}}, \Delta \text{MAE}_g^{\text{mode}}, \Delta r_g^{\text{mode}}, \Delta \rho_g^{\text{mode}}]. \quad (177)$$

Negative values for RMSE or MAE indicate lower numerical error under full-network adaptation relative to head-only adaptation, whereas positive values for Pearson or Spearman indicate greater numerical correspondence with the canonical reference.

The contrast is conditional on the objective configuration, the evaluated model state, and the local experimental protocol. Accordingly, it should be read as a within-diagnostic numerical contrast rather than as an estimate of a general causal effect attributable to adaptation strategy.

Importantly, the local diagnostic does not establish that a particular objective weighting or adaptation regime is intrinsically preferable. The observed changes are conditional on the finite set of evaluated configurations and the corresponding training trajectories.

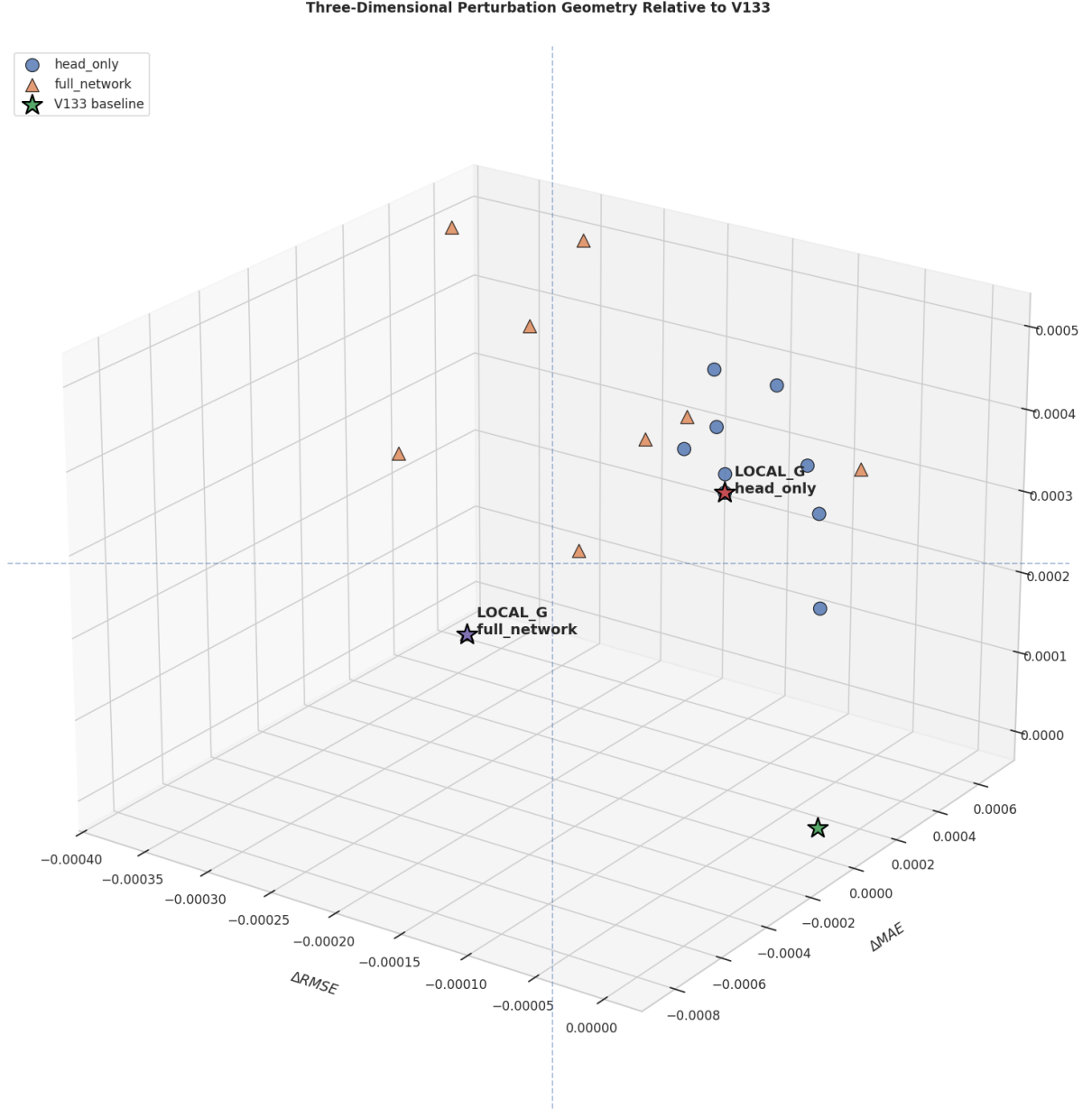


Figure 12: Three-dimensional perturbation geometry relative to the V133 control. Each evaluated configuration is represented by its change in corrupted RMSE, MAE, and Pearson correspondence relative to V133, with marker shape denoting the adaptation regime. The V133 control is represented at the zero-change reference point. The geometry is descriptive of local perturbation behaviour and is not interpreted as a causal response surface or as an independent selection criterion.

For candidate c , the metric displacement relative to the V133 reference is

$$\Delta \mathbf{m}_c = [\Delta \text{RMSE}_c, \Delta \text{MAE}_c, \Delta r_c], \quad (178)$$

with

$$\Delta \text{RMSE}_c = \text{RMSE}_c - \text{RMSE}_{\text{V133}}, \quad (179)$$

$$\Delta\text{MAE}_c = \text{MAE}_c - \text{MAE}_{\text{V133}}, \quad (180)$$

and

$$\Delta r_c = r_c - r_{\text{V133}}. \quad (181)$$

The V133 reference therefore corresponds to

$$\Delta\mathbf{m}_{\text{V133}} = \mathbf{0}. \quad (182)$$

Negative ΔRMSE or ΔMAE denotes a numerical reduction in the corresponding error metric relative to V133, whereas positive Δr denotes increased Pearson correspondence. These directions are interpreted separately and are not collapsed into a single scalar score.

For completeness, the local diagnostic contains a second correlation coordinate used in the broader metric vector. Its displacement is

$$\Delta\rho_c = \rho_c - \rho_{\text{V133}}. \quad (183)$$

Thus, the complete displacement representation can be written as

$$\Delta\mathbf{m}_c^* = [\Delta\text{RMSE}_c, \Delta\text{MAE}_c, \Delta r_c, \Delta\rho_c]. \quad (184)$$

The three-dimensional representation shown in Fig. 12 uses the first three coordinates explicitly; the Spearman displacement remains a separate metric and is not discarded by the three-dimensional projection.

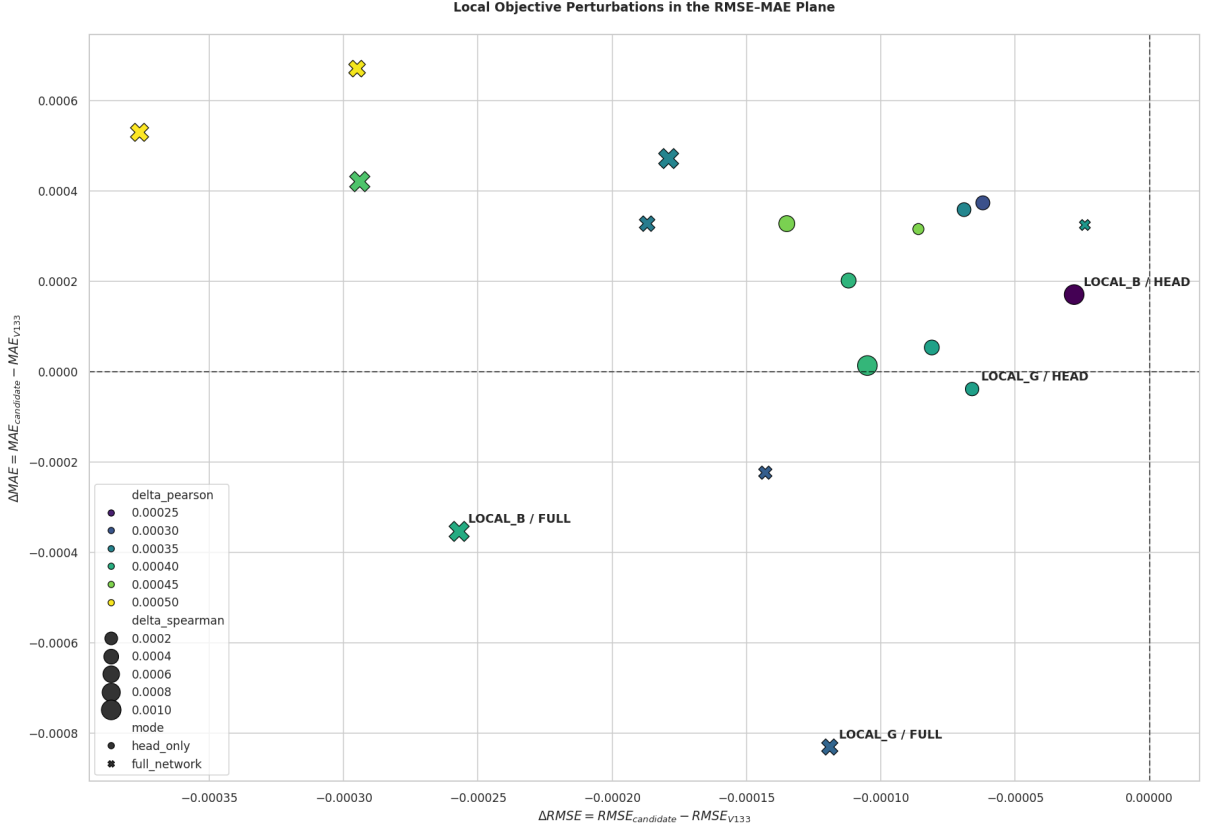


Figure 13: Local objective perturbations in the corrupted RMSE–MAE plane relative to the V133 control. The horizontal and vertical coordinates report changes in RMSE and MAE, respectively; additional visual encodings represent changes in Pearson and Spearman correspondence and the evaluated adaptation regime. The figure is used to characterize local trade-offs among validation metrics and does not constitute a post-hoc ranking or causal sensitivity analysis.

For a candidate c , the local perturbation coordinates are

$$\mathbf{p}_c = [\Delta RMSE_c, \Delta MAE_c]. \quad (185)$$

The V133 reference point is therefore

$$\mathbf{p}_{V133} = (0, 0). \quad (186)$$

The corresponding correlation changes are

$$\Delta r_c = r_c - r_{V133}, \quad (187)$$

and

$$\Delta \rho_c = \rho_c - \rho_{V133}. \quad (188)$$

This representation allows local metric trade-offs to be inspected directly. The four quadrants of the RMSE–MAE plane correspond to distinct combinations of error changes relative to V133, whereas the correlation quantities remain independently encoded rather than being absorbed into the horizontal or vertical coordinates.

The A15–A17 diagnostics therefore provide complementary descriptions of the same finite perturbation set. A15 characterizes the evaluated objective-weight simplex under the two

adaptation regimes, A21 decomposes the direct full-network-versus-head-only contrasts, A16 represents selected metric displacements relative to V133, and A17 resolves the local RMSE–MAE trade-off while retaining the correlation changes as separate quantities.

These analyses are descriptive of the evaluated local perturbation set. They do not establish that objective weights or adaptation regime causally determine the observed metric changes, and they do not replace the predefined architecture-selection procedure or the subsequent independent multi-seed confirmation.

The diagnostic nevertheless provides a finite empirical audit of local objective and adaptation perturbations. Within the recorded diagnostic runs, the objective configuration LOCAL_G produced a reduction in corrupted MAE under both adaptation regimes, with

$$\Delta\text{MAE}_{\text{LOCAL_G}}^{(\text{head})} = -0.000039 \quad (189)$$

and

$$\Delta\text{MAE}_{\text{LOCAL_G}}^{(\text{full})} = -0.000830. \quad (190)$$

These values are descriptive observations from the local diagnostic and are not used to declare LOCAL_G the final objective, nor to identify it as a universally optimal configuration.

The same diagnostic also recorded that all evaluated local objective–adaptation configurations satisfied the diagnostic guardrail check. This observation is treated only as a property of the finite diagnostic set; it does not supersede the formal multi-seed confirmation criterion used for the final S2 operating point.

The local diagnostic therefore serves as a methodological bridge between single-architecture inspection and confirmatory evaluation. It provides evidence that the numerical behaviour around V133 is not represented by a single undifferentiated metric direction, while preserving the separation between exploratory diagnostics and confirmatory model selection.

No model-selection decision for the final S2 checkpoint was made directly from A15–A17. The final S2 operating point was determined separately and was subsequently subjected to the independent multi-seed confirmation protocol.

3.4.7 Confirmatory Multi-Seed Evaluation

Following the preceding architecture and local diagnostic stages, the S2 configuration was subjected to an independent confirmatory retraining procedure using three seeds:

$$\{20260931, 20260932, 20260933\}. \quad (191)$$

Each seed was trained under the same model-selection protocol and evaluated on the full corrupted-validation benchmark using the predefined guardrails relative to V133. The confirmation criterion required the candidate to remain within the allowed RMSE, Pearson, and Spearman tolerances while improving corrupted MAE relative to the control.

All three confirmation seeds satisfied the predefined guardrails and all three reduced corrupted MAE relative to V133. The resulting confirmation metrics were:

Table 3: Final S2 multi-seed confirmation on corrupted validation.

Seed	RMSE	MAE	Pearson	Spearman	Guardrails
20260931	0.349045	0.196393	0.880235	0.849576	Pass
20260932	0.349288	0.196121	0.880109	0.849640	Pass
20260933	0.349023	0.196198	0.880231	0.850618	Pass

The confirmation therefore produced 3/3 guardrail passes and 3/3 MAE improvements relative to V133. The final operating point was fixed to seed 20260933 at epoch 6.

Its corrupted-validation performance was

$$\text{RMSE} = 0.349023, \quad \text{MAE} = 0.196198, \quad (192)$$

with profile-level correspondence

$$r_{\text{Pearson}} = 0.880231, \quad \rho_{\text{Spearman}} = 0.850618. \quad (193)$$

Relative to V133, the final S2 checkpoint therefore yielded

$$\Delta \text{RMSE} = 0.349023 - 0.348829 = +0.000194, \quad (194)$$

$$\Delta \text{MAE} = 0.196198 - 0.201426 = -0.005228, \quad (195)$$

$$\Delta r_{\text{Pearson}} = 0.880231 - 0.880498 = -0.000267, \quad (196)$$

and

$$\Delta \rho_{\text{Spearman}} = 0.850618 - 0.850461 = +0.000157. \quad (197)$$

The resulting metric changes demonstrate that the final S2 operating point should not be interpreted as a uniform improvement over V133 across every evaluation metric. Rather, the confirmed configuration represents a predefined multi-metric trade-off in which corrupted MAE improved across all three confirmation seeds while the final RMSE and correlation changes remained within the specified confirmation guardrails.

The three confirmation seeds constitute independent retraining runs of the same model-selection protocol. They are not independent participants and do not create additional participant-level observations. Their role is to assess stability of the observed operating point with respect to the training seed.

The confirmatory stage therefore should not be interpreted as a preregistered statistical trial or as a substitute for population-level inferential uncertainty estimation. It is a predefined stability verification procedure for the model-selection decision.

3.4.8 Final Checkpoint Selection and Freezing

The final checkpoint was frozen from seed 20260933 at epoch 6. The resulting Gateway is defined as

$$F_{\theta, \text{S2}}^{(120)} : \mathcal{X}_{120} \longrightarrow \mathbb{R}^{30}. \quad (198)$$

Its frozen operating configuration is

$$\boxed{120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30} \quad (199)$$

with 69,214 trainable parameters.

The frozen checkpoint is identified by the cryptographic digest

$$\text{SHA256}_{S_2} = 679c88d15552ee6efaf78cb0b3e288c09b3f00af53cf6d4100807830323e6992 \ . \quad (200)$$

The SHA-256 value serves as an artifact-integrity identifier for the frozen checkpoint. It verifies the identity of the stored artifact but does not, by itself, establish reproducibility of the training trajectory or validity of the scientific conclusions.

The final operating point is therefore fully specified by the tuple

$$\mathcal{P}_{S_2} = (\mathcal{A}_{S_2}, N_\theta, s, e, H) \ , \quad (201)$$

where $\mathcal{A}_{S_2}$ denotes the architecture $120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30$, $N_\theta = 69,214$ is the number of trainable parameters, $s = 20260933$ is the selected seed, $e = 6$ is the selected epoch, and H denotes the SHA-256 digest of the frozen checkpoint.

After confirmatory selection, the checkpoint was reloaded under strict architectural compatibility checks and frozen as the final IPIP-NEO-120 production benchmark. The independent TEST population remained outside architecture selection, objective perturbation diagnostics, confirmatory retraining, checkpoint selection, and final freezing.

The complete methodological sequence is therefore

$$\text{Architecture screening} \rightarrow \text{Local objective/adaptation diagnostics} \rightarrow \text{Multi-seed confirmation} \rightarrow \text{Checkpoint freeze} \quad (202)$$

The local perturbation analysis is consequently positioned as an intermediate descriptive stage rather than as a hidden optimisation loop. Its outputs are retained for transparency and mechanistic interpretation, while the final S2 operating point remains determined by the explicitly defined confirmatory procedure.

3.4.9 Confirmatory Multi-Seed Retraining

Following internal architecture tuning, the S2 configuration was subjected to independent confirmatory retraining under the same experimental protocol using three independently seeded training runs:

$$\mathcal{S}_{\text{confirm}} = \{20,260,931, 20,260,932, 20,260,933\} \ . \quad (203)$$

The purpose of this stage was to assess the stability of the selected architecture and metric trade-off with respect to stochastic variation in model training, rather than to create additional participant-level observations. The three runs therefore share the same instrument, training population, validation benchmark, architecture, objective specification, and model-selection protocol, while differing in their random seeds.

The confirmation criterion was defined relative to the V133 control through four simultaneous conditions. For confirmation seed s ,

$$\Delta\text{RMSE}_s = \text{RMSE}_s - \text{RMSE}_{\text{V133}} \leq 0.001, \quad (204)$$

$$\Delta r_{\text{Pearson},s} = r_{\text{Pearson},s} - r_{\text{Pearson,V133}} \geq -0.002, \quad (205)$$

$$\Delta \rho_{\text{Spearman},s} = \rho_{\text{Spearman},s} - \rho_{\text{Spearman,V133}} \geq -0.002, \quad (206)$$

and

$$\Delta\text{MAE}_s = \text{MAE}_s - \text{MAE}_{\text{V133}} < 0. \quad (207)$$

The predefined confirmation rule required at least two of the three confirmation runs to satisfy the complete set of guardrails and to reduce corrupted-validation MAE relative to V133. The observed result exceeded this minimum requirement: all three confirmation runs satisfied the guardrails and all three reduced corrupted-validation MAE.

The three confirmation runs produced the following corrupted-validation results:

Table 4: Confirmatory multi-seed evaluation of the IPIP-NEO-120 S2 Gateway.

Seed	RMSE	MAE	Pearson	Spearman	Guardrails
20,260,931	0.349045	0.196393	0.880235	0.849576	Pass
20,260,932	0.349288	0.196121	0.880109	0.849640	Pass
20,260,933	0.349023	0.196198	0.880231	0.850618	Pass

All three confirmation runs therefore satisfied the predefined RMSE, Pearson, and Spearman tolerances and achieved a strict reduction in corrupted-validation MAE relative to V133. The consistency of these results across the three independently seeded retraining runs provides evidence that the observed behaviour of S2 was not confined to a single realised optimisation trajectory.

This evidence is interpreted specifically as stability across the evaluated training seeds. It does not establish invariance to arbitrary initialisations, hardware environments, optimisation implementations, or untested random seeds, nor does it constitute a preregistered statistical trial or an inferential estimate of population-level uncertainty.

The confirmatory stage was performed after architecture screening and local objective/adaptation diagnostics and before production freezing. The independent IPIP-NEO-120 TEST population remained outside architecture selection, objective selection, hyperparameter tuning, confirmatory retraining, and checkpoint selection. The TEST benchmark was not loaded or used during this stage.

3.4.10 Final Checkpoint Selection and Freezing

The final operating point was designated from seed 20,260,933 at epoch 6 under the predefined confirmation and freezing protocol. The resulting Gateway contains 69,214 trainable parameters and implements the mapping

$$F_{\theta,\text{S2}}^{(120)} : \mathcal{X}_{120} \longrightarrow \mathbb{R}^{30}. \quad (208)$$

Its frozen architecture is

$$\boxed{120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30}. \quad (209)$$

Its corrupted-validation performance was

$$\text{RMSE} = 0.349023, \quad \text{MAE} = 0.196198, \quad (210)$$

with profile-level correspondence

$$r_{\text{Pearson}} = 0.880231, \quad \rho_{\text{Spearman}} = 0.850618. \quad (211)$$

Relative to the V133 baseline, the final S2 checkpoint changed the principal corrupted-validation metrics by

$$\Delta \text{RMSE} = 0.349023 - 0.348829 = +0.000194, \quad (212)$$

$$\Delta \text{MAE} = 0.196198 - 0.201426 = -0.005228, \quad (213)$$

$$\Delta r_{\text{Pearson}} = 0.880231 - 0.880498 = -0.000267, \quad (214)$$

and

$$\Delta \rho_{\text{Spearman}} = 0.850618 - 0.850461 = +0.000157. \quad (215)$$

The unrounded differences recorded for the frozen operating point are approximately +0.000193974 for RMSE, −0.005227696 for MAE, −0.000266882 for Pearson correspondence, and +0.000156515 for Spearman correspondence.

Accordingly, the final S2 checkpoint is not described as a uniform improvement over V133 across all metrics. Instead, the selected operating point represents a predefined multi-metric trade-off in which corrupted-validation MAE was reduced consistently across the three confirmation seeds while the changes in RMSE and profile-level correspondence remained within the specified confirmation tolerances.

The final architecture was therefore selected as a constrained operating point rather than as the unconstrained numerical minimiser of corrupted RMSE. This distinction is methodologically relevant because RMSE and MAE respond differently to the distribution of residual magnitudes. A reduction in squared error can coexist with a different pattern of absolute deviations, such that optimisation of one error functional does not uniquely determine the behaviour of the other.

The confirmatory results should consequently be interpreted as validation of the predefined operating-point criteria rather than as evidence that the S2 architecture dominates V133 under every metric.

3.4.11 Production Freezing and Integrity Control

After confirmatory selection, the seed-20,260,933 checkpoint was frozen as the final IPIP-NEO-120 production benchmark artifact. The frozen artifact is identified by its architecture, parameter count, training seed, selected epoch, model metadata, and cryptographic integrity record.

The sealed checkpoint is identified by the SHA-256 digest

$$\text{SHA256}_{\text{S}_2} = 679\text{c}88\text{d}15552\text{ee}6\text{efaf}78\text{cb}0\text{b}3\text{e}288\text{c}09\text{b}3\text{f}00\text{af}53\text{cf}6\text{d}4100807830323\text{e}6992 \quad . \quad (216)$$

The cryptographic digest provides an artifact-integrity identifier for the stored checkpoint. It establishes the identity of the referenced artifact but does not, by itself, establish reproducibility of the complete optimisation trajectory or validity of the scientific conclusions.

The production checkpoint was subsequently reloaded independently and checked for architectural compatibility, parameter identity, expected output dimensionality, and checkpoint integrity. The independent verification was performed without loading or evaluating the IPIP-NEO-120 TEST population.

The resulting model-development pathway is therefore

$$\text{initial architecture} \rightarrow \text{dimensional audit} \rightarrow \text{alternative architecture tests} \rightarrow \text{constrained architecture tuning} \rightarrow \quad (217)$$

The independent TEST benchmark constitutes a subsequent evaluation stage and is not part of this development pathway.

3.5 IPIP-NEO-300 Gateway

The IPIP-NEO-300 Gateway was developed through a staged architecture and objective-selection programme. The structured-output model initially explored during architecture screening is not the final production configuration. In particular, the earlier structured control

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\} \quad (218)$$

contained 121,054 trainable parameters and served as an intermediate architecture in the IPIP-NEO-300 development programme. It must therefore not be conflated with the final P83.25 production model.

The final IPIP-NEO-300 configuration is P83.25:

$$\boxed{300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}}. \quad (219)$$

The final model contains a shared 184-dimensional representation followed by five parallel six-dimensional output heads. Each head receives the same shared representation and produces six facet outputs:

$$H_k : \mathbb{R}^{184} \longrightarrow \mathbb{R}^6, \quad k = 1, \dots, 5. \quad (220)$$

Consequently,

$$5 \times 6 = 30, \quad (221)$$

and the complete model maps the 300-dimensional observation space to the common thirty-dimensional canonical reference space,

$$F_{\theta, \text{P83.25}}^{(300)} : \mathcal{X}_{300} \longrightarrow \mathbb{R}^{30}. \quad (222)$$

The five-head structure is an architectural grouping of output coordinates aligned with the five broad Five-Factor domains and their six corresponding facets. The heads are not treated

as statistically independent predictors: they receive a common learned representation and are jointly optimised through the shared loss.

The production model contains 220,782 trainable parameters. The parameter count is explicitly decomposed as

$$\begin{aligned}
 N_\theta &= (300 \times 344 + 344) + 2(344) \\
 &\quad + (344 \times 208 + 208) + 2(208) \\
 &\quad + (208 \times 184 + 184) + 2(184) \\
 &\quad + 5(184 \times 6 + 6) \\
 &= 220,782.
 \end{aligned} \tag{223}$$

The $2d$ terms correspond to the trainable scale and bias parameters of each d -dimensional Layer Normalization layer. GELU activations and dropout do not introduce additional trainable parameters.

The production implementation applies fixed affine input normalisation

$$\mathcal{N}(x) = \frac{x - 3}{2}, \tag{224}$$

followed by three sequential hidden transformations. For $\ell = 1, 2, 3$, each transformation consists of an affine map, Layer Normalization, GELU activation, and dropout with probability $p = 0.10$:

$$\mathbf{h}_1 = D_{0.10} [\text{GELU} (\text{LN}_1 (W_1 \mathcal{N}(x) + b_1))], \tag{225}$$

$$\mathbf{h}_2 = D_{0.10} [\text{GELU} (\text{LN}_2 (W_2 \mathbf{h}_1 + b_2))], \tag{226}$$

and

$$\mathbf{h}_3 = D_{0.10} [\text{GELU} (\text{LN}_3 (W_3 \mathbf{h}_2 + b_3))], \tag{227}$$

with

$$\mathbf{h}_3 \in \mathbb{R}^{184}. \tag{228}$$

The five output heads then operate on the same $\mathbf{h}_3$ representation:

$$\hat{\mathbf{y}}^{(k)} = W_k^{\text{out}} \mathbf{h}_3 + b_k^{\text{out}}, \quad k = 1, \dots, 5, \tag{229}$$

and the complete prediction vector is

$$\hat{Y} = \text{Concat} (\hat{\mathbf{y}}^{(1)}, \hat{\mathbf{y}}^{(2)}, \hat{\mathbf{y}}^{(3)}, \hat{\mathbf{y}}^{(4)}, \hat{\mathbf{y}}^{(5)}) \in \mathbb{R}^{30}. \tag{230}$$

No output activation function or output clipping operation is applied after the final linear heads in the production implementation.

The P83.25 objective is

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - r_{\text{Pearson}}), \tag{231}$$

where the Pearson component is computed at the participant-profile level across the thirty output facets rather than as a single pooled correlation across all observations and facets.

For observation n , let

$$\hat{y}_n = \frac{1}{30} \sum_{j=1}^{30} \hat{y}_{nj}, \quad \bar{y}_n = \frac{1}{30} \sum_{j=1}^{30} y_{nj}. \quad (232)$$

The profile-level Pearson correspondence for observation n is

$$r_n = \frac{\sum_{j=1}^{30} (\hat{y}_{nj} - \hat{y}_n)(y_{nj} - \bar{y}_n)}{\sqrt{\sum_{j=1}^{30} (\hat{y}_{nj} - \hat{y}_n)^2 + \varepsilon} \sqrt{\sum_{j=1}^{30} (y_{nj} - \bar{y}_n)^2 + \varepsilon}}, \quad (233)$$

with

$$\varepsilon = 10^{-8}. \quad (234)$$

The Pearson loss component used during optimisation is therefore

$$\mathcal{L}_{\text{Pearson}} = 1 - \frac{1}{N} \sum_{n=1}^N r_n, \quad (235)$$

and the complete production objective can equivalently be written as

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 \mathcal{L}_{\text{Pearson}}. \quad (236)$$

This definition is important because the objective explicitly encourages agreement in the shape of each thirty-dimensional predicted profile rather than only minimising pooled pointwise error.

The final P83.25 production model was retrained from the complete TRAIN population rather than using the 10,000-subject screening benchmark as the final fitting population. The production run comprised 101,772 TRAIN subjects and 14,553,396 training examples per epoch, corresponding to 143 views per subject:

$$143 = 3 + 7 \times 20. \quad (237)$$

Thus, each training subject contributed three clean views and twenty perturbed views for each of the seven corruption families.

The production optimisation used AdamW with batch size 8,192, learning rate 0.002, weight decay 0.0001, dropout probability 0.10, gradient-norm clipping at 1.0, BF16 mixed precision, and a maximum of 120 epochs. The minimum epoch count was five, the early-stopping patience was three epochs, and the minimum improvement threshold for replacement of the best corrupted validation RMSE was

$$\delta_{\min} = 10^{-4}. \quad (238)$$

The production run used global seed 20260920. The final P83.25 production configuration did not undergo the three-seed confirmation procedure applied to the IPIP-NEO-120 S2 configuration; its operating point was selected through the predefined production validation protocol.

The training data were presented through the predefined 143-view subject-isolation schedule. The scheduling mechanism ensured that a given subject was not duplicated within a batch and that each subject was exposed to its complete set of 143 views exactly once per epoch through the round-based schedule. This construction was retained to prevent repeated views of the same

participant from creating artificial within-batch replication.

The production validation benchmark contained

$$N_{\text{val,clean}} = 21,808 \quad (239)$$

clean observations and

$$N_{\text{val,corr}} = 338,444 \quad (240)$$

corrupted observations, for

$$N_{\text{val,total}} = 360,252. \quad (241)$$

Checkpoint selection was based exclusively on corrupted-validation RMSE. Clean RMSE, MAE, Pearson correspondence, and Spearman correspondence were recorded as validation diagnostics but were not used as the checkpoint selection criterion.

The official production checkpoint was selected at epoch 14, where the corrupted-validation RMSE was

$$\text{RMSE}_{\text{val,corr}} = 0.18483594655007943, \quad (242)$$

reported to six decimal places as 0.184836.

Training subsequently continued through epoch 17. Although the raw corrupted-validation RMSE at epoch 17 was 0.184769, the reduction relative to the epoch-14 best value was smaller than the predefined minimum improvement threshold of 10^{-4} . Epoch 17 therefore did not replace the epoch-14 best checkpoint, and early stopping terminated the production run after the predefined patience condition was reached.

The final production operating point is consequently defined by

$$\boxed{300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}} \quad (243)$$

with

$$N_{\theta} = 220,782, \quad \text{epoch} = 14, \quad \text{seed} = 20,260,920. \quad (244)$$

The corresponding production checkpoint was fixed before independent TEST evaluation. Its SHA-256 identity is

$$\text{SHA256}_{\text{P83.25}} = \text{f113f3b48b09f89deecce55788a59f257f31c09f12b17d3d2f657c0bd42fc287} . \quad (245)$$

The subsequent IPIP-NEO-300 TEST evaluation was conducted as a separate frozen stage. The independent TEST population remained outside training, architecture screening, objective screening, checkpoint selection, and retraining.

The distinction between the earlier 121,054-parameter structured control and the final 220,782-parameter P83.25 model is therefore maintained explicitly: the former belongs to the architecture-development sequence, whereas the latter is the frozen production configuration used for the independent IPIP-NEO-300 evaluation.

3.6 Objective Function

The Gateway is optimized against the canonical reference representation using instrument-specific objectives whose definitions are fixed within each development programme. The IPIP-NEO-120 and IPIP-NEO-300 configurations therefore do not share a single optimisation objective, and the objective used during an earlier development stage is not assumed to remain the objective of the final production checkpoint.

3.6.1 IPIP-NEO-120 S2 Objective

The final IPIP-NEO-120 S2 configuration uses a composite objective that combines pointwise squared error, absolute error, and profile-level correspondence:

$$\mathcal{L}_{120,S2} = 0.50 \mathcal{L}_{\text{MSE}} + 0.38 \mathcal{L}_{\text{MAE}} + 0.12 \mathcal{L}_{\text{Pearson}}. \quad (246)$$

For observation n , with thirty predicted outputs $\hat{\mathbf{y}}_n \in \mathbb{R}^{30}$ and corresponding canonical reference vector $\mathbf{y}_n^{\text{ref}} \in \mathbb{R}^{30}$, the pointwise mean squared error is

$$\mathcal{L}_{\text{MSE},n} = \frac{1}{30} \sum_{j=1}^{30} \left(\hat{y}_{nj} - y_{nj}^{\text{ref}} \right)^2, \quad (247)$$

and the corresponding mean absolute error is

$$\mathcal{L}_{\text{MAE},n} = \frac{1}{30} \sum_{j=1}^{30} \left| \hat{y}_{nj} - y_{nj}^{\text{ref}} \right|. \quad (248)$$

The profile-level Pearson component is expressed as a loss rather than as a maximisation term:

$$\mathcal{L}_{\text{Pearson},n} = 1 - r_n, \quad (249)$$

where r_n is the Pearson correspondence between the thirty-dimensional predicted and reference profiles for observation n .

Let

$$\bar{\hat{\mathbf{y}}}_n = \frac{1}{30} \sum_{j=1}^{30} \hat{y}_{nj}, \quad \bar{\mathbf{y}}_n^{\text{ref}} = \frac{1}{30} \sum_{j=1}^{30} y_{nj}^{\text{ref}}. \quad (250)$$

The profile-level Pearson coefficient used by the S2 objective is therefore

$$r_n = \frac{\sum_{j=1}^{30} \left(\hat{y}_{nj} - \bar{\hat{\mathbf{y}}}_n \right) \left(y_{nj}^{\text{ref}} - \bar{\mathbf{y}}_n^{\text{ref}} \right)}{\sqrt{\sum_{j=1}^{30} \left(\hat{y}_{nj} - \bar{\hat{\mathbf{y}}}_n \right)^2 + \varepsilon} \sqrt{\sum_{j=1}^{30} \left(y_{nj}^{\text{ref}} - \bar{\mathbf{y}}_n^{\text{ref}} \right)^2 + \varepsilon}}, \quad (251)$$

with

$$\varepsilon = 10^{-8}. \quad (252)$$

The batch-level Pearson loss is the mean of the observation-level losses:

$$\mathcal{L}_{\text{Pearson}} = \frac{1}{N} \sum_{n=1}^N \mathcal{L}_{\text{Pearson},n}. \quad (253)$$

Accordingly, the complete S2 training objective can equivalently be written as

$$\mathcal{L}_{120,\text{S2}} = 0.50 \mathcal{L}_{\text{MSE}} + 0.38 \mathcal{L}_{\text{MAE}} + 0.12 (1 - \bar{r}), \quad (254)$$

where

$$\bar{r} = \frac{1}{N} \sum_{n=1}^N r_n. \quad (255)$$

This formulation is important because the Pearson contribution is computed from the internal structure of each predicted thirty-dimensional profile rather than from a single correlation obtained by pooling all output coordinates across observations.

Pearson correspondence therefore contributes both to optimisation and to evaluation, whereas Spearman correspondence is retained as a complementary evaluation metric and is not part of the S2 training loss. The distinction between the training objective and the evaluation metrics is maintained throughout model development.

The final S2 objective was selected as part of the internal IPIP-120 tuning programme. It should not be interpreted as an independently optimal psychometric objective, nor as a theoretically privileged weighting of error and correspondence. Its role is operational: it defines the loss function used to train the evaluated S2 configuration.

3.6.2 IPIP-NEO-300 Production Objective

The IPIP-NEO-300 development programme passed through several objective stages. An earlier compact-architecture objective-screening experiment identified a criterion based on

$$\mathcal{L}_H = 0.90 \mathcal{L}_{\text{MSE}} + 0.10 \mathcal{L}_{\text{Pearson}}, \quad (256)$$

where

$$\mathcal{L}_{\text{Pearson}} = 1 - r_{\text{Pearson}}. \quad (257)$$

This criterion belongs to the historical objective-screening stage and should not be confused with the final production objective.

The final IPIP-NEO-300 production configuration uses the P83.25 objective:

$$\boxed{\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - r_{\text{Pearson}})}. \quad (258)$$

The corresponding weighting can also be expressed as

$$w_{\text{MSE}} = 0.1675, \quad w_{\text{Pearson}} = 0.8325, \quad w_{\text{MSE}} + w_{\text{Pearson}} = 1. \quad (259)$$

For observation n , the implementation computes the profile-level Pearson coefficient across the thirty output facets. The training objective is thus

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 \left(1 - \frac{1}{N} \sum_{n=1}^N r_n \right), \quad (260)$$

with each r_n calculated over the thirty coordinates of the corresponding predicted and canonical reference profiles.

The P83.25 objective therefore differs from the earlier H objective in both its numerical weighting and its development role. The earlier objective belonged to objective screening under a fixed compact architecture, whereas P83.25 was used for the subsequent structured-output architecture refinement and final production training.

The two instrument-specific objectives are consequently treated as separate experimental specifications:

$$\mathcal{L}_{120,S2} \neq \mathcal{L}_{P83.25}. \quad (261)$$

No objective weighting is transferred automatically between instruments.

3.7 Subject-Isolated Exposure Schedule

Training uses a Subject-Isolated Exposure Schedule (SIES) to control the presentation of repeated perturbation views associated with the same participant.

Each complete training participant is represented by 143 predefined views:

$$3 \text{ clean} + 7 \times 20 \text{ corrupted} = 143 \quad (262)$$

views per participant.

The seven corruption families represented in the exposure schedule are

$$\left\{ \begin{array}{l} \text{CLEAN,} \\ \text{ITEM_SWAP,} \\ \text{FACET_SWAP,} \\ \text{CATEGORY_PARTIAL_SWAP,} \\ \text{CATEGORY_SWAP,} \\ \text{ITEM_COLLAPSE,} \\ \text{CATEGORY_COLLAPSE,} \\ \text{RANDOM_ITEM_CHANGE} \end{array} \right\}. \quad (263)$$

The three clean views function as repeated clean observations associated with the same participant, whereas the twenty views allocated to each corruption family provide repeated perturbation realizations. These counts define the fixed exposure protocol used by the reported training runs and are not presented as theoretically optimal sampling ratios.

The SIES design is subject-isolated at batch level. Within a training batch, duplicate observations from the same underlying participant are prohibited. Thus, if $\mathcal{B}$ denotes a batch of participant indices,

$$|\mathcal{B}| = |\text{Unique}(\mathcal{B})|. \quad (264)$$

This constraint prevents multiple views of the same participant from being simultaneously presented within one optimisation batch and reduces an artificial form of within-batch dependence arising solely from repeated perturbation views.

For the final IPIP-NEO-300 P83.25 production run, the SIES implementation uses a fixed per-subject initial rank and a deterministic epoch-dependent subject permutation. Let

$$R_s \sim \text{DiscreteUniform} \{0, \dots, 142\} \quad (265)$$

denote the initial view rank assigned to subject s by the dedicated SIES random generator. For epoch e and round r ,

$$V_{s,e,r} = (R_s + r) \bmod 143. \quad (266)$$

The subject order for each round is generated using an epoch-dependent pseudo-random generator whose seed is

$$S_{\text{epoch}} = 20,261,697 + 1,000,003 e. \quad (267)$$

A new permutation is therefore generated for each round while preserving the subject-isolation constraint. Because r spans all values

$$r = 0, \dots, 142, \quad (268)$$

each subject receives each of its 143 views exactly once per epoch, although the ordering of the participants changes between rounds.

This construction separates two distinct sources of variation: the view assignment is deterministic conditional on the SIES seed, whereas the order in which participants are presented to the optimiser changes between rounds and epochs according to the predefined schedule.

For the complete IPIP-NEO-300 production training population,

$$N_{\text{TRAIN},300} = 101,772 \quad (269)$$

subjects contributed

$$101,772 \times 143 = 14,553,396 \quad (270)$$

training examples per epoch.

For the final IPIP-NEO-120 S2 production programme,

$$N_{\text{TRAIN},120} = 287,263 \quad (271)$$

subjects contributed

$$287,263 \times 143 = 41,078,609 \quad (272)$$

training examples per epoch.

The difference between these totals reflects the instrument-specific training population sizes and does not represent a difference in the SIES exposure rule itself.

The corruption family, severity level, donor identity, perturbation construction metadata, and any other generation metadata remain external to the predictive input. The Gateway receives the response vector, not the metadata describing how that vector was generated.

This distinction is important because SIES controls exposure and ordering but does not provide the model with information about the perturbation mechanism.

3.8 Architecture and Objective Selection

The Gateway configurations were not specified as arbitrary multilayer perceptrons independently of the data-generation protocol. Model development used staged experimental procedures in which architectural geometry and optimisation objective were treated as distinguishable factors.

For IPIP-NEO-300, an initial architecture-screening programme evaluated candidate structured-output models on a 10,000-subject screening subset while the independent TEST population remained untouched. The screening included compact, wider, balanced, and deeper candidate geometries under a common P83.25 objective.

The initial structured control was

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}, \quad (273)$$

with 121,054 trainable parameters.

This architecture belongs to the screening stage and is therefore retained as a development reference rather than identified with the final production model.

The subsequent refinement evaluated wider hidden geometries, including the principal WIDE configuration

$$300 \rightarrow 320 \rightarrow 192 \rightarrow 128 \rightarrow \{6, 6, 6, 6, 6\}, \quad (274)$$

which contained 187,806 trainable parameters.

The final refined architecture was

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (275)$$

with 220,782 trainable parameters.

The refined architecture was evaluated in an internal architecture-refinement benchmark before full-population production retraining. Its subsequent full production fit was trained from scratch on the complete TRAIN population and was therefore not obtained by simply reusing the parameters learned during the 10,000-subject screening stage.

The final P83.25 Gateway contains a shared trunk followed by five structured output heads:

$$B_\theta : \mathbb{R}^{300} \longrightarrow \mathbb{R}^{184}, \quad (276)$$

and

$$H_k : \mathbb{R}^{184} \longrightarrow \mathbb{R}^6, \quad k = 1, \dots, 5. \quad (277)$$

The complete prediction mapping is therefore

$$F_{\theta, \text{P83.25}}^{(300)} = \text{Concat} (H_1, H_2, H_3, H_4, H_5) \circ B_\theta \circ \mathcal{N}, \quad (278)$$

where the fixed input normalisation is

$$\mathcal{N}(x) = \frac{x - 3}{2}. \quad (279)$$

The five heads jointly produce

$$5 \times 6 = 30 \quad (280)$$

output coordinates, so that

$$\hat{\mathbf{Y}} = \text{Concat} \left(\hat{\mathbf{Y}}^{(1)}, \hat{\mathbf{Y}}^{(2)}, \hat{\mathbf{Y}}^{(3)}, \hat{\mathbf{Y}}^{(4)}, \hat{\mathbf{Y}}^{(5)} \right) \in \mathbb{R}^{30}. \quad (281)$$

The five heads are not independent models. They are jointly conditioned on the same learned 184-dimensional hidden representation and are optimised simultaneously through the common

P83.25 objective.

The structured five-head arrangement corresponds to the five broad Five-Factor domains represented by the thirty facet-level outputs. This organisation is an architectural partition of the output coordinates; it does not imply statistical independence between domains or between heads.

3.9 Final IPIP-NEO-300 Production Architecture

The final production trunk consists of three hidden transformations with dimensions 344, 208, and 184. Each hidden transformation applies an affine mapping, Layer Normalization, GELU activation, and dropout with probability $p = 0.10$.

Let

$$\mathbf{h}^{(344)} = D_{0.10} \left[\text{GELU} \left(\text{LN}_1 \left(\mathbf{W}^{(1)} \mathcal{N}(\mathbf{x}) + \mathbf{b}^{(1)} \right) \right) \right], \quad (282)$$

where

$$\mathbf{W}^{(1)} \in \mathbb{R}^{344 \times 300}, \quad \mathbf{b}^{(1)} \in \mathbb{R}^{344}. \quad (283)$$

The second hidden transformation is

$$\mathbf{h}^{(208)} = D_{0.10} \left[\text{GELU} \left(\text{LN}_2 \left(\mathbf{W}^{(2)} \mathbf{h}^{(344)} + \mathbf{b}^{(2)} \right) \right) \right], \quad (284)$$

with

$$\mathbf{W}^{(2)} \in \mathbb{R}^{208 \times 344}, \quad \mathbf{b}^{(2)} \in \mathbb{R}^{208}. \quad (285)$$

The third hidden transformation is

$$\mathbf{h}^{(184)} = D_{0.10} \left[\text{GELU} \left(\text{LN}_3 \left(\mathbf{W}^{(3)} \mathbf{h}^{(208)} + \mathbf{b}^{(3)} \right) \right) \right], \quad (286)$$

with

$$\mathbf{W}^{(3)} \in \mathbb{R}^{184 \times 208}, \quad \mathbf{b}^{(3)} \in \mathbb{R}^{184}. \quad (287)$$

The final five linear heads are

$$\hat{\mathbf{Y}}^{(k)} = \mathbf{W}_{\text{out}}^{(k)} \mathbf{h}^{(184)} + \mathbf{b}_{\text{out}}^{(k)}, \quad k = 1, \dots, 5, \quad (288)$$

where

$$\mathbf{W}_{\text{out}}^{(k)} \in \mathbb{R}^{6 \times 184}, \quad \mathbf{b}_{\text{out}}^{(k)} \in \mathbb{R}^6. \quad (289)$$

No output activation function is applied after the final linear heads, and no post-hoc output clipping is applied during production inference.

The exact trainable parameter count is

$$\begin{aligned} N_\theta &= (300 \times 344 + 344) + 2(344) \\ &\quad + (344 \times 208 + 208) + 2(208) \\ &\quad + (208 \times 184 + 184) + 2(184) \\ &\quad + 5(184 \times 6 + 6) \\ &= 220,782. \end{aligned} \quad (290)$$

The $2d$ terms correspond to the trainable scale and bias parameters of the three Layer Normalization layers. GELU and dropout introduce no trainable parameters.

For the first hidden layer, the bias vector therefore satisfies

$$\mathbf{b}^{(1)} \in \mathbb{R}^{344}, \quad (291)$$

so that the first-layer bias diagnostic contains 344 learned affine offsets.

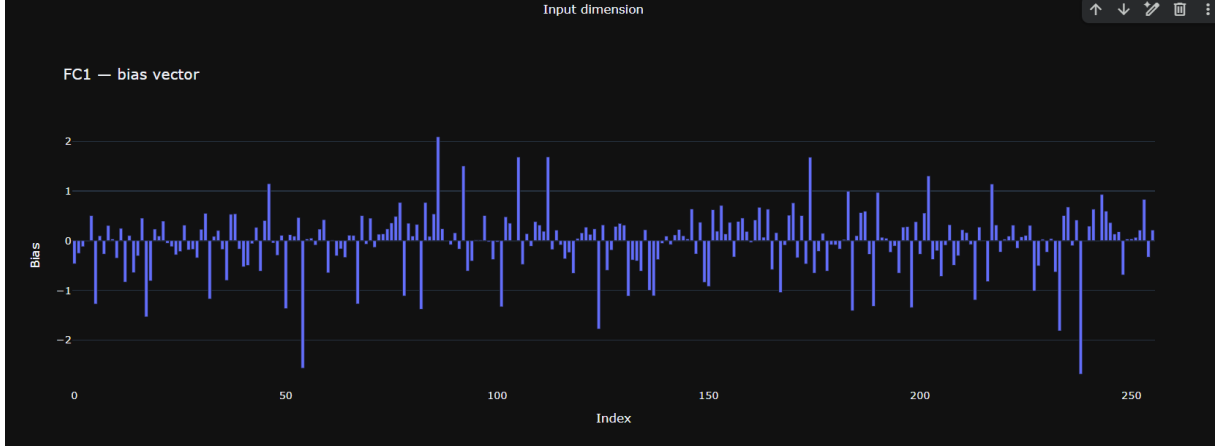


Figure 14: Learned bias vector of the first fully connected layer (FC1) of the IPIP-NEO-300 P83.25 production Gateway. Each bar corresponds to one of the 344 neurons in the first hidden representation. The distribution represents the learned affine offsets of the frozen production architecture and is used solely as a parameter-level diagnostic.

For neuron j in the first hidden representation, the corresponding pre-normalisation affine value is

$$z_j = \sum_{i=1}^{300} W_{ji}^{(1)} \mathcal{N}(x_i) + b_j^{(1)}, \quad j = 1, \dots, 344. \quad (292)$$

The bias distribution is interpreted solely as a parameter-level diagnostic. It is not treated as a direct measure of neuron importance, item importance, facet importance, or psychological meaning.

The hidden representations of the final production model have dimensions 344, 208, and 184, respectively.

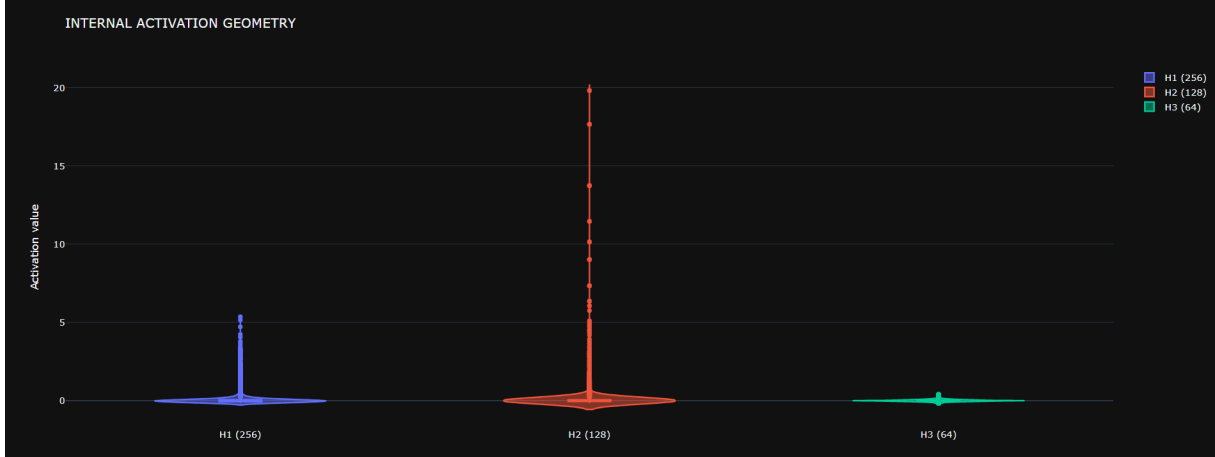


Figure 15: Internal activation distributions across the three hidden representations of the IPIP-NEO-300 P83.25 production Gateway. H1, H2, and H3 correspond to the 344-, 208-, and 184-dimensional hidden stages, respectively. The distributions characterise the numerical states produced by the frozen production architecture under the evaluated observations.

The activation distributions constitute descriptive diagnostics of the numerical state of the network. They are not interpreted as direct evidence of psychological constructs, feature importance, causal mechanisms, or functional necessity of individual hidden dimensions.

3.10 Representative Output Geometry

For a representative clean observation, the 30-dimensional Gateway prediction is compared directly with its canonical reference profile. Let

$$\hat{\mathbf{Y}}_{\text{Gateway}}^{\text{clean}} = F_{\theta, \text{P83.25}}^{(300)}(X_{\text{clean}}), \quad (293)$$

and let

$$\mathbf{Y}_{\text{Gold}} = g(X_{\text{clean}}) \in \mathbb{R}^{30} \quad (294)$$

denote the corresponding canonical reference representation.

The profile comparison provides a coordinate-level descriptive view of the relationship between the learned reconstruction and the fixed canonical reference.

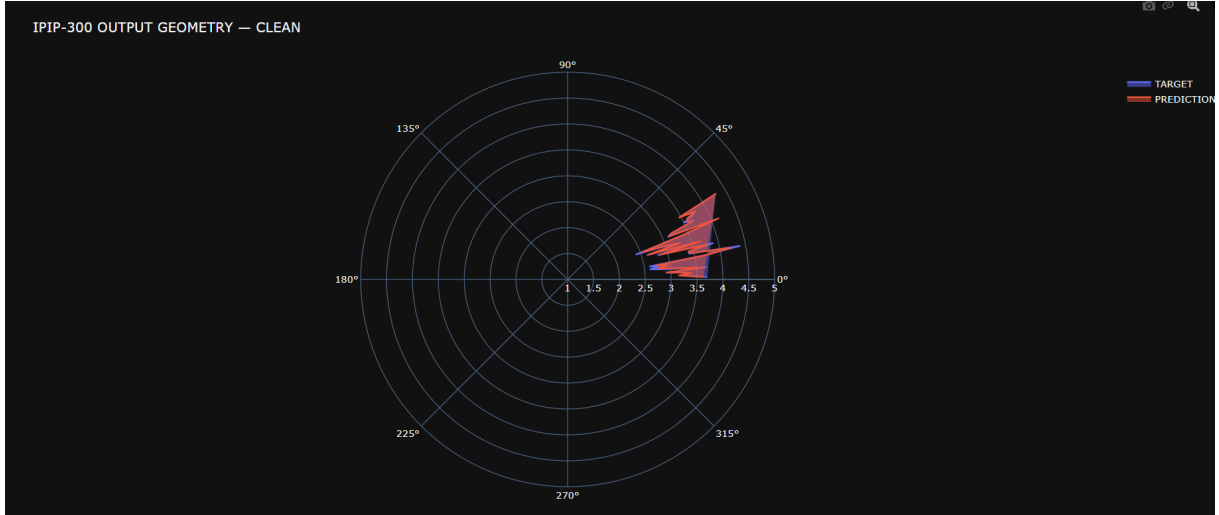


Figure 16: IPIP-NEO-300 output geometry for a representative clean observation. The P83.25 Gateway prediction is shown together with the canonical reference across the 30-dimensional psychometric representation. The visualization is descriptive and does not constitute an additional performance benchmark.

For the corresponding corrupted observation,

$$\hat{\mathbf{Y}}_{\text{Gateway}}^{\text{corr}} = F_{\theta, \text{P83.25}}^{(300)}(X_{\text{corr}}), \quad (295)$$

whereas the canonical reference remains fixed at

$$\mathbf{Y}_{\text{Gold}} = g(X_{\text{clean}}). \quad (296)$$

The reference is therefore not recomputed from X_{corr} . The clean and corrupted output profiles are compared against the same participant- specific canonical reference.

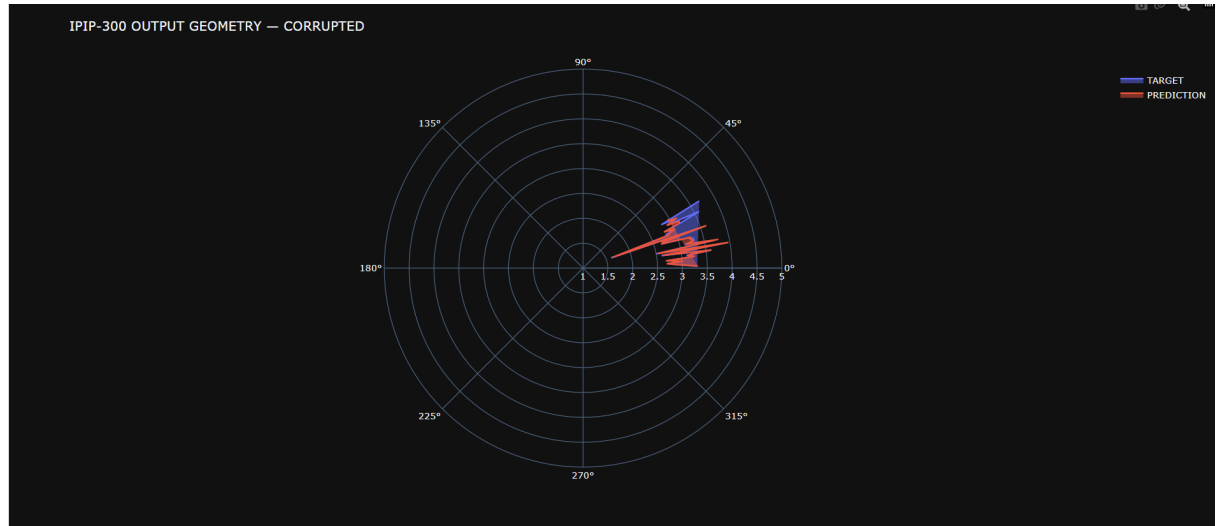


Figure 17: IPIP-NEO-300 output geometry for a representative corrupted observation. The P83.25 Gateway prediction is shown relative to the unchanged canonical reference across the 30-dimensional psychometric representation.

The clean and corrupted profiles provide complementary descriptive views of output-space behaviour. They do not constitute additional performance benchmarks, do not alter the model-

selection procedure, and do not replace the predefined aggregate error and correspondence metrics.

3.11 Neuron-Level Architectural Representation and Developmental Lineage

The neuron-level visualisations retained in this manuscript serve two complementary purposes. First, they provide an intuitive representation of the computational geometry of the Gateway. Second, and importantly, they preserve a visual record of the architectural development process through which the final IPIP-NEO-300 configuration was obtained.

The architectural development was sequential rather than static. The initial structured-control configuration used during the screening stage was

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}, \quad (297)$$

with 121,054 trainable parameters.

This compact architecture constituted an explicit development-stage reference and was used to investigate the behaviour of a structured five-head output configuration before subsequent width refinement. It therefore represents an earlier state of the research trajectory and is retained in the present manuscript as a documented architectural milestone rather than being identified with the final production checkpoint.

The subsequent development process explored wider hidden representations. Among these configurations, the principal WIDE architecture was

$$300 \rightarrow 320 \rightarrow 192 \rightarrow 128 \rightarrow \{6, 6, 6, 6, 6\}, \quad (298)$$

with 187,806 trainable parameters.

The final refined P83.25 production architecture subsequently adopted a different hidden geometry:

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (299)$$

with 220,782 trainable parameters.

Accordingly, the architectural states should be distinguished as

$$\text{structured control} \rightarrow \text{width refinement} \rightarrow \text{final P83.25 production model}. \quad (300)$$

This chronology is important for interpreting the retained visualisations. The figures generated during the earlier development stages were not retroactively modified when the production architecture changed. They are therefore intentionally preserved in their original form, including their original neuron counts and geometric structure. Their purpose is documentary: they provide an illustrative visual trace of the architectural trajectory followed during the investigation.

In particular, the historical neuron-level visualisations retained in Figures 18, 19, 22, and 23 correspond to the earlier compact structured-control representation rather than to the final P83.25 topology.

For the historical compact architecture, the number of layer-level scalar computational coordinates is

$$300 + 256 + 128 + 64 + 30 = 778. \quad (301)$$

The corresponding number of weighted affine inter-layer connections is

$$\begin{aligned}
C_{\text{historical}} &= 300 \times 256 + 256 \times 128 + 128 \times 64 + 64 \times 30 \\
&= 119,680.
\end{aligned} \tag{302}$$

These values define the structural quantities represented by the retained historical graph visualisations. The five output groups contain six scalar coordinates each, giving

$$5 \times 6 = 30 \tag{303}$$

output coordinates.

The historical graph quantities must therefore not be conflated with the corresponding quantities of the final P83.25 production architecture. For the final model, the authoritative structural specification is given directly by the mathematical architecture and parameter decomposition reported above, namely 1,066 layer-level coordinates and 218,544 weighted inter-layer connections.

The distinction can be stated explicitly as

$$\begin{aligned}
&\text{historical visualisation: } 300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 5 \times 6, \\
&\text{final production model: } 300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6.
\end{aligned} \tag{304}$$

Thus, the retained figures should be interpreted as visual documentation of the research timeline rather than as replacement specifications for the final production checkpoint.



Figure 18: Historical development-stage diagnostic of the first fully connected layer (FC1) of the IPIP-NEO-300 Gateway. The visualisation corresponds to the earlier compact structured-control architecture $300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 5 \times 6$, whose first hidden representation contained 256 neurons. Each bar represents the learned affine bias associated with one of those 256 hidden units. The figure is retained in its original form as a visual record of the architectural screening stage and is not presented as a depiction of the final P83.25 production topology.

For the historical first hidden representation, the corresponding affine pre-normalisation value for neuron j is

$$z_j = \sum_{i=1}^{300} W_{ji}^{(1)} \mathcal{N}(x_i) + b_j^{(1)}, \quad j = 1, \dots, 256. \tag{305}$$

The displayed bias distribution is therefore a parameter-level diagnostic of the historical development-stage model. It is not interpreted as a direct measure of neuron importance, item importance, facet importance, psychological meaning, or causal relevance.

The internal activation visualisation likewise belongs to the same developmental lineage. In the historical compact configuration, the three hidden representations were

$$256, \quad 128, \quad 64, \quad (306)$$

respectively.

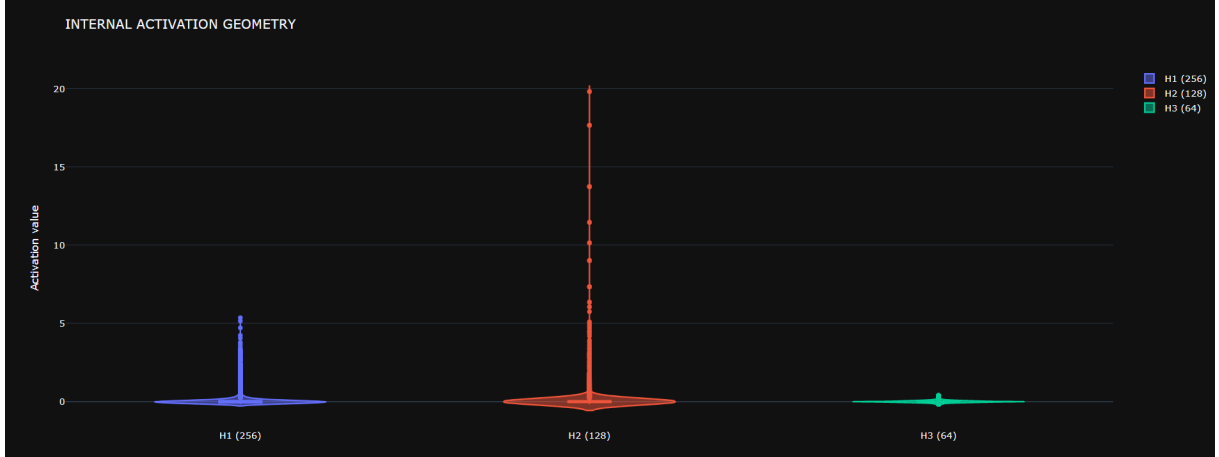


Figure 19: Historical internal activation visualisation from the IPIP-NEO-300 Gateway development process. The three hidden stages correspond to the 256-, 128-, and 64-dimensional representations of the earlier compact structured-control architecture. The figure is intentionally preserved as an illustrative record of the internal geometry examined during architectural screening and should not be read as a dimensionality diagram of the final P83.25 production model.

The activation distributions shown in this historical figure describe numerical states generated by the earlier network configuration. They are retained to document how intermediate representations were inspected during model development. They are not interpreted as direct evidence of psychological constructs, latent psychological dimensions, feature importance, causal mechanisms, or functional necessity of individual hidden coordinates.

3.12 Historical Output-Space Visualisations

The output-space figures provide a second component of the developmental record. They preserve representative visual examinations of Gateway reconstruction behaviour carried out during the earlier stages of the IPIP-NEO-300 investigation.

These visualisations should be distinguished from the aggregate quantitative TEST evaluation. Their function is descriptive and illustrative: they expose the geometry of representative thirty-dimensional output profiles and document how the reconstruction problem was inspected during development.

For a representative clean observation from the corresponding development stage, let

$$\hat{\mathbf{Y}}_{\text{Gateway}}^{\text{clean}} = F_{\theta}^{\text{hist}}(X_{\text{clean}}), \quad (307)$$

where F_{θ}^{hist} denotes the historical development-stage Gateway associated with the visualisation.

The corresponding canonical reference remains

$$\mathbf{Y}_{\text{Gold}} = g(X_{\text{clean}}) \in \mathbb{R}^{30}. \quad (308)$$

The resulting comparison provides a descriptive view of the relationship between the learned reconstruction and the fixed canonical reference.

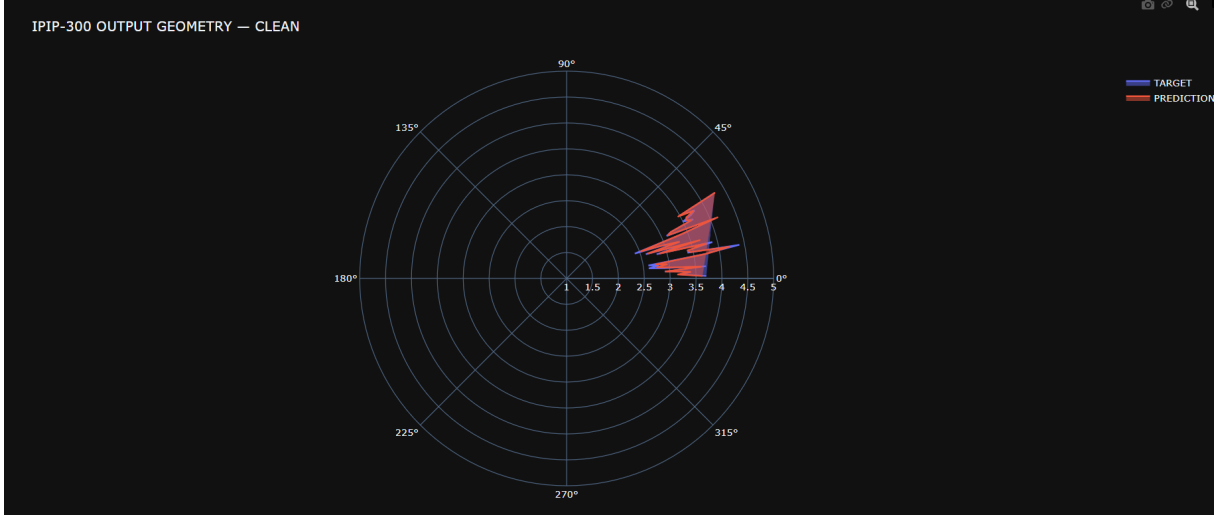


Figure 20: Historical IPIP-NEO-300 output-space visualisation for a representative clean observation. The figure belongs to the developmental and architectural screening phase of the Gateway study and is retained to illustrate the inspection of the thirty-dimensional reconstruction space at that stage. It constitutes a descriptive research-process visualisation rather than an additional quantitative performance benchmark or a direct representation of the final P83.25 production architecture.

For the corresponding corrupted observation, the same developmental-stage Gateway is evaluated against the participant-specific reference associated with the original clean observation:

$$\hat{\mathbf{Y}}_{\text{Gateway}}^{\text{corr}} = F_{\theta}^{\text{hist}}(X_{\text{corr}}), \quad (309)$$

while

$$\mathbf{Y}_{\text{Gold}} = g(X_{\text{clean}}). \quad (310)$$

The canonical reference is therefore not recomputed from the corrupted observation. The clean and corrupted examples remain linked to the same participant-specific canonical reference according to the study’s evaluation contract.

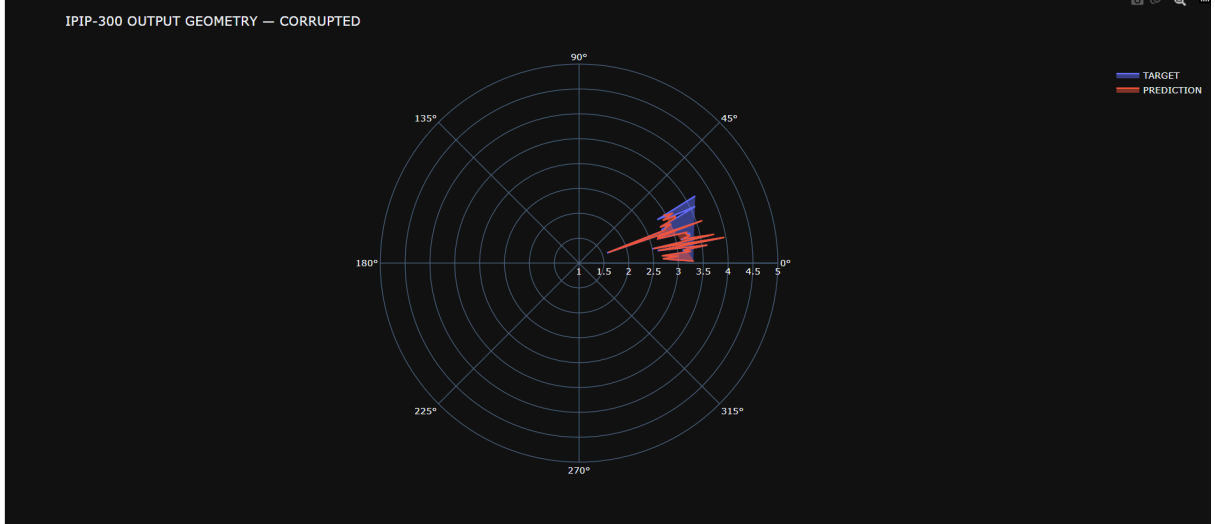


Figure 21: Historical IPIP-NEO-300 output-space visualisation for a representative corrupted observation. The figure records the developmental inspection of Gateway reconstruction under perturbation and is retained as part of the visual research timeline. The canonical reference remains associated with the original clean observation. The visualisation is descriptive and is not used as an independent performance estimate or as the definition of the final P83.25 production topology.

The clean and corrupted output figures therefore document two complementary developmental observations: behaviour under an unperturbed input and behaviour under a structured perturbation. Their inclusion preserves the historical progression of the investigation without introducing an additional evaluation protocol or altering the predefined model-selection procedure.

3.13 Historical Neuron-Level Graph Representation

The retained three-dimensional graph visualisation corresponds specifically to the earlier compact structured-control architecture:

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 30. \quad (311)$$

Its scalar layer-level coordinate count is

$$300 + 256 + 128 + 64 + 30 = 778, \quad (312)$$

and its weighted inter-layer connection count is

$$\begin{aligned} C_{\text{historical}} &= 300 \times 256 + 256 \times 128 + 128 \times 64 + 64 \times 30 \\ &= 119,680. \end{aligned} \quad (313)$$

These counts reproduce the structural specification represented by the historical graph visualisation and explain the scale of the graph displayed in the retained screenshot.

The graph should be interpreted as a reconstruction of the network topology used during the earlier development stage. It is not a visualisation of the final P83.25 checkpoint. The latter has the distinct topology $300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6$ and is specified formally elsewhere in this manuscript.

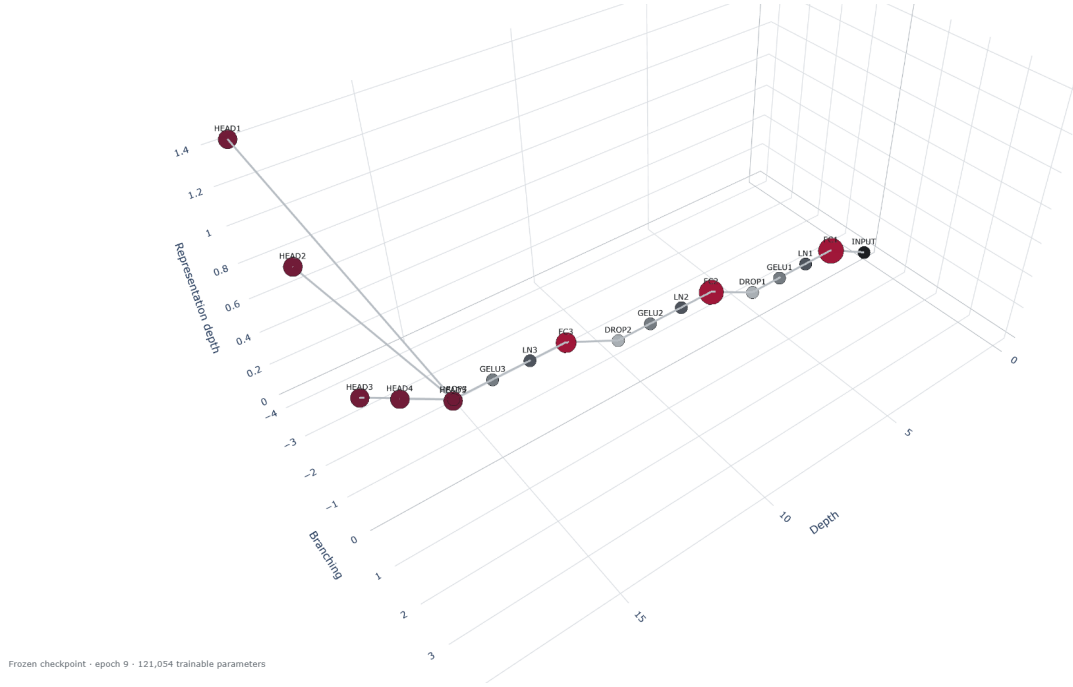
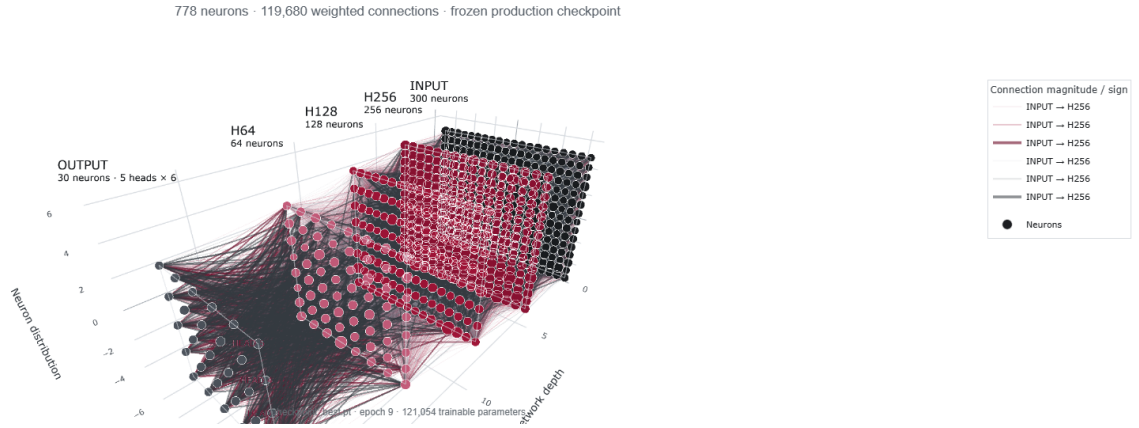


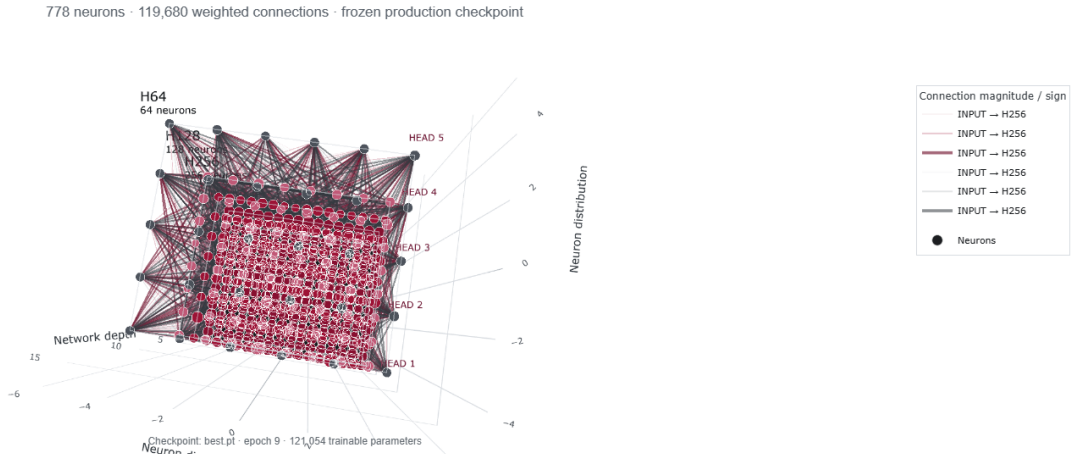
Figure 22: Historical three-dimensional graph representation of the IPIP-NEO-300 Gateway during the architectural development process. The retained visualisation corresponds to the compact structured-control topology with 300 input coordinates, hidden representations of 256, 128, and 64 units, and thirty structured output coordinates. The graph therefore contains 778 layer-level coordinates and 119,680 weighted inter-layer connections. The figure is preserved as an illustrative record of the research timeline and is not presented as the final P83.25 production architecture.

The graph is a structural representation of the earlier computational network. It is not an additional performance evaluation, does not constitute evidence for causal neural mechanisms, and does not provide an independent statistical observation. Its value is documentary and explanatory: it makes the progression from the initial compact architecture to the later refined production architecture visually traceable.

The same historical graph is retained from complementary viewing angles.



(a) Primary three-dimensional view of the historical compact structured-control Gateway architecture retained from the development stage.



(b) Complementary three-dimensional viewing angle of the same historical compact structured-control Gateway architecture.

Figure 23: Complementary views of the historical IPIP-NEO-300 Gateway graph retained from the architectural development process. Both panels depict the same computational graph associated with the $300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 5 \times 6$ structured-control configuration, comprising 778 layer-level coordinates and 119,680 weighted inter-layer connections. The two panels are not independent observations, independent model evaluations, or replications. Their purpose is solely to preserve different geometric views of the same developmental-stage architecture and to expose the research timeline through which the final production configuration was subsequently refined.

The preservation of these figures is intentional. Because the original development-stage screenshots constitute fixed research artefacts, the manuscript does not replace, redraw, or retroactively alter them to imitate the final production architecture. Instead, the mathematical specification of each architectural stage is stated explicitly, allowing the visual artefacts to remain faithful to the historical state in which they were generated.

This distinction also prevents a potential interpretive ambiguity for the reader. The figures document the evolution of the research programme, whereas the final production architecture, parameter count, training objective, checkpoint identity, and final evaluation results define the model used for the principal quantitative conclusions.

Accordingly, the visual lineage should be read chronologically as

$$\begin{array}{c}
300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 5 \times 6 \\
\downarrow \\
300 \rightarrow 320 \rightarrow 192 \rightarrow 128 \rightarrow 5 \times 6 \\
\downarrow \\
300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6
\end{array} \tag{314}$$

where the first stage corresponds to the historical visualisations retained in the present manuscript, the second stage represents subsequent width refinement, and the third stage is the final P83.25 production architecture.

The visualisations therefore function as an illustrative architectural timeline: they preserve how the model was examined, expanded, and refined during the investigation while leaving the final mathematical and quantitative specification unambiguous.

3.14 Inference Contract and Information Isolation

At inference time, the Gateway receives only the observed item-response vector:

$$X_{\text{input}} = X_{\text{corr}}. \tag{315}$$

No participant identifier, donor identifier, corruption-family label, severity code, perturbation-pattern identifier, or canonical reference vector is provided to the model.

The predictive information flow can therefore be expressed as

$$X_{\text{corr}} \xrightarrow{\mathcal{N}} [-1, 1]^d \xrightarrow{F_\theta} \hat{\mathbf{Y}} \in \mathbb{R}^{30}. \tag{316}$$

The normalisation mapping is

$$\mathcal{N}(x_i) = \frac{x_i - 3}{2}, \quad x_i \in \{1, 2, 3, 4, 5\}, \tag{317}$$

so that

$$x_i = 1 \mapsto -1, \quad x_i = 3 \mapsto 0, \quad x_i = 5 \mapsto 1. \tag{318}$$

The canonical reference representation enters the training procedure as the target and the evaluation procedure as the reference quantity. It is never provided as an inference input.

Likewise, the corruption metadata remain external to the predictive input. Consequently, the Gateway is not trained as an explicit corruption-family classifier and is not given direct access to the mechanism responsible for the observed perturbation.

The information-isolation contract can therefore be summarised as

$$\left\{ \begin{array}{l}
\text{Predictive input: } X_{\text{corr}}, \\
\text{Target during training: } Y_{\text{Gold}}, \\
\text{Corruption metadata: external,} \\
\text{Participant identity: external,} \\
\text{Donor identity: external,} \\
\text{Reference at inference: unavailable.}
\end{array} \right. \tag{319}$$

This separation is maintained because the objective of the Gateway is reconstruction from the observed response configuration rather than explicit identification of the perturbation process.

3.15 Output Range and Non-Clipping Policy

The Gateway does not apply post-hoc clipping of its outputs to the nominal questionnaire scoring interval.

Accordingly, the production model’s native output is retained as generated by the final linear output heads:

$$\hat{\mathbf{Y}} = F_{\theta}(X_{\text{corr}}), \quad (320)$$

with no final operation of the form

$$\hat{y}_j \leftarrow \min(5, \max(1, \hat{y}_j)). \quad (321)$$

Predictions outside the nominal $[1, 5]$ interval therefore remain observable and are retained in the model diagnostics rather than being concealed by a post-processing constraint.

This policy separates two quantities that would otherwise be conflated: the native numerical output of the learned model and a subsequently clipped version of that output. Primary reconstruction metrics are computed from the production predictions without post-hoc clipping.

The absence of clipping does not imply that every production prediction lies within the questionnaire range. Rather, it ensures that any such deviation is attributable to the learned model output itself and can be quantified explicitly.

The output range policy is consequently treated as part of the inference contract and is kept fixed between development and final evaluation.

4. Experimental Design

4.1 Instruments and Experimental Scope

The present study evaluates two complementary configurations of the IPIP-NEO family: the 300-item public-domain IPIP-NEO inventory and the 120-item short form developed by Johnson. The IPIP-NEO-120 was explicitly developed as a shorter form of the 300-item IPIP-NEO while retaining its thirty-facet Five-Factor Model organization, with four items per facet (**goldberg1999ipip**; **johnson2014ipipneo120**). The full IPIP-NEO configuration contains ten items per facet across the same thirty-facet organization (**goldberg1999ipip**).

The two configurations are treated as complementary instances of the same reconstruction problem rather than as identical architectural replications. Both preserve a thirty-dimensional reference representation, while differing in the dimensionality and redundancy of the observed item-response space. Accordingly,

$$d_{120} = 120, \quad d_{300} = 300, \quad p = 30, \quad (322)$$

where d denotes the input dimensionality and p denotes the dimensionality of the canonical reference representation.

The present empirical evaluation is restricted to the IPIP-NEO-120 and IPIP-NEO-300 configurations. The Gateway framework is designed as an extensible methodology for other established multidimensional personality inventories. Planned extensions include the HEXACO Personality Inventory–Revised (HEXACO-PI-R), particularly its 100-item and 200-item configurations, as well as other psychometrically documented instruments with established scoring frameworks (**lee2018hexaco**; **ashton2008hexaco**).

These extensions constitute future work and are not used to support any empirical claim reported in the present manuscript.

Table 5: Instrument and Gateway configurations evaluated in the present study.

Configuration	Reference basis	Items	Domains	Facets	Items/Facet	Gateway Parameters
IPIP-120	Johnson (2014) short form	120	5	30	4	69,214
IPIP-300	Goldberg (1999) IPIP-NEO	300	5	30	10	220,782

The two configurations therefore differ in observational dimensionality while sharing the same thirty-facet output structure. They should be interpreted as complementary evaluations of the reconstruction formulation, not as literal replications of a single architecture ([johnson2014ipipneo120](#)).

For both configurations, the deterministic comparator is the conventional instrument scoring procedure defined by the corresponding item keys and facet-item assignments. The Gateway receives the complete observed item-response vector and maps it to the same thirty-dimensional canonical reference space.

The exact architecture, parameter count, preprocessing procedure, random seeds, optimization settings, and model version are recorded in the reproducibility code and associated model metadata. The reported parameter counts therefore identify the evaluated model configurations rather than merely their nominal layer widths.

4.2 Traditional Scoring Baseline

Traditional scoring applies the established deterministic scoring rules of the corresponding instrument, including reverse scoring for negatively keyed items followed by aggregation within the designated facet-item sets. The same deterministic scoring implementation used to construct the canonical reference psychometric representation is used as the comparator.

For a clean observation,

$$\tilde{\mathbf{X}} = \mathbf{X}_{\text{clean}} \implies g(\tilde{\mathbf{X}}) = g(\mathbf{X}_{\text{clean}}) = . \quad (323)$$

Thus, exact agreement between traditional scoring and the canonical reference on clean observations is a property of the experimental construction rather than an empirical performance result.

For perturbed observations, the head-to-head comparison is instead defined by

$$= g(\tilde{\mathbf{X}}), \quad \hat{\mathbf{Y}} = \mathcal{F}_{\theta}(\tilde{\mathbf{X}}), \quad (324)$$

with both outputs evaluated against the same canonical reference

$$= g(\mathbf{X}_{\text{clean}}). \quad (325)$$

This establishes a direct comparison in which both methods receive the same observed input while differing in the transformation applied to that input.

4.3 Immutable Reference Construction

For every complete participant, the canonical reference psychometric representation is computed once from the original, unperturbed response vector:

$$= g(\mathbf{X}_{\text{clean}}). \quad (326)$$

The perturbation operator is subsequently applied only to the observed response vector:

$$\tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}}). \quad (327)$$

The reference representation is invariant under the subsequent perturbation:

$$\boxed{= g(\mathbf{X}_{\text{clean}}) \text{ and remains fixed for all } \tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}})}. \quad (328)$$

Consequently, the perturbation process cannot redefine, overwrite, or otherwise alter the participant-specific reference target. The target is neither recomputed from the perturbed vector nor modified according to the perturbation family, severity, donor identity, or realization-specific transformation.

The experimental information flow is therefore

$$\boxed{\mathbf{X}_{\text{clean}} \xrightarrow{g}} \quad \text{and} \quad \boxed{\mathbf{X}_{\text{clean}} \xrightarrow{\mathcal{C}_\phi} \tilde{\mathbf{X}}}. \quad (329)$$

The two reconstruction procedures then operate on the identical perturbed observation:

$$\boxed{\tilde{\mathbf{X}} \rightarrow \begin{cases} = g(\tilde{\mathbf{X}}), \\ \hat{\mathbf{Y}} = \mathcal{F}_\theta(\tilde{\mathbf{X}}) \end{cases} \rightarrow .} \quad (330)$$

This invariant is central to the experimental design. It ensures that the task being evaluated is reconstruction toward a participant-specific reference established before perturbation, rather than rescoring the perturbed response vector and treating the perturbed state as a new target.

4.4 Corruption Families

Seven perturbation families are evaluated:

1. ITEM_SWAP
2. FACET_SWAP
3. CATEGORY_PARTIAL_SWAP
4. CATEGORY_SWAP
5. ITEM_COLLAPSE
6. CATEGORY_COLLAPSE
7. RANDOM_ITEM_CHANGE

Each family is evaluated across a predefined severity structure. The resulting evaluation therefore distinguishes perturbation mechanism from perturbation magnitude and permits family-level and severity-level analysis.

4.5 IPIP-120 Validation and Independent Test

The IPIP-120 validation benchmark contains 61,556 clean observations and 338,444 corrupted observations.

The final independent IPIP-120 TEST benchmark contains 61,557 clean observations and 1,723,596 corrupted observations, for a total of 1,785,153 observations. The TEST population remained fully outside model training, hyperparameter selection, architecture selection, objective selection, and other model-development procedures.

4.6 IPIP-300 Validation and Independent Test

The IPIP-300 validation benchmark contains 21,808 clean observations and 338,444 corrupted observations, for a total of 360,252 observations.

An independent IPIP-300 TEST population was reserved during dataset construction and remained outside training, architecture selection, objective selection, hyperparameter tuning, and final checkpoint selection. After the production model and evaluation protocol had been fixed, the independent TEST benchmark was materialised, integrity-checked, and frozen for final evaluation.

The final independent IPIP-300 TEST benchmark contains 21,808 clean observations and 610,624 corrupted observations, for a total of 632,432 observations. The TEST benchmark was therefore evaluated only after the production configuration had been fixed, preserving its role as an independent held-out evaluation population.

4.7 Evaluation Metrics

The primary error metric is root mean squared error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i^{\text{ref}})^2}. \quad (331)$$

The secondary absolute error metric is mean absolute error:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i^{\text{ref}}|. \quad (332)$$

RMSE is designated as the primary robustness metric because the reconstruction task is particularly concerned with preventing large individual deviations between the reconstructed and canonical facet representations. Squaring the residual before averaging assigns progressively greater weight to larger errors, such that an unusually large reconstruction failure contributes disproportionately more to the aggregate error than a collection of small deviations. This property is useful when the objective is to identify and control potentially consequential reconstruction failures rather than to treat all absolute deviations as equally costly.

This consideration is especially relevant to psychometric reconstruction in contexts where predicted facet values may subsequently contribute to profile interpretation. A large error concentrated in one or more facets can alter the shape of the reconstructed multivariate profile more substantially than an equivalent number of small deviations distributed across observations. Accordingly, the squared-error emphasis of RMSE provides a conservative criterion for detecting such large reconstruction discrepancies.

The same rationale applies to the broader methodological context of clinical and medical prediction, where continuous prediction performance is commonly assessed through measures

of overall error and where the appropriate metric depends on the consequences associated with different types of prediction error. RMSE is therefore not assumed to be universally superior to MAE; rather, it is used here as the primary metric because the experimental objective places greater importance on controlling large reconstruction deviations. Clinical prediction guidance likewise emphasizes that model performance should be assessed using measures appropriate to the outcome and intended use rather than relying on a single universal metric (**collins2015tripod**).

MAE is retained as a complementary metric because it provides an immediately interpretable measure of the average absolute deviation and is less dominated by a relatively small number of large residuals. Consequently, the joint reporting of RMSE and MAE distinguishes two practically different properties of reconstruction quality: the control of large errors and the typical absolute magnitude of prediction error.

This distinction is particularly important for the present IPIP-NEO-120 S2 results. Relative to V133, the final S2 configuration produced a small increase in corrupted-validation RMSE of +0.000194 while reducing MAE by -0.005228 . The final configuration was therefore not selected because it minimized every error metric simultaneously, but because it preserved the primary RMSE robustness criterion within predefined guardrails while reducing the broader absolute-error burden. RMSE and MAE are consequently interpreted jointly rather than interchangeably.

Profile-level correspondence is evaluated using Pearson and Spearman correlation. Pearson correlation is used both as an evaluation metric and as a component of the composite training objective, whereas Spearman correlation is retained exclusively as an evaluation measure of rank-order correspondence.

4.8 Reproducibility Controls

The final IPIP-NEO-120 S2 Gateway was trained and confirmed under the following core configuration:

Table 6: Core training and reproducibility configuration for the final IPIP-NEO-120 S2 Gateway.

Parameter	Value
Architecture	120 $\rightarrow$ 192 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 30
Parameters	69,214
Optimizer	AdamW
Batch size	8,192
Learning rate	0.002
Weight decay	0.0001
Dropout	0.10
Gradient clipping	1.0
Mixed precision	BF16
MSE weight	0.50
MAE weight	0.38
Pearson weight	0.12
Confirmation seeds	20,260,931; 20,260,932; 20,260,933
Selected seed	20,260,933
Selected epoch	6
SIES rounds	143
Guardrail confirmation	3/3 seeds passed
MAE improvement	3/3 seeds
TEST exposure	Not loaded / not touched

For the IPIP-NEO-120 S2 configuration, the training objective is

$$\mathcal{L}_{120,S2} = 0.50\mathcal{L}_{\text{MSE}} + 0.38\mathcal{L}_{\text{MAE}} + 0.12\mathcal{L}_{\text{Pearson}}, \quad (333)$$

where

$$\mathcal{L}_{\text{Pearson}} = 1 - r_{\text{Pearson}}. \quad (334)$$

The final production checkpoint is frozen and identified by its architecture, selected seed, selected epoch, parameter count, model metadata, and cryptographic SHA-256 digest. The final IPIP-NEO-120 S2 checkpoint corresponds to seed 20,260,933, epoch 6, and the 69,214-parameter architecture reported above.

The three-seed confirmation procedure is part of the reproducibility protocol rather than an additional optimization stage. All three confirmation seeds satisfied the predefined RMSE, Pearson, and Spearman guardrails relative to V133 and achieved lower corrupted-validation MAE. The independent IPIP-NEO-120 TEST population remained outside architecture selection, objective selection, hyperparameter tuning, confirmatory retraining, and final checkpoint selection.

All seeds, architectural specifications, preprocessing rules, optimization parameters, objective weights, checkpoint metadata, and integrity hashes are retained in the reproducibility package.

5. Experimental Design

5.1 Instruments and Experimental Scope

The present study evaluates two complementary configurations of the IPIP-NEO family: the 300-item public-domain IPIP-NEO inventory and the 120-item short form developed by Johnson. The IPIP-NEO-120 was explicitly developed as a shorter form of the 300-item IPIP-NEO while retaining its thirty-facet Five-Factor Model organization, with four items per facet (**goldberg1999ipip**; **johnson2014ipipneo120**). The full IPIP-NEO configuration contains ten items per facet across the same thirty-facet organization (**goldberg1999ipip**).

The two configurations are treated as complementary instances of the same reconstruction problem rather than as identical architectural replications. Both preserve a thirty-dimensional reference representation, while differing in the dimensionality and redundancy of the observed item-response space. Accordingly,

$$d_{120} = 120, \quad d_{300} = 300, \quad p = 30, \quad (335)$$

where d denotes the input dimensionality and p denotes the dimensionality of the canonical reference representation.

The present empirical evaluation is restricted to the IPIP-NEO-120 and IPIP-NEO-300 configurations. The Gateway framework is designed as an extensible methodology for other established multidimensional personality inventories. Planned extensions include the HEXACO Personality Inventory–Revised (HEXACO-PI-R), particularly its 100-item and 200-item configurations, as well as other psychometrically documented instruments with established scoring frameworks (**lee2018hexaco**; **ashton2008hexaco**).

These extensions constitute future work and are not used to support any empirical claim reported in the present manuscript.

Table 7: Instrument and Gateway configurations evaluated in the present study.

Configuration	Reference basis	Items	Domains	Facets	Items/Facet	Gateway Parameters
IPIP-120	Johnson (2014) short form	120	5	30	4	69,214
IPIP-300	Goldberg (1999) IPIP-NEO	300	5	30	10	220,782

The two configurations therefore differ in observational dimensionality while sharing the same thirty-facet output structure. They should be interpreted as complementary evaluations of the reconstruction formulation, not as literal replications of a single architecture (**johnson2014ipipneo120**).

For both configurations, the deterministic comparator is the conventional instrument scoring procedure defined by the corresponding item keys and facet-item assignments. The Gateway receives the complete observed item-response vector and maps it to the same thirty-dimensional canonical reference space.

The exact architecture, parameter count, preprocessing procedure, random seeds, optimization settings, and model version are recorded in the reproducibility code and associated model metadata. The reported parameter counts therefore identify the evaluated model configurations rather than merely their nominal layer widths.

5.2 Traditional Scoring Baseline

Traditional scoring applies the established deterministic scoring rules of the corresponding instrument, including reverse scoring for negatively keyed items followed by aggregation within the designated facet-item sets. The same deterministic scoring implementation used to construct the canonical reference psychometric representation is used as the comparator.

For a clean observation,

$$\tilde{\mathbf{X}} = \mathbf{X}_{\text{clean}} \implies g(\tilde{\mathbf{X}}) = g(\mathbf{X}_{\text{clean}}) = . \quad (336)$$

Thus, exact agreement between traditional scoring and the canonical reference on clean observations is a property of the experimental construction rather than an empirical performance result.

For perturbed observations, the head-to-head comparison is instead defined by

$$= g(\tilde{\mathbf{X}}), \quad \hat{\mathbf{Y}} = \mathcal{F}_{\theta}(\tilde{\mathbf{X}}), \quad (337)$$

with both outputs evaluated against the same canonical reference

$$= g(\mathbf{X}_{\text{clean}}). \quad (338)$$

This establishes a direct comparison in which both methods receive the same observed input while differing in the transformation applied to that input.

5.3 Immutable Reference Construction

For every complete participant, the canonical reference psychometric representation is computed once from the original, unperturbed response vector:

$$= g(\mathbf{X}_{\text{clean}}). \quad (339)$$

The perturbation operator is subsequently applied only to the observed response vector:

$$\tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}}). \quad (340)$$

The reference representation is invariant under the subsequent perturbation:

$$\boxed{= g(\mathbf{X}_{\text{clean}}) \quad \text{and remains fixed for all } \tilde{\mathbf{X}} = \mathcal{C}_\phi(\mathbf{X}_{\text{clean}})}. \quad (341)$$

Consequently, the perturbation process cannot redefine, overwrite, or otherwise alter the participant-specific reference target. The target is neither recomputed from the perturbed vector nor modified according to the perturbation family, severity, donor identity, or realization-specific transformation.

The experimental information flow is therefore

$$\boxed{\mathbf{X}_{\text{clean}} \xrightarrow{g}} \quad \text{and} \quad \boxed{\mathbf{X}_{\text{clean}} \xrightarrow{\mathcal{C}_\phi} \tilde{\mathbf{X}}}. \quad (342)$$

The two reconstruction procedures then operate on the identical perturbed observation:

$$\boxed{\tilde{\mathbf{X}} \rightarrow \begin{cases} = g(\tilde{\mathbf{X}}), \\ \hat{\mathbf{Y}} = \mathcal{F}_\theta(\tilde{\mathbf{X}}) \end{cases} \rightarrow .} \quad (343)$$

This invariant is central to the experimental design. It ensures that the task being evaluated is reconstruction toward a participant-specific reference established before perturbation, rather than rescoring the perturbed response vector and treating the perturbed state as a new target.

5.4 Corruption Families

Seven perturbation families are evaluated:

1. ITEM_SWAP
2. FACET_SWAP
3. CATEGORY_PARTIAL_SWAP
4. CATEGORY_SWAP
5. ITEM_COLLAPSE
6. CATEGORY_COLLAPSE
7. RANDOM_ITEM_CHANGE

Each family is evaluated across a predefined severity structure. The resulting evaluation therefore distinguishes perturbation mechanism from perturbation magnitude and permits family-level and severity-level analysis.

5.5 IPIP-120 Validation and Independent Test

The IPIP-120 validation benchmark contains 61,556 clean observations and 338,444 corrupted observations.

The final independent IPIP-120 TEST benchmark contains 61,557 clean observations and 1,723,596 corrupted observations, for a total of 1,785,153 observations. The TEST population remained fully outside model training, hyperparameter selection, architecture selection, objective selection, and other model-development procedures.

5.6 IPIP-300 Validation and Independent Test

The IPIP-300 validation benchmark contains 21,808 clean observations and 338,444 corrupted observations, for a total of 360,252 observations.

An independent IPIP-300 TEST population was reserved during dataset construction and remained outside training, architecture selection, objective selection, hyperparameter tuning, and final checkpoint selection. After the production model and evaluation protocol had been fixed, the independent TEST benchmark was materialised, integrity-checked, and frozen for final evaluation.

The final independent IPIP-300 TEST benchmark contains 21,808 clean observations and 610,624 corrupted observations, for a total of 632,432 observations. The TEST benchmark was therefore evaluated only after the production configuration had been fixed, preserving its role as an independent held-out evaluation population.

5.7 Evaluation Metrics

The primary error metric is root mean squared error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i^{\text{ref}})^2}. \quad (344)$$

The secondary absolute error metric is mean absolute error:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i^{\text{ref}}|. \quad (345)$$

RMSE is designated as the primary robustness metric because the reconstruction task is particularly concerned with preventing large individual deviations between the reconstructed and canonical facet representations. Squaring the residual before averaging assigns progressively greater weight to larger errors, such that an unusually large reconstruction failure contributes disproportionately more to the aggregate error than a collection of small deviations. This property is useful when the objective is to identify and control potentially consequential reconstruction failures rather than to treat all absolute deviations as equally costly.

This consideration is especially relevant to psychometric reconstruction in contexts where predicted facet values may subsequently contribute to profile interpretation. A large error concentrated in one or more facets can alter the shape of the reconstructed multivariate profile more substantially than an equivalent number of small deviations distributed across observations. Accordingly, the squared-error emphasis of RMSE provides a conservative criterion for detecting such large reconstruction discrepancies.

The same rationale applies to the broader methodological context of clinical and medical prediction, where continuous prediction performance is commonly assessed through measures of overall error and where the appropriate metric depends on the consequences associated with different types of prediction error. RMSE is therefore not assumed to be universally superior to MAE; rather, it is used here as the primary metric because the experimental objective places greater importance on controlling large reconstruction deviations. Clinical prediction guidance likewise emphasizes that model performance should be assessed using measures appropriate to the outcome and intended use rather than relying on a single universal metric (**collins2015tripod**).

MAE is retained as a complementary metric because it provides an immediately interpretable measure of the average absolute deviation and is less dominated by a relatively small number of large residuals. Consequently, the joint reporting of RMSE and MAE distinguishes two practically

different properties of reconstruction quality: the control of large errors and the typical absolute magnitude of prediction error.

This distinction is particularly important for the present IPIP-NEO-120 S2 results. Relative to V133, the final S2 configuration produced a small increase in corrupted-validation RMSE of +0.000194 while reducing MAE by -0.005228 . The final configuration was therefore not selected because it minimized every error metric simultaneously, but because it preserved the primary RMSE robustness criterion within predefined guardrails while reducing the broader absolute-error burden. RMSE and MAE are consequently interpreted jointly rather than interchangeably.

Profile-level correspondence is evaluated using Pearson and Spearman correlation. Pearson correlation is used both as an evaluation metric and as a component of the composite training objective, whereas Spearman correlation is retained exclusively as an evaluation measure of rank-order correspondence.

5.8 Reproducibility Controls

The final IPIP-NEO-120 S2 Gateway was trained and confirmed under the following core configuration:

Table 8: Core training and reproducibility configuration for the final IPIP-NEO-120 S2 Gateway.

Parameter	Value
Architecture	120 $\rightarrow$ 192 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 30
Parameters	69,214
Optimizer	AdamW
Batch size	8,192
Learning rate	0.002
Weight decay	0.0001
Dropout	0.10
Gradient clipping	1.0
Mixed precision	BF16
MSE weight	0.50
MAE weight	0.38
Pearson weight	0.12
Confirmation seeds	20,260,931; 20,260,932; 20,260,933
Selected seed	20,260,933
Selected epoch	6
SIES rounds	143
Guardrail confirmation	3/3 seeds passed
MAE improvement	3/3 seeds
TEST exposure	Not loaded / not touched

For the IPIP-NEO-120 S2 configuration, the training objective is

$$\mathcal{L}_{120,S2} = 0.50\mathcal{L}_{\text{MSE}} + 0.38\mathcal{L}_{\text{MAE}} + 0.12\mathcal{L}_{\text{Pearson}}, \quad (346)$$

where

$$\mathcal{L}_{\text{Pearson}} = 1 - r_{\text{Pearson}}. \quad (347)$$

The final production checkpoint is frozen and identified by its architecture, selected seed, selected epoch, parameter count, model metadata, and cryptographic SHA-256 digest. The final

IPIP-NEO-120 S2 checkpoint corresponds to seed 20,260,933, epoch 6, and the 69,214-parameter architecture reported above.

The three-seed confirmation procedure is part of the reproducibility protocol rather than an additional optimization stage. All three confirmation seeds satisfied the predefined RMSE, Pearson, and Spearman guardrails relative to V133 and achieved lower corrupted-validation MAE. The independent IPIP-NEO-120 TEST population remained outside architecture selection, objective selection, hyperparameter tuning, confirmatory retraining, and final checkpoint selection.

All seeds, architectural specifications, preprocessing rules, optimization parameters, objective weights, checkpoint metadata, and integrity hashes are retained in the reproducibility package.

5.9 IPIP-NEO-300 P83.25 Production and Reproducibility Controls

The final IPIP-NEO-300 P83.25 configuration was subjected to a distinct production protocol after the preceding architecture-screening and objective- screening stages. The final production model was retrained from the complete TRAIN population rather than treating the 10,000-subject screening fit as the final fitted model. The final production configuration was

$$\boxed{300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}} \quad (348)$$

with five structured six-facet output heads and 220,782 trainable parameters. The corresponding production objective was

$$\boxed{\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - r_{\text{Pearson}})} \quad (349)$$

The production run used 101,772 TRAIN subjects, with 14,553,396 training examples per epoch and 143 predefined views per subject. The core production configuration was:

Table 9: Core training and reproducibility configuration for the final IPIP-NEO-300 P83.25 Gateway.

Parameter	Value
Architecture	$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6$
Parameters	220,782
Output structure	5 heads $\times$ 6 facets
Training subjects	101,772
Training examples/epoch	14,553,396
Views/subject	143
Optimizer	AdamW
Batch size	8,192
Learning rate	0.002
Weight decay	0.0001
Dropout	0.10
Gradient clipping	1.0
Mixed precision	BF16
MSE weight	0.1675
Pearson weight	0.8325
Maximum epochs	120
Minimum epochs	5
Early-stop patience	3
Minimum improvement	10^{-4}
Global seed	20,260,920
SIES seed	20,261,697
SIES rounds	143
Validation clean	21,808
Validation corrupted	338,444
Validation total	360,252
Selected epoch	14
Selection criterion	Corrupted-validation RMSE
Selected validation RMSE	0.1848359466
TEST exposure during production	Reserved / untouched

The final production fit was trained from scratch on the complete TRAIN population. The preceding 10,000-subject architecture-screening result was used for development and architecture refinement only and was not treated as the final fitted model. This separation preserves the distinction between internal development evidence and the full-population production checkpoint.

The validation benchmark used for checkpoint selection contained 21,808 clean and 338,444 corrupted observations, for a total of 360,252 observations. The checkpoint criterion was fixed to corrupted-validation RMSE. Clean-validation performance, Pearson correspondence, Spearman correspondence, and the independent TEST benchmark were not used as alternative checkpoint-selection criteria.

The selected checkpoint was epoch 14, with

$$\text{RMSE}_{\text{val,corr}} = 0.18483594655007943. \quad (350)$$

Training subsequently continued through epoch 17. The later decrease in corrupted-validation RMSE did not satisfy the predefined minimum-improvement threshold of 10^{-4} required to replace the epoch-14 best checkpoint. Early stopping therefore terminated the production run after epoch 17 while retaining epoch 14 as the official production checkpoint.

The final production artifact was generated from the full TRAIN-population run at epoch 14. Its recorded SHA-256 digest is

$$\text{f113f3b48b09f89deecce55788a59f257f31c09f12b17d3d2f657c0bd42fc287}. \quad (351)$$

The checkpoint identity is therefore defined jointly by the architecture, 220,782 trainable parameters, P83.25 objective, full-TRAIN production fit, selected epoch, and cryptographic digest rather than by the nominal layer widths alone.

Unlike the IPIP-NEO-120 S2 configuration, the final P83.25 production run was not accompanied by a three-seed confirmatory selection procedure. The reported IPIP-NEO-300 production checkpoint is therefore a single predefined production fit selected under the stated validation criterion. No multi-seed confirmation claim is made for P83.25.

The subsequent independent IPIP-NEO-300 TEST evaluation was treated as a separate locked stage. The frozen TEST population contains 21,808 clean observations and 610,624 corrupted observations, for a total of 632,432 observations. The TEST benchmark remained outside training, architecture screening, objective screening, checkpoint selection, and retraining. The final production and TEST stages can therefore be represented as the following sequence:

$$\begin{aligned} &\text{development and screening} \rightarrow \text{full-TRAIN production retraining} \rightarrow \text{validation checkpoint selection} \\ &\quad \rightarrow \text{checkpoint freezing} \rightarrow \text{independent TEST evaluation} \end{aligned} \quad (352)$$

This ordering is maintained to prevent information from the independent TEST population from influencing the fitted production checkpoint or any preceding model-development decision.

6. Results

6.1 IPIP-120 Validation

The IPIP-NEO-120 robustness evaluation was conducted on a frozen validation benchmark comprising 61,556 clean observations and 338,444 structured-perturbed observations, for a total of 400,000 evaluated observations. The benchmark contains seven perturbation families and thirteen evaluated severity values. Because these validation observations were also used during the model-development process, the validation benchmark is not treated as an independent generalization test. The historical Gateway, V133 control, and final S2 results are therefore reported as successive model-development stages on the same frozen validation benchmark.

The validation observations should also not be interpreted as 400,000 independent participants. Clean observations and repeated perturbation instances may originate from the same underlying participant according to the predefined exposure structure. The reported aggregate metrics therefore describe the evaluated observation set and its paired reconstruction conditions rather than an independently sampled participant population of 400,000 individuals.

The historical Original Gateway corresponds to the compact architecture

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30, \quad (353)$$

with 58,910 trainable parameters and the composite objective

$$\mathcal{L}_H = 0.90\mathcal{L}_{\text{MSE}} + 0.10\mathcal{L}_{\text{Pearson}}. \quad (354)$$

The subsequent V133 control retained the same architecture and parameter count but represents the refined control checkpoint used for final S2 confirmation. Its corrupted-validation metrics were $\text{RMSE} = 0.348829$, $\text{MAE} = 0.201426$, $\text{Pearson} = 0.880498$, and $\text{Spearman} = 0.850461$.

The final S2 configuration replaces the 64-dimensional bottleneck with a 128-dimensional third hidden representation:

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (355)$$

contains 69,214 trainable parameters, and uses the final objective

$$\mathcal{L}_{120, \text{S2}} = 0.50\mathcal{L}_{\text{MSE}} + 0.38\mathcal{L}_{\text{MAE}} + 0.12\mathcal{L}_{\text{Pearson}}, \quad (356)$$

where

$$\mathcal{L}_{\text{Pearson}} = 1 - r_{\text{Pearson}}. \quad (357)$$

Here, the Pearson component is evaluated at the profile level across the thirty output facets for each observation rather than by pooling all facet coordinates across observations into a single correlation coefficient. The same distinction between coordinate-level error and profile-level correspondence is maintained throughout the reported evaluation.

The final production operating point corresponds to seed 20,260,933 and epoch 6. The final S2 operating point was retained only after confirmatory retraining across seeds 20,260,931, 20,260,932, and 20,260,933. The predefined confirmation guardrails required

$$\Delta \text{RMSE} \leq 0.001, \quad (358)$$

$$\Delta \text{Pearson} \geq -0.002, \quad (359)$$

$$\Delta \text{Spearman} \geq -0.002, \quad (360)$$

together with strict improvement in corrupted-validation MAE:

$$\Delta \text{MAE} < 0. \quad (361)$$

All three confirmatory seeds satisfied the predefined guardrails, and all three reduced corrupted-validation MAE relative to V133. The observed confirmation outcome was therefore 3/3 guardrail passes and 3/3 MAE improvements. The final operating point was fixed to seed 20,260,933 at epoch 6. These seed-level repetitions are treated as reproducibility and stability checks under the predefined protocol, not as independent participant-level replications or as a preregistered inferential statistical trial.

The final S2 corrupted-validation performance relative to V133 was

$$\Delta \text{RMSE} = +0.000194, \quad (362)$$

$$\Delta \text{MAE} = -0.005228, \quad (363)$$

$$\Delta \text{Pearson} = -0.000267, \quad (364)$$

and

$$\Delta\text{Spearman} = +0.000157. \quad (365)$$

The corresponding unrounded audit values were $\Delta\text{RMSE} = +0.000193974$, $\Delta\text{MAE} = -0.005227696$, $\Delta\text{Pearson} = -0.000266882$, and $\Delta\text{Spearman} = +0.000156515$. The displayed six-decimal values reported throughout this section are rounded representations of those underlying validation results.

On clean observations, conventional deterministic scoring reproduced the canonical reference psychometric representation exactly, with

$$\text{RMSE} = 0 \quad (366)$$

and maximum absolute deviation equal to zero by numerical audit. This agreement is a construction property of the reference-generating scoring procedure and is therefore not treated as an empirical advantage in the robustness comparison.

The global head-to-head validation results across the historical and final IPIP-120 model stages are shown below.

Table 10: IPIP-NEO-120 clean-validation performance across the historical Gateway, V133 control, and final S2 checkpoint.

Method	RMSE	MAE	Pearson	Spearman
Traditional	0.000000	0.000000	1.000000	1.000000
Original Gateway	0.136102	0.092990	0.986004	0.985641
V133 control	0.123152	0.080640	0.988704	0.974316
S2 final	0.114294	0.072383	0.990228	0.974334

Table 11: IPIP-NEO-120 corrupted-validation performance across the historical Gateway, V133 control, and final S2 checkpoint.

Method	RMSE	MAE	Pearson	Spearman
Traditional	0.462274	0.181624	0.818282	0.816762
Original Gateway	0.347833	0.210576	0.891710	0.875251
V133 control	0.348829	0.201426	0.880498	0.850461
S2 final	0.349023	0.196198	0.880231	0.850618

The corrupted-validation comparison is the principal quantitative basis for the model-development interpretation reported here. The clean-validation results are retained as a complementary diagnostic, whereas the corrupted validation benchmark directly evaluates reconstruction under the structured perturbation conditions for which the Gateway was developed.

Relative to conventional scoring, the historical Original Gateway reduced corrupted-input RMSE by

$$0.462274 - 0.347833 = 0.114441, \quad (367)$$

corresponding to a relative reduction of

$$\frac{0.462274 - 0.347833}{0.462274} \times 100 = 24.756\%. \quad (368)$$

The V133 control reduced corrupted RMSE relative to conventional scoring by

$$0.462274 - 0.348829 = 0.113445, \quad (369)$$

or 24.541% relative reduction. Its MAE was 0.201426, corresponding to a 10.903% increase over conventional scoring.

The final S2 checkpoint reduced corrupted RMSE relative to conventional scoring by

$$0.462274 - 0.349023 = 0.113251, \quad (370)$$

corresponding to a relative reduction of

$$\frac{0.462274 - 0.349023}{0.462274} \times 100 = 24.499\%. \quad (371)$$

Its corrupted MAE was 0.196198, corresponding to an 8.024% increase over conventional scoring, while Pearson and Spearman correlations increased by 0.061949 and 0.033856, respectively.

The percentage reductions reported above are calculated from the six-decimal values displayed in Table 11. They are therefore reported as rounded descriptive quantities rather than as substitutes for calculations performed from the underlying unrounded evaluation arrays.

The progression from the Original Gateway to V133 and subsequently to S2 should not be interpreted as monotonic improvement in every metric. Relative to the Original Gateway, V133 changed RMSE by +0.000996 but reduced MAE by 0.009150. Relative to V133, the final S2 checkpoint changed RMSE by +0.000194 while reducing MAE by 0.005228. The model-development pathway therefore represents successive controlled changes in the error distribution rather than a sequence in which every metric increases or decreases monotonically.

The final S2 checkpoint should accordingly not be characterised as uniformly better than V133 across all reported metrics. Its retention reflects the predefined multi-metric confirmation procedure: a small increase in corrupted-validation RMSE was accepted within the specified guardrail while corrupted-validation MAE improved and the profile-level Pearson and Spearman measures remained within the corresponding predefined tolerances.

6.1.1 Mathematical Error Decomposition

For the scalar reconstruction residuals, let

$$e_i = \hat{y}_i - y_i^{\text{ref}}, \quad (372)$$

where the index i ranges over the complete set of evaluated prediction-reference coordinate pairs, i.e. the evaluated observations and their thirty facet coordinates. Let N denote the total number of such scalar residuals.

The squared-error mean is

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N e_i^2 = \text{RMSE}^2, \quad (373)$$

whereas

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |e_i|. \quad (374)$$

Because $e_i^2 = |e_i|^2$, the second central moment of the absolute residuals can be written directly in terms of the two reported error metrics:

$$\text{Var}(|e|) = \text{RMSE}^2 - \text{MAE}^2. \quad (375)$$

Consequently,

$$\text{RMSE}^2 = \text{MAE}^2 + \text{Var}(|e|), \quad (376)$$

which also establishes the inequality

$$\text{RMSE} \geq \text{MAE}, \quad (377)$$

with equality only when the absolute residual magnitude is constant across the evaluated residual set.

This identity provides a useful mathematical interpretation of the final selection. Under the common residual aggregation used for the reported global RMSE and MAE metrics, the empirical squared-error quantities obtained from the displayed six-decimal values are

$$\begin{aligned} \text{MSE}_{\text{Traditional}} &= 0.462274^2 \approx 0.213697, \\ \text{MSE}_{\text{Original}} &= 0.347833^2 \approx 0.120988, \\ \text{MSE}_{\text{V133}} &= 0.348829^2 \approx 0.121682, \\ \text{MSE}_{\text{S2}} &= 0.349023^2 \approx 0.121817. \end{aligned} \quad (378)$$

The approximation symbols are intentional: these values are reconstructed from the rounded RMSE values reported in the table rather than calculated directly from the underlying unrounded residual arrays.

Thus, relative to conventional scoring, the final S2 checkpoint reduces the empirical squared-error quantity derived from the displayed metrics by approximately

$$\frac{0.213697 - 0.121817}{0.213697} \times 100 = 42.996\%. \quad (379)$$

Relative to V133, the final S2 checkpoint increases the corresponding squared-error quantity by approximately 0.111% while decreasing MAE by approximately 2.595%. This quantifies the final model-selection trade-off in the same error units used by the training objective.

For the final S2 checkpoint,

$$\text{RMSE}^2 - \text{MAE}^2 = 0.349023^2 - 0.196198^2 = 0.083323, \quad (380)$$

whereas for V133 the corresponding quantity is approximately 0.081109. Again, these values are derived from the rounded metrics displayed in the results table.

The final S2 configuration therefore traded a small increase in the dispersion-sensitive component of the scalar residual distribution for a reduction in the mean absolute deviation. This is consistent with the predefined multi-metric selection procedure and does not imply that S2 minimizes every possible summary of the residual distribution.

Importantly, the error decomposition above describes the aggregate coordinate-level residual distribution only. It does not replace the profile-level Pearson or Spearman analyses, and it does not imply that the variance of absolute residuals corresponds to a distinct latent psychological quantity. It is used solely as a mathematical decomposition of the reported reconstruction errors.

6.1.2 Final Multi-Seed Confirmation

The final S2 checkpoint was confirmed using three independent retraining runs under distinct random seeds:

$$\mathcal{S} = \{20,260,931, 20,260,932, 20,260,933\}. \quad (381)$$

The term “independent” here refers to independent stochastic training trajectories under the same predefined experimental protocol; it does not denote independent participant samples or independent validation datasets.

For each seed, the confirmation guardrails required

$$\Delta\text{RMSE} \leq 0.001, \quad (382)$$

$$\Delta r_{\text{Pearson}} \geq -0.002, \quad (383)$$

$$\Delta \rho_{\text{Spearman}} \geq -0.002, \quad (384)$$

together with strict improvement in corrupted-validation MAE relative to V133:

$$\Delta\text{MAE} < 0. \quad (385)$$

All three seeds satisfied all criteria.

Table 12: Final IPIP-NEO-120 S2 confirmatory multi-seed validation.

Seed	RMSE	MAE	Pearson	Spearman	Guardrails	MAE improved
20,260,931	0.349045	0.196393	0.880235	0.849576	Pass	Yes
20,260,932	0.349288	0.196121	0.880109	0.849640	Pass	Yes
20,260,933	0.349023	0.196198	0.880231	0.850618	Pass	Yes

Across these three runs, the mean corrupted RMSE was 0.349119 with a sample standard deviation of 0.000147, while mean MAE was 0.196237 with a sample standard deviation of 0.000140. Pearson correlation had mean 0.880192 and standard deviation 0.000072, and Spearman correlation had mean 0.849945 and standard deviation 0.000584. These seed-level dispersions are reported as reproducibility diagnostics rather than as participant-level uncertainty intervals.

The corresponding observed seed-level ranges were

$$\text{Range}(\text{RMSE}) = 0.349288 - 0.349023 = 0.000265, \quad (386)$$

$$\text{Range}(\text{MAE}) = 0.196393 - 0.196121 = 0.000272, \quad (387)$$

$$\text{Range}(\text{Pearson}) = 0.880235 - 0.880109 = 0.000126, \quad (388)$$

and

$$\text{Range}(\text{Spearman}) = 0.850618 - 0.849576 = 0.001042. \quad (389)$$

These quantities provide a direct descriptive measure of the spread observed across the three

confirmation trajectories. They are not participant-level confidence intervals and should not be interpreted as population-level sampling uncertainty.

The confirmation procedure therefore produced 3/3 guardrail passes and 3/3 corrupted-validation MAE improvements relative to V133. The final operating point was fixed to seed 20,260,933 at epoch 6.

$$\text{RMSE}_{\text{S2, val, corr}} = 0.349023, \quad \text{MAE}_{\text{S2, val, corr}} = 0.196198. \quad (390)$$

The corresponding profile-level correlations were

$$r_{\text{Pearson}} = 0.880231, \quad \rho_{\text{Spearman}} = 0.850618. \quad (391)$$

Relative to V133, the final S2 checkpoint yielded

$$\Delta \text{RMSE} = 0.349023 - 0.348829 = +0.000194, \quad (392)$$

$$\Delta \text{MAE} = 0.196198 - 0.201426 = -0.005228, \quad (393)$$

$$\Delta \text{Pearson} = 0.880231 - 0.880498 = -0.000267, \quad (394)$$

and

$$\Delta \text{Spearman} = 0.850618 - 0.850461 = +0.000157. \quad (395)$$

The unrounded audit values corresponding to these rounded differences were +0.000193974, −0.005227696, −0.000266882, and +0.000156515, respectively. The rounded values are reported in the main results tables for readability, whereas the unrounded quantities define the corresponding checkpoint-level audit.

6.2 Historical-to-Final Interpretation

The historical Original Gateway, V133 control, and final S2 checkpoint should therefore be interpreted as three distinct stages of the same IPIP-120 development program. The Original Gateway established the initial robustness signal relative to conventional scoring. V133 served as the refined control used to diagnose and quantify the limitations of the original architecture. S2 subsequently altered the internal representation from 120 → 192 → 128 → 64 → 30 to 120 → 192 → 128 → 128 → 30 and was retained only after three-seed confirmation.

The final S2 result is consequently best characterized as a constrained multi-metric operating point: relative to V133, it accepts a RMSE increase of 0.000194 in exchange for a MAE reduction of 0.005228, while maintaining correlation measures within the predefined confirmation regime.

This interpretation is deliberately narrower than a claim of uniform metric improvement. The final S2 checkpoint did not minimize every reported metric relative to V133. Its retention was determined by the predefined guardrail procedure, under which RMSE remained within the allowed increase, Pearson and Spearman remained within their allowed tolerances, and corrupted-validation MAE improved across all three confirmation seeds.

6.3 Corruption-Family Analysis

The following family-level analysis reports the historical Original Gateway results retained from the initial IPIP-120 validation benchmark. It is presented as a historical reference rather than as

a claim that the final S2 checkpoint reproduces every family-specific result of the original model.

The historical robustness pattern was not confined to the aggregate validation metric. The Original Gateway achieved lower RMSE than conventional scoring in all seven evaluated perturbation families. Relative RMSE reductions ranged from 7.091% for CATEGORY_PARTIAL_SWAP to 44.793% for CATEGORY_COLLAPSE.

The family-specific results are shown in Table 13.

Table 13: IPIP-NEO-120 corrupted-validation RMSE by perturbation family.

Perturbation family	Traditional RMSE	MLP Gateway RMSE	Relative RMSE reduction
ITEM_SWAP	0.311312	0.288860	7.212%
FACET_SWAP	0.355175	0.313356	11.774%
CATEGORY_PARTIAL_SWAP	0.283323	0.263232	7.091%
CATEGORY_SWAP	0.637445	0.517382	18.835%
ITEM_COLLAPSE	0.338333	0.272785	19.374%
CATEGORY_COLLAPSE	0.733339	0.404853	44.793%
RANDOM_ITEM_CHANGE	0.365996	0.300759	17.824%

For a family f , the reported relative RMSE reduction is defined as

$$\Delta_{\text{rel},f} = \frac{\text{RMSE}_{\text{Traditional},f} - \text{RMSE}_{\text{Gateway},f}}{\text{RMSE}_{\text{Traditional},f}} \times 100. \quad (396)$$

Thus, the family-level quantities describe relative changes within each perturbation family and should not be interpreted as a simple decomposition of the global RMSE into seven equally weighted components. The aggregate corrupted-validation RMSE is computed over the complete evaluated residual set, whereas family-level RMSE values are calculated within their respective subsets.

Thus, within the defined perturbation protocol, the historical aggregate RMSE reduction was observed across all seven corruption families rather than being confined to a single perturbation mechanism. The magnitude of the reduction varied substantially across families, indicating that the response to corruption depended on the structure of the applied transformation.

6.4 IPIP-NEO-120 Independent Frozen TEST

The independent IPIP-NEO-120 TEST benchmark constitutes a separate evaluation stage from the frozen validation benchmark used during model development. The final TEST population contains 61,557 clean observations and 1,723,596 corrupted observations, for a total of 1,785,153 evaluated observations.

$$61,557 + 1,723,596 = 1,785,153. \quad (397)$$

The corrupted observations correspond to the predefined perturbation structure associated with the held-out participants. The TEST population was not used for architecture screening, objective selection, hyperparameter tuning, confirmatory retraining, checkpoint selection, or any other model-development decision. The final S2 checkpoint had already been fixed before the independent TEST evaluation.

The final evaluated S2 model is the frozen

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30 \quad (398)$$

architecture with 69,214 trainable parameters, seed 20,260,933, and epoch 6.

The TEST evaluation also retained the distinction between clean reconstruction and robustness under perturbation. On clean observations, conventional scoring exactly reproduces the canonical reference by construction:

$$\text{RMSE}_{\text{Traditional, clean}} = 0, \quad (399)$$

$$\text{MAE}_{\text{Traditional, clean}} = 0, \quad (400)$$

$$r_{\text{Pearson, Traditional, clean}} = 1, \quad (401)$$

and

$$\rho_{\text{Spearman, Traditional, clean}} = 1. \quad (402)$$

The corresponding frozen S2 clean-TEST metrics were

$$\text{RMSE}_{\text{S2, clean}} = 0.114098616, \quad (403)$$

$$\text{MAE}_{\text{S2, clean}} = 0.072518252, \quad (404)$$

$$r_{\text{Pearson, S2, clean}} = 0.990334181, \quad (405)$$

and

$$\rho_{\text{Spearman, S2, clean}} = 0.983815665. \quad (406)$$

The non-zero S2 clean-TEST error does not represent a failure of the Traditional reference construction. It reflects approximation error of the learned Gateway when reconstructing the canonical representation even in the absence of a response perturbation. Accordingly, clean-TEST results are reported as a model-behaviour diagnostic and are not used to claim robustness to corruption.

The principal independent robustness comparison is the corrupted TEST subpopulation:

Table 14: Final IPIP-NEO-120 S2 evaluation on the independent frozen corrupted TEST benchmark.

Method	RMSE	MAE	Pearson	Spearman
Traditional	0.579328515	0.287756126	0.715059733	0.712384880
S2 final	0.428448503	0.264785287	0.829916061	0.798439021

The absolute metric differences are

$$\Delta \text{RMSE} = 0.428448503 - 0.579328515 = -0.150880012, \quad (407)$$

$$\Delta \text{MAE} = 0.264785287 - 0.287756126 = -0.022970839, \quad (408)$$

$$\Delta\text{Pearson} = 0.829916061 - 0.715059733 = +0.114856328, \quad (409)$$

and

$$\Delta\text{Spearman} = 0.798439021 - 0.712384880 = +0.086054141. \quad (410)$$

The corresponding relative RMSE change is

$$\frac{0.428448503 - 0.579328515}{0.579328515} \times 100 = -26.043947\%. \quad (411)$$

Equivalently, the RMSE is reduced by approximately 26.044%.

The corresponding relative MAE change is

$$\frac{0.264785287 - 0.287756126}{0.287756126} \times 100 = -7.982745\%. \quad (412)$$

The relative Pearson change is

$$\frac{0.829916061 - 0.715059733}{0.715059733} \times 100 = 16.062480\%, \quad (413)$$

and the relative Spearman change is

$$\frac{0.798439021 - 0.712384880}{0.712384880} \times 100 = 12.079726\%. \quad (414)$$

These relative correlation changes are presented as descriptive ratios of the observed correlation coefficients and should not be interpreted as percentages of explained variance.

6.4.1 Independent-TEST Error Decomposition for S2

For the corrupted TEST residuals, let

$$e_i = \hat{y}_i - y_i^{\text{ref}}. \quad (415)$$

Under the same scalar-residual aggregation used for the global RMSE and MAE,

$$\text{MSE} = \text{RMSE}^2. \quad (416)$$

Using the displayed TEST metrics, the squared-error quantities are approximately

$$\text{MSE}_{\text{Traditional,TEST}} = 0.579328515^2 \approx 0.335621528, \quad (417)$$

and

$$\text{MSE}_{\text{S2,TEST}} = 0.428448503^2 \approx 0.183568120. \quad (418)$$

The corresponding absolute-residual second moments are therefore

$$\text{Var}(|e|)_{\text{Traditional}} \approx 0.335621528 - 0.287756126^2 = 0.252817940, \quad (419)$$

and

$$\text{Var}(|e|)_{\text{S2}} \approx 0.183568120 - 0.264785287^2 = 0.113456872. \quad (420)$$

Thus, relative to Traditional scoring, the squared-error quantity derived from the displayed RMSE decreases by approximately

$$\frac{0.335621528 - 0.183568120}{0.335621528} \times 100 = 45.305022\%. \quad (421)$$

This reduction is numerically larger than the 26.043947% RMSE reduction because RMSE is the square root of the underlying mean squared error. The two percentages therefore describe different scales of the same error quantity and should not be treated as interchangeable performance measures.

The TEST paired comparison additionally evaluated each corrupted prediction against its corresponding participant-specific Gold reference. Among the corrupted TEST observations, S2 was closer to Gold than Traditional scoring for 72.9250% of perturbation instances, Traditional scoring was closer for 27.0750%, and no ties were observed.

$$72.9250\% + 27.0750\% + 0\% = 100\%. \quad (422)$$

This paired statistic is a row-level comparison and should not be interpreted as an independent participant-level probability, nor as a replacement for the aggregate error and correspondence metrics.

6.4.2 IPIP-120 Independent TEST Family Analysis

The final S2 corrupted TEST also showed lower RMSE than Traditional scoring in each of the seven evaluated perturbation families:

Table 15: Final IPIP-NEO-120 S2 corrupted-TEST RMSE by perturbation family.

Perturbation family	Traditional RMSE	S2 RMSE	Relative RMSE reduction
ITEM_SWAP	0.454519	0.396420	12.783%
FACET_SWAP	0.459663	0.394646	14.144%
CATEGORY_PARTIAL_SWAP	0.305406	0.272947	10.628%
CATEGORY_SWAP	0.718169	0.591406	17.651%
ITEM_COLLAPSE	0.500320	0.366206	26.806%
CATEGORY_COLLAPSE	0.886608	0.495586	44.103%
RANDOM_ITEM_CHANGE	0.534827	0.410012	23.337%

For each family f , the relative reduction is calculated as

$$\Delta_{\text{rel},f}^{\text{TEST}} = \frac{\text{RMSE}_{\text{Traditional},f} - \text{RMSE}_{\text{S2},f}}{\text{RMSE}_{\text{Traditional},f}} \times 100. \quad (423)$$

The largest family-level reduction in the frozen S2 TEST occurred for CATEGORY_COLLAPSE, where RMSE changed from 0.886608 to 0.495586. The corresponding relative reduction was approximately 44.103%.

The family-level result is important because the global S2 improvement is not restricted to a single perturbation family. At the same time, the family-level magnitudes differ materially, demonstrating that the effect of reconstruction depends on the structure of the perturbation mechanism.

The family table is descriptive of the predefined TEST conditions. It does not establish that the Gateway is uniformly more robust to arbitrary perturbations outside the evaluated families.

6.5 IPIP-NEO-300 Independent Frozen TEST

The final IPIP-NEO-300 production model was the P83.25 configuration

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (424)$$

with 220,782 trainable parameters and the objective

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - \rho_{\text{Pearson}}). \quad (425)$$

The production checkpoint was selected at epoch 14 using corrupted-validation RMSE as the predefined checkpoint-selection criterion. The selected corrupted-validation RMSE was

$$\text{RMSE}_{\text{val,corr}} = 0.18483594655007943, \quad (426)$$

reported in the main results as 0.184836.

Training continued through epoch 17, but the additional reduction in corrupted-validation RMSE did not satisfy the predefined minimum improvement threshold of 10^{-4} required to replace the epoch-14 best checkpoint. Accordingly, epoch 14, rather than the final epoch executed, defines the production checkpoint evaluated on TEST.

The corresponding checkpoint is identified by the production SHA-256 digest

$$\text{f113f3b48b09f89deecce55788a59f257f31c09f12b17d3d2f657c0bd42fc287} . \quad (427)$$

Unlike the final IPIP-NEO-120 S2 configuration, the P83.25 production result was not subjected to the same three-seed confirmatory protocol. The P83.25 result is therefore interpreted as the outcome of its fixed full-TRAIN production trajectory followed by locked checkpoint selection and independent TEST evaluation, rather than as a three-seed confirmed operating point.

The independent IPIP-NEO-300 TEST contains 21,808 held-out participants and 632,432 evaluated observations:

$$21,808 + 610,624 = 632,432. \quad (428)$$

The 610,624 corrupted observations correspond to the predefined perturbation conditions associated with the held-out participant population. In the corrupted TEST evaluation, the seven perturbation families and four perturbation levels generate

$$7 \times 4 = 28 \quad (429)$$

family-level perturbation blocks, giving

$$21,808 \times 28 = 610,624 \quad (430)$$

corrupted observations.

The independent IPIP-NEO-300 TEST population was frozen separately from model development. The final checkpoint was fixed before TEST evaluation, and the TEST data were not used for training, architecture screening, objective screening, checkpoint selection, or retraining.

On clean TEST observations, conventional scoring again reproduces the canonical Gold representation by construction:

$$\text{RMSE}_{\text{Traditional, clean}} = 0, \quad \text{MAE}_{\text{Traditional, clean}} = 0, \quad (431)$$

$$r_{\text{Pearson, Traditional, clean}} = 1, \quad \rho_{\text{Spearman, Traditional, clean}} = 1. \quad (432)$$

The final P83.25 clean-TEST metrics were

$$\text{RMSE}_{\text{P83.25, clean}} = 0.114098616, \quad (433)$$

$$\text{MAE}_{\text{P83.25, clean}} = 0.072518252, \quad (434)$$

$$r_{\text{Pearson, P83.25, clean}} = 0.990334181, \quad (435)$$

and

$$\rho_{\text{Spearman, P83.25, clean}} = 0.983815665. \quad (436)$$

The principal P83.25 robustness evaluation is performed on the corrupted TEST observations.

Table 16: Final IPIP-NEO-300 P83.25 evaluation on the independent frozen corrupted TEST benchmark.

Method	RMSE	MAE	Pearson	Spearman
Traditional	0.290377488	0.139889697	0.756625041	0.747784084
P83.25 E14	0.223052573	0.137993711	0.833130759	0.810581771

The absolute changes are

$$\Delta \text{RMSE} = 0.223052573 - 0.290377488 = -0.067324915, \quad (437)$$

$$\Delta \text{MAE} = 0.137993711 - 0.139889697 = -0.001895986, \quad (438)$$

$$\Delta \text{Pearson} = 0.833130759 - 0.756625041 = +0.076505718, \quad (439)$$

and

$$\Delta \text{Spearman} = 0.810581771 - 0.747784084 = +0.062797687. \quad (440)$$

The relative RMSE change is

$$\frac{0.223052573 - 0.290377488}{0.290377488} \times 100 = -23.185308\%. \quad (441)$$

Thus, the corrupted-TEST RMSE is reduced by approximately 23.185%.

The relative MAE change is

$$\frac{0.137993711 - 0.139889697}{0.139889697} \times 100 = -1.355344\%. \quad (442)$$

The relative Pearson change is

$$\frac{0.833130759 - 0.756625041}{0.756625041} \times 100 = 10.111444\%, \quad (443)$$

and the relative Spearman change is

$$\frac{0.810581771 - 0.747784084}{0.747784084} \times 100 = 8.397837\%. \quad (444)$$

The P83.25 TEST therefore exhibits simultaneous reductions in the two reported error measures and increases in the two profile-level correspondence measures. These are descriptive properties of the locked corrupted TEST benchmark and should not be generalized beyond the specified perturbation protocol and evaluated instrument without additional external validation.

6.5.1 Independent-TEST Error Decomposition for P83.25

Using the same residual definition,

$$e_i = \hat{y}_i - y_i^{\text{ref}}, \quad (445)$$

the scalar mean squared error associated with the reported RMSE is

$$\text{MSE} = \text{RMSE}^2. \quad (446)$$

For Traditional scoring,

$$\text{MSE}_{\text{Traditional,TEST}} = 0.290377488^2 \approx 0.084319086, \quad (447)$$

whereas for P83.25 E14,

$$\text{MSE}_{\text{P83.25,TEST}} = 0.223052573^2 \approx 0.049752450. \quad (448)$$

The corresponding second central moments of absolute residual magnitude are

$$\text{Var}(|e|)_{\text{Traditional}} \approx 0.084319086 - 0.139889697^2 = 0.064749958, \quad (449)$$

and

$$\text{Var}(|e|)_{\text{P83.25}} \approx 0.049752450 - 0.137993711^2 = 0.030710186. \quad (450)$$

The squared-error quantity derived from the reported RMSE therefore changes by approximately

$$\frac{0.084319086 - 0.049752450}{0.084319086} \times 100 = 40.995031\%. \quad (451)$$

Thus, the approximately 23.185% RMSE reduction corresponds to an approximately 40.995% reduction in the underlying mean squared error. The difference between the two percentages is a mathematical consequence of the square-root transformation defining RMSE.

6.6 Paired TEST Comparison Across Final Configurations

For each corrupted TEST observation, let

$$d_{\text{method}} = d\left(\hat{\mathbf{Y}}_{\text{method}}, \mathbf{Y}_{\text{Gold}}\right) \quad (452)$$

denote the reconstruction error under the predefined observation-level comparison metric used for the paired analysis. The paired result records which method yields the smaller distance to the same participant-specific Gold reference.

For IPIP-NEO-120,

$$p_{S2} = 72.9250\%, \quad p_{\text{Traditional}} = 27.0750\%, \quad p_{\text{tie}} = 0\%. \quad (453)$$

For IPIP-NEO-300,

$$p_{P83.25} = 69.9270\%, \quad p_{\text{Traditional}} = 30.0730\%, \quad p_{\text{tie}} = 0\%. \quad (454)$$

In both cases,

$$p_{\text{method 1}} + p_{\text{method 2}} + p_{\text{tie}} = 100\%. \quad (455)$$

These paired proportions are complementary to, rather than substitutes for, the aggregate RMSE, MAE, Pearson, and Spearman statistics. They operate at the perturbation-instance level and should not be interpreted as participant-level inferential probabilities.

6.7 Cross-Instrument Independent TEST Summary

The two frozen corrupted-TEST evaluations can be summarised using the same four aggregate metrics while retaining the fact that the two Gateway architectures were developed independently for their respective input dimensionalities.

Table 17: Independent corrupted-TEST summary for the final IPIP-NEO-120 S2 and IPIP-NEO-300 P83.25 configurations.

Metric	IPIP-120 Traditional	IPIP-120 S2	IPIP-300 Traditional	IPIP-300 P83.25 E14
RMSE	0.579328515	0.428448503	0.290377488	0.223052573
MAE	0.287756126	0.264785287	0.139889697	0.137993711
Pearson	0.715059733	0.829916061	0.756625041	0.833130759
Spearman	0.712384880	0.798439021	0.747784084	0.810581771

The final corrupted-TEST reductions in RMSE were 26.044% for IPIP-120 S2 and 23.185% for IPIP-300 P83.25. The corresponding MAE reductions were 7.983% and 1.355%, respectively. The direction of the aggregate metric changes was consistent across the two instrument-specific final evaluations: lower RMSE, lower MAE, higher Pearson correspondence, and higher Spearman correspondence relative to the respective Traditional scoring comparator.

This cross-instrument consistency should nevertheless be interpreted within the common IPIP-NEO framework used in the present study. The two experiments do not constitute literal replications with identical architectures, parameter counts, input dimensionalities, or training objectives. Rather, they represent complementary evaluations of the same reconstruction formulation under two instrument configurations.

The IPIP-120 S2 configuration was subjected to explicit three-seed confirmation, whereas P83.25 E14 corresponds to its single full-production training trajectory and locked validation-based checkpoint-selection procedure. The two evidence structures are therefore not identical and should not be represented as though they were.

Finally, the independent TEST results are bounded to the predefined semi-synthetic perturbation protocol, the two evaluated IPIP-NEO configurations, and the corresponding held-out

populations. They do not by themselves establish clinical validity, diagnostic utility, psychometric equivalence across external instruments, causal mechanisms of reconstruction, or robustness to perturbation processes outside those explicitly evaluated.

6.7.1 Final S2 Family-Level Delta Analysis

The final S2 checkpoint was additionally compared with the V133 control within each perturbation family. The family-level RMSE difference is defined as

$$\Delta_f^{\text{RMSE}} = \text{RMSE}_{f,\text{S2}} - \text{RMSE}_{f,\text{V133}}, \quad (456)$$

and the corresponding MAE difference is

$$\Delta_f^{\text{MAE}} = \text{MAE}_{f,\text{S2}} - \text{MAE}_{f,\text{V133}}. \quad (457)$$

The final S2 checkpoint reduced MAE relative to V133 in all seven corruption families. RMSE decreased in four families and increased in three, showing that the final architecture did not uniformly minimize every family-specific error criterion. By convention in this analysis, a negative delta denotes lower error for S2 relative to V133, whereas a positive delta denotes higher error. The family-level deltas are descriptive within each family-specific subset and are not averaged arithmetically to reconstruct the global validation metric, because the family subsets need not contribute equally to the aggregate observation set.

Table 18: Final IPIP-NEO-120 S2 family-level deltas relative to V133.

Perturbation family	ΔRMSE	ΔMAE
ITEM_SWAP	-0.000655	-0.004456
FACET_SWAP	+0.000205	-0.005612
CATEGORY_PARTIAL_SWAP	-0.001262	-0.006713
CATEGORY_SWAP	+0.003168	-0.002225
ITEM_COLLAPSE	-0.000289	-0.004107
CATEGORY_COLLAPSE	-0.001111	-0.004926
RANDOM_ITEM_CHANGE	+0.000313	-0.003541

The family-level results therefore provide a more precise interpretation of the final selection: the principal MAE reduction is distributed across all seven perturbation families, whereas RMSE behavior remains heterogeneous. This is consistent with the global conclusion that the S2 checkpoint is a controlled multi-metric operating point rather than a universal minimizer of all error functions.

6.8 Severity-Stratified Analysis

The following severity-stratified values likewise describe the historical Original Gateway evaluation. The final S2 checkpoint is subsequently compared with V133 by severity to distinguish the effect of the final architectural change from the original Gateway-to-Traditional comparison.

The Original Gateway also achieved lower RMSE at every evaluated severity value. Across the thirteen reported severity levels, relative RMSE reductions ranged from 4.825% to 30.703%.

The smallest relative improvement occurred at severity 6 (4.825%), whereas the largest occurred at severity 5 (30.703%). The corresponding reductions at severity values 1, 2, 3, 4, 10, 12, 15, 24, 36, 48, and 60 were also positive.

This result indicates that the aggregate robustness effect was not produced by a single extreme corruption regime. Lower Gateway RMSE was observed across the complete set of reported severity values used in the IPIP-120 validation protocol. At the same time, the variation in relative reduction across severity levels indicates that robustness was not constant as perturbation severity changed.

6.8.1 Final S2 Severity-Level Delta Analysis

An analogous comparison was performed across the thirteen severity values. For severity s ,

$$\Delta_s^{\text{RMSE}} = \text{RMSE}_{s,S2} - \text{RMSE}_{s,V133}, \quad (458)$$

and

$$\Delta_s^{\text{MAE}} = \text{MAE}_{s,S2} - \text{MAE}_{s,V133}. \quad (459)$$

The final S2 checkpoint reduced MAE relative to V133 at nine of thirteen severity values. The four exceptions occurred at severities 24, 36, 48, and 60, where MAE increased. RMSE decreased at severities 1 and 2 and increased at the remaining reported severities.

Table 19: Final IPIP-NEO-120 S2 severity-level deltas relative to V133.

Severity	ΔRMSE	ΔMAE
1	-0.002692	-0.006914
2	-0.000931	-0.006021
3	+0.000941	-0.004476
4	+0.002704	-0.002031
5	+0.002612	-0.001429
6	+0.000246	-0.004187
10	+0.002557	-0.002503
12	+0.001908	-0.001631
15	+0.003360	-0.000712
24	+0.002668	+0.000391
36	+0.002022	+0.000660
48	+0.002016	+0.001068
60	+0.001998	+0.001327

The severity analysis shows that the S2 MAE benefit is concentrated below the highest evaluated corruption severities. Accordingly, the global MAE improvement should not be interpreted as uniform across the entire severity range. The explicit high-severity reversals are retained as part of the final result. Across the thirteen severity strata, MAE decreased at nine levels and increased at four levels, whereas RMSE decreased at two levels and increased at eleven levels. The largest RMSE decrease occurred at severity 1 (-0.002692), while the largest RMSE increase occurred at severity 15 ($+0.003360$). The largest MAE decrease occurred at severity 1 (-0.006914), while the largest MAE increase occurred at severity 60 ($+0.001327$). These counts and extrema are descriptive and should not be interpreted as weighted averages across the severity distribution.

6.9 Facet-Level Analysis

The historical aggregate robustness effect was also distributed across the thirty output facets. The MLP Gateway achieved lower RMSE than conventional scoring for all thirty facets evaluated under the corrupted-validation benchmark.

Relative facet-level RMSE reductions ranged from 10.404% to 43.819%. The largest reduction was observed for C1_SelfEfficacy (43.819%), whereas the smallest was observed for O6_Liberalism (10.404%). The complete facet-level results are retained in the machine-readable benchmark output.

The observation that all thirty evaluated facets exhibited lower RMSE should be distinguished from equality of effect magnitude: the relative reductions varied across facets, indicating heterogeneous reconstruction performance within the output representation. The facet-level statement refers specifically to the historical Original Gateway comparison retained in this subsection; no final-S2 facet-level superiority is inferred from it.

6.10 Row-Level Paired Analysis

A historical row-level paired comparison was conducted across all 338,444 corrupted validation observations. The MLP Gateway produced lower row-wise RMSE than conventional scoring in 57.135% of evaluated rows, whereas conventional scoring produced lower RMSE in 42.865%; no exact ties were observed.

The mean row-level RMSE difference, defined as

$$\Delta_i = RMSE_{i, \text{Gateway}} - RMSE_{i, \text{Traditional}}, \quad (460)$$

was -0.060243 . A two-sided Wilcoxon signed-rank test applied to the row-level paired differences yielded $p < 10^{-300}$, and the corresponding paired Cohen’s d was -0.363 .

These inferential quantities are reported as row-level supplementary statistics rather than participant-level population inference because participant identifiers were not available in the evaluated validation export. Accordingly, they should not be interpreted as evidence based on independent participant-level observations, nor should the reported row-level significance be treated as a substitute for subject-level inference.

6.11 Output-Range Diagnostic

The Original Gateway was evaluated without post-hoc clipping of its predictions to the nominal $[1, 5]$ scoring interval. Under corrupted validation, the fraction of output elements falling outside this interval was 0.00101740 (approximately 0.102%). This diagnostic is reported explicitly rather than being concealed through output clipping.

The presence of a small fraction of out-of-range predictions indicates that the Gateway was evaluated in its unconstrained numerical output state. No post-processing operation was introduced that could otherwise alter the reported RMSE, MAE, Pearson, or Spearman values.

6.12 IPIP-120 Independent TEST

The independent IPIP-NEO-120 TEST benchmark was generated after completion of model development and was evaluated only after the final S2 checkpoint had been frozen. The TEST population comprises 61,557 clean observations and 1,723,596 corrupted observations, for a total of 1,785,153 evaluated observations. The benchmark additionally contains 31,316 TEST donor subjects and 28 family-level corruption blocks.

$$61,557 + 1,723,596 = 1,785,153. \quad (461)$$

The frozen S2 checkpoint corresponds to seed 20,260,933, epoch 6, the $120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30$ architecture, and 69,214 trainable parameters. Its recorded checkpoint SHA-256 digest is

$$679c88d15552ee6efaf78cb0b3e288c09b3f00af53cf6d4100807830323e6992. \quad (462)$$

The independent TEST population was not loaded or touched during the S2 multi-seed confirmation and was not used for architecture selection, objective selection, hyperparameter tuning, checkpoint selection, or retraining. The TEST evaluation therefore represents a separate held-out stage following the completion of model development.

On clean TEST observations, Traditional scoring reproduces the canonical Gold reference by construction. Accordingly,

$$\text{RMSE}_{\text{Traditional,clean}} = 0, \quad \text{MAE}_{\text{Traditional,clean}} = 0, \quad (463)$$

$$r_{\text{Pearson,Traditional,clean}} = 1, \quad \rho_{\text{Spearman,Traditional,clean}} = 1. \quad (464)$$

The frozen S2 clean-TEST metrics were

$$\text{RMSE}_{\text{S2,clean}} = 0.114098616, \quad \text{MAE}_{\text{S2,clean}} = 0.072518252, \quad (465)$$

$$r_{\text{Pearson,S2,clean}} = 0.990334181, \quad \rho_{\text{Spearman,S2,clean}} = 0.983815665. \quad (466)$$

The principal robustness analysis is performed on the corrupted TEST subpopulation, because clean TEST observations do not contain a perturbation from which robustness can be assessed.

Table 20: Final IPIP-NEO-120 S2 head-to-head performance on the independent frozen corrupted TEST benchmark.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	0.579328515	0.287756126	0.715059733	0.712384880
S2 final	0.428448503	0.264785287	0.829916061	0.798439021

The absolute differences are

$$\Delta \text{RMSE} = 0.428448503 - 0.579328515 = -0.150880012, \quad (467)$$

$$\Delta \text{MAE} = 0.264785287 - 0.287756126 = -0.022970839, \quad (468)$$

$$\Delta \text{Pearson} = 0.829916061 - 0.715059733 = +0.114856328, \quad (469)$$

and

$$\Delta \text{Spearman} = 0.798439021 - 0.712384880 = +0.086054141. \quad (470)$$

The corresponding relative RMSE change is

$$\frac{0.428448503 - 0.579328515}{0.579328515} \times 100 = -26.043947\%. \quad (471)$$

Thus, the corrupted-TEST RMSE is reduced by approximately 26.044%. The relative MAE change is

$$\frac{0.264785287 - 0.287756126}{0.287756126} \times 100 = -7.982745\%. \quad (472)$$

The relative Pearson change is

$$\frac{0.829916061 - 0.715059733}{0.715059733} \times 100 = 16.062480\%, \quad (473)$$

and the relative Spearman change is

$$\frac{0.798439021 - 0.712384880}{0.712384880} \times 100 = 12.079726\%. \quad (474)$$

These relative correlation changes are descriptive ratios of the observed correlation coefficients and should not be interpreted as percentages of explained variance.

6.12.1 Independent-TEST Error Decomposition for S2

For a scalar corrupted-TEST residual

$$e_i = \hat{y}_i - y_i^{\text{ref}}, \quad (475)$$

the aggregate squared error satisfies

$$\text{MSE} = \text{RMSE}^2. \quad (476)$$

Using the displayed corrupted-TEST RMSE values, the corresponding mean squared errors are approximately

$$\text{MSE}_{\text{Traditional}} = 0.579328515^2 \approx 0.335621528, \quad (477)$$

and

$$\text{MSE}_{\text{S2}} = 0.428448503^2 \approx 0.183568120. \quad (478)$$

The second central moment of the absolute residual magnitude follows from

$$\text{Var}(|e|) = \text{RMSE}^2 - \text{MAE}^2. \quad (479)$$

Therefore,

$$\text{Var}(|e|)_{\text{Traditional}} \approx 0.335621528 - 0.287756126^2 = 0.252817940, \quad (480)$$

whereas

$$\text{Var}(|e|)_{\text{S2}} \approx 0.183568120 - 0.264785287^2 = 0.113456872. \quad (481)$$

The squared-error quantity derived from the reported RMSE values therefore changes by approximately

$$\frac{0.183568120 - 0.335621528}{0.335621528} \times 100 = -45.305022\%. \quad (482)$$

The approximately 26.044% RMSE reduction and approximately 45.305% reduction in the corresponding MSE describe different scales of the same error quantity and must not be conflated.

At the perturbation-instance level, S2 was closer to the same participant- specific Gold reference than Traditional scoring for 72.9250% of corrupted TEST observations, Traditional scoring was closer for 27.0750%, and no ties were observed:

$$72.9250\% + 27.0750\% + 0\% = 100\%. \quad (483)$$

This paired statistic is evaluated at the perturbation-instance level and is not an independent participant-level probability or a replacement for the aggregate error metrics.

6.12.2 IPIP-120 Independent TEST Family Analysis

The final S2 corrupted-TEST RMSE was lower than Traditional scoring in all seven evaluated perturbation families:

Table 21: Final IPIP-NEO-120 S2 corrupted-TEST RMSE by perturbation family.

Perturbation family	Traditional RMSE	S2 RMSE	Relative reduction
ITEM_SWAP	0.454519	0.396420	12.783%
FACET_SWAP	0.459663	0.394646	14.144%
CATEGORY_PARTIAL_SWAP	0.305406	0.272947	10.628%
CATEGORY_SWAP	0.718169	0.591406	17.651%
ITEM_COLLAPSE	0.500320	0.366206	26.806%
CATEGORY_COLLAPSE	0.886608	0.495586	44.103%
RANDOM_ITEM_CHANGE	0.534827	0.410012	23.337%

For family f , the relative reduction is defined by

$$\Delta_{f,\text{rel}} = \frac{\text{RMSE}_{f,\text{Traditional}} - \text{RMSE}_{f,\text{S2}}}{\text{RMSE}_{f,\text{Traditional}}} \times 100. \quad (484)$$

The family-level reductions range from approximately 10.628% for CATEGORY_PARTIAL_SWAP to 44.103% for CATEGORY_COLLAPSE. These family subsets are not assumed to have equal cardinality, so the family-level relative reductions should not be averaged to reconstruct the global 26.044% RMSE reduction.

The family-level TEST result therefore provides a stratified robustness check showing that the global S2 result is not attributable to a single corruption family. It does not establish uniform robustness to perturbations outside the seven predefined families.

6.13 IPIP-NEO-300 Independent TEST

The final IPIP-NEO-300 production checkpoint was the P83.25 configuration with architecture

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (485)$$

220,782 trainable parameters, and objective

$$\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - \rho_{\text{Pearson}}). \quad (486)$$

The validation checkpoint-selection criterion was corrupted-validation RMSE. The selected checkpoint was epoch 14 with

$$\text{RMSE}_{\text{val,corr}} = 0.18483594655007943, \quad (487)$$

reported as 0.184836 in summary tables. Training subsequently continued through epoch 17, but the additional reduction did not satisfy the predefined minimum improvement threshold of 10^{-4} . Epoch 14 therefore remained the official production checkpoint.

The checkpoint is identified by the recorded SHA-256 digest

$$\text{f113f3b48b09f89deecce55788a59f257f31c09f12b17d3d2f657c0bd42fc287} . \quad (488)$$

Unlike the final IPIP-NEO-120 S2 configuration, P83.25 was not subjected to the same three-seed confirmatory retraining protocol. The P83.25 result therefore represents its fixed full-TRAIN production trajectory followed by validation- based checkpoint selection and independent frozen-TEST evaluation. It should not be described as a three-seed-confirmed result.

The independent IPIP-NEO-300 TEST contains 21,808 held-out participants and 632,432 evaluated observations, comprising 21,808 clean and 610,624 corrupted observations:

$$21,808 + 610,624 = 632,432. \quad (489)$$

The corrupted subset is structured as seven perturbation families evaluated at four perturbation levels:

$$7 \times 4 = 28, \quad (490)$$

and therefore

$$21,808 \times 28 = 610,624 \quad (491)$$

corrupted perturbation instances.

The final P83.25 checkpoint was fixed before TEST evaluation. The independent TEST population was not used for training, architecture screening, objective screening, checkpoint selection, or retraining.

The principal P83.25 robustness evaluation is conducted on the corrupted TEST subpopulation.

Table 22: Final IPIP-NEO-300 P83.25 E14 head-to-head performance on the independent frozen corrupted TEST benchmark.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	0.290377488	0.139889697	0.756625041	0.747784084
P83.25 E14	0.223052573	0.137993711	0.833130759	0.810581771

The absolute differences are

$$\Delta\text{RMSE} = 0.223052573 - 0.290377488 = -0.067324915, \quad (492)$$

$$\Delta\text{MAE} = 0.137993711 - 0.139889697 = -0.001895986, \quad (493)$$

$$\Delta\text{Pearson} = 0.833130759 - 0.756625041 = +0.076505718, \quad (494)$$

and

$$\Delta\text{Spearman} = 0.810581771 - 0.747784084 = +0.062797687. \quad (495)$$

The relative RMSE change is

$$\frac{0.223052573 - 0.290377488}{0.290377488} \times 100 = -23.185308\%. \quad (496)$$

The corresponding relative MAE change is

$$\frac{0.137993711 - 0.139889697}{0.139889697} \times 100 = -1.355344\%. \quad (497)$$

The relative Pearson change is

$$\frac{0.833130759 - 0.756625041}{0.756625041} \times 100 = 10.111444\%, \quad (498)$$

and the relative Spearman change is

$$\frac{0.810581771 - 0.747784084}{0.747784084} \times 100 = 8.397837\%. \quad (499)$$

The changes are reported as descriptive differences on the frozen corrupted TEST benchmark. They do not constitute probability estimates or claims of performance on unobserved populations.

6.13.1 Independent-TEST Error Decomposition for P83.25

For the corrupted P83.25 TEST residuals,

$$e_i = \hat{y}_i - y_i^{\text{ref}}, \quad (500)$$

with

$$\text{MSE} = \text{RMSE}^2. \quad (501)$$

The corresponding mean squared errors derived from the displayed RMSE values are approximately

$$\text{MSE}_{\text{Traditional}} = 0.290377488^2 \approx 0.084319086, \quad (502)$$

and

$$\text{MSE}_{\text{P83.25}} = 0.223052573^2 \approx 0.049752450. \quad (503)$$

The corresponding second central moments of absolute residual magnitude are

$$\text{Var}(|e|)_{\text{Traditional}} \approx 0.084319086 - 0.139889697^2 = 0.064749958, \quad (504)$$

and

$$\text{Var}(|e|)_{\text{P83.25}} \approx 0.049752450 - 0.137993711^2 = 0.030710186. \quad (505)$$

The squared-error quantity derived from the reported RMSE values therefore changes by approximately

$$\frac{0.049752450 - 0.084319086}{0.084319086} \times 100 = -40.995031\%. \quad (506)$$

The approximately 23.185% RMSE reduction and approximately 40.995% reduction in the corresponding MSE are mathematically different summaries and should not be conflated.

At the perturbation-instance level, P83.25 E14 was closer to Gold than Traditional scoring for 69.9270% of corrupted TEST observations, Traditional scoring was closer for 30.0730%, and no ties were observed.

$$69.9270\% + 30.0730\% + 0\% = 100\%. \quad (507)$$

This paired statistic is a perturbation-instance-level comparison and should not be interpreted as an independent participant-level probability.

6.14 Cross-Instrument Independent TEST Summary

The final frozen corrupted-TEST evaluations can be summarised using the same four aggregate metrics while retaining the distinction between the two instrument-specific production programs.

Table 23: Independent corrupted-TEST summary for the final IPIP-NEO-120 S2 and IPIP-NEO-300 P83.25 configurations.

Metric	IPIP-120 Traditional	IPIP-120 S2	IPIP-300 Traditional	IPIP-300 P83.25 E14
RMSE	0.579328515	0.428448503	0.290377488	0.223052573
MAE	0.287756126	0.264785287	0.139889697	0.137993711
Pearson	0.715059733	0.829916061	0.756625041	0.833130759
Spearman	0.712384880	0.798439021	0.747784084	0.810581771

Relative to Traditional scoring, the final corrupted-TEST RMSE reduction was 26.043947% for IPIP-120 S2 and 23.185308% for IPIP-300 P83.25. The corresponding MAE reductions were 7.982745% and 1.355344%, respectively. Pearson and Spearman increased in both instrument-specific TEST evaluations.

This cross-instrument consistency is descriptive rather than a claim of architectural equivalence or universal external validity. The two production models differ in input dimensionality, architecture, objective, parameter count, and model-selection history. In addition, the IPIP-120 final S2 result has three-seed confirmatory retraining, whereas P83.25 E14 does not. The two results therefore provide complementary evidence within the same IPIP-NEO reconstruction framework rather than statistically identical replications.

The independent TEST results remain bounded to the specified held-out populations and semi-synthetic perturbation protocol. They do not establish clinical validity, diagnostic utility, causal mechanisms, or robustness to perturbation families not represented in the locked evaluation protocol.

6.15 IPIP-300 Validation

The IPIP-NEO-300 validation benchmark contains 21,808 clean observations and 338,444 corrupted observations, yielding

$$21,808 + 338,444 = 360,252 \quad (508)$$

evaluated validation observations. The corrupted validation subset was generated under the same locked seven-family perturbation protocol defined for the study, whilst the clean subset provides the corresponding unperturbed reference condition. Traditional scoring reproduces the canonical Gold representation exactly on clean observations by construction.

The final production model evaluated on this benchmark is the P83.25 Gateway, whose selected architecture is

$$\boxed{300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}} \quad (509)$$

with five structured output heads, each producing six facet values, for a total of 30 reconstructed facet dimensions. The model contains 220,782 trainable parameters.

The complete parameter count is obtained as

$$\begin{aligned} N_\theta &= (300 \times 344 + 344) + (2 \times 344) \\ &\quad + (344 \times 208 + 208) + (2 \times 208) \\ &\quad + (208 \times 184 + 184) + (2 \times 184) \\ &\quad + 5(184 \times 6 + 6) \\ &= 220,782. \end{aligned} \quad (510)$$

The terms proportional to twice each hidden-layer width correspond to the trainable affine scale and bias parameters of the associated LayerNorm operations. The five final affine output heads contribute

$$5(184 \times 6 + 6) = 5 \times 1,110 = 5,550 \quad (511)$$

trainable parameters.

The production objective was the composite loss

$$\boxed{\mathcal{L}_{\text{P83.25}} = 0.1675 \mathcal{L}_{\text{MSE}} + 0.8325 (1 - \rho_{\text{Pearson}})}. \quad (512)$$

Thus, the optimisation assigns a larger coefficient to profile-level Pearson correspondence than to the element-wise MSE component. The model nevertheless reconstructs the original 30-dimensional facet representation directly rather than introducing an additional reduced output space.

For an input vector $X \in \mathbb{R}^{300}$, the production implementation first applies the fixed input normalisation

$$X' = \mathcal{N}(X) = \frac{X - 3}{2}. \quad (513)$$

The nonlinear trunk may therefore be represented as

$$H_1 = \mathcal{D}_{0.10} [\text{GELU} (\text{LN} (W_1 X' + b_1))] , \quad (514)$$

$$H_2 = \mathcal{D}_{0.10} [\text{GELU} (\text{LN} (W_2 H_1 + b_2))] , \quad (515)$$

and

$$H_3 = \mathcal{D}_{0.10} [\text{GELU} (\text{LN} (W_3 H_2 + b_3))], \quad (516)$$

where

$$H_1 \in \mathbb{R}^{344}, \quad H_2 \in \mathbb{R}^{208}, \quad H_3 \in \mathbb{R}^{184}. \quad (517)$$

Each structured output head is then defined by

$$\hat{Y}^{(k)} = W_k^{\text{out}} H_3 + b_k^{\text{out}}, \quad k \in \{1, \dots, 5\}, \quad (518)$$

with

$$\hat{Y}^{(k)} \in \mathbb{R}^6. \quad (519)$$

The complete 30-dimensional reconstruction is

$$\hat{Y} = \text{Concat} (\hat{Y}^{(1)}, \hat{Y}^{(2)}, \hat{Y}^{(3)}, \hat{Y}^{(4)}, \hat{Y}^{(5)}) \in \mathbb{R}^{30}. \quad (520)$$

For each validation observation n , Pearson profile correspondence is computed across the 30 reconstructed facets. Let

$$\hat{\bar{y}}_n = \frac{1}{30} \sum_{j=1}^{30} \hat{y}_{nj}, \quad \bar{y}_n = \frac{1}{30} \sum_{j=1}^{30} y_{nj}. \quad (521)$$

The observation-level Pearson coefficient is

$$r_n = \frac{\sum_{j=1}^{30} (\hat{y}_{nj} - \hat{\bar{y}}_n) (y_{nj} - \bar{y}_n)}{\sqrt{\sum_{j=1}^{30} (\hat{y}_{nj} - \hat{\bar{y}}_n)^2 + \varepsilon} \sqrt{\sum_{j=1}^{30} (y_{nj} - \bar{y}_n)^2 + \varepsilon}}, \quad (522)$$

with

$$\varepsilon = 10^{-8}. \quad (523)$$

The corresponding Pearson profile loss is

$$\mathcal{L}_{\text{Pearson}} = 1 - \frac{1}{N} \sum_{n=1}^N r_n. \quad (524)$$

The element-wise mean squared error is

$$\mathcal{L}_{\text{MSE}} = \frac{1}{30N} \sum_{n=1}^N \sum_{j=1}^{30} (\hat{y}_{nj} - y_{nj})^2. \quad (525)$$

The reported aggregate reconstruction metrics are defined by

$$\text{RMSE} = \sqrt{\frac{1}{30N} \sum_{n=1}^N \sum_{j=1}^{30} (\hat{y}_{nj} - y_{nj})^2}, \quad (526)$$

and

$$\text{MAE} = \frac{1}{30N} \sum_{n=1}^N \sum_{j=1}^{30} |\hat{y}_{nj} - y_{nj}|. \quad (527)$$

The final production fit was retrained from scratch on the complete TRAIN population of 101,772 subjects. Each subject contributed 143 predefined views per epoch, yielding

$$101,772 \times 143 = 14,553,396 \quad (528)$$

training examples per epoch.

The production optimisation configuration was:

Table 24: Core production and reproducibility configuration for the final IPIP-NEO-300 P83.25 Gateway.

Parameter	Value
Architecture	$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6$
Trainable parameters	220,782
Output structure	5 heads $\times$ 6 facets
TRAIN subjects	101,772
Views per subject	143
Training examples per epoch	14,553,396
Optimizer	AdamW
Batch size	8,192
Learning rate	0.002
Weight decay	0.0001
Dropout	0.10
Gradient clipping	1.0
Mixed precision	BF16
MSE weight	0.1675
Pearson weight	0.8325
Maximum epochs	120
Minimum epochs	5
Early-stop patience	3
Minimum improvement	10^{-4}
Global seed	20,260,920
SIES seed	20,261,697
SIES rounds	143
Validation clean	21,808
Validation corrupted	338,444
Validation total	360,252
Checkpoint criterion	Corrupted-validation RMSE
Selected epoch	14
TEST exposure during production	Reserved / untouched

The independent TEST population remained reserved and untouched throughout production training. The checkpoint-selection criterion was fixed in advance to the minimum corrupted-validation RMSE. Clean-validation RMSE, clean correspondence, corrupted MAE, Pearson correspondence, Spearman correspondence, and the independent TEST benchmark were not used as alternative checkpoint selection criteria.

The selected checkpoint was epoch 14, with

$$\text{RMSE}_{\text{val,corr}} = 0.18483594655007943. \quad (529)$$

The training process subsequently continued through epoch 17. At epoch 17, the observed corrupted-validation RMSE was approximately

$$\text{RMSE}_{\text{val,corr}}^{(17)} = 0.184769. \quad (530)$$

The corresponding numerical change relative to the epoch-14 checkpoint was

$$\Delta \text{RMSE}_{14 \rightarrow 17} = 0.184769 - 0.18483594655007943 \approx -0.000066947. \quad (531)$$

Because

$$|\Delta \text{RMSE}_{14 \rightarrow 17}| < 10^{-4}, \quad (532)$$

the later checkpoint did not satisfy the predefined minimum-improvement criterion. Early stopping therefore terminated training at epoch 17 while retaining epoch 14 as the official production checkpoint. The training run completed 17 epochs, but epoch 14 remained the frozen production artifact.

The selected checkpoint is identified by the recorded SHA-256 digest

$$\text{f113f3b48b09f89deecce55788a59f257f31c09f12b17d3d2f657c0bd42fc287} . \quad (533)$$

The checkpoint identity is therefore jointly specified by the architecture, parameter count, P83.25 objective, full-TRAIN production fit, selected epoch, and cryptographic digest.

Unlike the final IPIP-NEO-120 S2 configuration, the P83.25 production result was not subjected to a three-seed confirmatory retraining protocol. The IPIP-NEO-300 result therefore represents a single predefined full-TRAIN production trajectory followed by validation-based checkpoint selection and independent frozen-TEST evaluation. No multi-seed confirmation claim is made for P83.25.

6.15.1 Head-to-Head Corrupted-Validation Performance

The principal validation comparison was performed on the 338,444 corrupted observations. The final epoch-14 P83.25 checkpoint produced

$$\text{RMSE}_{\text{P83.25}} = 0.18483594655007943, \quad (534)$$

$$\text{MAE}_{\text{P83.25}} = 0.105477, \quad (535)$$

with

$$\rho_{\text{Pearson,P83.25}} = 0.902523 \quad (536)$$

and

$$\rho_{\text{Spearman,P83.25}} = 0.892732. \quad (537)$$

The corresponding Traditional-scoring reference values were

$$\text{RMSE}_{\text{Traditional}} = 0.240123, \quad (538)$$

$$\text{MAE}_{\text{Traditional}} = 0.092047, \quad (539)$$

$$\rho_{\text{Pearson,Traditional}} = 0.842268, \quad (540)$$

and

$$\rho_{\text{Spearman,Traditional}} = 0.839497. \quad (541)$$

Table 25: Final IPIP-NEO-300 P83.25 E14 head-to-head performance on the corrupted validation benchmark.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	0.240123	0.092047	0.842268	0.839497
P83.25 E14	0.184836	0.105477	0.902523	0.892732

Traditional scoring has an exact clean-reference identity condition on the clean validation subset. For the P83.25 checkpoint, the recorded clean validation RMSE at the selected epoch was

$$\text{RMSE}_{\text{val,clean}} = 0.060923, \quad (542)$$

whereas Traditional scoring reproduces Gold exactly on clean observations by construction. The absolute differences between P83.25 and Traditional on corrupted validation are

$$\Delta \text{RMSE} = 0.18483594655007943 - 0.240123 = -0.05528705345, \quad (543)$$

$$\Delta \text{MAE} = 0.105477 - 0.092047 = +0.013430, \quad (544)$$

$$\Delta \text{Pearson} = 0.902523 - 0.842268 = +0.060255, \quad (545)$$

and

$$\Delta \text{Spearman} = 0.892732 - 0.839497 = +0.053235. \quad (546)$$

The relative corrupted-RMSE reduction is

$$\Delta_{\text{RMSE,rel}} = \frac{0.240123 - 0.18483594655007943}{0.240123} \times 100 = 23.024472\%. \quad (547)$$

The corresponding relative MAE change is

$$\Delta_{\text{MAE,rel}} = \frac{0.105477 - 0.092047}{0.092047} \times 100 = 14.590372\%. \quad (548)$$

Thus, the validation result exhibits a non-uniform metric pattern: P83.25 has lower RMSE and higher Pearson and Spearman correspondence, whereas Traditional scoring has the lower MAE on this validation benchmark.

The relative changes in correlation are

$$\Delta_{\text{Pearson,rel}} = \frac{0.902523 - 0.842268}{0.842268} \times 100 = 7.153899\%, \quad (549)$$

and

$$\Delta_{\text{Spearman,rel}} = \frac{0.892732 - 0.839497}{0.839497} \times 100 = 6.341297\%. \quad (550)$$

6.15.2 Validation Error Decomposition

Because

$$\text{MSE} = \text{RMSE}^2, \quad (551)$$

the corresponding squared-error quantities implied by the reported validation RMSE values are

$$\text{MSE}_{\text{Traditional}} = 0.240123^2 \approx 0.057659055, \quad (552)$$

and

$$\text{MSE}_{\text{P83.25}} = 0.18483594655007943^2 \approx 0.034164328. \quad (553)$$

The corresponding relative reduction in this squared-error quantity is

$$\frac{0.034164328 - 0.057659055}{0.057659055} \times 100 \approx -40.747681\%. \quad (554)$$

Therefore, the approximately 23.024% RMSE reduction corresponds to an approximately 40.748% reduction in the derived mean-squared error. These quantities remain distinct because RMSE is expressed on the original error scale whereas MSE is expressed on the squared-error scale.

A second decomposition follows from

$$\text{Var}(|e|) = \mathbb{E}[e^2] - (\mathbb{E}[|e|])^2 = \text{RMSE}^2 - \text{MAE}^2. \quad (555)$$

For Traditional scoring,

$$\text{Var}(|e|)_{\text{Traditional}} \approx 0.057659055 - 0.092047^2 = 0.049186405, \quad (556)$$

whereas for P83.25,

$$\text{Var}(|e|)_{\text{P83.25}} \approx 0.034164328 - 0.105477^2 = 0.023038930. \quad (557)$$

The resulting change is approximately

$$\frac{0.023038930 - 0.049186405}{0.049186405} \times 100 = -53.159964\%. \quad (558)$$

This decomposition is descriptive rather than inferential. It characterises the second-moment structure implied by the reported aggregate error metrics and does not constitute a separate statistical test.

6.15.3 Corruption-Family Validation Analysis

The final epoch-14 production checkpoint was also evaluated separately across the seven predefined corruption families. Traditional scoring was evaluated on the same frozen validation observations and therefore provides a common deterministic reference against which the P83.25 outputs are compared.

Table 26: Corrupted-validation RMSE by perturbation family for Traditional scoring and the final P83.25 E14 checkpoint.

Corruption family	Traditional RMSE	P83.25 RMSE	Relative change
ITEM_SWAP	0.123056	0.124162	−0.899%
FACET_SWAP	0.211145	0.177554	15.909%
CATEGORY_PARTIAL_SWAP	0.170146	0.141357	16.920%
CATEGORY_SWAP	0.382800	0.294410	23.090%
ITEM_COLLAPSE	0.126446	0.121360	4.022%
CATEGORY_COLLAPSE	0.363239	0.229590	36.794%
RANDOM_ITEM_CHANGE	0.143088	0.134574	5.950%

For family f , the relative RMSE reduction is defined as

$$\Delta_{f,\text{rel}} = \frac{\text{RMSE}_{f,\text{Traditional}} - \text{RMSE}_{f,\text{P83.25}}}{\text{RMSE}_{f,\text{Traditional}}} \times 100. \quad (559)$$

Under this definition, P83.25 has lower RMSE than Traditional scoring in six of the seven predefined corruption families. ITEM_SWAP is the sole family in which the P83.25 RMSE is higher, by approximately 0.001106, corresponding to a relative change of approximately −0.899%. The largest observed family-level RMSE reduction occurs for CATEGORY_COLLAPSE at approximately 36.794%.

These family-level results show that the aggregate validation behaviour is not uniform across corruption mechanisms. The magnitude and direction of the difference depend on the perturbation family, and the results therefore remain bounded to the predefined corruption protocol.

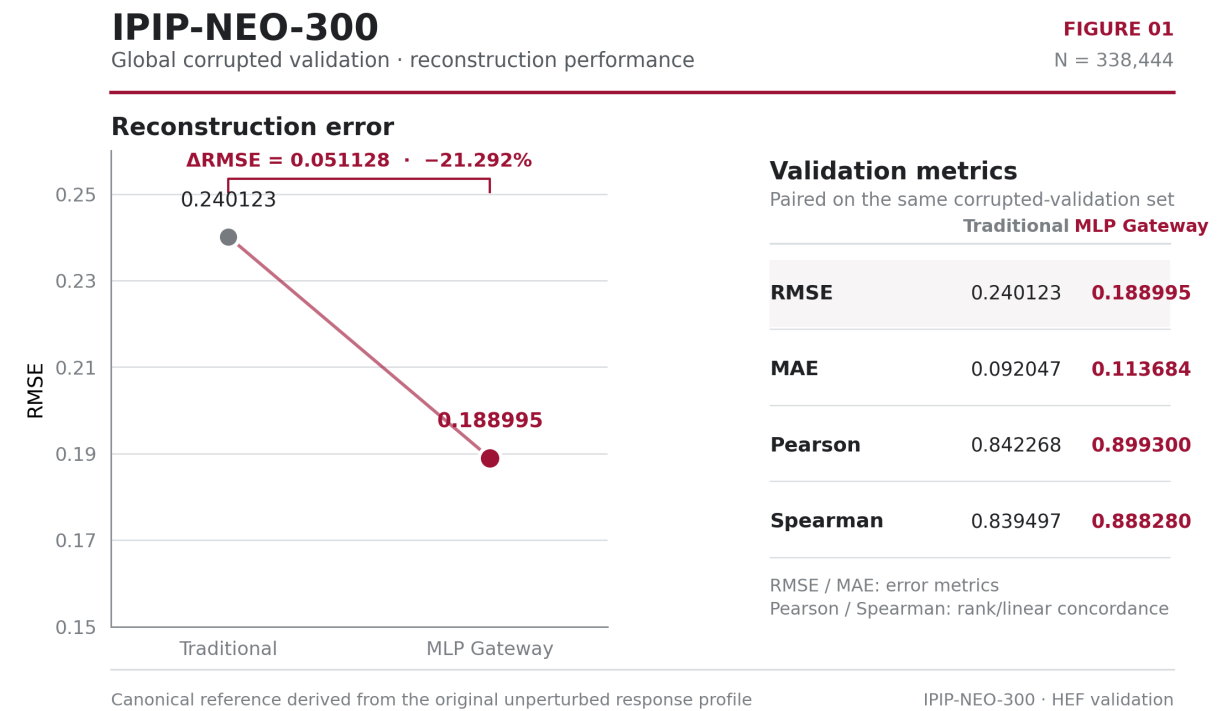


Figure 24: Historical global corrupted-validation visualisation retained from the earlier IPIP-NEO-300 development benchmark. The figure is preserved as a documentary research-stage artefact and is not treated as an additional evaluation condition for the final P83.25 production checkpoint.

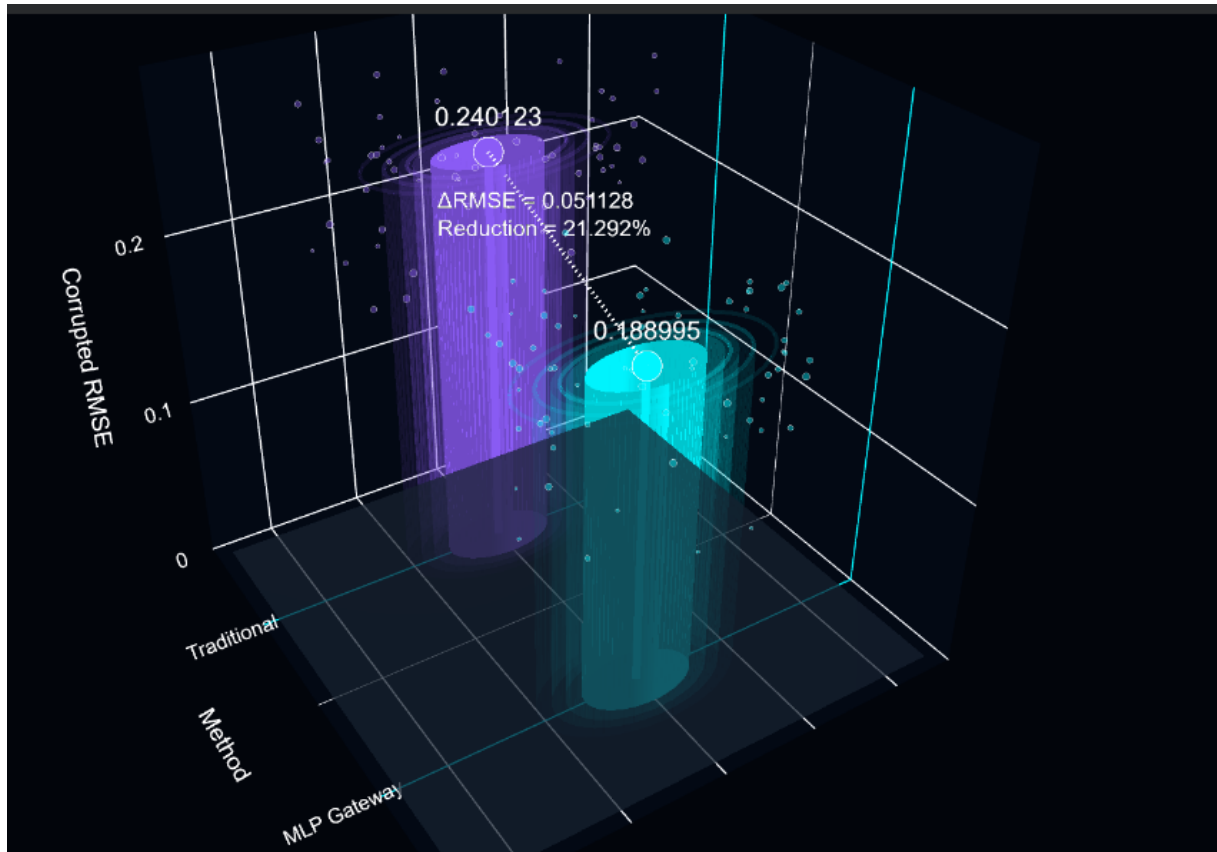


Figure 25: Historical three-dimensional visualisation of the global corrupted validation comparison for IPIP-NEO-300. The visualisation is retained unchanged as part of the documented development record and represents the same aggregate validation quantities rather than a separate benchmark.

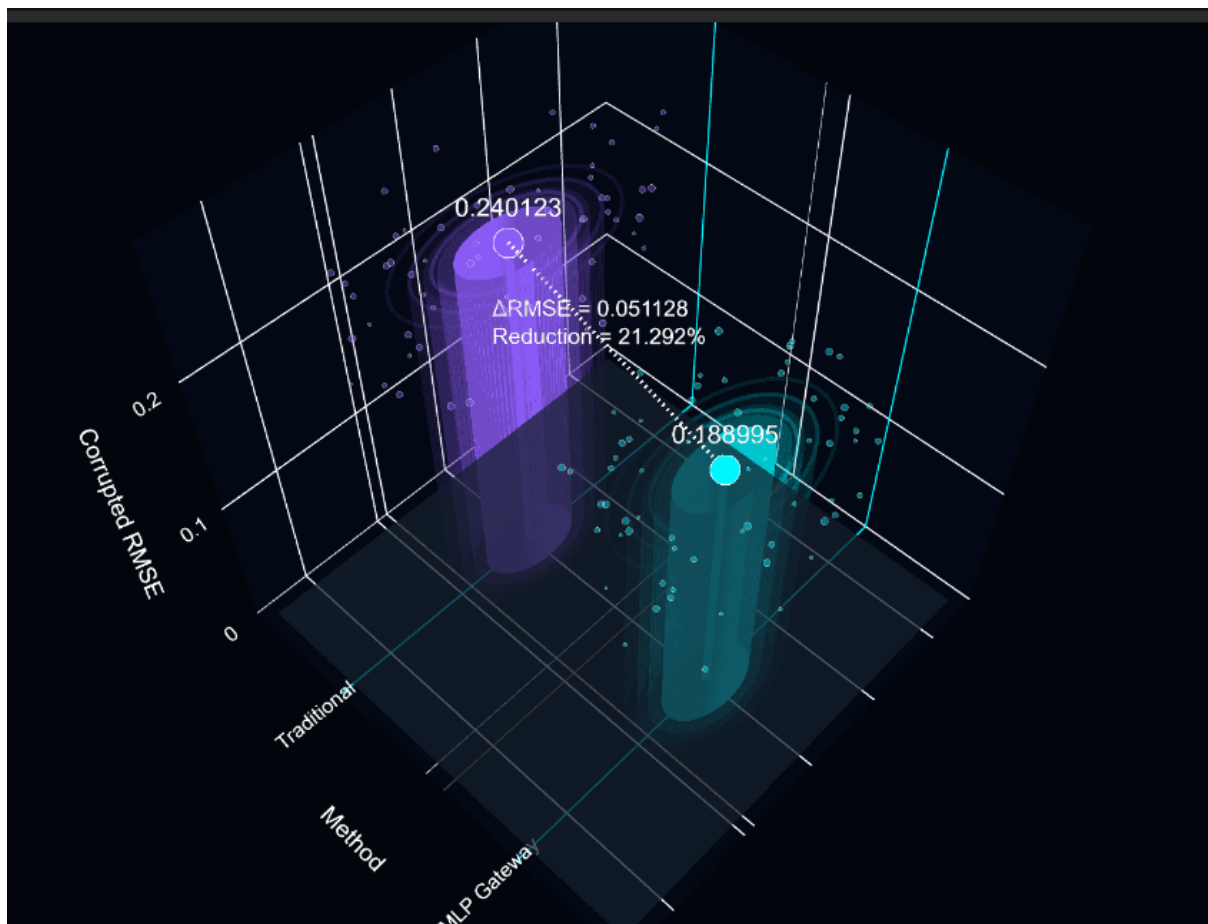


Figure 26: Historical complementary three-dimensional visualisation of the IPIP-NEO-300 corrupted-validation comparison. The figure is retained unchanged for research-history continuity and does not define the final P83.25 production performance.

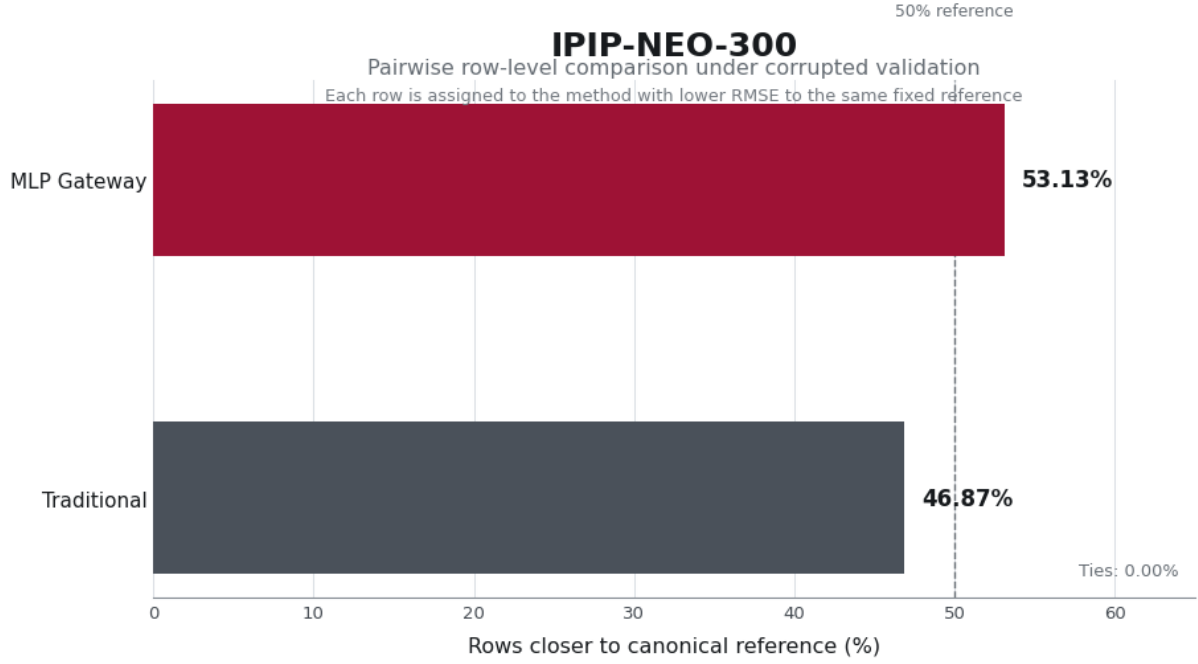


Figure 27: Historical paired row-level visualisation from the earlier IPIP-NEO-300 development validation benchmark. The figure is retained as a documentary artefact and is not used as a separate statistical evaluation of the final P83.25 checkpoint.

6.15.4 Independent IPIP-NEO-300 TEST

The independent IPIP-NEO-300 TEST was generated separately and frozen before final evaluation. It contains 21,808 held-out participants and

$$21,808 + 610,624 = 632,432 \quad (560)$$

total evaluated observations, comprising 21,808 clean observations and 610,624 corrupted observations.

The corrupted TEST subset is structured as seven corruption families evaluated at four predefined perturbation levels:

$$7 \times 4 = 28 \quad (561)$$

perturbation blocks, and therefore

$$21,808 \times 28 = 610,624 \quad (562)$$

corrupted perturbation instances.

The final production checkpoint was fixed before TEST evaluation. The independent TEST population was not used for training, architecture screening, objective screening, checkpoint selection, or retraining.

The principal TEST robustness analysis is conducted on the 610,624 corrupted observations because this subset corresponds to the controlled perturbation regime in which the observed corrupted response vector is evaluated against the same participant-specific Gold representation for both Traditional scoring and the learned Gateway.

Table 27: Final IPIP-NEO-300 P83.25 E14 head-to-head performance on the independent frozen corrupted TEST benchmark.

Method	Corrupted RMSE	Corrupted MAE	Pearson	Spearman
Traditional	0.290377488	0.139889697	0.756625041	0.747784084
P83.25 E14	0.223052573	0.137993711	0.833130759	0.810581771

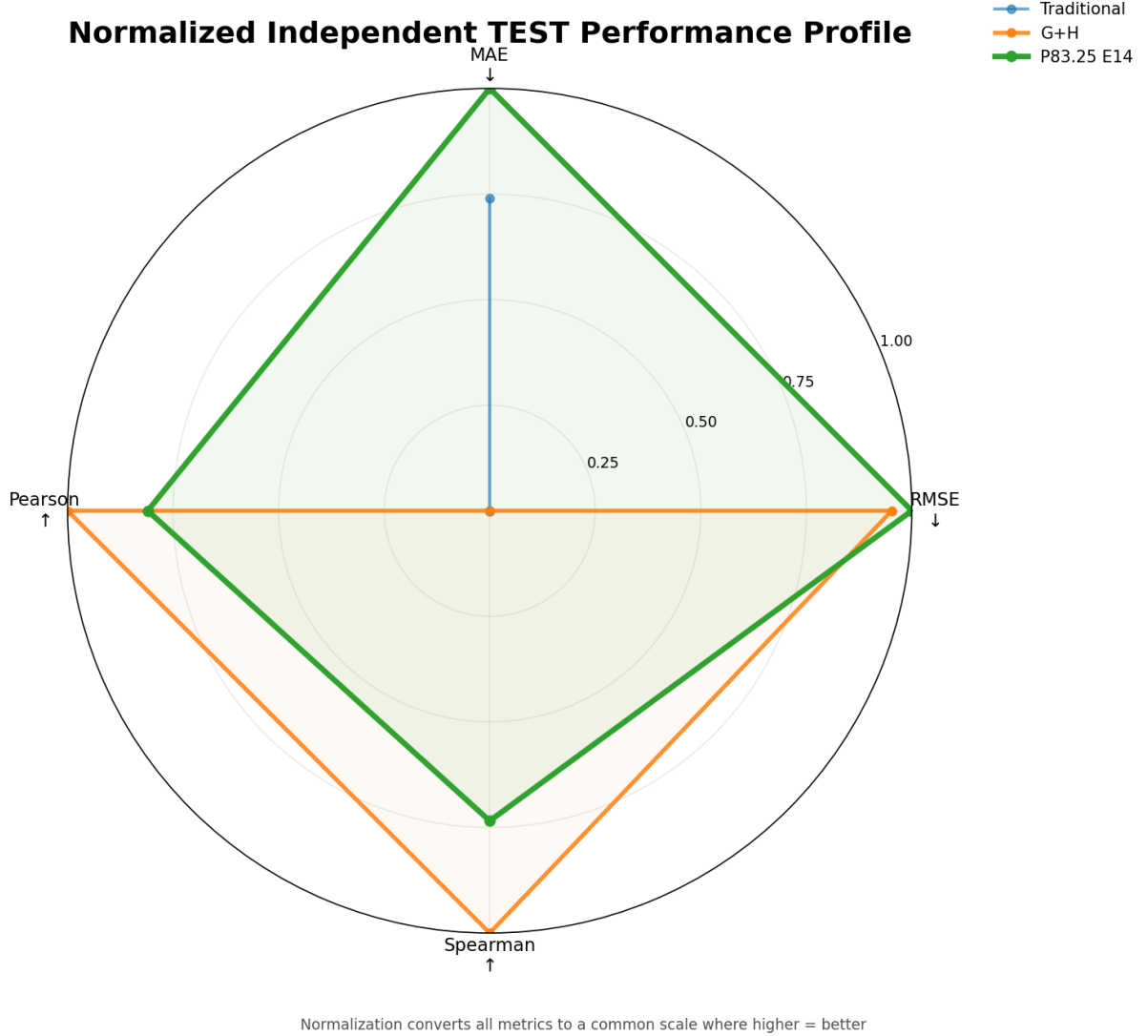


Figure 28: Normalised independent corrupted-TEST performance profile for IPIP-NEO-300. The visualisation places RMSE and MAE on error-oriented axes whose direction is inverted for display so that larger radial values indicate more favourable visual position, whilst Pearson and Spearman retain their direct higher-is-better orientation. The normalisation is used only for visual comparison and does not alter the reported numerical metrics. The P83.25 E14 checkpoint is the final production model; G+H is shown only as a historical development comparator and was not used as the final production checkpoint.

The absolute differences are

$$\Delta\text{RMSE} = 0.223052573 - 0.290377488 = -0.067324915, \quad (563)$$

$$\Delta\text{MAE} = 0.137993711 - 0.139889697 = -0.001895986, \quad (564)$$

$$\Delta\text{Pearson} = 0.833130759 - 0.756625041 = +0.076505718, \quad (565)$$

and

$$\Delta\text{Spearman} = 0.810581771 - 0.747784084 = +0.062797687. \quad (566)$$

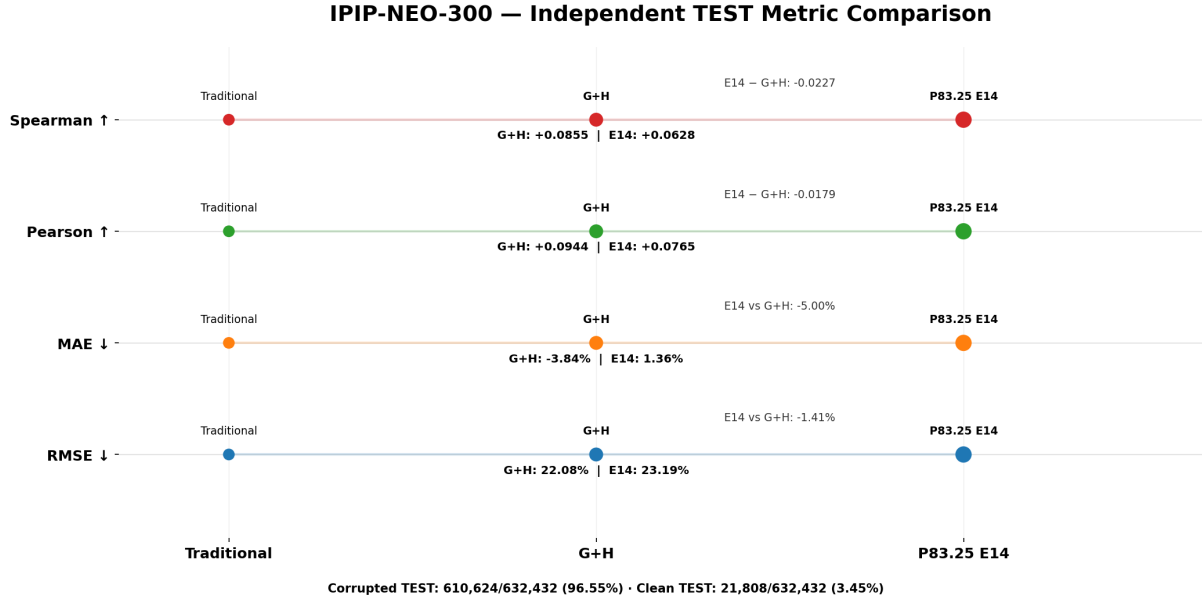


Figure 29: Metric-level comparison on the independent corrupted IPIP-NEO-300 TEST benchmark. The figure visualises the observed values of RMSE, MAE, Pearson, and Spearman for Traditional scoring, the historical G+H development comparator, and the final P83.25 E14 checkpoint. The annotated differences are descriptive comparisons against Traditional scoring. G+H is included solely to document the development trajectory and does not constitute an alternative final production model.

The corresponding relative RMSE reduction is

$$\Delta_{\text{RMSE,rel}} = \frac{0.290377488 - 0.223052573}{0.290377488} \times 100 = 23.185308\%. \quad (567)$$

The relative MAE reduction is

$$\Delta_{\text{MAE,rel}} = \frac{0.139889697 - 0.137993711}{0.139889697} \times 100 = 1.355344\%. \quad (568)$$

The relative changes in Pearson and Spearman correspondence are

$$\Delta_{\text{Pearson,rel}} = \frac{0.833130759 - 0.756625041}{0.756625041} \times 100 = 10.111444\%, \quad (569)$$

and

$$\Delta_{\text{Spearman,rel}} = \frac{0.810581771 - 0.747784084}{0.747784084} \times 100 = 8.397837\%. \quad (570)$$

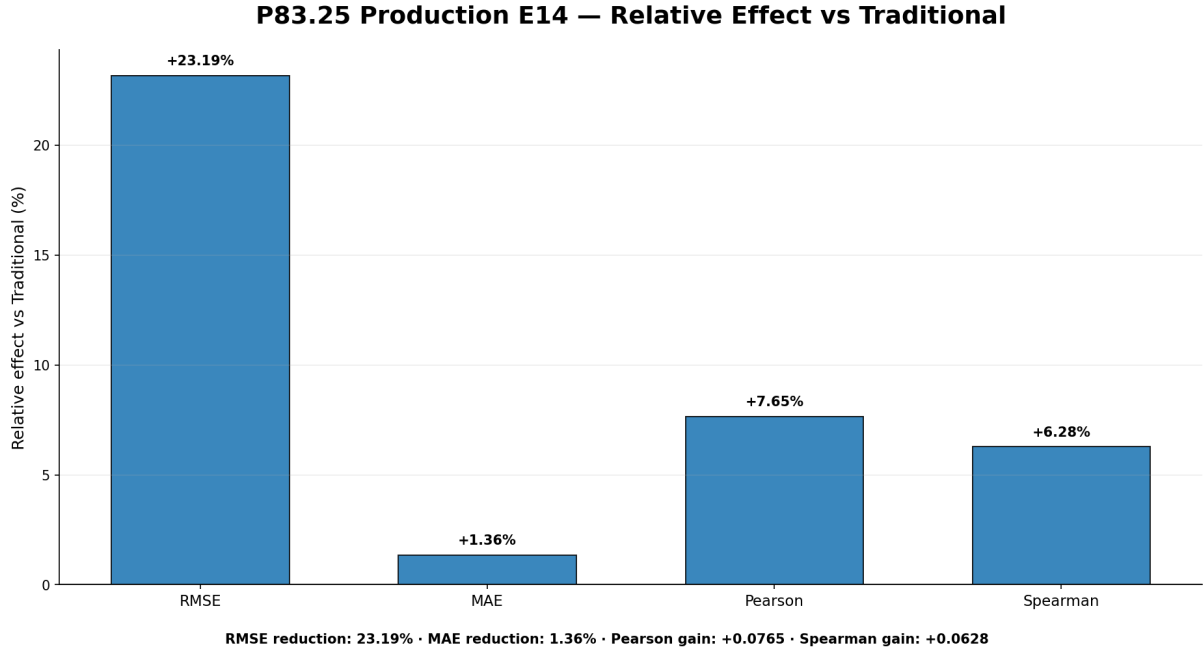


Figure 30: Relative effect of the final P83.25 E14 checkpoint with respect to Traditional scoring on the independent corrupted IPIP-NEO-300 TEST benchmark. The error metrics are expressed as relative reductions, whereas Pearson and Spearman are expressed as relative changes in correspondence. G+H is not used to define the final P83.25 result and is not part of the final checkpoint-selection procedure.

Thus, on the independent corrupted TEST benchmark, P83.25 E14 produces lower reported RMSE and MAE values together with higher Pearson and Spearman correspondence values than Traditional scoring. These changes are descriptive differences on the frozen benchmark and are not interpreted as probability estimates or claims of performance on unobserved populations.

6.15.5 Independent TEST Error Decomposition

The squared-error quantities implied by the independent TEST RMSE values are

$$\text{MSE}_{\text{Traditional}} = 0.290377488^2 \approx 0.084319086, \quad (571)$$

and

$$\text{MSE}_{\text{P83.25}} = 0.223052573^2 \approx 0.049752450. \quad (572)$$

The resulting relative reduction in the derived mean-squared error is

$$\frac{0.049752450 - 0.084319086}{0.084319086} \times 100 = -40.995031\%. \quad (573)$$

The corresponding absolute-error variance quantities are

$$\text{Var}(|e|)_{\text{Traditional}} \approx 0.084319086 - 0.139889697^2 = 0.064749958, \quad (574)$$

and

$$\text{Var}(|e|)_{\text{P83.25}} \approx 0.049752450 - 0.137993711^2 = 0.030710186. \quad (575)$$

Therefore, the corresponding relative change is approximately

$$\frac{0.030710186 - 0.064749958}{0.064749958} \times 100 = -52.795\%. \quad (576)$$

The approximately 23.185% RMSE reduction and the approximately 40.995% reduction in the corresponding MSE are mathematically distinct summaries and should not be conflated.

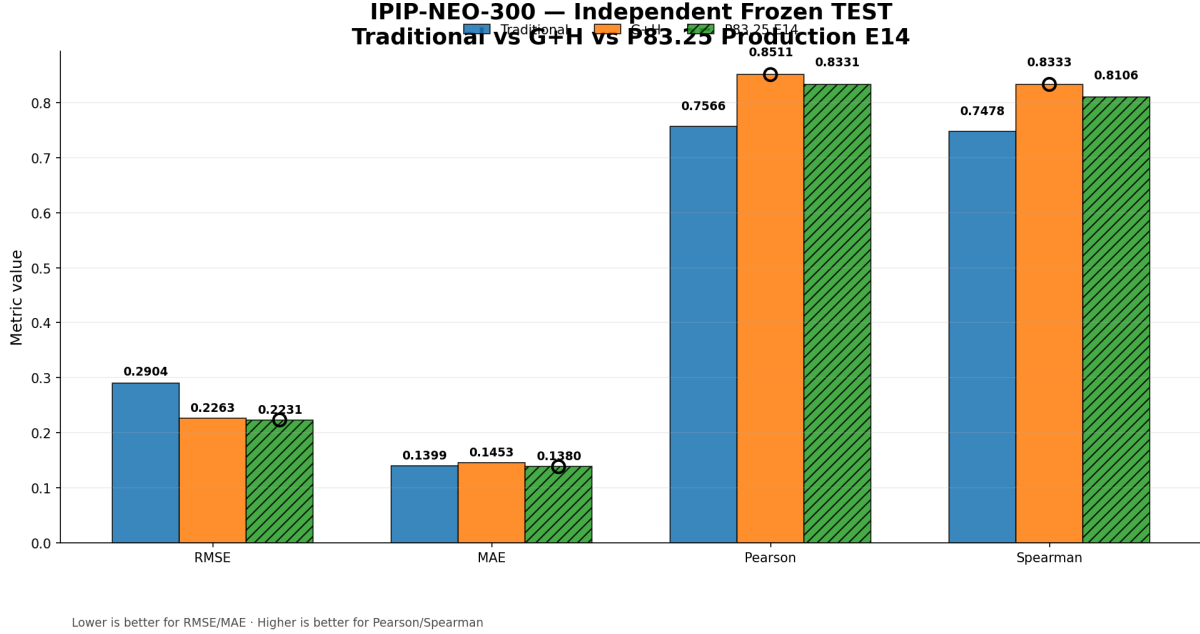


Figure 31: Absolute metric comparison on the independent frozen IPIP-NEO-300 TEST benchmark. The figure displays the corrupted-TEST values for Traditional scoring, the historical G+H development comparator, and the final P83.25 E14 checkpoint. Lower values represent lower reconstruction error for RMSE and MAE, whereas higher values represent greater correspondence for Pearson and Spearman. G+H is shown only for documented developmental context and does not constitute the final production checkpoint.

At the perturbation-instance level, P83.25 E14 was closer to Gold than Traditional scoring for 69.9270% of corrupted TEST observations, whereas Traditional scoring was closer for 30.0730%; no ties were observed.

$$69.9270\% + 30.0730\% + 0\% = 100\%. \quad (577)$$

This paired statistic is an observation-level comparison over predefined perturbation instances and is not interpreted as an independent participant-level probability, because multiple perturbation instances may belong to the same held-out participant.

6.15.6 Interpretive Boundaries of the IPIP-NEO-300 Result

The IPIP-NEO-300 validation and TEST results provide evidence for the behaviour of the P83.25 reconstruction mapping under the specified seven-family semi-synthetic perturbation protocol. They do not establish performance under arbitrary missingness mechanisms, naturally occurring response noise, adversarial perturbations outside the benchmark design, alternative psychometric instruments, clinical populations, or independently defined psychological ground truth.

The retained Gold representation is an operational facet-level reference computed from the original clean response vector. Accordingly, the reported reductions in reconstruction error

quantify agreement with that fixed reference representation rather than direct recovery of an independently observed psychological state.

Likewise, the correlation values quantify correspondence between the predicted and Gold-defined 30-dimensional facet profiles. They should not be interpreted as establishing causal equivalence between learned model coordinates and psychological constructs beyond the predefined scoring representation.

The final IPIP-NEO-300 result is therefore bounded to the following sequence:

development and screening $\rightarrow$ full-TRAIN production retraining $\rightarrow$ validation checkpoint selection $\rightarrow$ checkpoint selection (578)

This ordering ensures that the independent TEST population remains outside the fitted production model and outside the model-development decision path.

6.16 Cross-Instrument Results

The final IPIP-NEO-120 S2 configuration and the final IPIP-NEO-300 P83.25 configuration were evaluated under their respective frozen validation protocols. Both experiments use the same Gold-defined 30-dimensional facet representation and the same seven structured perturbation families, but they differ in input dimensionality, architecture, optimisation objective, parameterisation, checkpoint-selection history, and confirmation procedure.

For the IPIP-NEO-120 experiment, the final S2 checkpoint corresponds to architecture $120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30$, whereas the final IPIP-NEO-300 production checkpoint corresponds to $300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6$. The IPIP-NEO-120 S2 result was additionally subjected to a predefined three-seed confirmation procedure, whereas the P83.25 result represents a single full-TRAIN production trajectory followed by validation-based checkpoint selection and independent frozen-TEST evaluation. The two results should therefore be interpreted as complementary instrument-specific experiments rather than statistically identical replications.

Table 28: Final cross-instrument corrupted-validation summary for the IPIP-NEO-120 S2 and IPIP-NEO-300 P83.25 configurations.

Configuration	Traditional RMSE	Final Gateway RMSE	Relative RMSE reduction	Gateway checkpoint
IPIP-120 S2	0.462274	0.349023	24.499%	S2, seed 20260933
IPIP-300 P83.25	0.240123	0.184836	23.024%	E14

For each instrument m , the relative corrupted-validation RMSE reduction is defined as

$$\Delta_{\text{RMSE,rel}}^{(m)} = \frac{\text{RMSE}_{\text{Traditional}}^{(m)} - \text{RMSE}_{\text{Gateway}}^{(m)}}{\text{RMSE}_{\text{Traditional}}^{(m)}} \times 100. \quad (579)$$

For IPIP-NEO-120 S2,

$$\Delta_{\text{RMSE,rel}}^{(120)} = \frac{0.462274 - 0.349023}{0.462274} \times 100 = 24.499\% \quad (580)$$

to the displayed precision, whereas for IPIP-NEO-300 P83.25 E14,

$$\Delta_{\text{RMSE,rel}}^{(300)} = \frac{0.240123 - 0.18483594655007943}{0.240123} \times 100 = 23.024472\%. \quad (581)$$

The complete corrupted-validation metric profiles are

Table 29: Final corrupted-validation metric profiles across the two instrument-specific production configurations.

Configuration	Method	RMSE	MAE	Pearson	Spearman
IPIP-120	Traditional	0.462274	0.181624	0.818282	0.816762
IPIP-120	S2	0.349023	0.196198	0.880231	0.850618
IPIP-300	Traditional	0.240123	0.092047	0.842268	0.839497
IPIP-300	P83.25 E14	0.184836	0.105477	0.902523	0.892732

The two final validation programmes therefore exhibit the same directional pattern for three of the four aggregate metrics: the learned Gateway has lower corrupted RMSE and higher Pearson and Spearman correspondence than Traditional scoring in both instrument configurations. Traditional scoring retains the lower corrupted MAE in both cases.

For IPIP-NEO-120, the absolute changes relative to Traditional scoring are

$$\Delta\text{RMSE}_{120} = 0.349023 - 0.462274 = -0.113251, \quad (582)$$

$$\Delta\text{MAE}_{120} = 0.196198 - 0.181624 = +0.014574, \quad (583)$$

$$\Delta\text{Pearson}_{120} = 0.880231 - 0.818282 = +0.061949, \quad (584)$$

and

$$\Delta\text{Spearman}_{120} = 0.850618 - 0.816762 = +0.033856. \quad (585)$$

For IPIP-NEO-300 P83.25 E14,

$$\Delta\text{RMSE}_{300} = 0.18483594655007943 - 0.240123 = -0.05528705345, \quad (586)$$

$$\Delta\text{MAE}_{300} = 0.105477 - 0.092047 = +0.013430, \quad (587)$$

$$\Delta\text{Pearson}_{300} = 0.902523 - 0.842268 = +0.060255, \quad (588)$$

and

$$\Delta\text{Spearman}_{300} = 0.892732 - 0.839497 = +0.053235. \quad (589)$$

The corresponding MAE changes demonstrate that the lower RMSE achieved by the learned Gateway should not be interpreted as uniform improvement across all loss functions. The validation evidence instead exhibits a metric-specific trade-off that is consistent with the distinct optimisation objectives and model-selection protocols used in the two instrument-specific experiments.

The cross-instrument results are not pooled into a single effect size. Such pooling would combine different input spaces, parameterisations, validation populations, production trajectories, and checkpoint-selection procedures. The reported percentages therefore remain instrument-specific descriptive quantities.

The cross-instrument comparison also does not establish architectural equivalence. The IPIP-120 S2 Gateway and IPIP-300 P83.25 Gateway were not required to share hidden dimensions,

parameter counts, objective weighting, or training trajectories. Their commonality is instead defined at the level of the reconstruction problem, Gold target definition, perturbation protocol, and 30-dimensional output representation.

6.17 Corruption-Family Analysis

Family-specific analysis was performed for the final IPIP-NEO-300 P83.25 E14 checkpoint to determine whether the aggregate validation behaviour is concentrated in a single perturbation family or persists across the seven predefined corruption mechanisms.

The seven families are

$$\mathcal{F} = \{\text{ITEM_SWAP}, \text{FACET_SWAP}, \text{CATEGORY_PARTIAL_SWAP}, \text{CATEGORY_SWAP}, \text{ITEM_COLLAPSE}, \text{CATEGORY_COLLAPSE}, \text{RANDOM_ITEM_CHANGE}\} \quad (590)$$

For each family $f \in \mathcal{F}$, the family-specific RMSE is computed on the corresponding frozen subset of corrupted validation observations against the same participant-specific Gold target.

Table 30: Final IPIP-NEO-300 P83.25 E14 corrupted-validation RMSE by perturbation family.

Corruption family	Traditional RMSE	P83.25 E14 RMSE	Absolute change	Relative reduction
ITEM_SWAP	0.123056	0.124162	+0.001106	−0.899%
FACET_SWAP	0.211145	0.177554	−0.033591	15.909%
CATEGORY_PARTIAL_SWAP	0.170146	0.141357	−0.028789	16.920%
CATEGORY_SWAP	0.382800	0.294410	−0.088390	23.090%
ITEM_COLLAPSE	0.126446	0.121360	−0.005086	4.022%
CATEGORY_COLLAPSE	0.363239	0.229590	−0.133649	36.794%
RANDOM_ITEM_CHANGE	0.143088	0.134574	−0.008514	5.950%

For family f , the absolute RMSE difference is

$$\Delta\text{RMSE}_f = \text{RMSE}_{f,\text{P83.25}} - \text{RMSE}_{f,\text{Traditional}}, \quad (591)$$

and the corresponding relative reduction is

$$\Delta_{f,\text{rel}} = \frac{\text{RMSE}_{f,\text{Traditional}} - \text{RMSE}_{f,\text{P83.25}}}{\text{RMSE}_{f,\text{Traditional}}} \times 100. \quad (592)$$

The final P83.25 checkpoint has lower corrupted-validation RMSE in six of the seven predefined families. The sole exception is **ITEM_SWAP**, for which the P83.25 RMSE is approximately 0.001106 higher than the Traditional reference. This exception is retained explicitly and is not removed from the analysis.

The largest family-specific reduction occurs for **CATEGORY_COLLAPSE**, at approximately 36.794%. The remaining families exhibit intermediate reductions ranging from approximately 4.022% to 23.090%.

The family-level results therefore show that the aggregate validation difference is heterogeneous with respect to perturbation mechanism. The results do not support a claim of uniform robustness across all possible response perturbations; they support a bounded robustness statement over the seven families explicitly represented in the locked protocol.

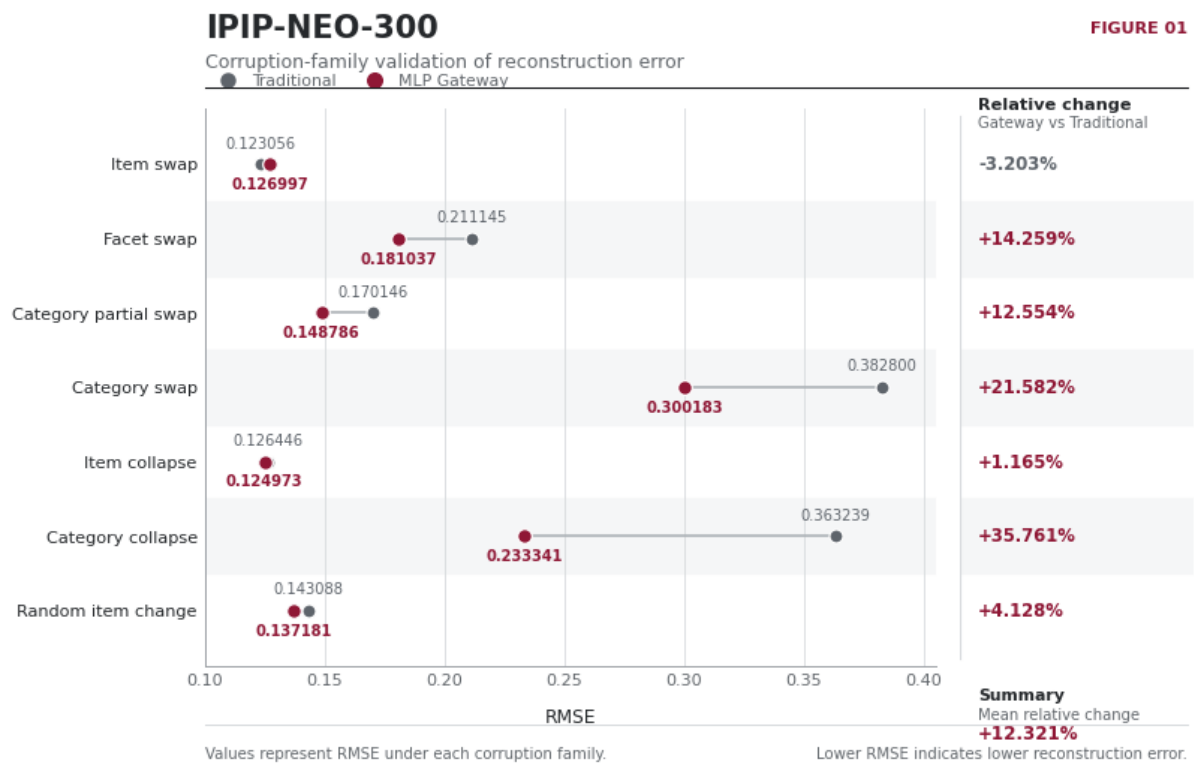


Figure 32: Historical family-level corrupted-validation RMSE visualisation retained from the earlier IPIP-NEO-300 development analysis. The figure provides a visual record of family-specific reconstruction behaviour during the earlier development programme. The final P83.25 E14 family-level numerical results are reported independently in Table 30; the historical figure is not treated as a second measurement of the final checkpoint.

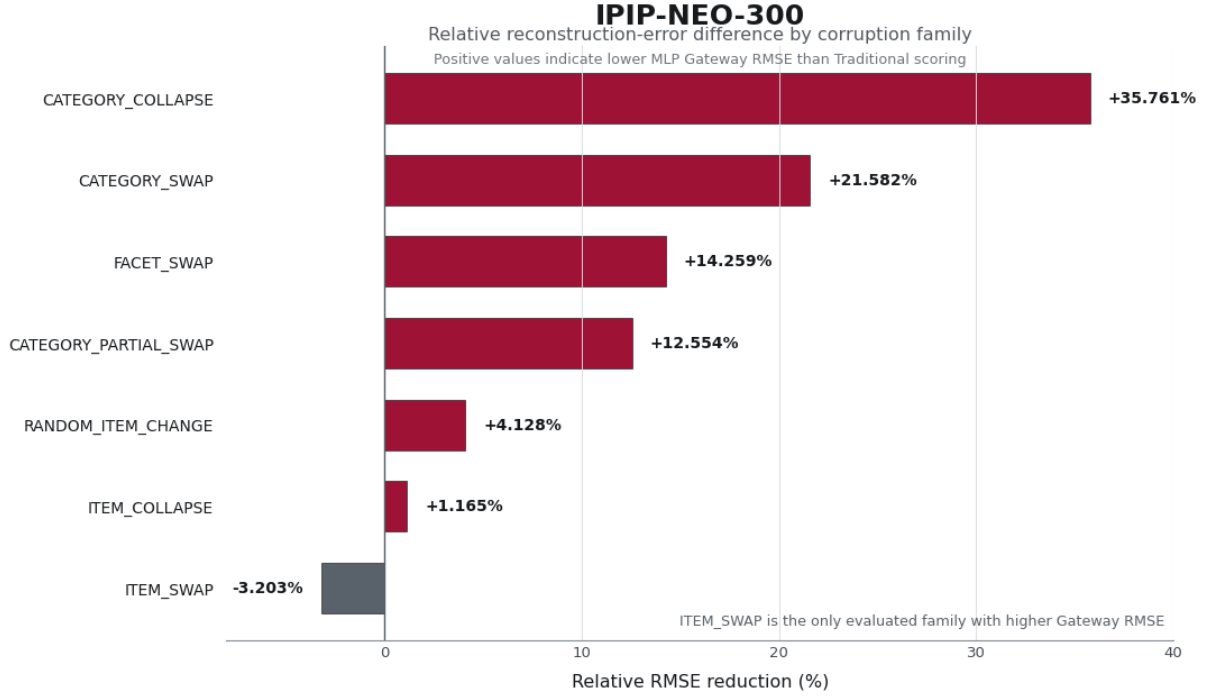


Figure 33: Historical relative corrupted-RMSE reduction by IPIP-NEO-300 corruption family. The visualisation is retained unchanged as part of the development record. Its plotted values correspond to an earlier development configuration and are not used to define the final P83.25 E14 numerical result.

6.18 Severity Analysis

Perturbation severity provides a second stratification axis through which robustness can be examined. The locked benchmark contains thirteen predefined severity values,

$$\mathcal{S} = \{1, 2, 3, 4, 5, 6, 10, 12, 15, 24, 36, 48, 60\}. \quad (593)$$

The severity-specific subsets are construction-defined evaluation strata rather than independent participant samples. Multiple severity conditions may therefore correspond to the same held-out participant.

The final P83.25 production record used for checkpoint selection establishes the global corrupted-validation RMSE and the family-level validation quantities reported above. It does not provide a frozen E14 severity table sufficient to support a new production-level claim about the number of severity levels at which P83.25 outperforms Traditional scoring.

Accordingly, no severity-level winner count, monotonicity claim, or severity-specific P83.25 effect size is inferred from the historical development plots. This distinction prevents earlier architecture-screening results from being silently promoted to final production evidence.

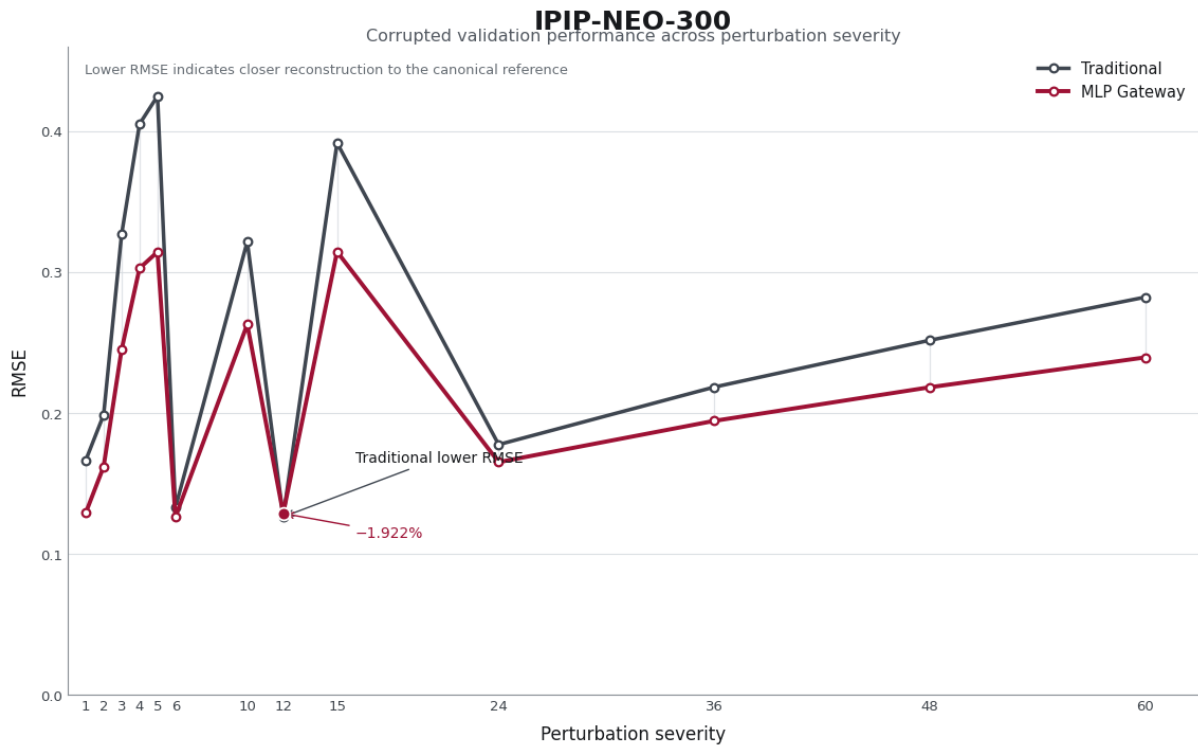


Figure 34: Historical corrupted-validation RMSE across the predefined IPIP-NEO-300 perturbation severity values. The figure is retained unchanged as a development-stage visualisation. It is not treated as a numerical severity analysis of the final P83.25 E14 checkpoint.

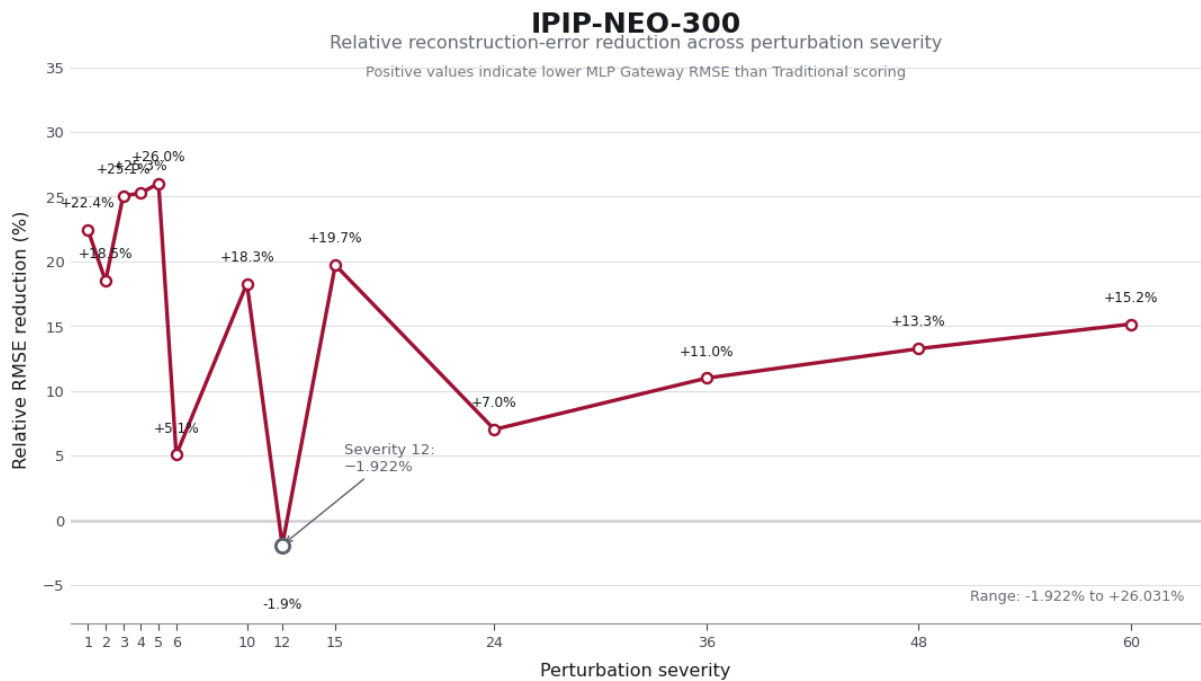


Figure 35: Historical relative corrupted-RMSE change across IPIP-NEO-300 perturbation severity. The figure is retained for provenance of the development programme and is not used to establish a final P83.25 severity-level effect.

The absence of a final E14 severity table in the production record does not invalidate the

global or family-level results. It instead establishes an explicit boundary around the claims made from the available frozen numerical record.

6.19 Facet-Level Analysis

The 30-dimensional output representation permits facet-resolved inspection of the reconstructed Gold-defined profile. Such an analysis can be informative about whether aggregate error differences are concentrated in a small subset of outputs or distributed across the output space.

For the final P83.25 E14 production checkpoint, however, the verified frozen results record used for the present manuscript does not contain a complete facet-by-facet numerical table corresponding to the final checkpoint. Accordingly, no new claim that P83.25 improves on Traditional scoring in all 30 facets is inferred from an earlier development visualisation.

The final aggregate evidence remains fully defined by the global validation metrics and family-level results reported above. Any facet-level interpretation beyond those quantities would require the corresponding final-checkpoint machine-readable facet table.

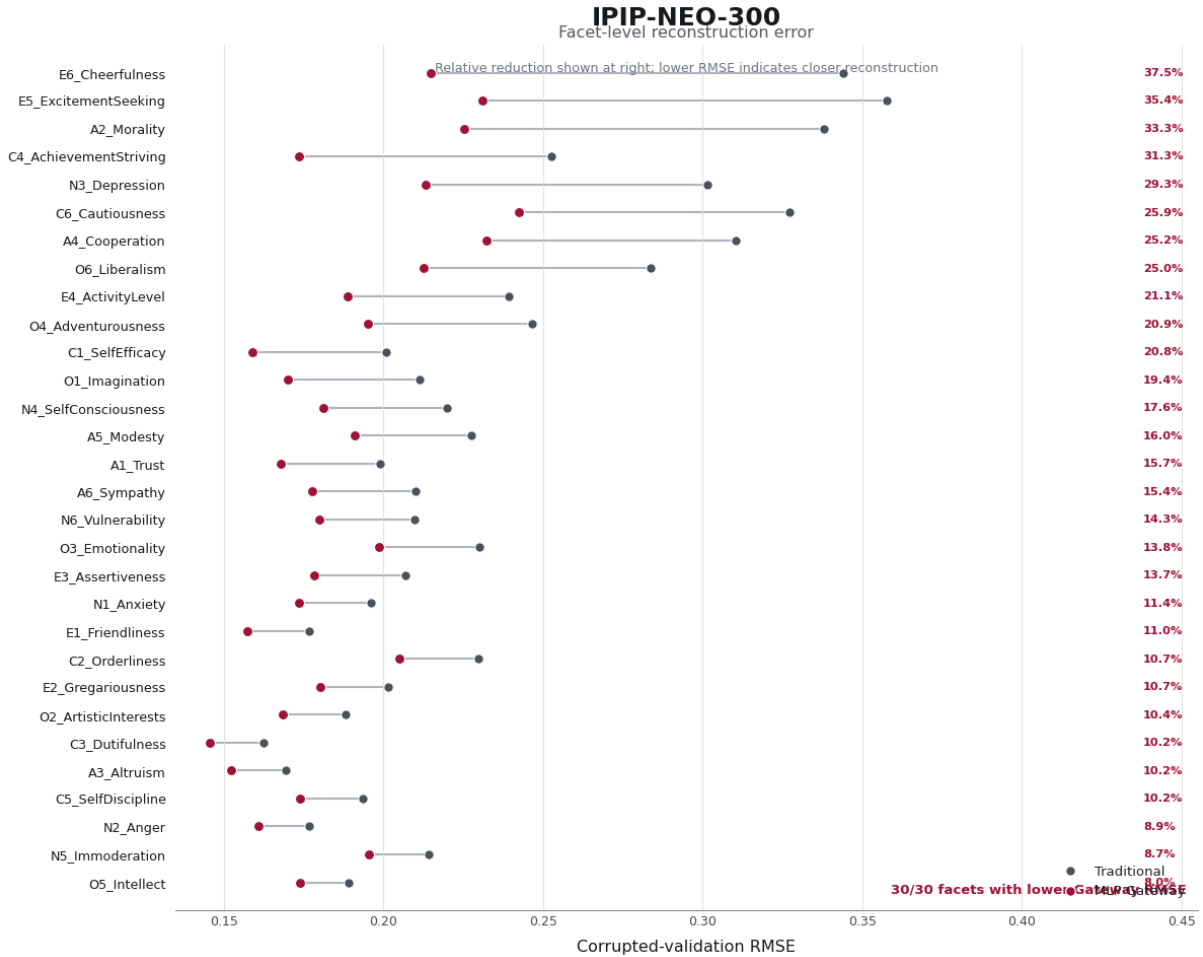


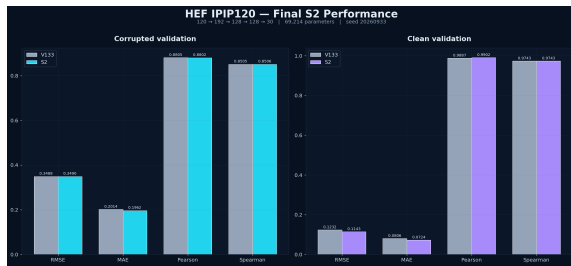
Figure 36: Historical facet-level reconstruction-RMSE visualisation for the 30-dimensional IPIP-NEO-300 Gold-defined output representation. The figure is retained unchanged as a development-stage inspection of facet-specific behaviour. It is not treated as an independent numerical evaluation of the final P83.25 E14 checkpoint.

The historical family-, severity-, and facet-level visualisations are therefore retained for complete research provenance, while the final production conclusions are restricted to the

numerical records associated with the selected P83.25 E14 checkpoint and the independently frozen evaluation protocol.

6.20 Extended S2 Validation Gallery

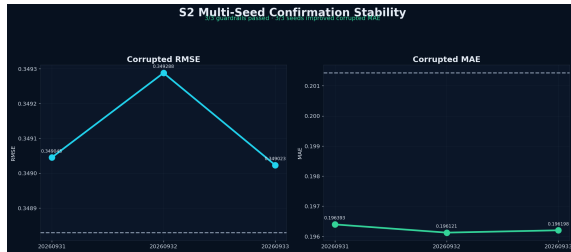
The following gallery provides extended visual diagnostics for the frozen IPIP-NEO-120 S2 operating point. Each panel is generated from the same frozen evaluation state used for the corresponding quantitative analysis. The panels are presented primarily as a visual audit; detailed numerical interpretation is retained in the main Results and Error Analysis sections.



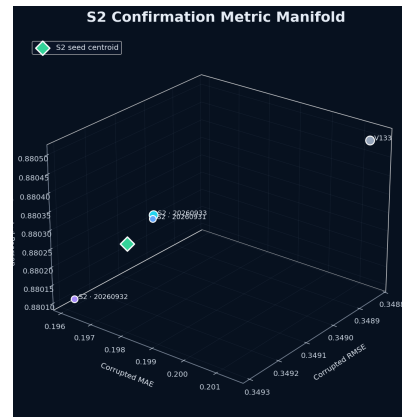
A29



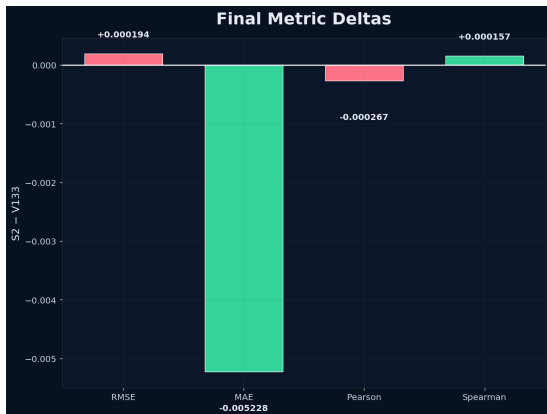
A31



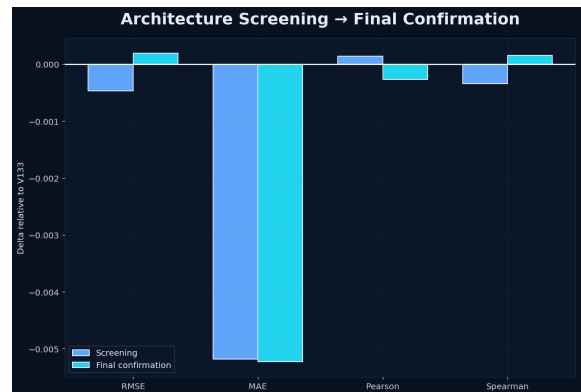
A33



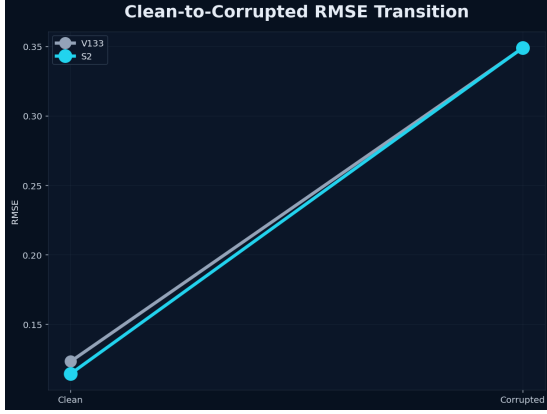
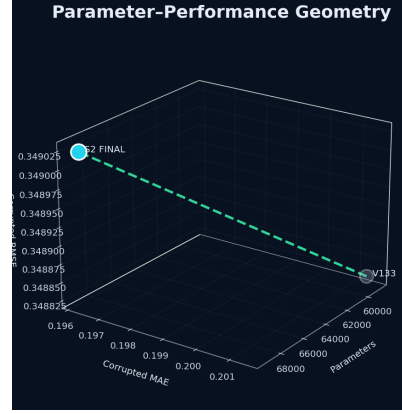
A35



A37



A39

**A41****A42**

The gallery contains the retained diagnostic panels A29, A31, A33, A35, A37, A39, A41, and A42, providing complementary visual views of the frozen $120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30$ IPIP-NEO-120 S2 operating point. The gallery is intended as an extended visual audit; the corresponding quantitative results, model-selection procedure, and statistical interpretations are reported separately in the main manuscript.

The complete facet-level results are retained in the machine-readable benchmark record, with the plotted comparison providing the principal visual summary of the 30-dimensional reconstruction behavior.

7. Representation and Geometric Analysis

The mechanistic laboratory evaluates whether structured input corruption is accompanied by measurable changes in the learned internal representation and whether these changes are empirically associated with output-space displacement and reconstruction error.

The analysis is organized as a sequence of distinct, non-equivalent measurements:

input perturbation $\rightarrow$ latent displacement $\rightarrow$ geometric diagnostics $\rightarrow$ output-space displacement $\rightarrow$ reconstruction error (594)

This sequence is descriptive rather than causal. In particular, latent displacement, geometric deformation, output-space displacement, and reconstruction error are treated as distinct quantities and are not used interchangeably.

7.1 IPIP-NEO-120 S2 Mechanistic and Geometric Audit

The final IPIP-NEO-120 S2 Gateway was subjected to a dedicated mechanistic audit against the V133 control in order to examine whether the architectural change from

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30 \quad (595)$$

to

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30 \quad (596)$$

was accompanied by measurable changes in the internal numerical representation. The audit was performed on the frozen validation configuration and did not modify the model-selection procedure. The independent IPIP-120 TEST population was not loaded or touched during this analysis.

The S2 checkpoint corresponds to seed 20260933, epoch 6, and 69,214 trainable parameters. Its state dictionary was loaded under strict architectural compatibility checks, and the frozen artifact is identified by the SHA-256 digest

$$679c88d15552ee6efaf78cb0b3e288c09b3f00af53cf6d4100807830323e6992. \quad (597)$$

The mechanistic comparison therefore concerns two explicitly defined numerical models: the 58,910-parameter V133 control and the 69,214-parameter S2 final configuration.

7.1.1 Hidden Representation Dimensionality

Let

$$\mathbf{h}^{(192)}, \quad \mathbf{h}^{(128,\text{pre})}, \quad \mathbf{h}^{(128,\text{wide})} \quad (598)$$

denote the corresponding hidden representations of S2, and let $\mathbf{h}^{(192)}$, $\mathbf{h}^{(128)}$, and $\mathbf{h}^{(64)}$ denote the corresponding V133 representations. Effective dimensionality was evaluated using the entropy effective rank

$$R_{\text{eff}} = \exp \left(- \sum_{i:p_i>0} p_i \log p_i \right), \quad (599)$$

where

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j} \quad (600)$$

and λ_i are the eigenvalues of the centered covariance matrix of the corresponding representation.

The measured effective ranks were

Table 31: Effective representational dimensionality of the V133 and final S2 hidden states.

Model	Representation	Nominal dimension	R_{eff}	R_{95}
V133	H192	192	59.057	75
V133	H128	128	32.954	45
V133	H64	64	20.179	24
S2	H192	192	61.684	76
S2	H128-pre	128	35.746	45
S2	H128-wide	128	41.141	57

Here R_{95} denotes the smallest number of leading spectral components accounting for at least 95% of the observed variance. The transition from the V133 terminal H64 representation to the S2 terminal H128-wide representation increased the entropy effective rank from approximately 20.179 to 41.141, an absolute increase of 20.962 effective dimensions and a relative increase of approximately 103.9%. The corresponding R_{95} increased from 24 to 57, an increase of 33 components or approximately 137.5%. These quantities do not imply that all 128 coordinates are independently informative, nor do they establish that effective dimensionality alone caused the observed validation behavior. They provide a numerical description of how much of the nominal

representation is occupied by the learned activation geometry under the evaluated validation distribution.

For a collection of hidden representations $\mathbf{H} \in \mathbb{R}^{n \times d}$, let the centered representation matrix be

$$\mathbf{H}_c = \mathbf{H} - \mathbf{1}\boldsymbol{\mu}^\top, \quad \boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{h}_i, \quad (601)$$

and define the empirical covariance matrix

$$\boldsymbol{\Sigma}_H = \frac{1}{n-1} \mathbf{H}_c^\top \mathbf{H}_c. \quad (602)$$

Let its non-negative eigenvalues be $\lambda_1, \dots, \lambda_d$, ordered as $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$. The normalized spectral mass is

$$p_j = \frac{\lambda_j}{\sum_{k=1}^d \lambda_k}, \quad (603)$$

with $\sum_j p_j = 1$ whenever the total variance is non-zero. The effective dimensionality is then expressed through the entropy-based effective rank

$$r_{\text{eff}}(\boldsymbol{\Sigma}_H) = \exp \left(- \sum_{j=1}^d p_j \log p_j \right), \quad (604)$$

while the participation ratio is

$$r_{\text{PR}}(\boldsymbol{\Sigma}_H) = \frac{\left(\sum_{j=1}^d \lambda_j \right)^2}{\sum_{j=1}^d \lambda_j^2}. \quad (605)$$

The cumulative spectral dimension required to explain a fraction $q \in (0, 1)$ of the total variance is defined as

$$R_q = \min \left\{ m : \frac{\sum_{j=1}^m \lambda_j}{\sum_{j=1}^d \lambda_j} \geq q \right\}. \quad (606)$$

These quantities provide a mathematical description of the internal geometry without requiring a graphical embedding. In particular, r_{eff} and r_{PR} characterize the dispersion of activation variance across the nominal coordinates, while R_q measures how many principal directions are required to explain a prescribed fraction of the empirical variance. None of these quantities is interpreted as a proof that the corresponding coordinates are independently informative or causally responsible for the observed validation behavior.

7.1.2 Common-Path Representation Drift

Matched clean and corrupted observations were used to quantify representational movement before the terminal output mapping. For a representation $\mathbf{h}$ and its matched corrupted counterpart $\tilde{\mathbf{h}}$, relative Euclidean displacement is defined as

$$D_{\text{rel}} = \frac{\|\tilde{\mathbf{h}} - \mathbf{h}\|_2}{\|\mathbf{h}\|_2 + \epsilon}. \quad (607)$$

Angular similarity is evaluated separately by cosine similarity,

$$C = \frac{\mathbf{h} \cdot \tilde{\mathbf{h}}}{\|\mathbf{h}\|_2 \|\tilde{\mathbf{h}}\|_2}. \quad (608)$$

The common first two representations exhibited mean cosine similarities of approximately 0.97079 at H192 and 0.92706 at H128 when the matched V133 and S2 paths were compared under corruption. Mean absolute representation differences were approximately 0.02350 at H192 and 0.03227 at H128. These quantities show that the models retain strongly related early representations while allowing measurable divergence before the terminal representation.

For the terminal hidden state, the S2 H128-wide representation had a mean activation norm of approximately 1.89486, compared with approximately 0.60123 for the V133 H64 representation. The preceding S2 H128-pre representation had a mean norm of approximately 1.35032. Norm differences are reported as numerical diagnostics and are not interpreted as direct indicators of model quality.

7.1.3 Weight Spectral Structure

The singular-value structure of the learned affine transformations was also examined. For a weight matrix $W \in \mathbb{R}^{m \times n}$ with singular values

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0, \quad r = \text{rank}(W), \quad (609)$$

the spectral condition number is

$$\kappa(W) = \frac{\sigma_1}{\sigma_r}, \quad (610)$$

for non-zero σ_r . This quantity measures the spread between the largest and smallest non-zero singular directions and is therefore reported as a numerical conditioning diagnostic rather than a performance metric.

The normalized singular-value spectrum is defined by

$$q_j = \frac{\sigma_j}{\sum_{k=1}^r \sigma_k}, \quad (611)$$

which allows an effective spectral rank to be written as

$$r_{\text{eff}}^{(W)} = \exp \left(- \sum_{j=1}^r q_j \log q_j \right). \quad (612)$$

The resulting diagnostics showed that V133 had approximate condition numbers of 158.1, 151.4, 400.31, and 2.79 for FC1, FC2, FC3, and the output layer, respectively, while the corresponding S2 values were approximately 89.66, 115.38, 36,105.54, and 2.61. The S2 FC3 condition number is therefore much larger than that of V133 and is retained explicitly as a spectral diagnostic rather than being interpreted as evidence of improved numerical conditioning.

At the same time, the effective singular-value rank of the third affine layer increased from approximately 11.292 for V133 to 24.666 for S2. Thus, the wider S2 transformation simultaneously exhibits a substantially richer effective singular-value structure and a more ill-conditioned spectrum. These properties are distinct and should not be conflated.

$$\Delta r_{\text{eff}}^{(W_3)} = r_{\text{eff}, \text{S2}}^{(W_3)} - r_{\text{eff}, \text{V133}}^{(W_3)} = 24.666 - 11.292 = 13.374. \quad (613)$$

The corresponding condition-number ratio is

$$\frac{\kappa_{S2}(W_3)}{\kappa_{V133}(W_3)} = \frac{36105.54}{400.31} \approx 90.19, \quad (614)$$

which makes explicit that the increase in effective spectral rank occurred simultaneously with a substantially larger singular-value spread. The two diagnostics therefore describe different properties of the learned linear transformation.

7.1.4 Corruption-Family Linear Probe

To examine whether hidden states preserve information associated with the corruption structure, a linear probe was trained to classify the seven corruption families from the corresponding hidden representations. The probe was used solely as a representation diagnostic and was not part of Gateway training.

The balanced accuracies were

Table 32: Linear corruption-family probe accuracy for V133 and S2 internal representations.

Representation	V133	S2
H192	0.369705	0.368735
H128 / H128-pre	0.344524	0.345202
Terminal hidden state	0.227407	0.266031

For $K = 7$ corruption families, balanced accuracy is defined as the unweighted mean of the class-wise recalls,

$$\text{BA} = \frac{1}{K} \sum_{k=1}^K \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k}. \quad (615)$$

At the terminal hidden stage, balanced probe accuracy increased from 0.227407 for V133 H64 to 0.266031 for S2 H128-wide, an absolute increase of 0.038624. This diagnostic is consistent with the effective-rank analysis in showing that the wider terminal representation retained more information associated with the evaluated corruption structure. It does not establish that the representation encodes a psychologically meaningful corruption variable, nor does it imply that increased probe accuracy is intrinsically desirable.

The relative terminal probe change can be written as

$$\Delta_{\text{BA}} = 0.266031 - 0.227407 = 0.038624, \quad (616)$$

and, relative to the V133 terminal value,

$$\Delta_{\text{BA,rel}} = \frac{0.266031 - 0.227407}{0.227407} \approx 0.1698, \quad (617)$$

corresponding to an approximately 16.98% relative increase in the measured probe accuracy. This is a representation diagnostic only and is not interpreted as a measure of psychometric validity.

7.1.5 Terminal Representation Gradient Sensitivity

Local sensitivity of the loss to an internal representation was evaluated using the gradient norm. For representation $\mathbf{h}$,

$$G(\mathbf{h}) = \left\| \frac{\partial \mathcal{L}}{\partial \mathbf{h}} \right\|_2. \quad (618)$$

The mean gradient norms and 95th percentiles were measured as follows:

Table 33: Gradient sensitivity of hidden representations under the evaluated corrupted validation conditions.

Representation	Mean G	95th percentile
V133 H192	0.000069	0.000162
V133 H128	0.000190	0.000491
V133 H64	0.000164	0.000387
S2 H192	0.000069	0.000165
S2 H128-pre	0.000129	0.000330
S2 H128-wide	0.000047	0.000110

For a sample set of size n , the empirical mean sensitivity is

$$\overline{G} = \frac{1}{n} \sum_{i=1}^n G(\mathbf{h}_i), \quad (619)$$

while the empirical 95th percentile is

$$G_{0.95} = Q_{0.95}(G(\mathbf{h}_1), \dots, G(\mathbf{h}_n)), \quad (620)$$

where $Q_{0.95}$ denotes the empirical 0.95 quantile.

At the terminal hidden stage, the S2 H128-wide representation exhibited a mean gradient norm approximately 71.3% lower than the V133 H64 representation and a 95th-percentile gradient approximately 71.6% lower. These relative changes are obtained from

$$\Delta_{\text{mean}} = 1 - \frac{0.000047}{0.000164} \approx 0.7134, \quad (621)$$

and

$$\Delta_{0.95} = 1 - \frac{0.000110}{0.000387} \approx 0.7158. \quad (622)$$

This is a local numerical sensitivity diagnostic and should not be interpreted as a formal proof of global stability or robustness.

7.1.6 S2 Mechanistic Interpretation Boundaries

The representation audit provides several convergent numerical observations: S2 preserves a wider terminal hidden state, occupies a larger effective representation, retains greater terminal corruption-family probe separability, and exhibits lower terminal gradient sensitivity under the evaluated conditions. At the same time, the S2 third affine transformation is more spectrally ill-conditioned than the corresponding V133 transformation. These observations therefore do not support a single scalar explanation of the architecture change.

The complete audit can be represented as a multidimensional diagnostic vector

$$\mathcal{A} = \left[r_{\text{eff}}, R_q, r_{\text{eff}}^{(W)}, \kappa(W), \text{BA}, \overline{G}, G_{0.95} \right], \quad (623)$$

where each component is evaluated at the corresponding hidden or affine representation. The audit therefore remains a vector of independently interpretable diagnostics rather than being reduced to a single composite “quality” score.

For the terminal representation, the principal V133–S2 numerical changes may be summarized directly by

$$\Delta r_{\text{eff}} = 41.141 - 20.179 = 20.962, \quad (624)$$

$$\frac{41.141 - 20.179}{20.179} \approx 1.039, \quad (625)$$

corresponding to approximately a 103.9% increase in the measured effective rank, together with

$$\Delta R_{95} = 57 - 24 = 33 \quad (626)$$

and

$$\frac{57 - 24}{24} = 1.375, \quad (627)$$

corresponding to a 137.5% increase in the number of principal directions required to account for 95% of the evaluated variance.

The appropriate interpretation is that the S2 architecture changes the numerical geometry of the final hidden representation in measurable ways that are consistent with the preservation of information before the 30-dimensional output projection. They do not establish a causal mechanism for the observed validation trade-off, nor do they imply that effective rank, probe accuracy, or gradient sensitivity is a direct proxy for psychometric validity.

8. Representation and Geometric Analysis

8.1 IPIP-NEO-300 Mechanistic Scope and Model Identity

The IPIP-NEO-300 mechanistic laboratory was established as a diagnostic stage of the research programme to examine whether structured response corruption was accompanied by measurable changes in the numerical representation generated by the Gateway, and whether those changes were empirically associated with displacement in the output space and with reconstruction error.

The mechanistic analysis is explicitly observational. It does not assume that a geometric change in a hidden representation constitutes a causal mechanism for the corresponding reconstruction effect. The analysis therefore maintains a strict distinction between representation displacement, geometric deformation, output-space displacement, and reconstruction error.

The diagnostic sequence is

$$\text{input perturbation} \rightarrow \text{latent displacement} \rightarrow \text{geometric diagnostics} \rightarrow \text{output-space displacement} \rightarrow \text{reconstruction error} \quad (628)$$

These quantities are evaluated separately. No geometric diagnostic is treated as a surrogate for psychometric validity, and no association identified in the mechanistic laboratory is interpreted as a causal relation without an independent causal design.

A critical model-identity distinction is required for the IPIP-NEO-300 analysis. The final production checkpoint used for the independent evaluation is P83.25, with architecture

$$\boxed{300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}} \quad (629)$$

and 220,782 trainable parameters. Its production checkpoint was selected at epoch 14 under the corrupted-validation RMSE criterion. The final production model is therefore not architecturally equivalent to the earlier compact structured-output model used during development. The production specification and checkpoint identity are fixed independently of the retained mechanistic visualisations. *The final P83.25 architecture is the authoritative model definition for final performance claims.*

The geometric analyses presented in this section deliberately precede the final P83.25 architecture. They belong to an earlier stage of the investigation in which the compact internal geometry of the Gateway was examined directly. Their purpose was diagnostic rather than confirmatory: the analyses were used to inspect how the internal representation responded to structured corruption and to determine whether the compact hidden geometry warranted architectural reconsideration.

This distinction is important because the historical geometric analyses and the final architecture-selection programme answer different questions. The former characterise what was numerically observable in the earlier model; the latter tests whether alternative internal geometries produce different empirical behaviour under a controlled benchmark.

8.2 From Mechanistic Observation to Architectural Refinement

The initial structured IPIP-NEO-300 Gateway used the compact configuration

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}, \quad (630)$$

with 121,054 trainable parameters.

This configuration was deliberately structured as a shared trunk followed by five six-dimensional output heads. The five heads jointly reconstruct the 30-dimensional facet representation while sharing the same internal representation before the structured output projection.

The early mechanistic laboratory was conducted on this compact internal geometry. The analysis examined matched clean and corrupted representations rather than treating the network as an opaque input–output mapping. The purpose was to determine whether corruption-induced changes were observable inside the computational pathway itself, and, more specifically, whether the nominal hidden dimensionality appeared sufficient to characterise the numerical response of the model without introducing an unjustified architectural assumption.

The resulting analysis did not, by itself, select a new architecture and it was not used as a hidden optimisation criterion. Instead, it generated an architectural hypothesis: the internal geometry should be revisited rather than assuming that the compact bottleneck was necessarily the appropriate representation for the reconstruction problem.

This distinction is expressed explicitly as

$$\text{mechanistic observation} \rightarrow \text{architectural hypothesis} \rightarrow \text{controlled architecture screening}. \quad (631)$$

Thus, the geometric laboratory served as a discovery and diagnosis stage, whereas the subsequent architecture-screening benchmark served as the empirical test of the proposed architectural changes.

The architecture-screening programme was conducted on a 10,000-subject development subset with 143 predefined views per subject, yielding

$$10,000 \times 143 = 1,430,000 \quad (632)$$

screening examples for a single SIES cycle. The independent TEST population remained untouched throughout this development programme.

The screening explored compact, wider, balanced, and deeper internal geometries. The architectural trajectory can therefore be represented as

$$\begin{array}{lcl} \text{compact structured control} & 300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 5 \times 6 & \\ \downarrow & & \\ \text{wider structured candidates} & 300 \rightarrow 320 \rightarrow 192 \rightarrow 128 \rightarrow 5 \times 6 & (633) \\ \downarrow & & \\ \text{refined P83.25 architecture} & 300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6. & \end{array}$$

The principal wide candidate therefore represented a deliberate increase in internal representational capacity rather than an arbitrary change of layer widths. The subsequent refinement moved beyond a uniformly wider design and tested a more specific hidden-dimensional geometry.

The refined configuration

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\} \quad (634)$$

contained 220,782 trainable parameters.

On the internal architecture-refinement benchmark, this configuration produced

$$\text{RMSE} = 0.220199, \quad (635)$$

$$\text{MAE} = 0.147551, \quad (636)$$

$$\rho_{\text{Pearson}} = 0.861929, \quad (637)$$

and

$$\rho_{\text{Spearman}} = 0.842994. \quad (638)$$

Relative to the principal WIDE development configuration, the corresponding changes were

$$\Delta \text{RMSE} = -0.003109, \quad (639)$$

$$\Delta \text{MAE} = -0.004251, \quad (640)$$

$$\Delta \text{Pearson} = +0.002119, \quad (641)$$

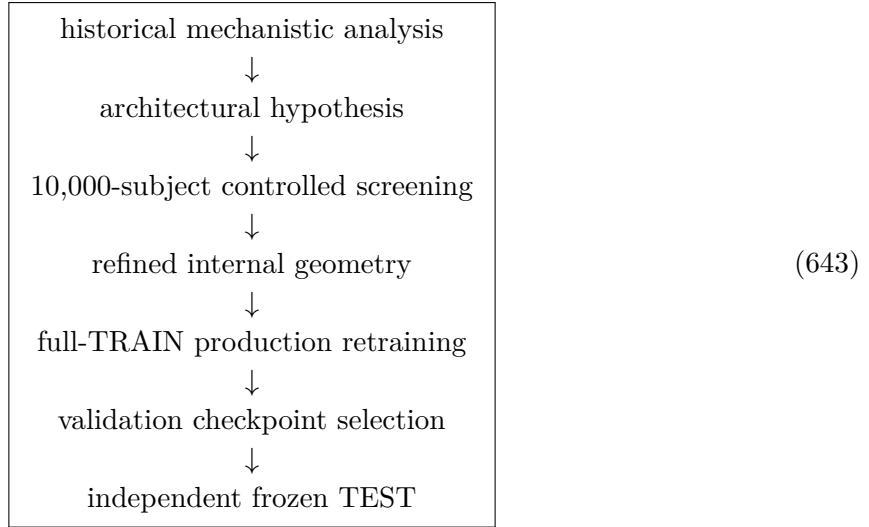
and

$$\Delta \text{Spearman} = +0.002648. \quad (642)$$

These values belong exclusively to the internal architecture-refinement benchmark. They

are not the final independent TEST results and are not used as a substitute for the subsequent full-TRAIN production fit.

The architecture-selection process therefore has a clear provenance:



This ordering is important. The final P83.25 model was not obtained by retroactively redefining the historical geometric analysis as evidence from the final architecture, nor was the 10,000-subject screening run treated as the final fitted model.

Instead, the refined architecture was subsequently retrained from scratch on the complete TRAIN population, producing the final P83.25 production checkpoint used for independent evaluation.

8.3 Historical Development Model and Provenance of the Geometric Analyses

Several retained IPIP-NEO-300 geometric visualisations were generated during the earlier architectural development programme using the compact structured representation with hidden dimensions

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}. \quad (644)$$

Those representations are denoted below by

$$\mathbf{h}^{(256)}, \quad \mathbf{h}^{(128)}, \quad \mathbf{h}^{(64)}. \quad (645)$$

They are retained because they constitute an actual part of the research trajectory and preserve the state of the internal model before the subsequent architectural change.

The corresponding figures were generated before the transition to the final P83.25 internal geometry. Their visual content is therefore intentionally preserved without retroactive alteration. They are not being presented as images of the final architecture, nor are their historical dimensions being relabelled to match P83.25.

This distinction is operationally enforced throughout the present section:

development-stage geometric result $\neq$ final P83.25 production result

 . (646)

Historical geometric results are therefore not transferred to P83.25 by architectural analogy, visual similarity, shared input dimensionality, or shared output dimensionality.

Their value is methodological and historical: they document how the internal representation behaved before the architecture was changed and therefore provide the observational context in which the subsequent architectural refinement was conceived.

8.4 Development-Stage Latent Representation Geometry

The retained development-stage mechanistic analysis evaluates the hidden representations of the earlier structured-output IPIP-NEO-300 configuration at dimensions 256, 128, and 64. These representations are treated as observational numerical states of that development-stage Gateway.

Matched clean and corrupted inputs are paired at the observation level so that representation displacement can be measured under controlled perturbation.

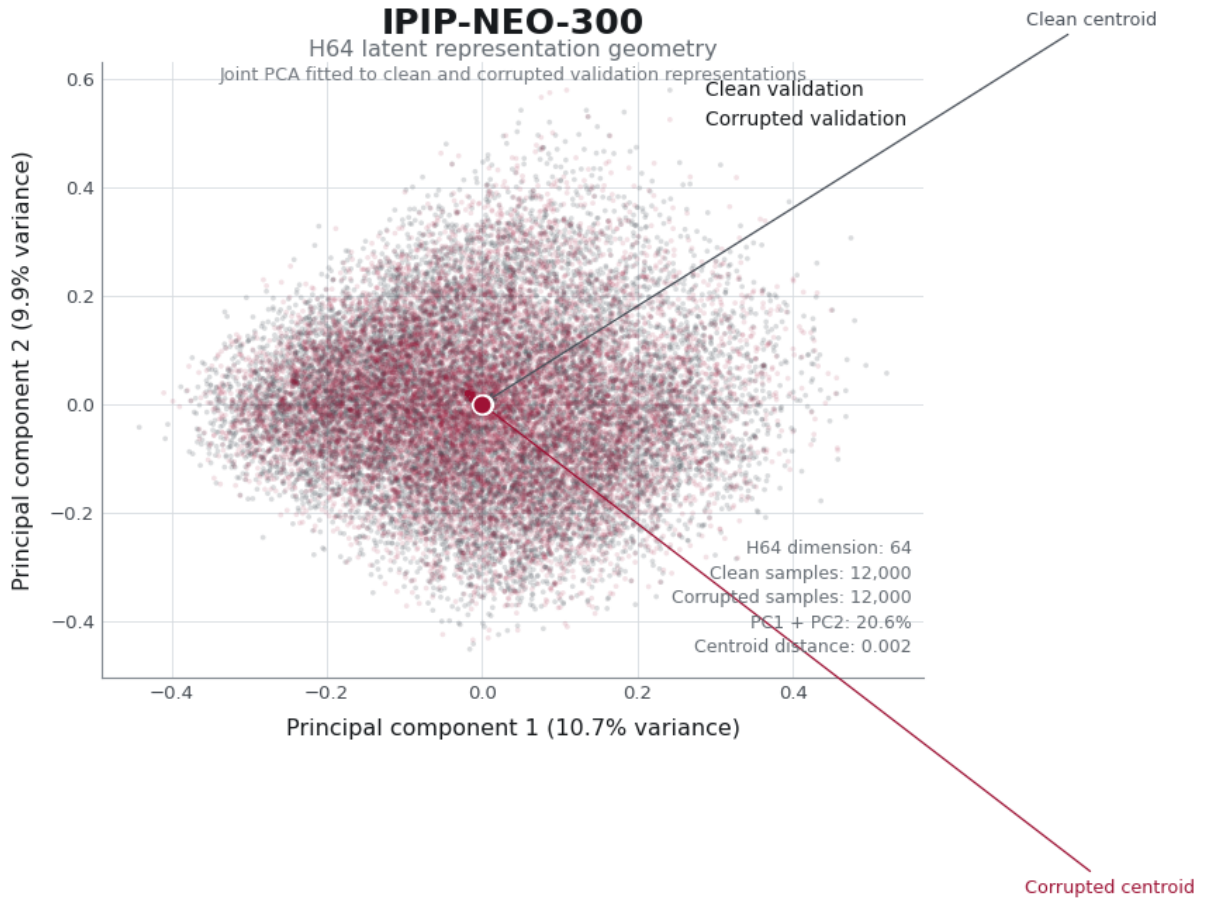


Figure 37: Development-stage latent representation geometry of the earlier IPIP-NEO-300 structured-output Gateway. The visualisation corresponds to the historical hidden dimensions 256, 128, and 64 and is retained unchanged as a documentary record of the internal-model analysis performed before the subsequent architectural refinement. It is not a visualisation of the final P83.25 E14 hidden topology.

The historical visualisation is therefore interpreted as architectural provenance and mechanistic evidence from the pre-P83.25 development model. It is not treated as direct evidence about the numerical geometry of the final production checkpoint.

8.5 Latent Displacement

For a matched clean representation $\mathbf{h}^{(d)}$ and corrupted representation $\tilde{\mathbf{h}}^{(d)}$, where $d \in \{256, 128, 64\}$, the relative Euclidean displacement is defined as

$$D_{\text{rel}}^{(d)} = \frac{\|\tilde{\mathbf{h}}^{(d)} - \mathbf{h}^{(d)}\|_2}{\|\mathbf{h}^{(d)}\|_2 + \epsilon}, \quad (647)$$

where $\epsilon > 0$ is a numerical stabiliser.

The quantity $D_{\text{rel}}^{(d)}$ measures the magnitude of representation movement relative to the norm of the corresponding clean representation. It does not measure prediction error and does not indicate whether the resulting movement is beneficial or detrimental.

Angular alignment is evaluated separately through cosine similarity,

$$C^{(d)} = \frac{\mathbf{h}^{(d)} \cdot \tilde{\mathbf{h}}^{(d)}}{\|\mathbf{h}^{(d)}\|_2 \|\tilde{\mathbf{h}}^{(d)}\|_2}, \quad (648)$$

for non-zero representation norms.

Euclidean displacement and angular alignment are therefore retained as distinct geometric descriptors. A representation can exhibit a substantial change in magnitude with comparatively small angular change, or conversely substantial angular reorganisation without a proportionally large relative norm change.

The paired displacement measures are interpreted only within the specified semi-synthetic perturbation protocol. They are not interpreted as measures of psychological change.

8.6 Spectral Structure and Effective Representation Dimensionality

For a collection of hidden representations $\mathbf{H} \in \mathbb{R}^{n \times d}$, the centred representation matrix is

$$\mathbf{H}_c = \mathbf{H} - \mathbf{1}\boldsymbol{\mu}^\top, \quad \boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{h}_i. \quad (649)$$

The empirical covariance matrix is

$$\boldsymbol{\Sigma}_H = \frac{1}{n-1} \mathbf{H}_c^\top \mathbf{H}_c. \quad (650)$$

Let its non-negative eigenvalues be ordered as

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0. \quad (651)$$

Provided that the total variance is non-zero, the normalised spectral mass is

$$p_j = \frac{\lambda_j}{\sum_{k=1}^d \lambda_k}. \quad (652)$$

The entropy-based effective rank is

$$R_{\text{eff}} = \exp \left(- \sum_{j:p_j>0} p_j \log p_j \right). \quad (653)$$

The participation ratio is

$$R_{\text{PR}} = \frac{\left(\sum_{j=1}^d \lambda_j\right)^2}{\sum_{j=1}^d \lambda_j^2}. \quad (654)$$

The number of principal directions required to explain a proportion $q \in (0, 1)$ of the observed variance is

$$R_q = \min \left\{ m : \frac{\sum_{j=1}^m \lambda_j}{\sum_{j=1}^d \lambda_j} \geq q \right\}. \quad (655)$$

These quantities describe the empirical distribution of variance across representation coordinates. They do not imply that the corresponding coordinates are independently informative, nor do they establish a causal relationship between effective dimensionality and reconstruction performance.

In particular,

$$R_{\text{eff}} \neq d \quad (656)$$

in general. The nominal hidden dimension therefore must not be reported as equivalent to the effective dimensionality occupied by the activation cloud.

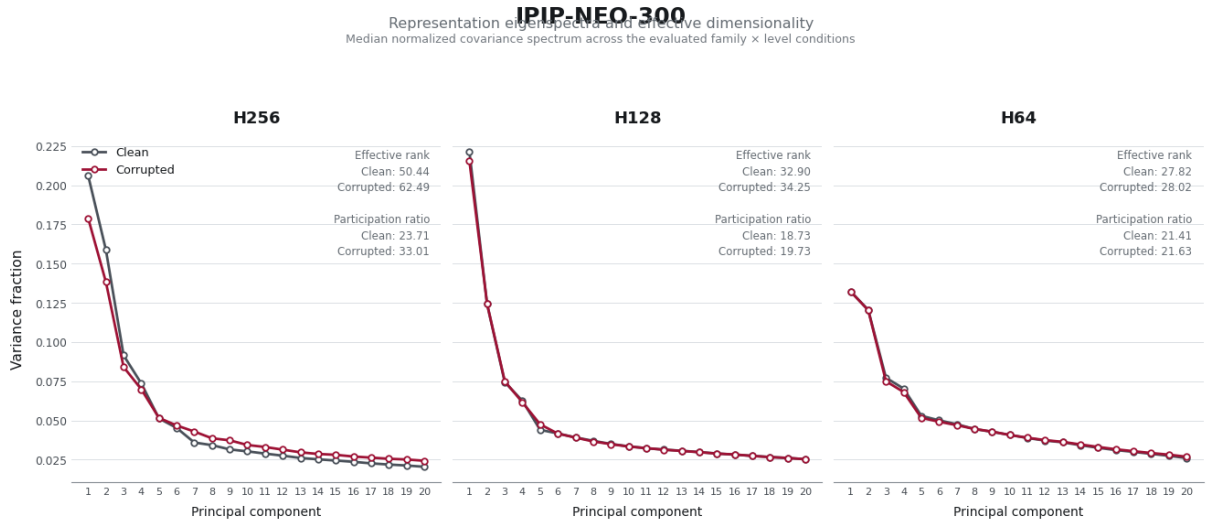


Figure 38: Development-stage covariance spectrum and effective representational dimensionality of the earlier IPIP-NEO-300 Gateway. The figure is retained unchanged as a historical numerical diagnostic of the pre-P83.25 hidden representations and is not interpreted as an eigenspectrum of the final P83.25 E14 hidden states.

8.7 Spectral-Diffusion Geometry

A complementary development-stage analysis characterises the computational graph using a diffusion operator. Let P denote the normalised diffusion operator defined on the graph representation, with eigenpairs satisfying

$$P\psi_\ell = \lambda_\ell \psi_\ell. \quad (657)$$

At diffusion time t , a diffusion-coordinate representation can be written as

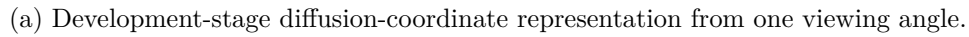
$$\Phi_t(v) = \left(\lambda_1^t \psi_1(v), \lambda_2^t \psi_2(v), \dots, \lambda_q^t \psi_q(v) \right), \quad (658)$$

where v denotes a graph element and the retained eigencomponents define the reported embedding.

For the documented visualisation, the first three reported diffusion coordinates are represented as

$$\Phi(v) = (\phi_1(v), \phi_2(v), \phi_3(v)). \quad (659)$$

The diffusion coordinates are induced by the spectral properties of the specified graph operator. They are not physical spatial coordinates and do not imply a physical geometry of the neural network.



The diffusion coordinates are therefore interpreted as a geometric embedding of the historical computational graph rather than as physical spatial locations. Their role in the research programme was diagnostic: they provided another view of the numerical organisation of the pre-P83.25 model and contributed to the broader assessment of whether the compact internal geometry warranted further architectural investigation.

For a frozen Gateway mapping F_θ , the input Jacobian is

144

$$J(\mathbf{x}) = \frac{\partial F_\theta(\mathbf{x})}{\partial \mathbf{x}}. \quad (660)$$

Its operator norm is

$$\|J(\mathbf{x})\|_2 = \sigma_{\max}(J(\mathbf{x})), \quad (661)$$

where $\sigma_{\max}$ denotes the largest singular value. This quantity characterises the maximum first-order amplification of an infinitesimal input perturbation in the local linearisation of the mapping.

For a matrix A , stable rank is defined as

$$\text{srnk}(A) = \frac{\|A\|_F^2}{\|A\|_2^2}. \quad (662)$$

Stable rank and operator norm describe different properties. Stable rank characterises the distribution of matrix energy relative to the dominant singular direction, whereas the operator norm measures the magnitude of the largest singular direction. Neither is a reconstruction-performance metric.

All such local sensitivity diagnostics are interpreted as properties of the evaluated development-stage numerical mapping. They do not constitute proofs of global Lipschitz behaviour or global robustness, and they are not silently reassigned to the P83.25 E14 checkpoint.

8.9 Persistent Homology

The development-stage latent representation is additionally examined through persistent homology to characterise topological features over a range of distance scales.

For a filtered point cloud, each finite persistence feature is represented by a birth–death pair

$$(b_i, d_i), \quad (663)$$

with lifetime

$$\ell_i = d_i - b_i. \quad (664)$$

The analysis distinguishes connected components through H_0 and one-dimensional structures through H_1 .

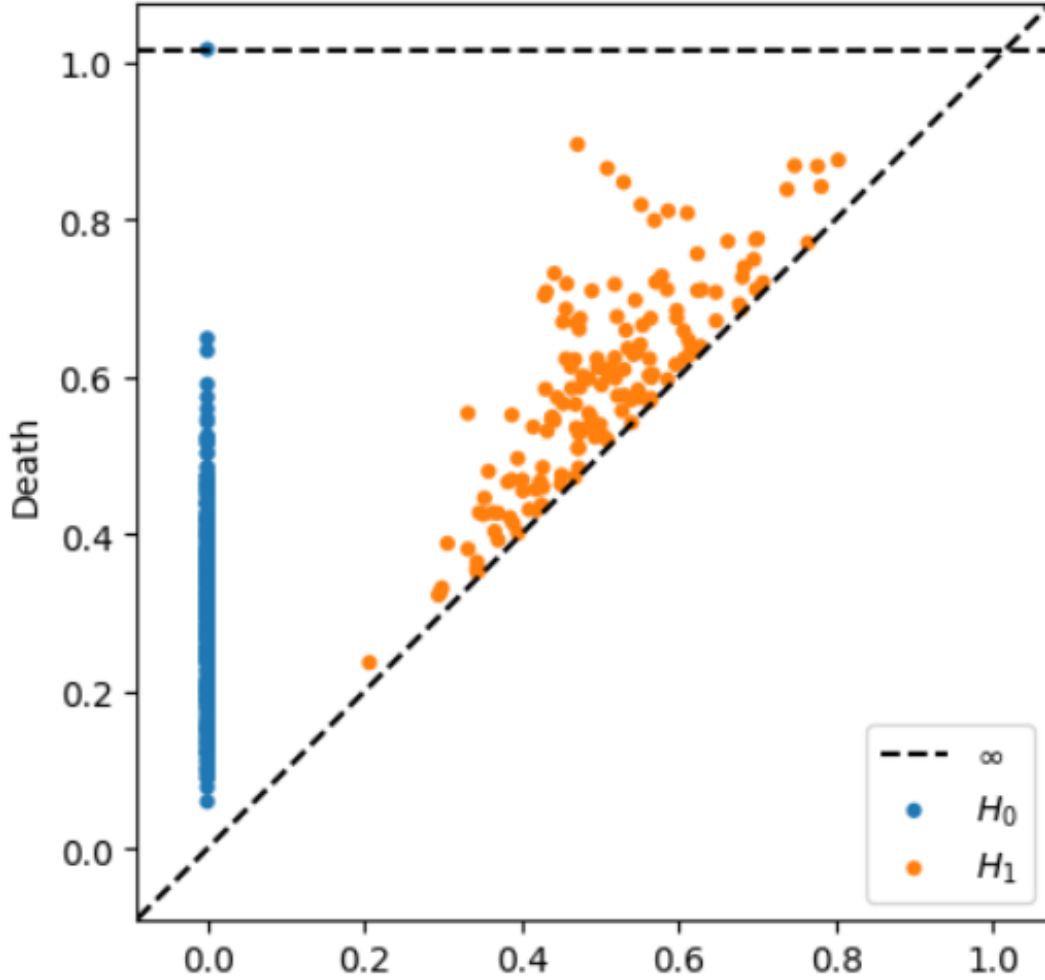


Figure 40: Development-stage persistence diagram for the analysed IPIP-NEO-300 latent representation. Features are represented in birth–death coordinates and separated by homological dimension. The visualisation characterises the topology of the numerical representation generated by the earlier development-stage mapping; it does not establish a topology of psychological constructs and is not presented as a final P83.25 E14 topology.

Persistent homology is therefore treated as a complementary numerical description of the analysed historical representation and not as evidence of a psychological manifold. Its presence in the manuscript documents the breadth of the mechanistic analysis undertaken before the final architecture was established.

8.10 Centered Kernel Alignment

Linear Centered Kernel Alignment (CKA) is used to compare the structural organisation of matched clean and corrupted development-stage representations.

For centred representation matrices A and B ,

$$\text{CKA}_{\text{lin}}(A, B) = \frac{\|A^T B\|_F^2}{\|A^T A\|_F \|B^T B\|_F}. \quad (665)$$

Accordingly,

$$\text{CKA}_{256}, \quad \text{CKA}_{128}, \quad \text{CKA}_{64} \quad (666)$$

measure the structural similarity between matched clean and corrupted representations at the three analysed development-stage hidden layers.

CKA is a representation-alignment statistic. It is not reconstruction error, does not constitute a measure of psychometric validity, and should not be interpreted as equivalent to cosine similarity between individual observations.

8.11 Local-Neighbourhood Preservation

Local geometric preservation is evaluated in the development-stage terminal representation. For a representation $\mathbf{h}_i$, nearest neighbours are defined using cosine similarity after excluding the observation itself.

The selected neighbourhood sizes are

$$k \in \{5, 10, 20\}. \quad (667)$$

Let $\mathcal{N}_k^{\text{clean}}(i)$ and $\mathcal{N}_k^{\text{corr}}(i)$ denote the corresponding clean and corrupted neighbourhood sets. Their overlap is

$$O_k(i) = \frac{|\mathcal{N}_k^{\text{clean}}(i) \cap \mathcal{N}_k^{\text{corr}}(i)|}{k}. \quad (668)$$

This statistic measures local representational preservation under the specified perturbation. It does not establish preservation of an external identity or psychological construct.

8.12 Output-Space Deformation

Output-space displacement is explicitly separated from reconstruction error.

Let

$$\mathbf{Y}_{\text{Gateway}}^{\text{clean}}, \quad \mathbf{Y}_{\text{Gateway}}^{\text{corr}} \quad (669)$$

denote the Gateway outputs generated from matched clean and corrupted observations. Their relative displacement is

$$D_{\text{out}}^{\text{Gateway}} = \frac{\|\mathbf{Y}_{\text{Gateway}}^{\text{corr}} - \mathbf{Y}_{\text{Gateway}}^{\text{clean}}\|_2}{\|\mathbf{Y}_{\text{Gateway}}^{\text{clean}}\|_2 + \epsilon}. \quad (670)$$

The analogous Traditional-scoring quantity is

$$D_{\text{out}}^{\text{Traditional}} = \frac{\|\mathbf{Y}_{\text{Traditional}}^{\text{corr}} - \mathbf{Y}_{\text{Traditional}}^{\text{clean}}\|_2}{\|\mathbf{Y}_{\text{Traditional}}^{\text{clean}}\|_2 + \epsilon}. \quad (671)$$

These quantities measure movement in output space relative to the clean output. They are not equivalent to error with respect to Gold. A method can produce a large output displacement while remaining close to Gold, or a small output displacement while remaining systematically biased.

8.13 Final P83.25 Representation Audit Boundary

The final P83.25 production checkpoint has hidden representations at dimensions

$$\mathbf{h}^{(344)}, \quad \mathbf{h}^{(208)}, \quad \mathbf{h}^{(184)}, \quad (672)$$

corresponding to the production architecture

$$\mathbb{R}^{300} \rightarrow \mathbb{R}^{344} \rightarrow \mathbb{R}^{208} \rightarrow \mathbb{R}^{184} \rightarrow \mathbb{R}^6 \times 5 \rightarrow \mathbb{R}^{30}. \quad (673)$$

Consequently, any representation-level quantity presented as a final P83.25 mechanistic result must be computed from these production hidden states and from the frozen E14 checkpoint. Historical measurements obtained from $\mathbf{h}^{(256)}$, $\mathbf{h}^{(128)}$, or $\mathbf{h}^{(64)}$ must not be reassigned to $\mathbf{h}^{(344)}$, $\mathbf{h}^{(208)}$, or $\mathbf{h}^{(184)}$.

This establishes the following provenance rule:

$$\boxed{\text{historical representation metric} \Rightarrow \text{historical model only}} \quad (674)$$

and

$$\boxed{\text{P83.25 mechanistic claim} \Rightarrow \text{P83.25 E14 hidden states only}}. \quad (675)$$

The current frozen production record establishes the P83.25 architecture, checkpoint identity, training history, and independent TEST results, but the verified production record used for the present manuscript does not establish a corresponding frozen numerical table for every mechanistic diagnostic listed above. Accordingly, no numerical CKA, neighbourhood-overlap, persistent-homology, Jacobian, effective-rank, or output-displacement result is promoted to a final P83.25 claim unless its computation is explicitly tied to the E14 checkpoint.

This restriction is intentional. It prevents development-stage geometric measurements from being silently reinterpreted as mechanistic properties of the final production model.

The distinction can be represented formally as

$$\boxed{\begin{array}{ll} \text{historical geometry} & \rightarrow \text{architectural discovery} \\ \text{architecture screening} & \rightarrow \text{P83.25 refinement} \\ \text{full-TRAIN retraining} & \rightarrow \text{final E14 checkpoint} \\ \text{E14 checkpoint} & \rightarrow \text{independent TEST.} \end{array}} \quad (676)$$

Thus, the mechanistic laboratory and the final benchmark should not be read as two measurements of the same frozen model. They are sequential components of the research programme: the former documents the internal behaviour of the pre-P83.25 architecture and contributed to the architectural discovery process; the latter establishes the performance of the final production checkpoint.

8.14 Mechanistic Interpretation Boundaries

The mechanistic laboratory provides numerical descriptions of the internal computational states and their response to controlled perturbation. The resulting diagnostics are not interpreted as evidence that the hidden representation corresponds directly to a psychological construct.

The following quantities are therefore kept analytically distinct:

$$\{D_{\text{rel}}, C, R_{\text{eff}}, R_{\text{PR}}, R_q, \text{CKA}, O_k, \|J(\mathbf{x})\|_2, \text{srank}(J), D_{\text{out}}, \text{RMSE}, \text{MAE}\}. \quad (677)$$

No composite “mechanistic score” is formed from these quantities.

The representation analysis is also not used to retroactively justify checkpoint selection. Model selection was completed through the predefined validation protocol before the independent

TEST benchmark was evaluated. Mechanistic diagnostics therefore function as a documented development and interpretation layer rather than as hidden optimisation criteria.

The architectural change itself should likewise not be interpreted as having been mathematically determined by a single geometric diagnostic. The historical analysis provided convergent numerical observations that motivated re-examination of the compact internal geometry; the subsequent controlled architecture screening was the empirical stage in which alternative geometries were compared. This separation preserves the distinction between scientific observation, engineering hypothesis, and empirical model selection.

Finally, because several perturbation conditions can be constructed from the same held-out participant, observation-level geometric measurements are not automatically independent participant-level observations. Any inferential analysis that treats participant identity as the clustering unit must preserve that dependence structure explicitly.

The appropriate interpretation is therefore that the mechanistic laboratory characterises measurable properties of the learned numerical mapping during the development of the IPIP-NEO-300 Gateway. The retained analyses document the internal behaviour of the pre-P83.25 architecture and provide part of the research evidence that motivated the subsequent architectural refinement. They do not establish a causal explanation for the final performance differences, universal robustness, clinical validity, or a direct geometric equivalence between neural representations and psychological structure.

The final P83.25 architecture and its independent TEST results remain separately defined by the frozen production checkpoint and the locked evaluation protocol.

9. Error Analysis

The error analysis is conducted against the same immutable canonical reference representation used throughout the experimental protocol. The reference is computed exclusively from the original clean response vector and is therefore fixed before corruption is introduced. For every corrupted observation, Traditional scoring and the learned Gateway receive the same corrupted input and are evaluated against the same participant-specific canonical target.

This section has a descriptive and comparative role. It does not constitute a second model-selection procedure, does not retroactively modify any selected checkpoint, and does not redefine the validation protocol after observing the independent TEST results.

A central distinction is maintained throughout the analysis:

$$\boxed{\text{development} \rightarrow \text{model selection} \rightarrow \text{checkpoint freezing} \rightarrow \text{independent TEST} \rightarrow \text{error analysis}} \quad (678)$$

Consequently, quantities derived from validation data are interpreted as validation-stage measurements, whereas quantities derived from the frozen TEST partitions are interpreted as independent final evaluation measurements.

The primary comparisons considered in this section are therefore:

$$\{\text{Traditional, V133, IPIP-NEO-120 S2, IPIP-NEO-300 P83.25}\}, \quad (679)$$

with the important qualification that V133 is a development control for IPIP-NEO-120, S2 is the frozen IPIP-NEO-120 production operating point, and P83.25 is the frozen IPIP-NEO-300 production checkpoint.

9.1 Reference, Error Definition, and Evaluation Units

Let

$$\mathbf{y}_i^{\text{Gold}} = g(\mathbf{x}_{i,\text{clean}}) \in \mathbb{R}^{30} \quad (680)$$

denote the immutable canonical reference vector for participant i .

For method m , let

$$\hat{\mathbf{y}}_i^{(m)} = (\hat{y}_{i1}^{(m)}, \dots, \hat{y}_{i30}^{(m)}) \quad (681)$$

denote the corresponding reconstructed or conventionally scored output.

The coordinate-level absolute reconstruction error is

$$e_{ij}^{(m)} = |\hat{y}_{ij}^{(m)} - y_{ij}^{\text{Gold}}|. \quad (682)$$

The canonical target is never recomputed from the corrupted observation:

$$\mathbf{y}_i^{\text{Gold}} \neq g(\mathbf{x}_{i,\text{corr}}). \quad (683)$$

Instead, for Traditional scoring,

$$\mathbf{y}_i^{\text{Traditional}} = g(\mathbf{x}_{i,\text{corr}}), \quad (684)$$

whereas for a learned Gateway,

$$\mathbf{y}_i^{\text{Gateway}} = F_{\theta}(\mathbf{x}_{i,\text{corr}}). \quad (685)$$

This distinction is essential because the conventional scorer is intentionally evaluated on the corrupted response itself, whereas the Gateway is evaluated as a reconstruction mapping toward the immutable pre-corruption reference.

The primary coordinate-level summary metrics are

$$\text{MAE}^{(m)} = \frac{1}{30N} \sum_{i=1}^N \sum_{j=1}^{30} e_{ij}^{(m)}, \quad (686)$$

and

$$\text{RMSE}^{(m)} = \sqrt{\frac{1}{30N} \sum_{i=1}^N \sum_{j=1}^{30} (e_{ij}^{(m)})^2}. \quad (687)$$

RMSE and MAE are deliberately retained as separate primary quantities. MAE measures the average absolute magnitude of reconstruction error, whereas RMSE weights larger residuals more strongly. Consequently, a reduction in one metric does not imply a corresponding reduction in the other.

Pearson correlation and Spearman rank correlation are retained as complementary association measures:

$$r_{\text{Pearson}} = \text{corr}(\hat{\mathbf{y}}, \mathbf{y}^{\text{Gold}}), \quad (688)$$

and

$$\rho_{\text{Spearman}} = \text{corr}(\text{rank}(\hat{\mathbf{y}}), \text{rank}(\mathbf{y}^{\text{Gold}})). \quad (689)$$

These coefficients describe association with the canonical reference and are not interpreted as clinical validity coefficients or as measures of causal psychological correspondence.

9.2 Separation of Developmental and Final Error Analyses

The error analysis uses two distinct evidentiary layers.

First, the validation analysis is used to characterise the operating points that were available during model development and selection. Second, the independent TEST analysis is used only after the production checkpoints have been fixed and the corresponding TEST partitions have been frozen.

This distinction can be represented as

$$\boxed{\text{validation error} \neq \text{independent TEST error}} \quad (690)$$

and more specifically,

$$\boxed{\text{validation} \rightarrow \text{selection} \rightarrow \text{freeze} \rightarrow \text{TEST}} \quad (691)$$

rather than

$$\boxed{\text{TEST} \rightarrow \text{selection} \rightarrow \text{retraining}}. \quad (692)$$

No TEST result is therefore used to justify a prior architecture decision, objective choice, checkpoint choice, or training intervention.

9.3 IPIP-NEO-120 Validation Operating Points

For IPIP-NEO-120, the V133 development control architecture is

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30, \quad (693)$$

with 58,910 trainable parameters.

The final S2 production architecture is

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (694)$$

with 69,214 trainable parameters.

The final S2 operating point corresponds to seed 20260933 at epoch 6. Its selection followed the predefined architecture-screening and confirmation procedure, including confirmation seeds

$$\{20260931, 20260932, 20260933\}. \quad (695)$$

All three confirmation seeds satisfied the predefined guardrails and reduced corrupted-validation MAE relative to V133. The TEST partition was not used in this procedure.

The principal aggregate corrupted-validation values are:

Table 34: Corrupted-validation error comparison for Traditional scoring, V133, and the final IPIP-NEO-120 S2 Gateway.

Method	RMSE	MAE	Pearson r	Spearman ρ
Traditional	0.462274	0.181624	0.818282	0.816762
V133	0.348829	0.201426	0.880498	0.850461
S2	0.349023	0.196198	0.880231	0.850618

Relative to Traditional scoring, the learned Gateway configurations exhibit lower corrupted-validation RMSE but higher MAE. This is a metric-specific difference and should not be compressed into a single undifferentiated performance statement.

Within the learned Gateway family, the V133-to-S2 comparison is more specific. Using the reported values,

$$\Delta \text{RMSE}_{\text{S2-V133}} = 0.349023 - 0.348829 = 0.000194, \quad (696)$$

while

$$\Delta \text{MAE}_{\text{S2-V133}} = 0.196198 - 0.201426 = -0.005228. \quad (697)$$

Using the unrounded validation values,

$$\Delta \text{RMSE}_{\text{S2-V133}} = +0.000193974, \quad (698)$$

and

$$\Delta \text{MAE}_{\text{S2-V133}} = -0.005227696. \quad (699)$$

The relative change in MAE is

$$\Delta \text{MAE}_{\text{rel}} = \frac{0.196198304 - 0.201426000}{0.201426000} \approx -0.02595, \quad (700)$$

corresponding to an approximately 2.60% reduction in corrupted-validation MAE.

The corresponding correlation changes are

$$\Delta r_{\text{S2-V133}} = 0.880231118 - 0.880498000 = -0.000266882, \quad (701)$$

and

$$\Delta \rho_{\text{S2-V133}} = 0.850617515 - 0.850461000 = +0.000156515. \quad (702)$$

Accordingly, S2 is not described as a uniform improvement across all validation metrics. Rather, the selected operating point constitutes a controlled multi-metric trade-off in which corrupted-validation MAE decreases materially, RMSE changes only slightly, and rank correlation remains closely preserved.

This interpretation is based on the predefined selection procedure and not on a retrospective reweighting of the reported metrics.

9.4 IPIP-NEO-120 Validation Error Localization Across Corruption Families

The V133-to-S2 comparison can be decomposed by corruption family while holding the canonical reference and evaluation procedure constant.

The validation-stage changes in RMSE and MAE are:

Table 35: IPIP-NEO-120 V133-to-S2 validation differences by corruption family. Values are S2 minus V133; negative values indicate lower error for S2.

Corruption family	ΔRMSE	ΔMAE
ITEM_SWAP	-0.000655	-0.004456
FACET_SWAP	+0.000205	-0.005612
CATEGORY_PARTIAL_SWAP	-0.001262	-0.006713
CATEGORY_SWAP	+0.003168	-0.002225
ITEM_COLLAPSE	-0.000289	-0.004107
CATEGORY_COLLAPSE	-0.001111	-0.004926
RANDOM_ITEM_CHANGE	+0.000313	-0.003541

Across the seven corruption families, S2 has lower validation MAE in all seven families. In contrast, lower RMSE is observed in four families and higher RMSE in three. This distribution is consistent with the aggregate trade-off already identified above and demonstrates why family-level analysis is informative beyond a single global metric.

A corresponding severity-level comparison provides an additional description. Across the thirteen evaluated severity levels, S2 has lower MAE than V133 at nine levels and higher MAE at four levels. RMSE is lower at two levels and higher at eleven levels.

The largest observed validation RMSE decrease occurs at severity 1:

$$\Delta\text{RMSE}_{s=1} = -0.002692, \quad (703)$$

while the largest increase occurs at severity 15:

$$\Delta\text{RMSE}_{s=15} = +0.003360. \quad (704)$$

For MAE, the largest decrease occurs at severity 1:

$$\Delta\text{MAE}_{s=1} = -0.006914, \quad (705)$$

whereas the largest increase occurs at severity 60:

$$\Delta\text{MAE}_{s=60} = +0.001327. \quad (706)$$

These severity-level quantities remain descriptive. They do not constitute independent inferential tests because multiple perturbation observations can derive from the same participant.

9.5 Independent Frozen TEST: IPIP-NEO-120

After the S2 operating point was frozen, the independent IPIP-NEO-120 TEST partition was evaluated without retraining, checkpoint modification, or architecture modification.

The frozen TEST contains

$$N_{\text{clean}} = 61,557, \quad (707)$$

$$N_{\text{corr}} = 1,723,596, \quad (708)$$

and

$$N_{\text{total}} = 1,785,153 \quad (709)$$

evaluated observations.

On the clean TEST subset, Traditional scoring is identical to Gold by construction. The final S2 Gateway gives

Table 36: Independent frozen IPIP-NEO-120 TEST results on clean observations.

Method	RMSE	MAE	Pearson r	Spearman ρ
Traditional	0.000000	0.000000	1.000000	1.000000
S2	0.114099	0.072518	0.990334	0.983816

The clean-case comparison is primarily a diagnostic anchor rather than the robustness target of the experiment. The decisive comparison is the corrupted TEST partition, where the conventional scorer is exposed to the structured perturbations and the Gateway is evaluated on its reconstruction capability.

The corrupted TEST results are:

Table 37: Independent frozen IPIP-NEO-120 TEST results on corrupted observations.

Method	RMSE	MAE	Pearson r	Spearman ρ
Traditional	0.579329	0.287756	0.715060	0.712385
S2	0.428449	0.264785	0.829916	0.798439

The absolute TEST differences are

$$\Delta\text{RMSE} = 0.428448503 - 0.579328515 = -0.150880012, \quad (710)$$

$$\Delta\text{MAE} = 0.264785287 - 0.287756126 = -0.022970839, \quad (711)$$

$$\Delta r = 0.829916061 - 0.715059733 = +0.114856328, \quad (712)$$

and

$$\Delta\rho = 0.798439021 - 0.712384880 = +0.086054141. \quad (713)$$

The corresponding relative RMSE reduction is

$$\Delta\text{RMSE}_{\text{rel}} = \frac{0.428448503 - 0.579328515}{0.579328515} \approx -0.26043947, \quad (714)$$

corresponding to a 26.043947% reduction.

The relative MAE reduction is

$$\Delta\text{MAE}_{\text{rel}} = \frac{0.264785287 - 0.287756126}{0.287756126} \approx -0.07982745, \quad (715)$$

corresponding to a 7.982745% reduction.

The correlation changes should primarily be interpreted on their absolute coefficient scale. The increase in Pearson correlation is +0.114856328 and the increase in Spearman correlation is +0.086054141.

9.6 Second-Moment Error Decomposition

Because RMSE is the square root of the mean squared error, the squared RMSE provides the empirical second moment of the signed residual magnitude under the reported coordinate-level aggregation:

$$\text{MSE}^{(m)} = \left(\text{RMSE}^{(m)}\right)^2. \quad (716)$$

The variance of the absolute residual magnitude can therefore be written as

$$\text{Var}\left(|E^{(m)}|\right) = \left(\text{RMSE}^{(m)}\right)^2 - \left(\text{MAE}^{(m)}\right)^2, \quad (717)$$

provided the reported RMSE and MAE are computed under the same aggregation. For the corrupted IPIP-NEO-120 TEST partition,

$$\text{MSE}_{\text{Traditional}} = 0.579328515^2 \approx 0.335621528, \quad (718)$$

while

$$\text{MSE}_{\text{S2}} = 0.428448503^2 \approx 0.183568120. \quad (719)$$

The corresponding absolute-error variances are approximately

$$\text{Var}\left(|E_{\text{Traditional}}|\right) \approx 0.252817940, \quad (720)$$

and

$$\text{Var}\left(|E_{\text{S2}}|\right) \approx 0.113456872. \quad (721)$$

The reduction in the second error moment is therefore

$$\Delta\text{MSE}_{\text{rel}} = 1 - \frac{0.183568120}{0.335621528} \approx 0.45305022, \quad (722)$$

corresponding to approximately a 45.305022% reduction.

This decomposition is useful because it shows that the substantial reduction in RMSE is associated with a large reduction in the squared-error second moment. It does not, however, by itself identify a particular causal mechanism for the reduction.

9.7 Paired Row-Level Comparison

For each evaluated observation, a scalar row-level RMSE is defined as

$$r_i^{(m)} = \sqrt{\frac{1}{30} \sum_{j=1}^{30} \left(\hat{y}_{ij}^{(m)} - y_{ij}^{\text{Gold}}\right)^2}. \quad (723)$$

For two methods a and b , the paired row-level difference is

$$\Delta r_i^{(a-b)} = r_i^{(a)} - r_i^{(b)}. \quad (724)$$

For direct Traditional–S2 comparison on the IPIP-NEO-120 corrupted TEST partition, the empirical proportion of evaluated rows for which S2 has lower row-level RMSE is

$$P_{\text{S2} < \text{Traditional}} = 0.729250, \quad (725)$$

corresponding to 72.9250% of evaluated rows. Traditional scoring has lower row-level RMSE for 27.0750% of rows, and no ties were observed.

This statistic is descriptive rather than inferential. Because multiple corruption conditions can be generated from the same held-out participant, the evaluated rows cannot automatically be treated as independent participant-level observations.

The appropriate clustering unit for any future participant-level inferential analysis is therefore the participant from whom the repeated perturbation conditions were generated.

9.8 IPIP-NEO-120 Independent TEST Error by Corruption Family

The frozen IPIP-NEO-120 TEST enables a direct decomposition of robustness by the seven predefined corruption families.

Table 38: IPIP-NEO-120 corrupted TEST RMSE by corruption family.

Corruption family	Traditional RMSE	S2 RMSE	Relative RMSE reduction
ITEM_SWAP	0.454519	0.396420	12.7825%
FACET_SWAP	0.459663	0.394646	14.1445%
CATEGORY_PARTIAL_SWAP	0.305406	0.272947	10.6281%
CATEGORY_SWAP	0.718169	0.591406	17.6509%
ITEM_COLLAPSE	0.500320	0.366206	26.8056%
CATEGORY_COLLAPSE	0.886608	0.495586	44.1031%
RANDOM_ITEM_CHANGE	0.534827	0.410012	23.3375%

S2 has lower TEST RMSE than Traditional scoring in all seven corruption families. The reduction is not uniform across families. The largest relative reduction occurs for CATEGORY_COLLAPSE, whereas the smallest occurs for CATEGORY_PARTIAL_SWAP.

These family-level differences are descriptive conditional properties of the specified perturbation protocol. They do not establish that a corruption family is intrinsically more or less difficult outside the experimental distribution.

9.9 Error Tails

Upper-tail behavior is evaluated from the empirical distribution of row-level reconstruction errors. This analysis complements aggregate RMSE and MAE by describing the magnitude and frequency of unusually large row-level errors.

Let

$$\hat{F}^{(m)}(t) = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(r_i^{(m)} \leq t) \quad (726)$$

denote the empirical cumulative distribution function.

The empirical upper-tail quantile at probability $p \in (0, 1)$ is

$$Q_p^{(m)} = \inf \left\{ t : \hat{F}^{(m)}(t) \geq p \right\}. \quad (727)$$

The analysis uses the prespecified set of tail probabilities

$$\mathcal{P}_{\text{tail}} = \{p_1, p_2, \dots, p_K\}. \quad (728)$$

The exact numerical values and quantile implementation are fixed by the computational protocol. This specification is retained because different software implementations can use different interpolation conventions (**hyndman1996quantiles**).

For a predefined high-error threshold τ , the tail exceedance proportion is

$$\pi_{\tau}^{(m)} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(r_i^{(m)} > \tau). \quad (729)$$

Thus, upper-tail analysis distinguishes at least two quantities:

$$\{Q_p^{(m)}, \pi_{\tau}^{(m)}\}, \quad (730)$$

where the first describes the magnitude of errors at a specified quantile and the second describes the frequency with which errors exceed a specified threshold.

Neither quantity is interpreted as a clinical or psychometric validity estimate.

For direct paired comparison,

$$\Delta r_i^{(a-b)} = r_i^{(a)} - r_i^{(b)} \quad (731)$$

is evaluated across matched observations. This paired construction allows the direction of the difference to be preserved rather than reducing the analysis to two unrelated empirical distributions.

9.10 Output Range

The learned Gateway is evaluated without output clipping. Predictions outside the nominal scoring interval are therefore retained rather than projected back into the admissible domain.

For the 30-dimensional facet representation,

$$\mathcal{Y} = [1, 5]^{30}. \quad (732)$$

This interval follows from the five-point response scale and the canonical facet-aggregation procedure (**goldberg1999pip; johnson2014pipneo120**).

For predicted coordinate $\hat{y}_{ij}$, an out-of-range indicator is

$$I_{ij}^{\text{out}} = \mathbb{I}(\hat{y}_{ij} < 1 \vee \hat{y}_{ij} > 5). \quad (733)$$

The overall out-of-range rate is

$$R_{\text{out}}^{(m)} = \frac{1}{30N} \sum_{i=1}^N \sum_{j=1}^{30} I_{ij}^{\text{out}}. \quad (734)$$

The corresponding mean out-of-range magnitude is

$$M_{\text{out}}^{(m)} = \frac{1}{30N} \sum_{i=1}^N \sum_{j=1}^{30} [\max(0, 1 - \hat{y}_{ij}) + \max(0, \hat{y}_{ij} - 5)]. \quad (735)$$

Thus,

$$R_{\text{out}}^{(m)} \quad (736)$$

measures the frequency of violations of the nominal output interval, whereas

$$M_{\text{out}}^{(m)} \quad (737)$$

measures their average magnitude.

Because no clipping is applied, these quantities characterise the unconstrained learned mapping rather than a post-processed prediction.

For a direct comparison between two frozen operating points a and b ,

$$\Delta R_{\text{out}}^{(a-b)} = R_{\text{out}}^{(a)} - R_{\text{out}}^{(b)}, \quad (738)$$

and

$$\Delta M_{\text{out}}^{(a-b)} = M_{\text{out}}^{(a)} - M_{\text{out}}^{(b)}. \quad (739)$$

Numerical output-range comparisons are reported only where the corresponding predictions were explicitly archived from the evaluated frozen operating points. No missing output-range quantity is inferred from RMSE, MAE, or from a different checkpoint.

9.11 Failure Localization

Failure localization is formulated as a conditional decomposition of reconstruction error over experimentally defined groups. These groups include corruption families, severity levels, affected input items, affected input facets, and output facets where the corresponding construction metadata are available.

The purpose is to identify where error is concentrated within the experimental distribution rather than to infer a causal source of error.

For observation i and output facet j ,

$$e_{ij}^{(m)} = \left| \hat{y}_{ij}^{(m)} - y_{ij}^{\text{Gold}} \right|. \quad (740)$$

For a diagnostic group G , let $\mathcal{I}_G$ denote the set of observations belonging to that group.

The group-specific MAE is

$$\text{MAE}_G^{(m)} = \frac{1}{30 |\mathcal{I}_G|} \sum_{i \in \mathcal{I}_G} \sum_{j=1}^{30} e_{ij}^{(m)}. \quad (741)$$

The group-specific RMSE is

$$\text{RMSE}_G^{(m)} = \sqrt{\frac{1}{30 |\mathcal{I}_G|} \sum_{i \in \mathcal{I}_G} \sum_{j=1}^{30} \left(e_{ij}^{(m)} \right)^2}. \quad (742)$$

For an output facet j ,

$$\text{MAE}_j^{(m)} = \frac{1}{N} \sum_{i=1}^N e_{ij}^{(m)}, \quad (743)$$

and

$$\text{RMSE}_j^{(m)} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(e_{ij}^{(m)} \right)^2}. \quad (744)$$

To separate error concentration from simple group frequency, an error-mass statistic is defined as

$$S_G^{(m)} = \frac{\sum_{i,j} \mathbb{I}_G(i,j) \left(e_{ij}^{(m)}\right)^2}{\sum_{i,j} \left(e_{ij}^{(m)}\right)^2}. \quad (745)$$

Here $\mathbb{I}_G(i,j)$ indicates whether the corresponding observation–output pair belongs to diagnostic group G .

The joint use of

$$\{\text{MAE}_G, \text{RMSE}_G, S_G\} \quad (746)$$

therefore distinguishes average error magnitude, sensitivity to large residuals, and the fraction of total squared-error mass associated with the specified group.

For direct comparison of two learned Gateway configurations,

$$\Delta \text{MAE}_G^{(a-b)} = \text{MAE}_G^{(a)} - \text{MAE}_G^{(b)}, \quad (747)$$

and

$$\Delta \text{RMSE}_G^{(a-b)} = \text{RMSE}_G^{(a)} - \text{RMSE}_G^{(b)}. \quad (748)$$

When the reference quantity is non-zero, the corresponding relative MAE change is

$$\Delta \text{MAE}_{G,\text{rel}}^{(a-b)} = \frac{\text{MAE}_G^{(a)} - \text{MAE}_G^{(b)}}{\text{MAE}_G^{(b)}}. \quad (749)$$

These quantities allow the architecture change to be examined at the same family, severity, input-item, input-facet, and output-facet levels without changing the reference definition.

However, group-specific error concentration must not be interpreted as causal attribution. A group can exhibit high error because of its intrinsic construction, because it is associated with other perturbation properties, or because of its frequency in the evaluation distribution.

Accordingly,

$$\boxed{\text{error localization} \neq \text{causal attribution}}. \quad (750)$$

9.12 Input-Side and Output-Side Localization

When perturbation metadata identify the affected input coordinates, error can be further summarized conditionally on the affected item or facet.

For an affected input group A ,

$$\text{MAE}_A^{(m)} = \frac{1}{30 |\mathcal{I}_A|} \sum_{i \in \mathcal{I}_A} \sum_{j=1}^{30} e_{ij}^{(m)}. \quad (751)$$

Likewise,

$$\text{RMSE}_A^{(m)} = \sqrt{\frac{1}{30 |\mathcal{I}_A|} \sum_{i \in \mathcal{I}_A} \sum_{j=1}^{30} \left(e_{ij}^{(m)}\right)^2}. \quad (752)$$

The analogous output-facet quantities are already defined above.
This separation permits the analysis to distinguish

$$\text{where corruption was introduced} \quad (753)$$

from

$$\text{where reconstruction error was expressed.} \quad (754)$$

The two locations need not coincide, particularly in a nonlinear mapping with cross-coordinate predictive structure.

Such cross-facet predictive behaviour remains an empirical property of the learned mapping. It is not interpreted as evidence that the model has learned a causal or semantic equivalence between psychological facets.

9.13 IPIP-NEO-120 Error-Analysis Interpretation Boundary

The IPIP-NEO-120 S2 error analysis is anchored to the frozen production operating point

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (755)$$

with 69,214 trainable parameters, seed 20260933, and epoch 6.

The validation quantities are therefore historical operating-point diagnostics, whereas the corrupted TEST quantities are the independent final evaluation measurements.

The error-analysis stage does not alter:

$$\{\text{architecture, seed, checkpoint, selection criterion, confirmation rule}\}. \quad (756)$$

In particular,

$$\boxed{\text{error analysis} \not\rightarrow \text{post-hoc checkpoint selection}}. \quad (757)$$

The purpose of this separation is to ensure that the final TEST comparison remains interpretable as an evaluation of a frozen operating point rather than as a continuation of model development.

9.14 IPIP-NEO-300 P83.25: Final Production Error Analysis

The IPIP-NEO-300 final production model is P83.25.

Its architecture is

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (758)$$

with 220,782 trainable parameters.

The final production checkpoint was selected at epoch 14 according to the corrupted-validation RMSE criterion. Unlike IPIP-NEO-120 S2, the present IPIP-NEO-300 production record is a single-seed production result and is not described as a three-seed confirmation study.

The final P83.25 checkpoint was frozen before evaluation of the independent TEST partition. The frozen IPIP-NEO-300 TEST contains

$$N_{\text{clean}} = 21,808, \quad (759)$$

$$N_{\text{corr}} = 610,624, \quad (760)$$

and

$$N_{\text{total}} = 632,432 \quad (761)$$

evaluated observations.

The corrupted TEST comparison is:

Table 39: Independent frozen IPIP-NEO-300 TEST results for Traditional scoring and the final P83.25 Gateway.

Method	RMSE	MAE	Pearson r	Spearman ρ
Traditional	0.290377	0.139890	0.756625	0.747784
P83.25	0.223053	0.137994	0.833131	0.810582

The corresponding absolute changes are

$$\Delta\text{RMSE} = 0.223052573 - 0.290377488 = -0.067324915, \quad (762)$$

$$\Delta\text{MAE} = 0.137993711 - 0.139889697 = -0.001895986, \quad (763)$$

$$\Delta r = 0.833130759 - 0.756625041 = +0.076505718, \quad (764)$$

and

$$\Delta\rho = 0.810581771 - 0.747784084 = +0.062797687. \quad (765)$$

The relative RMSE reduction is

$$\Delta\text{RMSE}_{\text{rel}} = \frac{0.223052573 - 0.290377488}{0.290377488} \approx -0.23185308, \quad (766)$$

corresponding to a 23.185308% reduction.

The relative MAE reduction is

$$\Delta\text{MAE}_{\text{rel}} = \frac{0.137993711 - 0.139889697}{0.139889697} \approx -0.01355344, \quad (767)$$

corresponding to a 1.355344% reduction.

As in the IPIP-120 analysis, these changes are interpreted on their respective metric scales rather than collapsed into a single ranking.

9.15 IPIP-NEO-300 Second-Moment Error Decomposition

For the corrupted IPIP-NEO-300 TEST partition,

$$\text{MSE}_{\text{Traditional}} = 0.290377488^2 \approx 0.084319086, \quad (768)$$

whereas

$$\text{MSE}_{\text{P83.25}} = 0.223052573^2 \approx 0.049752450. \quad (769)$$

The corresponding absolute-error variances are approximately

$$\text{Var}(|E_{\text{Traditional}}|) \approx 0.064749958, \quad (770)$$

and

$$\text{Var}(|E_{\text{P83.25}}|) \approx 0.030710186. \quad (771)$$

The reduction in squared error is

$$\Delta\text{MSE}_{\text{rel}} = 1 - \frac{0.049752450}{0.084319086} \approx 0.40995031, \quad (772)$$

corresponding to approximately a 40.995031% reduction.

As with IPIP-120, the difference between the RMSE reduction and the smaller MAE reduction indicates that the two metrics are capturing different properties of the residual distribution. The result is not interpreted as proof of a unique mechanism of error reduction.

9.16 IPIP-NEO-300 Paired Row-Level Comparison

Using the same row-level RMSE definition,

$$r_i^{(m)} = \sqrt{\frac{1}{30} \sum_{j=1}^{30} \left(\hat{y}_{ij}^{(m)} - y_{ij}^{\text{Gold}} \right)^2}, \quad (773)$$

the proportion of corrupted TEST rows for which P83.25 has lower row-level RMSE than Traditional scoring is

$$P_{\text{P83.25} < \text{Traditional}} = 0.699270, \quad (774)$$

corresponding to 69.9270% of evaluated rows.

Traditional scoring has lower row-level RMSE for 30.0730% of rows, and no ties were observed.

This statistic remains descriptive. The existence of repeated perturbation conditions means that the observation-level rows do not automatically constitute independent participant-level replicates.

9.17 Clean-Input Integrity as a Separate Diagnostic

Clean-input results are maintained separately from corrupted-input robustness because the experimental question differs in the two conditions.

Under clean input, Traditional scoring reproduces the canonical target by construction:

$$\mathbf{y}_{\text{clean}}^{\text{Traditional}} = \mathbf{y}^{\text{Gold}}. \quad (775)$$

Consequently,

$$\text{RMSE}_{\text{Traditional, clean}} = 0, \quad \text{MAE}_{\text{Traditional, clean}} = 0. \quad (776)$$

This identity should not be interpreted as a learned-model benchmark. It is a property of the canonical scoring construction.

For the learned models, clean-input error instead measures the deviation of the learned mapping from the canonical scoring reference under an uncorrupted input. It therefore provides

a useful fidelity diagnostic but does not answer the primary robustness question posed by the corrupted evaluation.

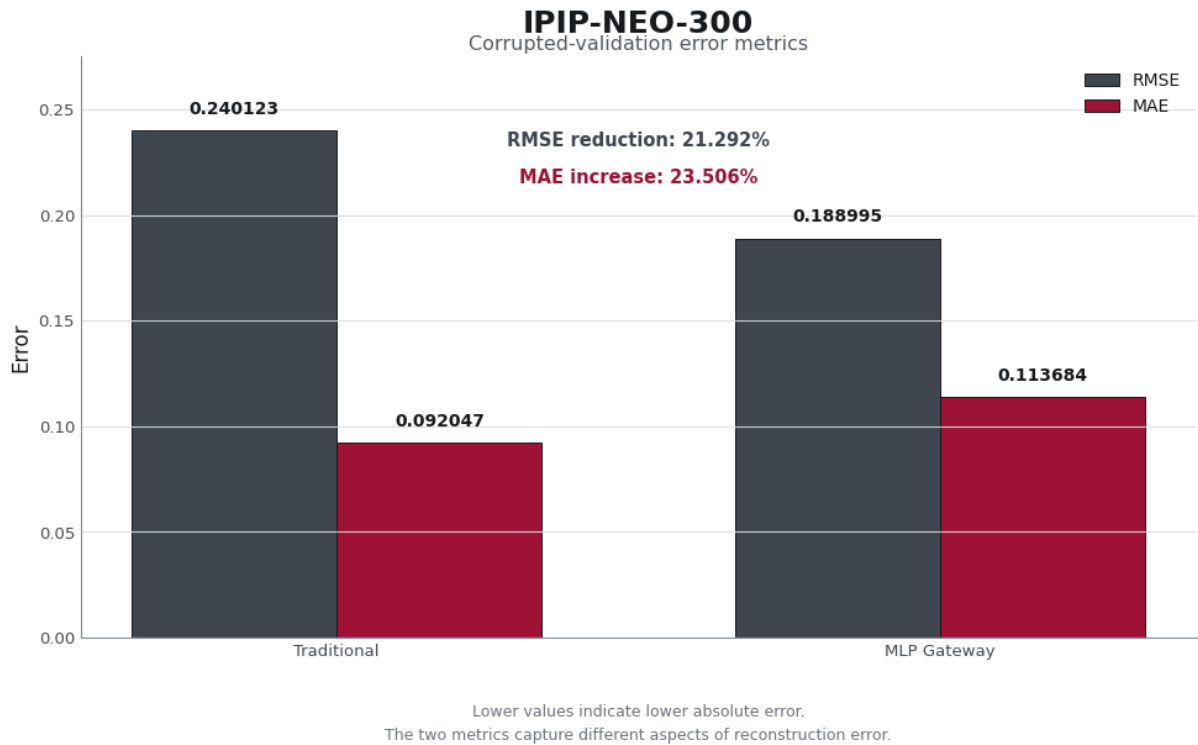
9.18 Figures and Graphical Error Diagnostics

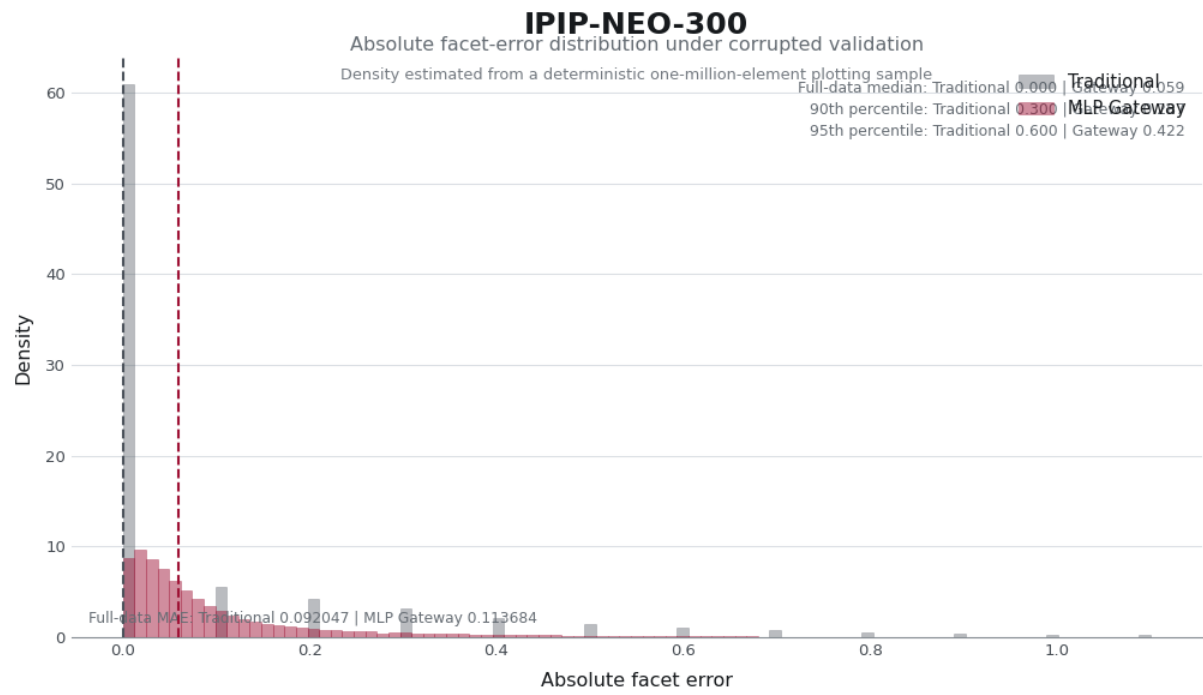
The graphical error diagnostics retained in this section are treated as visual complements to the numerical summaries rather than as independent statistical evidence.

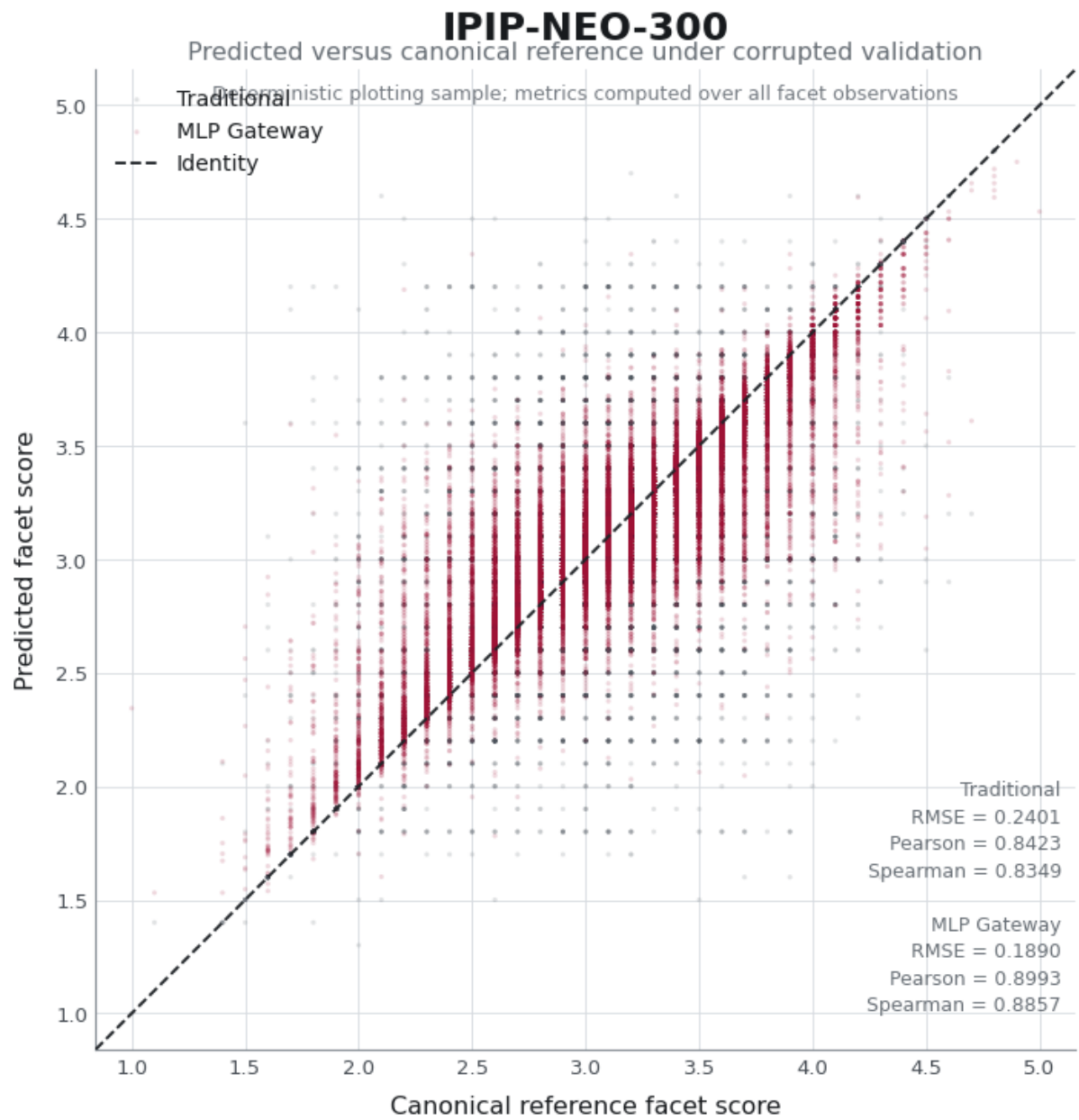
The retained graphical analyses are:

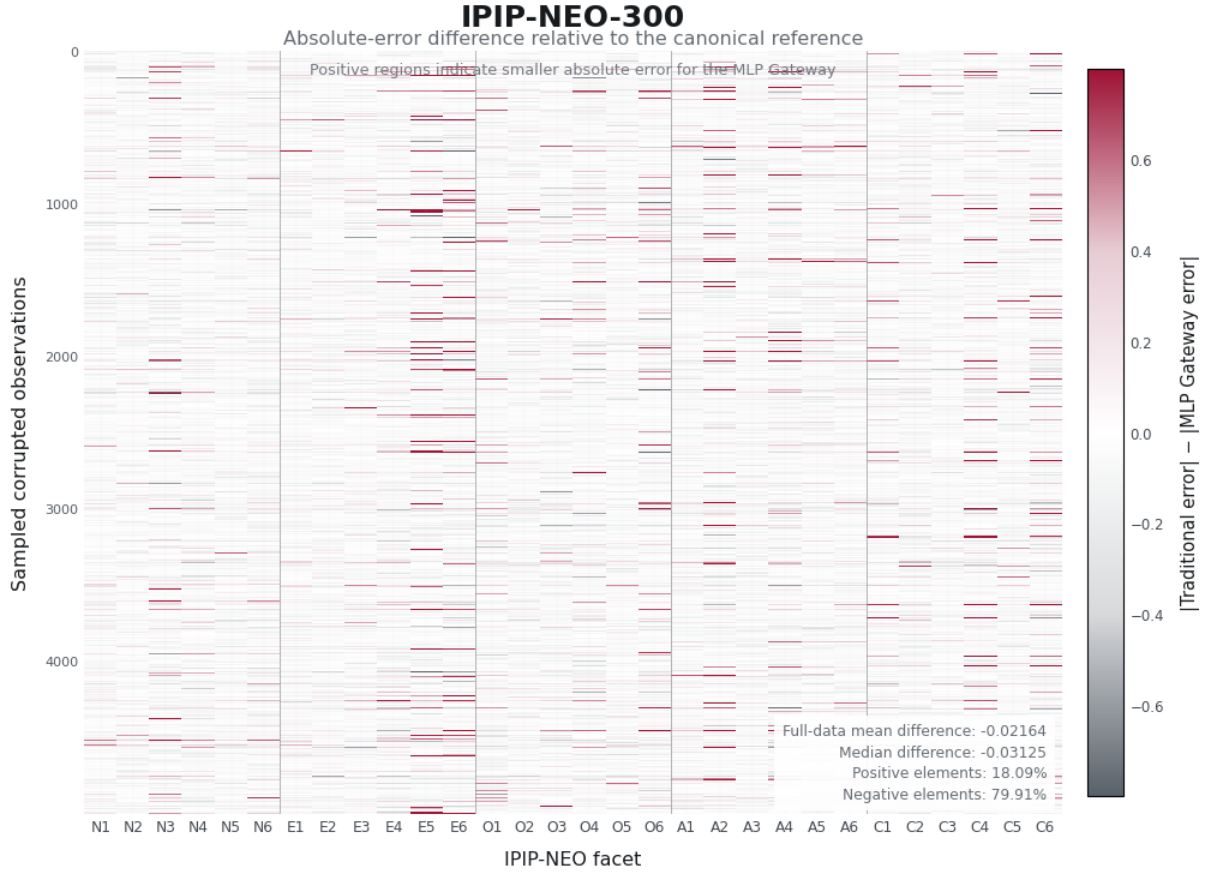
{aggregate error distributions, error-tail visualisations, output-range diagnostics, family-level comparisons} .
(777)

Each figure is interpreted according to the frozen numerical quantity that it visualises. A graphical difference is not treated as a separate observation and is not counted as an additional independent statistical test.









The figures therefore serve as descriptive representations of the numerical results reported in the corresponding tables and equations.

9.19 Dependence Structure and Inferential Caution

The experimental design intentionally generates repeated perturbation conditions for held-out participants. Consequently, several evaluated observations may share the same underlying participant and clean reference.

Formally, if participant s receives a set of perturbation views

$$\{v_{s,1}, v_{s,2}, \dots, v_{s,K}\}, \quad (778)$$

then

$$\text{Gold}_{s,v_{s,1}} = \text{Gold}_{s,v_{s,2}} = \dots = \text{Gold}_{s,v_{s,K}}, \quad (779)$$

even though the corrupted inputs differ.

The resulting observations are therefore repeated experimental conditions within participant rather than independent participants.

This distinction does not invalidate descriptive aggregation across the evaluation set. It does, however, constrain inferential interpretation.

Accordingly,

$$\boxed{\text{observation-level frequency} \neq \text{participant-level inferential evidence}}. \quad (780)$$

Any future hypothesis test, confidence interval, or hierarchical model intended to generalise to participants should account explicitly for participant-level dependence and repeated perturbation

structure.

9.20 Absence of Post-Hoc Error-Based Model Selection

A central methodological safeguard is that the error analysis does not feed back into the model-selection process.

For IPIP-NEO-120, S2 was selected and confirmed before the independent TEST partition was evaluated. For IPIP-NEO-300, P83.25 was selected at epoch 14 according to the predefined corrupted-validation RMSE criterion and then frozen before independent TEST evaluation.

Thus,

$$\boxed{\text{TEST error} \not\rightarrow \text{architecture modification}} \quad (781)$$

and

$$\boxed{\text{TEST error} \not\rightarrow \text{checkpoint re-selection}}. \quad (782)$$

This is especially important for the family-level and tail analyses because disaggregated results can reveal attractive or unattractive subgroups after the fact. Such observations are reported descriptively and do not justify retroactive optimisation against the TEST partition.

9.21 Cross-Instrument Error Interpretation

The IPIP-NEO-120 and IPIP-NEO-300 experiments are treated as separate instrument-specific evaluations.

The corresponding production models differ in architecture, parameter count, training history, and evaluation population:

Table 40: Final production configurations and independent corrupted-TEST evaluation frames.

	IPIP-NEO-120 S2	IPIP-NEO-300 P83.25
Architecture	120→192→128→128→30	300→344→208→184→5×6
Parameters	69,214	220,782
Production epoch	6	14
Corrupted TEST observations	1,723,596	610,624

Accordingly, the two experiments should not be interpreted as if they were matched replications of a single architecture.

Instead, each provides instrument-specific evidence under its respective corruption protocol.

No numerical performance quantity from IPIP-NEO-120 is used to infer a corresponding IPIP-NEO-300 value, and no IPIP-NEO-300 value is substituted into the IPIP-NEO-120 analysis.

9.22 Global Interpretation Boundary

Across all reported analyses, the error quantities are interpreted as conditional empirical properties of the evaluated systems under the specified semi-synthetic perturbation distribution.

They do not constitute:

$$\{\text{clinical diagnosis, psychological ground truth, causal effect, universal robustness, population-level participant i} \quad (783)$$

The analyses also do not establish that the learned reconstruction represents the latent psychological state of a participant in a broader ontological sense. The evaluated target is the operational 30-dimensional canonical facet representation obtained from the original clean response vector.

Accordingly,

$$\boxed{\text{canonical reconstruction accuracy} \neq \text{independent measurement of psychological truth}}. \quad (784)$$

Similarly,

$$\boxed{\text{robustness under the specified corruption protocol} \neq \text{robustness to all possible real-world missingness, noise, etc.}}. \quad (785)$$

The reported results are therefore bounded to the experimental design, including the response domain, corruption families, severity structure, participant population, scoring rule, and frozen evaluation protocol.

9.23 Final Error-Analysis Summary

For IPIP-NEO-120, the final S2 operating point exhibits a controlled validation-stage trade-off relative to V133, with lower corrupted-validation MAE and only a small increase in RMSE. The independent frozen TEST then provides the final robustness comparison against Traditional scoring.

For IPIP-NEO-300, the final production checkpoint is P83.25, not S2. Its independent TEST evaluation is reported separately and is based on the frozen $300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6$ architecture selected through the preceding controlled development process.

The final interpretation is therefore organised according to the following provenance structure:

$$\boxed{\begin{aligned} &\text{validation} \rightarrow \text{model selection} \rightarrow \text{checkpoint freeze} \\ &\rightarrow \text{independent TEST} \rightarrow \text{aggregate error} \rightarrow \text{tail and failure localisation} \end{aligned}} \quad (786)$$

The aggregate metrics, second-moment decomposition, paired row-level analysis, corruption-family decomposition, severity analysis, output-range diagnostics, and error-tail analysis are consequently interpreted as complementary descriptions of the same frozen evaluation framework.

No single metric is treated as sufficient to characterise the system.

In particular, RMSE, MAE, Pearson correlation, Spearman correlation, row-level paired error, tail quantiles, output-range violations, and group-specific localisation answer different empirical questions:

$$\{\text{RMSE}, \text{MAE}, r_{\text{Pearson}}, \rho_{\text{Spearman}}, Q_p, \pi_\tau, R_{\text{out}}, M_{\text{out}}, \text{MAE}_G, \text{RMSE}_G, S_G\}. \quad (787)$$

No composite “error score” is constructed from these quantities, and no overall ranking is assigned to the evaluated methods.

The error analysis therefore remains a transparent decomposition of observed behaviour under the locked experimental conditions rather than a retrospective mechanism for selecting or promoting a model.

10. Discussion

10.1 Principal Methodological Interpretation

The empirical results support a specific methodological observation: under the controlled perturbation protocols evaluated here, a learned nonlinear mapping can use information from the complete questionnaire response vector to reconstruct the canonical pre-corruption facet representation with lower aggregate error than direct deterministic scoring in the independent corrupted TEST evaluations.

This distinction is central to the formulation of the problem. Traditional scoring computes each facet from its designated item set, whereas the Gateway receives the complete response vector. For facet j , the conventional mapping can therefore be represented schematically as

$$y_j^{\text{Traditional}} = g_j(\mathbf{x}_{J_j}), \quad (788)$$

where J_j denotes the item subset assigned to facet j . By contrast, the Gateway computes

$$\hat{y}_j^{\text{Gateway}} = F_{\theta,j}(\mathbf{x}_{1:d}), \quad (789)$$

allowing the learned mapping to use predictive information contained in the full questionnaire response vector.

The observed results are therefore consistent with the central hypothesis that information predictive of a corrupted facet can be distributed beyond the designated items used by the conventional scoring rule. In this sense, the Gateway is not merely learning a nonlinear replacement for the same localised calculation. It is learning a reconstruction mapping over the joint response space.

The most plausible interpretation within the present experimental framework is that the Gateway is exploiting distributed statistical regularities across items and facets that are predictive of the canonical facet representation. The observed consistency across the evaluated corruption conditions provides empirical support for this interpretation. At the same time, the present evidence does not establish that the learned representation constitutes a complete or causal model of psychological structure. The experiment directly tests reconstruction of an operational canonical representation derived from questionnaire responses.

Accordingly, the relevant distinction is

$$\boxed{\text{distributed statistical structure} \neq \text{demonstrated causal psychological structure}}. \quad (790)$$

The former is directly compatible with the observed reconstruction behaviour; the latter would require an independent experimental design and external psychological or clinical criteria.

10.2 Why Full-Vector Access Matters

The difference in information access between Traditional scoring and the Gateway provides a direct explanation for why reconstruction is possible in the first place.

Under the conventional scoring rule, a corruption affecting the designated items of a facet directly alters the information used to compute that facet. The conventional scorer has no mechanism for using information outside that item set to reconstruct the missing or distorted signal.

The Gateway instead operates on

$$\mathbf{x}_{\text{corr}} \in \mathbb{R}^d, \quad (791)$$

and therefore has access to all observed questionnaire coordinates.

The corresponding conceptual contrast is

$$\text{local deterministic scoring} \longleftrightarrow \text{global nonlinear reconstruction}. \quad (792)$$

The empirical results suggest that this additional information access is not merely theoretical. Under the evaluated perturbation distributions, the Gateway was able to reconstruct the canonical target more accurately than the conventional corrupted-input score on the final independent TEST evaluations.

The family-level results further indicate that the usefulness of distributed information depends on perturbation structure. The reconstruction gain is therefore not equivalent to uniform invariance. Instead, the results are more consistent with a learned conditional recovery process whose effectiveness varies according to which response coordinates and structures have been altered.

This distinction is important for interpreting the system as a reconstruction model rather than as a universally corruption-invariant scorer.

10.3 Error Behaviour on the Independent TEST Evaluations

The independent corrupted TEST results provide the clearest evidence for the behaviour of the final frozen operating points.

For IPIP-NEO-120, the final S2 Gateway reduced corrupted TEST RMSE from

$$0.579328515 \quad (793)$$

for Traditional scoring to

$$0.428448503, \quad (794)$$

corresponding to a relative RMSE reduction of approximately 26.04%.

MAE also decreased, from

$$0.287756126 \quad (795)$$

to

$$0.264785287, \quad (796)$$

corresponding to a relative reduction of approximately 7.98%.

The corresponding Pearson correlation increased from

$$0.715059733 \quad (797)$$

to

$$0.829916061, \quad (798)$$

while Spearman correlation increased from

0.712384880 (799)
to

0.798439021. (800)

For IPIP-NEO-300, the final P83.25 Gateway reduced corrupted TEST RMSE from

0.290377488 (801)
to

0.223052573, (802)
corresponding to a relative reduction of approximately 23.19%.
MAE decreased from

0.139889697 (803)
to

0.137993711, (804)
while Pearson correlation increased from

0.756625041 (805)
to

0.833130759 (806)
and Spearman correlation increased from

0.747784084 (807)
to

0.810581771. (808)

Consequently, the final independent TEST results for both evaluated instruments show the same qualitative direction across the principal reported metrics: lower RMSE, lower MAE, higher Pearson correlation, and higher Spearman correlation for the final learned Gateway relative to Traditional scoring under the specified corrupted-input conditions.

This result is materially different from the earlier validation-stage V133-to-S2 trade-off discussed below. The final TEST comparison is therefore not appropriately described as a simple RMSE–MAE trade-off.

final corrupted TEST: Gateway RMSE ↓, MAE ↓, Pearson ↑, Spearman ↑ (809)

The agreement across these metrics provides a more complete description of the final TEST behaviour than any individual error statistic considered in isolation.

10.4 The IPIP-NEO-120 S2 Architectural Refinement

The IPIP-NEO-120 experiment provides a controlled example of architectural refinement within a single instrument.

The V133 control architecture was

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30, \quad (810)$$

with 58,910 trainable parameters, whereas the final S2 architecture was

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (811)$$

with 69,214 trainable parameters.

The final S2 operating point corresponds to seed 20260933 at epoch 6 and was retained after the predefined confirmation procedure.

On the corrupted validation benchmark, the unrounded S2-to-V133 differences were

$$\Delta \text{RMSE}_{\text{S2-V133}} = +0.000193974, \quad (812)$$

$$\Delta \text{MAE}_{\text{S2-V133}} = -0.005227696, \quad (813)$$

$$\Delta r_{\text{S2-V133}} = -0.000266882, \quad (814)$$

and

$$\Delta \rho_{\text{S2-V133}} = +0.000156515. \quad (815)$$

Thus, the S2 validation result should not be characterised as a uniform improvement over V133. It represents a controlled architectural trade-off under the predefined validation guardrails: MAE decreases materially, RMSE increases only slightly, and rank correlation remains numerically close to the V133 control.

The confirmation procedure was repeated using seeds

$$\{20260931, 20260932, 20260933\}, \quad (816)$$

and all three evaluated seeds satisfied the predefined guardrails and reduced corrupted-validation MAE relative to V133.

This provides evidence of stability under the specified confirmation procedure. It does not establish invariance to arbitrary random initialisations, optimisation procedures, training distributions, or perturbation distributions.

The independent TEST evaluation then provides the more consequential result: the frozen S2 operating point produced simultaneous reductions in both RMSE and MAE and increases in both correlation measures relative to Traditional scoring. The validation trade-off and TEST outcome therefore answer different questions and should not be conflated.

10.5 IPIP-NEO-300: From Geometric Observation to P83.25

The IPIP-NEO-300 program provides a distinct and more explicit example of architecture discovery.

The initial structured-output Gateway used the compact configuration

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow \{6, 6, 6, 6, 6\}, \quad (817)$$

with 121,054 trainable parameters.

Before the final architecture was established, this model was subjected to a dedicated mechanistic laboratory examining its internal numerical geometry. The analyses included latent displacement, covariance spectra and effective rank, spectral-diffusion geometry, local Jacobian and stable-rank diagnostics, persistent homology, CKA, local-neighbourhood preservation, and output-space deformation.

These analyses were not treated as proofs that the compact architecture was incorrect, nor were they used as hidden optimisation objectives. Their role was diagnostic. They provided convergent numerical observations concerning how the pre-P83.25 representation responded to structured corruption and motivated a more specific architectural question:

Is the compact internal geometry sufficiently expressive for the reconstruction problem? (818)

The resulting scientific workflow can therefore be expressed as

$$\boxed{\text{mechanistic observation} \rightarrow \text{architectural hypothesis} \rightarrow \text{controlled architecture screening}} \quad (819)$$

This distinction is particularly important because the retained geometric figures were generated before the P83.25 internal architectural change. Their hidden dimensions of 256, 128, and 64 describe the historical development-stage model and are not representations of the final 344-208-184 production topology.

The subsequent screening programme provided the empirical test of the architectural hypothesis. The search progressed through wider and structurally different candidate geometries before reaching the refined configuration

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}. \quad (820)$$

The refined configuration contained 220,782 trainable parameters. On the internal architecture-refinement benchmark, it produced

$$\text{RMSE} = 0.220199, \quad (821)$$

$$\text{MAE} = 0.147551, \quad (822)$$

$$r = 0.861929, \quad (823)$$

and

$$\rho = 0.842994. \quad (824)$$

Relative to the principal WIDE development configuration, the corresponding changes were

$$\Delta\text{RMSE} = -0.003109, \quad (825)$$

$$\Delta\text{MAE} = -0.004251, \quad (826)$$

$$\Delta r = +0.002119, \quad (827)$$

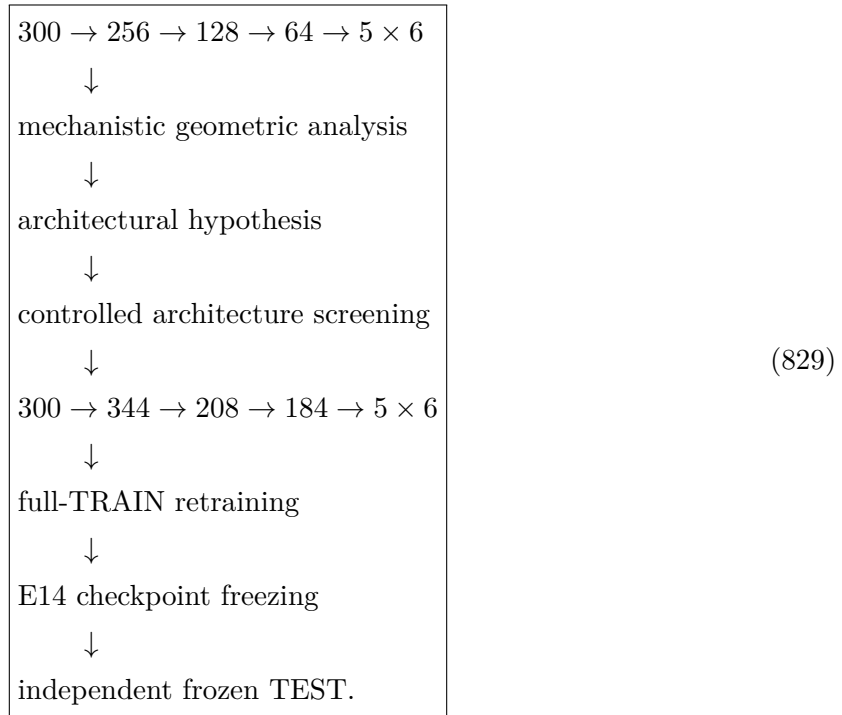
and

$$\Delta\rho = +0.002648. \quad (828)$$

These values belong to the architecture-refinement benchmark and not to the final independent TEST evaluation.

The refined architecture was subsequently retrained from scratch on the complete TRAIN population rather than being treated as a continuation of the screening fit. The production checkpoint was selected at epoch 14 according to the predefined corrupted-validation RMSE criterion and was then frozen for the independent TEST evaluation.

The P83.25 development history can therefore be represented as



The historical geometric figures therefore constitute documentary evidence of the research process through which the architectural question arose. They should not be interpreted as obsolete representations of P83.25 or as direct mechanistic measurements of the final production hidden states.

This provenance is scientifically useful because it distinguishes discovery from confirmation. The geometry generated the architectural question; the controlled screening evaluated candidate answers; the final frozen TEST evaluated the selected production architecture.

10.6 What the P83.25 Result Does and Does Not Establish

The IPIP-NEO-300 results provide evidence that the refined internal geometry selected through the controlled development process can support lower corrupted-input reconstruction error than Traditional scoring on the independent TEST population.

The result should not, however, be interpreted as evidence that the exact 344-208-184 topology is universally optimal.

Architectural optimality has not been established over the space of all possible neural architectures, widths, depths, objectives, optimisation procedures, or training distributions.

The appropriate statement is narrower:

P83.25 was the selected production architecture under the specified development and validation protocol. (830)

Likewise, the historical geometric analysis does not establish that the observed properties of the 256-128-64 representation mathematically required the final 344-208-184 structure. Rather, those observations motivated an empirically testable architectural hypothesis whose subsequent screening produced the refined configuration.

This distinction preserves the difference between scientific discovery and post-hoc explanation.

10.7 Cross-Instrument Interpretation

The IPIP-NEO-120 and IPIP-NEO-300 experiments are complementary but are not direct architectural replications.

IPIP-NEO-120 evolved from V133 to the final S2 architecture

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (831)$$

whereas IPIP-NEO-300 evolved through its own development trajectory toward

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6. \quad (832)$$

The two programs therefore demonstrate that the reconstruction framework can be instantiated successfully with instrument-specific architectures, but they do not establish the existence of a single universal internal geometry.

This is an important distinction. The general methodological principle may be shared:

$$\text{corrupted full response vector} \rightarrow \text{learned reconstruction} \rightarrow \text{canonical facet representation}, \quad (833)$$

while the architecture through which that mapping is implemented may depend on the dimensionality, structure, redundancy, and empirical characteristics of the instrument.

Accordingly, the existence of successful IPIP-120 and IPIP-300 configurations should not be interpreted as evidence that one architecture can simply be transferred unchanged to another psychometric instrument.

10.8 Distributed Information and the Role of Sample Diversity

The present results also raise a broader empirical hypothesis concerning the amount and diversity of response information available during training.

If predictive structure is distributed across questionnaire coordinates, then a training population containing a wider range of response configurations may allow the model to learn a richer approximation of those conditional relationships.

In conceptual form,

greater population diversity $\rightarrow$ broader observed response configurations $\rightarrow$ potentially richer reconstruction m
(834)

A larger and more diverse participant population could therefore plausibly improve generalisation and potentially improve reconstruction performance, particularly if the additional observations expand coverage of response patterns that are underrepresented in the current training population.

This should nevertheless be regarded as a testable hypothesis rather than an automatic consequence of increasing sample size. More participants do not guarantee better point performance if the additional observations are redundant, poorly sampled, or drawn from a distribution that does not expand relevant response diversity.

The stronger methodological prediction is therefore that performance should depend not only on sample size but also on the diversity and coverage of the response manifold represented during training.

Future experiments should consequently report both population size and relevant distributional diversity rather than treating sample size alone as a complete proxy for representational coverage.

10.9 From Semi-Synthetic Perturbation to Human Perturbation

The current robustness evidence is deliberately bounded to the semi-synthetic perturbation protocol used in the present study.

This design provides substantial experimental control because the corruption mechanism, family, severity, and construction parameters can be fixed and reproduced. It nevertheless does not reproduce every way in which real human respondents may generate low-quality, incomplete, inconsistent, or otherwise atypical questionnaire responses.

A natural next experimental stage would therefore be to construct a prospective human perturbation benchmark in which the perturbation source is the participant rather than an exclusively algorithmic transformation.

Such a study could include matched repeated assessments under different controlled response conditions, including rested and fatigued sessions, time-constrained administration, low-attention or low-engagement conditions, controlled omission patterns, low-information response behaviour, and deliberately induced response-collapse conditions at the item or domain level.

The corresponding experimental structure could include

{clean human response, fatigued response, controlled low-information response, domain-level collapse, repeated
(835)

Where scientifically and ethically appropriate, future datasets could also include separately characterized participant populations with clinically relevant conditions, including attention-related populations such as participants with clinically established ADHD, together with appropriate comparison groups.

The purpose would not be to reinterpret such groups as “corrupted” humans. Rather, the purpose would be to determine whether reconstruction behaviour observed under semi-synthetic perturbations remains informative when response variation arises from genuine human conditions.

A particularly important extension would compare three distinct sources of variation:

algorithmic corruption $\leftrightarrow$ human within-person variation $\leftrightarrow$ clinically characterized variation. (836)

Agreement across these sources would provide substantially stronger evidence for ecological relevance than any one perturbation protocol alone.

However, even a successful human-perturbation study would not by itself establish clinical utility. A genuinely clinical validation programme would also require independent clinical endpoints, clinically meaningful decision criteria, appropriately recruited populations, prespecified analysis plans, and prospective evaluation under the intended use conditions.

10.10 Canonical Gold as an Operational Reference

The interpretation of the results depends critically on the meaning of the Gold representation.

The target is defined operationally as

$$\mathbf{Y}_{\text{Gold}} = g(\mathbf{X}_{\text{clean}}). \quad (837)$$

This representation is immutable and is computed exclusively from the original clean response vector.

It should not, however, be interpreted as an independently observed psychological truth.

The distinction is

$$\boxed{\mathbf{Y}_{\text{Gold}} = \text{canonical operational reference}} \quad (838)$$

rather than

$$\boxed{\mathbf{Y}_{\text{Gold}} = \text{independently observed psychological ground truth}}. \quad (839)$$

This limitation is not a weakness specific to the Gateway. It defines the scope of the present experimental question. The study asks whether a learned mapping can reconstruct the operational canonical representation from a structuredly perturbed response vector.

Claims about psychological validity beyond that operational target require independent psychometric constructs, longitudinal outcomes, behavioural criteria, clinical measures, or other external validation instruments.

10.11 Geometric Analysis and Mechanistic Interpretation

The geometric analysis provides an additional layer of evidence by examining whether structured corruption is accompanied by measurable changes in the internal numerical representations of the development-stage Gateway.

The observed analyses include latent displacement, spectral structure, effective dimensionality, diffusion geometry, Jacobian sensitivity, stable rank, persistent homology, CKA, neighbourhood preservation, and output-space deformation.

These quantities provide multiple numerical descriptions of how the earlier Gateway representation changed under corruption.

Their interpretation nevertheless remains observational.

The relevant distinction is

$$\boxed{\text{geometric association} \neq \text{demonstrated causal mechanism}}. \quad (840)$$

Furthermore, most of the retained mechanistic visualisations belong to the pre-P83.25 development model. They therefore describe the model state that preceded the architectural refinement rather than the final E14 production checkpoint.

Their scientific role is consequently best understood as part of the architectural discovery process:

$$\text{representation observation} \rightarrow \text{architectural question} \rightarrow \text{controlled empirical test}. \quad (841)$$

This is stronger than treating the historical analyses merely as illustrative figures, but more defensible than retroactively claiming that they directly explain the final P83.25 mechanism.

A complete mechanistic interpretation of P83.25 would require the same diagnostics to be recomputed from the frozen E14 hidden states. Until such analyses are independently established, the historical 256-128-64 measurements remain development-stage evidence.

10.12 Perturbation-Specific Robustness Rather Than Universal Robustness

The results demonstrate robustness under the evaluated corruption protocol, but they do not demonstrate universal robustness.

Formally,

$$\boxed{\text{robustness under evaluated perturbations} \neq \text{robustness to all real-world response errors}}. \quad (842)$$

The seven corruption families represent a controlled experimental distribution rather than an exhaustive catalogue of possible respondent behaviour.

The family-level differences observed on the frozen TEST partitions further demonstrate that reconstruction difficulty is structured rather than uniform.

This supports a conditional formulation:

$$\text{Performance} = f(\text{architecture, instrument, corruption family, severity, response distribution}). \quad (843)$$

The Gateway should therefore be described as robust within the evaluated experimental domain rather than as inherently or universally robust.

10.13 Repeated Perturbations and Statistical Independence

The experimental construction also imposes an important inferential boundary.

Multiple corruption conditions may be generated from the same held-out participant. Consequently, the observations generated from one participant do not automatically constitute statistically independent participant-level replicates.

If participant s contributes multiple perturbation views,

$$\{v_{s,1}, v_{s,2} \dots, v_{s,K}\}, \quad (844)$$

then these observations share the same underlying clean participant-specific reference.

The appropriate distinction is therefore

$$\boxed{\text{repeated perturbation observations} \neq \text{independent participants}}. \quad (845)$$

This does not invalidate the aggregate descriptive TEST results. It does mean that future participant-level inferential analyses should explicitly model the within-participant dependence structure rather than treating every corrupted row as an independent draw from the population.

10.14 Implications for Real-World and Clinical Translation

The present study deliberately stops short of clinical claims.

No patient cohort, clinical endpoint, diagnosis, treatment decision, prospective clinical workflow, or clinical utility assessment was evaluated. The current evidence therefore concerns computational reconstruction of an operational psychometric representation under controlled response perturbations.

A future translational programme could proceed in stages.

First, the semi-synthetic perturbation benchmark could be complemented by prospectively collected human-response perturbation data.

Second, the training population could be expanded to increase both sample size and diversity of response patterns.

Third, independently characterized populations could be incorporated to test whether reconstruction behaviour remains stable under clinically meaningful heterogeneity.

Finally, clinical validation would require external endpoints that are not defined by the same questionnaire scoring rule used to construct the Gold target.

This final distinction is essential:

$$\boxed{\text{reconstruction of a psychometric reference} \neq \text{demonstrated clinical utility}}. \quad (846)$$

Accordingly, the present results should be viewed as establishing a computational reconstruction capability under specified experimental conditions, while the clinical question remains an independent empirical programme.

10.15 Overall Interpretation

Taken together, the experiments support a coherent methodological picture.

First, the Gateway can exploit information from the complete response vector that is unavailable to the conventional localised scoring rule.

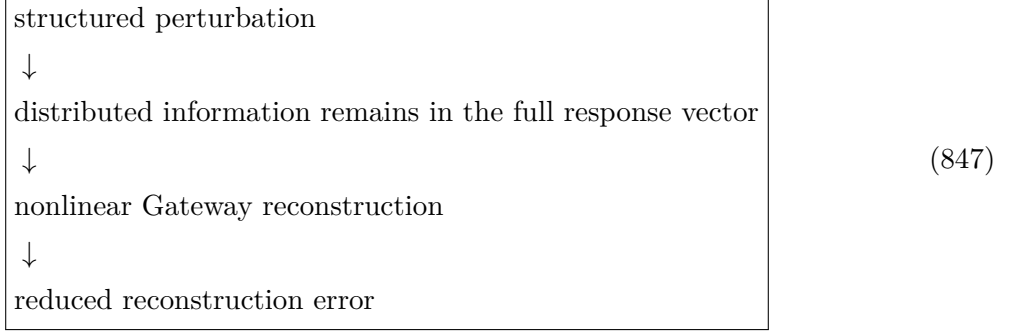
Second, the independent corrupted TEST results show that the final frozen Gateway configurations for both IPIP-NEO-120 and IPIP-NEO-300 achieve lower aggregate reconstruction error and higher reported correlation with the canonical reference than Traditional scoring under the specified perturbation protocol.

Third, the architecture-selection histories are instrument-specific. IPIP-120 developed from V133 to S2 through controlled refinement and multi-seed confirmation, whereas IPIP-300 progressed from a compact structured architecture through mechanistic investigation and controlled architecture screening to P83.25.

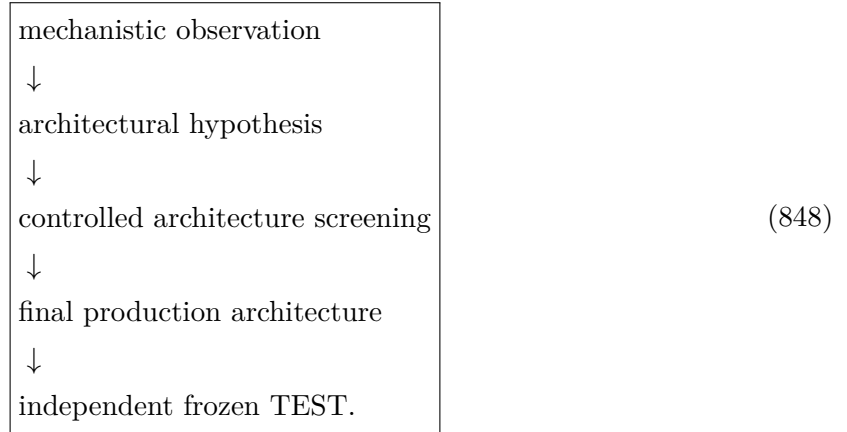
Fourth, the retained geometric analyses provide a documented account of how the architectural question arose in the IPIP-NEO-300 programme. They are therefore part of the scientific provenance of P83.25, while remaining distinct from mechanistic measurements of the final E14 checkpoint.

Fifth, the results are compatible with the hypothesis that the Gateway is learning distributed statistical structure across questionnaire coordinates that supports reconstruction of the canonical facet representation.

The conceptual chain is therefore



and, for the architectural discovery process,



The strongest conclusion supported by the present evidence is therefore not that the Gateway has recovered an ultimate psychological representation, nor that P83.25 is universally optimal, nor that the system is universally robust.

Rather, the study demonstrates that, under the specified questionnaire, canonical scoring rule, semi-synthetic corruption families, training distribution, architecture-selection procedure, and frozen evaluation protocol, a learned nonlinear reconstruction mapping can exploit distributed information in the complete response vector to recover the predefined canonical facet representation more accurately than direct deterministic scoring under the evaluated corrupted conditions.

The results consequently motivate a broader research programme in which the same reconstruction principle is tested with larger and more diverse human populations, prospectively collected human perturbations, clinically characterized cohorts where appropriate, and independent external criteria.

Such extensions would determine whether the computational regularities observed here persist beyond the semi-synthetic experimental domain and whether the reconstruction framework can ultimately support applications requiring stronger ecological, psychometric, or clinical validity.

11. Limitations

1. The perturbation protocol is controlled and semi-synthetic. It uses observed response patterns from real participants as donor material for selected transformations, while other perturbation families are generated through deterministic or controlled transformations. This design provides reproducibility and experimental control, but it does not establish robustness to every real-world mechanism of missingness, fatigue, careless responding, inconsistent responding, measurement error, response inattention, or adversarial behaviour.

2. The present perturbation framework therefore should not be interpreted as a direct substitute for naturally occurring human response degradation. In particular, algorithmically induced corruption does not reproduce all temporal, cognitive, behavioural, or contextual factors that may influence questionnaire responses during genuine human administration.
3. The canonical reference representation is operationally defined by established deterministic scoring of the original, unperturbed response vector. It should not be interpreted as an independently observed latent psychological truth, an independently validated mental-state measurement, or a clinical ground truth.
4. The present study therefore establishes reconstruction agreement with a canonical psychometric reference rather than independent psychological validity. Demonstrating validity beyond this operational target would require external psychometric constructs, behavioural criteria, longitudinal outcomes, independently measured clinical variables, or other external reference criteria.
5. The IPIP-NEO-120 and IPIP-NEO-300 configurations use different Gateway architectures and should not be treated as a single repeated architecture experiment. The final IPIP-NEO-120 S2 configuration is $120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30$, while the preceding V133 control is $120 \rightarrow 192 \rightarrow 128 \rightarrow 64 \rightarrow 30$. The IPIP-NEO-300 production model is P83.25 with architecture $300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6$. These architecture histories therefore constitute instrument-specific development programmes rather than evidence for a universal architecture.
6. The final IPIP-NEO-120 S2 operating point was selected through architecture screening followed by independent three-seed confirmation under predefined guardrails. The confirmation seeds were 20260931, 20260932, and 20260933. All three evaluated seeds satisfied the predefined confirmation criteria. This provides evidence of stability under the specified confirmation protocol, but does not establish invariance to all possible random initialisations, optimisation schedules, hyperparameter configurations, training distributions, or perturbation distributions.
7. The final IPIP-NEO-300 P83.25 production result is a single-seed production evaluation. Its final checkpoint was selected at epoch 14 under the predefined corrupted-validation RMSE criterion and subsequently frozen for the independent TEST evaluation. Unlike IPIP-NEO-120 S2, the present P83.25 production result should therefore not be described as a multi-seed confirmation study.
8. Validation model selection and independent TEST evaluation are distinct stages. The final TEST partitions were not used for architecture screening, objective selection, checkpoint selection, retraining, or post-hoc model modification. Consequently, the reported TEST results should be interpreted as evaluation of frozen operating points rather than as another stage of model optimisation.
9. Multiple corrupted observations may originate from the same held-out participant. These repeated perturbation conditions constitute repeated experimental observations rather than independent participants. Aggregate descriptive metrics remain valid for the defined evaluation set, but participant-level inferential procedures would need to model the within-participant dependence explicitly.
10. The Gateway currently does not provide calibrated predictive uncertainty, normative percentile estimates, validated confidence intervals for individual reconstructions, or clinically validated

decision thresholds. The point estimates reported in the present study should therefore not be interpreted as uncertainty-aware clinical measurements.

11. Output values are not clipped to the nominal $[1, 5]$ interval in the reported production inference. This preserves the unconstrained behaviour of the learned mapping, but it also means that some predicted coordinates may lie outside the conventional scoring domain.
12. The present study evaluates reconstruction agreement with the canonical reference representation rather than clinical validity, diagnostic performance, treatment response, clinical decision support, or prospective decision utility. No patient cohort, clinical endpoint, diagnostic decision, treatment decision, or prospective clinical workflow was evaluated.
13. Population, language, demographic, geographic, institutional, and external-site generalisation have not been established by these experiments alone. The observed performance is conditional on the sampled populations, instruments, response distributions, preprocessing, perturbation families, and evaluation protocols used in the present study.
14. The present sample size and response diversity may not fully represent the broader variability of questionnaire response configurations. Larger and more heterogeneous participant populations could provide broader coverage of response patterns and may improve generalisation, but this remains an empirical hypothesis rather than a guaranteed consequence of increasing sample size.
15. The mechanistic and geometric analyses are descriptive diagnostics of the evaluated numerical representations. Effective rank, spectral structure, representation displacement, diffusion geometry, Jacobian sensitivity, stable rank, persistent homology, CKA, neighbourhood-preservation measures, output-space deformation, and related quantities do not by themselves establish causal mechanisms, psychometric validity, clinical validity, or global robustness.
16. Most of the retained IPIP-NEO-300 mechanistic visualisations were generated during the pre-P83.25 architectural development stage. Their hidden dimensions correspond to the historical 256-128-64 internal topology and therefore must not be interpreted as direct measurements of the final P83.25 E14 hidden representations. They document the architectural discovery process and motivated an empirically testable architectural hypothesis, but they do not constitute mechanistic measurements of P83.25 unless the corresponding diagnostics are recomputed from the frozen E14 checkpoint.
17. The architectural refinement process does not establish that P83.25 is globally optimal over all possible neural architectures, layer widths, depths, output parameterisations, objectives, optimisation procedures, or training schedules. P83.25 should instead be understood as the selected production architecture under the specified development, screening, and validation protocol.
18. The present evidence is bounded to the evaluated semi-synthetic perturbation families and severity conditions. It does not establish robustness to arbitrary real-world questionnaire degradation. In particular, the current study does not directly evaluate matched human perturbation conditions such as naturally occurring fatigue, sustained inattention, time pressure, within-person inconsistency, experimentally induced low-information responding, or clinically characterized response variation.
19. A clinically oriented extension would require a separate prospective human-perturbation programme in which the source of response variation is the participant rather than an exclusively

algorithmic transformation. Such a programme could include repeated clean assessments, fatigued assessments, time-constrained administration, controlled low-attention conditions, omission patterns, response-collapse conditions, and other ethically and scientifically justified perturbation states. The present study does not yet provide evidence for performance under these human perturbation conditions.

20. The transition to additional psychometric instrument families and questionnaire tests has been intentionally deferred to a future version of the research programme. Extending the framework to instruments beyond the currently evaluated IPIP-NEO configurations, including the planned HEXACO-100 extension and other future questionnaire families, requires additional data acquisition, instrument-specific scoring verification, architecture screening, training, validation, and independent TEST evaluation. These extensions were not incorporated into the present version because of practical time and resource constraints rather than because the corresponding research questions were considered resolved.
21. The limitation concerning additional instruments also applies to any claim of cross-instrument architectural generality. No architecture, objective trade-off, validation result, or mechanistic conclusion from IPIP-NEO-120 or IPIP-NEO-300 should be assumed to transfer unchanged to another psychometric instrument without a new instrument-specific evaluation.
22. The present study is therefore intentionally scoped as a first controlled validation stage rather than a complete deployment-level evaluation programme. Future versions can expand the experimental space along several independent dimensions, including larger and more diverse participant populations, prospectively collected human perturbations, additional psychometric instruments, independent external criteria, multi-site sampling, and clinically characterized cohorts where appropriate.
23. Finally, even successful future validation on human perturbation datasets or additional instruments would not by itself establish clinical utility. A genuinely clinical validation programme would additionally require independently defined clinical endpoints, clinically meaningful decision criteria, prespecified analysis plans, appropriate recruitment and comparison groups, prospective evaluation, and assessment under the intended use conditions.

12. Future Studies

The present study establishes a controlled computational framework for reconstructing an operational psychometric reference from structuredly perturbed questionnaire responses. Although the current evaluation is bounded to the investigated IPIP-NEO configurations and the specified perturbation protocols, the underlying reconstruction principle is not intrinsically tied to a single questionnaire, a single item count, or a single output architecture.

The central formulation is sufficiently general to permit application to other instruments:

$$\mathbf{X}_{\text{corr}} \in \mathcal{X}_d \quad \rightarrow \quad F_{\theta}(\mathbf{X}_{\text{corr}}) = \hat{\mathbf{Y}} \in \mathbb{R}^p, \quad (849)$$

where both the observation dimensionality d and the target dimensionality p may be defined by the scoring structure of the instrument under study.

This formulation suggests that the principal methodological idea is broader than any one architecture. The learned system does not require the target instrument to possess the exact same number of items, facets, or hidden dimensions used in the present experiments. Instead,

the framework requires a well-defined response space, an operational canonical reference, and a perturbation process through which reconstruction can be evaluated.

Accordingly, a major direction for future research is to determine how far the observed reconstruction principle generalises across psychometric instrument families.

12.1 Cross-Instrument Generalisation

The results obtained with both IPIP-NEO-120 and IPIP-NEO-300 provide an initial basis for considering broader applicability of the framework. The two experiments differ substantially in input dimensionality and production architecture, yet both instantiate the same fundamental computational principle:

complete corrupted response vector $\rightarrow$ learned nonlinear reconstruction $\rightarrow$ canonical facet representation. (850)

The recurrence of this structure across two instrument configurations is consistent with the hypothesis that the framework captures a general computational problem rather than a property specific to one particular questionnaire implementation.

The next question is therefore not whether the same architecture should be copied from one instrument to another, but whether the same reconstruction principle remains effective when the instrument changes.

This can be expressed as the distinction

$$\boxed{\text{methodological generality} \neq \text{architectural invariance}}. \quad (851)$$

The present results provide evidence relevant to the former while leaving the latter open.

A particularly important future objective is therefore to test whether instrument-specific architectures can consistently discover comparable reconstruction advantages when the response dimensionality, facet structure, item allocation, scoring rules, and redundancy patterns are changed.

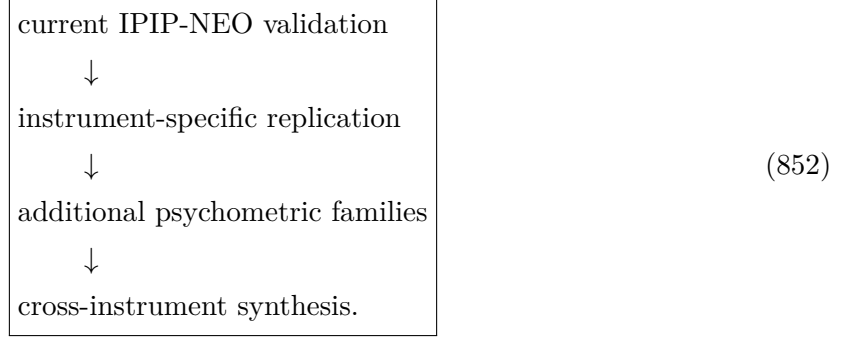
If comparable results were subsequently observed across substantially different instruments, this would provide progressively stronger evidence that the Gateway principle is broadly applicable across questionnaire families.

12.2 Expansion to Additional Psychometric Families

The transition to additional psychometric families has intentionally been reserved for a future version of the research programme. This decision is practical rather than conceptual: each new instrument requires dedicated scoring verification, data preparation, perturbation construction, architecture screening, training, validation, and independent TEST evaluation. The current study therefore does not attempt to compress multiple instrument families into a single underpowered experimental stage.

The planned extension includes the HEXACO-100 framework together with additional questionnaire families in subsequent versions.

The future experimental structure can be represented as



The principal scientific question will be whether the reconstruction advantage persists when the instrument possesses substantially different construct organisations, item-to-facet assignments, response distributions, and scoring geometries.

No cross-instrument result will be assumed a priori. Each new family will be treated as an independent experimental programme with its own canonical reference, perturbation protocol, model-development procedure, and frozen TEST evaluation.

12.3 Why Broad Applicability Is a Particularly Promising Hypothesis

The prospect of broad applicability is scientifically compelling because the Gateway does not depend on a fixed psychometric meaning for its hidden units. Its fundamental operation is the learned approximation of a mapping from an observed response vector to a predefined canonical representation.

The general form is

$$F_{\theta} : \mathcal{X}_d \rightarrow \mathbb{R}^p. \quad (853)$$

The dimensions d and p can, in principle, be adapted to the structure of the instrument.

This makes the framework conceptually compatible with questionnaires that differ in item count, facet count, domain organisation, and scoring structure. The internal architecture would then be treated as an empirical design variable rather than as a fixed universal topology.

The observed success of two different IPIP-NEO configurations therefore provides an encouraging initial signal for broader applicability. It is scientifically plausible that the same computational principle could operate across additional psychometric families whenever the instrument contains sufficient distributed statistical information to support reconstruction.

However, the present evidence does not justify assigning a numerical probability to universal applicability, nor does it establish that every questionnaire contains the degree of redundancy required for successful reconstruction.

The appropriate future hypothesis is therefore:

broad cross-instrument applicability is a strong and testable research hypothesis

(854)

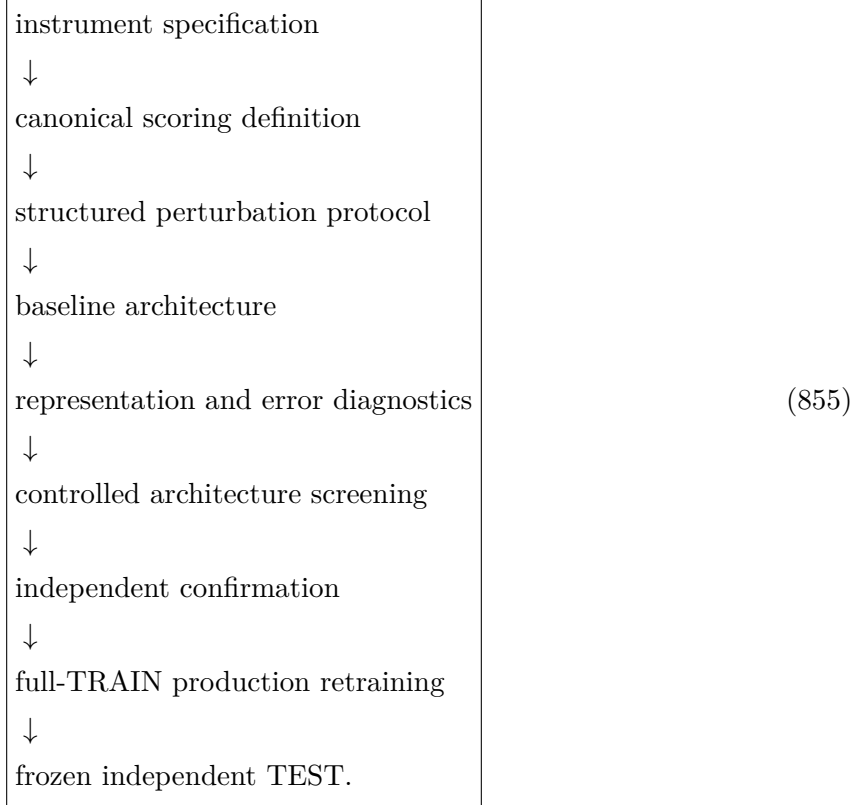
rather than a demonstrated universal property.

The distinction is important because a framework can be conceptually general while the magnitude of its empirical benefit may depend strongly on the statistical structure of each instrument.

12.4 Instrument-Specific Architecture Discovery

Future studies should preserve the instrument-specific architecture discovery principle established by the present work.

For a new instrument $\mathcal{I}$, the corresponding development process can be represented as



This procedure avoids assuming that the architecture discovered for IPIP-NEO-120 or IPIP-NEO-300 is necessarily appropriate for another instrument.

It also preserves the methodological distinction between architectural discovery and final performance evaluation.

12.5 Larger and More Diverse Participant Populations

Another major future direction is expansion of both participant count and response diversity.

If useful predictive information is distributed across questionnaire coordinates, broader sampling may expose the model to a larger range of joint response configurations.

A useful conceptual hypothesis is

greater response diversity $\rightarrow$ broader conditional coverage $\rightarrow$ potentially stronger generalisation. (856)

This prediction is especially relevant when the present training population does not uniformly cover the response space.

Future studies should therefore examine sample size together with distributional coverage, rather than interpreting participant count alone as sufficient evidence of representational completeness.

A particularly valuable design would compare progressively larger training populations while holding the independent TEST population and evaluation protocol fixed.

This would permit empirical estimation of whether reconstruction performance continues to improve, saturates, or changes as response diversity increases.

12.6 Prospective Human Perturbation Studies

The most important methodological extension beyond the current semi-synthetic framework is the evaluation of naturally generated human response variation.

Future studies could collect repeated questionnaire administrations from the same participants under controlled conditions such as rested and fatigued states, time-constrained administration, controlled reductions in attention, low-engagement conditions, omission patterns, and other ethically justified response perturbations.

A prospective design could therefore compare

$$\{\text{clean baseline, human perturbed response, repeated clean control}\}. \quad (857)$$

This would allow the reconstruction framework to be evaluated against variation produced by the participant rather than by a synthetic corruption operator alone.

The distinction is important because the current semi-synthetic benchmark establishes controlled reconstruction behaviour, whereas prospective human studies would test ecological relevance.

A particularly informative future design would combine algorithmically constructed perturbations with matched within-participant human perturbations:

$$\boxed{\text{algorithmic corruption} \leftrightarrow \text{human within-person variation}}. \quad (858)$$

Agreement between these conditions would provide stronger evidence that the controlled corruption families capture response phenomena relevant to actual questionnaire administration.

12.7 Clinically Characterised Populations

A later research stage could investigate whether the reconstruction framework remains informative in clinically characterised populations.

Such studies would need to be designed independently of the present canonical reconstruction benchmark and should not treat clinically characterized participants as intrinsically “corrupted” observations.

Instead, clinically relevant populations could be used to test whether the relationship between questionnaire response patterns and reconstruction behaviour changes under clinically meaningful heterogeneity.

Where scientifically and ethically appropriate, this could include populations characterized by attentional, cognitive, behavioural, or other clinically relevant conditions, together with appropriately recruited comparison populations.

The principal question would be whether the learned reconstruction mapping remains stable when the response distribution shifts in ways that are potentially underrepresented in the current training data.

Such experiments would require independent ethical oversight, appropriate participant characterization, and external criteria beyond the questionnaire scoring rule itself.

12.8 External Psychometric and Behavioural Validation

A further extension is to evaluate the reconstructed representation against external criteria that are not used to construct the canonical Gold target.

The current framework evaluates

$$\hat{\mathbf{Y}} \text{ against } \mathbf{Y}_{\text{Gold}}, \quad (859)$$

whereas future validation could additionally examine relationships with independently measured behavioural, longitudinal, psychometric, or clinical variables.

Such analyses would address a fundamentally different question:

Does improved reconstruction of the canonical representation $\rightarrow$ improve agreement with independent external criteria? (860)

A positive answer would substantially strengthen the interpretation of the Gateway beyond reconstruction fidelity alone.

12.9 Multi-Site and Cross-Population Replication

Future versions should also examine whether the observed behaviour persists across independent sites, populations, languages, and sampling frames.

A useful progression would be

single population $\rightarrow$ independent population $\rightarrow$ multi-site replication $\rightarrow$ cross-population evaluation. (861)

Such replication would distinguish robustness to the perturbation mechanism from robustness to changes in the underlying participant distribution.

This is particularly important for evaluating the broader applicability hypothesis because an architecture may perform strongly within one sampling frame while requiring adaptation when the response distribution changes.

12.10 Scaling Toward a General Reconstruction Framework

If future instrument-specific studies repeatedly demonstrate the same qualitative phenomenon, the research programme could progress from individual instrument models toward a more general reconstruction framework.

The long-term conceptual objective would be to determine whether a common learning framework can accommodate multiple instruments while allowing each instrument to retain its own canonical scoring definition and architecture.

A possible abstract formulation is

$$F_{\theta, \mathcal{I}} : \mathcal{X}_{d_{\mathcal{I}}} \rightarrow \mathbb{R}^{p_{\mathcal{I}}}, \quad (862)$$

where $\mathcal{I}$ denotes the instrument and the dimensions $d_{\mathcal{I}}$ and $p_{\mathcal{I}}$ are instrument-specific.

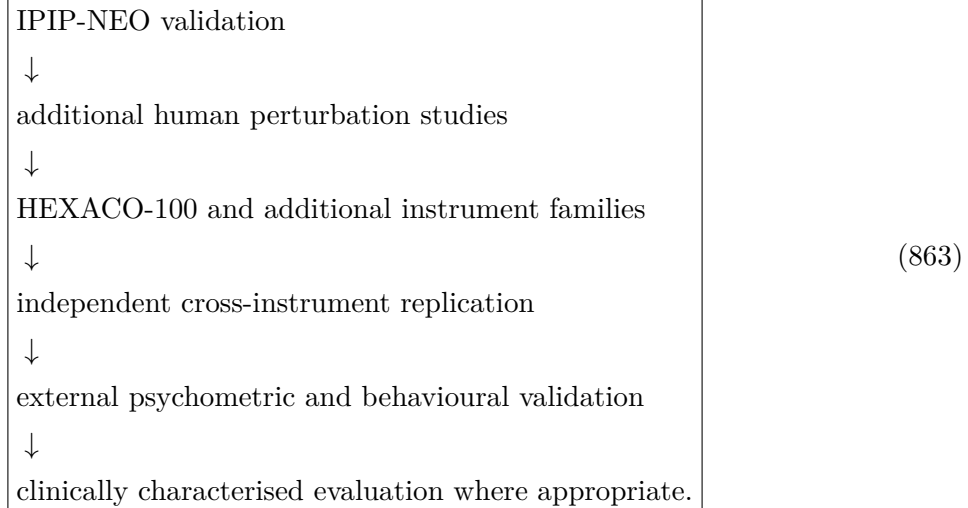
The scientific question would then become whether the reconstruction principle is shared across instruments even when the learned parameters and internal architectures differ.

This represents a substantially stronger and more general hypothesis than simply transferring one trained network between questionnaires.

12.11 Planned Scope of the Next Research Version

Because the present study was constrained by practical time and resource availability, the transition to additional psychometric families is reserved for a future update rather than being compressed into the current manuscript.

The next version is therefore intended to extend the present framework in a controlled sequence:



The deferred status of these studies should not be interpreted as evidence against the broader applicability of the framework. Rather, it reflects the deliberate decision to preserve experimental depth and validation quality within the current scope instead of expanding the number of instruments before each can be evaluated with an equivalently controlled methodology.

12.12 Long-Term Research Hypothesis

The long-term hypothesis motivating the programme is that structured questionnaire responses contain distributed statistical information that can be used to reconstruct canonical representations after substantial structured perturbation.

In its broadest form,

$$\boxed{\text{distributed response information} \rightarrow \text{learned reconstruction} \rightarrow \text{cross-instrument applicability}} \quad (864)$$

is the central hypothesis to be tested across future instrument families.

The present IPIP-NEO-120 and IPIP-NEO-300 results provide an encouraging foundation for that programme because the same reconstruction principle has already been instantiated across two different questionnaire dimensionalities and two different production architectures.

Nevertheless, universal applicability remains an open empirical question.

The decisive evidence would require repeated successful replication across substantially different psychometric families, populations, perturbation mechanisms, and external evaluation criteria.

The future research programme is therefore not based on the assumption that universality has already been established. It is based on the stronger and more scientifically useful proposition that the present results justify testing that possibility systematically.

promising general principle $\rightarrow$ systematic cross-instrument testing $\rightarrow$ empirical determination of generality (865)
--

The transition to additional instrument families, including HEXACO-100, will therefore be undertaken as a subsequent research phase once sufficient time, data, computational resources, and instrument-specific validation capacity are available.

13. Intended Use and Ethical Scope

The present work is intended as a methodological study of robust reconstruction in questionnaire-derived representations. The experiments evaluate whether a learned nonlinear mapping can reconstruct an operational canonical reference representation from structuredly perturbed questionnaire responses. The present evidence does not justify autonomous diagnosis, clinical triage, treatment recommendation, or other medical decision-making.

For the evaluated IPIP-NEO-120 and IPIP-NEO-300 configurations, the learned systems reconstruct the canonical pre-corruption reference representation defined by the experimental scoring procedure. This target is an operational reference derived from the original clean response vector and should therefore be interpreted as a computational reconstruction target rather than as an independently observed psychological latent state or clinical ground truth.

The present IPIP-NEO-120 S2 and IPIP-NEO-300 P83.25 production systems are research models evaluated under the specified perturbation distributions and frozen TEST protocols. Their reported performance should not be interpreted as evidence of general clinical validity, diagnostic accuracy, or readiness for deployment in clinical decision-making.

The intended use of the current system is therefore limited to research applications involving computational reconstruction, robustness analysis, representation modelling, and controlled evaluation of questionnaire-derived targets. Any use beyond these purposes requires independent validation under the corresponding intended-use conditions.

Future extensions to additional psychometric families, including HEXACO-100 and other questionnaire instruments, are reserved for subsequent research versions. Their potential applicability should be evaluated independently, with instrument-specific scoring verification, architecture development, validation, and frozen independent TEST evaluation. No present result should be assumed to transfer automatically to an instrument that has not yet been independently evaluated.

Likewise, future studies involving naturally occurring human response variation or clinically characterized populations should independently establish population validity, measurement properties, uncertainty estimation, privacy protections, informed-consent procedures where applicable, governance requirements, and prospective utility before any clinically oriented use is considered.

In particular, a future clinical programme would require external clinical or behavioural criteria that are not defined solely by the same questionnaire scoring rule used to construct the canonical reference. Reconstruction accuracy against the canonical reference should therefore not be treated as a substitute for clinical validity.

The project also recognises the possibility that the underlying reconstruction principle may prove broadly applicable across psychometric instrument families. That possibility is a future research hypothesis rather than a current deployment claim. Demonstrating such generality will require systematic cross-instrument replication across substantially different questionnaire structures, populations, and response distributions.

The open-science objective of the project is to facilitate independent scrutiny, reproduction, extension, and validation rather than to bypass professional psychometric, ethical, regulatory, or clinical standards. Any future translation toward applied or clinical settings should therefore be preceded by the corresponding scientific, ethical, governance, and regulatory evaluation.

14. Reproducibility and Open Science

The study is designed around a frozen model-and-analysis contract intended to make the computational provenance of each reported result independently auditable. Reproducibility is therefore defined at multiple levels, including model architecture, trainable-parameter count, optimisation configuration, random seeds, checkpoint identity, validation-selection rules, perturbation construction, canonical-reference generation, TEST isolation, analysis code, and machine-readable result archives.

The public research package will contain source code, configuration files, model metadata, reproducibility documentation, machine-readable results, figure-generation code, analysis utilities, environment specifications, frozen evaluation manifests where distributable, and cryptographic hashes for frozen artifacts.

The reproducibility contract distinguishes between

$$\boxed{\text{model identity} + \text{data provenance} + \text{evaluation protocol} + \text{analysis specification}} \quad (866)$$

rather than treating checkpoint files alone as sufficient evidence of reproducibility.

14.1 Final IPIP-NEO-120 S2 Checkpoint

For the final IPIP-NEO-120 S2 operating point, the reproducibility record identifies the production architecture

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (867)$$

with 69,214 trainable parameters, seed 20260933, and epoch 6.

The SHA-256 digest of the frozen production checkpoint is

$$\text{SHA256}_{S_2} = 679c88d15552ee6efaf78cb0b3e288c09b3f00af53cf6d4100807830323e6992. \quad (868)$$

The cryptographic identity provides a deterministic artifact-integrity check. It does not, by itself, establish that the training procedure has been independently reproduced.

The S2 reproducibility record also retains the three confirmation seeds

$$\{20260931, 20260932, 20260933\}, \quad (869)$$

together with the predefined guardrails and confirmation rule used to establish the final operating point.

14.2 Final IPIP-NEO-300 P83.25 Checkpoint

For the final IPIP-NEO-300 production model, the reproducibility record identifies P83.25 with architecture

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6, 6\}, \quad (870)$$

with 220,782 trainable parameters.

The production checkpoint was selected at epoch 14 according to the predefined corrupted-validation RMSE criterion and subsequently frozen for the independent TEST evaluation.

The SHA-256 digest of the frozen P83.25 production checkpoint is

$$\text{SHA256}_{\text{P83.25}} = \text{f113f3b48b09f89deecce55788a59f257f31c09f12b17d3d2f657c0bd42fc287}. \quad (871)$$

The P83.25 production result is a single-seed production evaluation and is therefore not represented as a multi-seed confirmation study.

The production training record additionally preserves the deterministic SIES configuration, including the global seed, SIES seed, view schedule, epoch-dependent seed construction, batch configuration, optimisation hyperparameters, gradient-clipping rule, numerical precision, and stopping criteria required to reproduce the training environment.

14.3 Frozen TEST Integrity

Reproducibility extends beyond the model checkpoint to the independent TEST partitions used for final evaluation.

The TEST protocol preserves the separation

$$\boxed{\text{TRAIN} \cap \text{VALIDATION} \cap \text{TEST} = \emptyset} \quad (872)$$

at the participant-allocation level defined by the experimental protocol.

The frozen TEST records retain the corresponding canonical Gold targets, corrupted inputs, perturbation metadata, participant identifiers where permitted, dataset structure, integrity checks, and evaluation manifests required to reconstruct the reported comparisons.

For IPIP-NEO-300, the frozen TEST generation record identifies source route `/content/HEF_IPIP300_TEST_1` and generation seed 20261220, together with the recorded structural and integrity checks.

The final evaluation contract preserves the distinction

$$\boxed{\text{model input} = X_{\text{corr}}, \quad \text{evaluation target} = Y_{\text{Gold}}} \quad (873)$$

with Gold generated exclusively from the corresponding clean participant response.

The TEST partition is therefore treated as a frozen evaluation artifact rather than as a mutable collection that can be regenerated selectively after observing model performance.

14.4 Validation and Model-Selection Provenance

The reproducibility package will preserve the distinction between model development and final independent evaluation.

For each production operating point, the record will identify:

$$\{\text{candidate architectures, selection metric, guardrails, confirmation procedure, selected checkpoint, TEST-free d} \quad (874)$$

This is particularly important for IPIP-NEO-120 S2, where architecture screening was followed by independent three-seed confirmation, and for IPIP-NEO-300 P83.25, where architecture

screening was followed by full-TRAIN production retraining and corrupted-validation checkpoint selection.

The reproducibility package will therefore preserve the provenance chain

$$\boxed{\text{development} \rightarrow \text{selection} \rightarrow \text{confirmation where applicable} \rightarrow \text{checkpoint freeze} \rightarrow \text{independent TEST}}. \quad (875)$$

No TEST result is used to reconstruct or modify this sequence.

14.5 Code Availability

The reproducibility repository will contain the final IPIP-NEO-120 and IPIP-NEO-300 evaluation code together with training and analysis utilities, configuration files, environment specifications, deterministic seeds where applicable, preprocessing definitions, metric implementations, checkpoint loading procedures, and figure-generation code.

Where separate development and production scripts exist, the package will preserve their versioned identities rather than merging them into a single post-hoc implementation.

The code will therefore distinguish explicitly between development-stage analysis and final production evaluation.

14.6 Model Availability

The released production model files will be accompanied by architecture metadata, trainable-parameter counts, training metadata, checkpoint selection information, and SHA-256 digests so that users can verify byte-level artifact identity.

The production identity will distinguish at minimum:

$$\{\text{instrument, architecture, parameter count, seed, epoch, checkpoint hash}\}. \quad (876)$$

Historical development checkpoints and exploratory candidates, where retained, will be labelled as development artifacts and will not be presented as equivalent to the final production models.

14.7 Dataset Provenance

Original instrument materials and derived perturbation datasets will be documented separately.

The provenance record will distinguish:

$$\left\{ \begin{array}{l} \text{source instrument data,} \\ \text{clean response representations,} \\ \text{canonical Gold targets,} \\ \text{semi-synthetic perturbation data,} \\ \text{donor material used for selected perturbations,} \\ \text{corruption metadata,} \\ \text{model predictions,} \\ \text{derived evaluation tables.} \end{array} \right. \quad (877)$$

This separation is intended to prevent transformed evaluation material from being confused with the original questionnaire observations.

Corruption metadata, including family, severity, donor configuration, and construction parameters, are retained as experimental metadata and are not treated as predictive model inputs unless explicitly specified by a future experiment.

14.8 Canonical Gold Provenance

The canonical target generation procedure is itself part of the reproducibility contract.

For each clean response vector,

$$Y_{\text{Gold}} = g(X_{\text{clean}}). \quad (878)$$

The reproducibility package will preserve the scoring definitions required to regenerate this target, including item scoring, reverse-scored items, facet membership, facet aggregation, and instrument-specific dimensional structure.

The Gold target will never be regenerated from corrupted inputs during evaluation.

This provides an explicit reproducibility invariant:

$$\boxed{Y_{\text{Gold}} = g(X_{\text{clean}}) \quad \text{for all reported evaluation conditions}}. \quad (879)$$

14.9 Perturbation Provenance

The semi-synthetic perturbation framework will be documented at the level of corruption family, severity, construction mechanism, donor material where applicable, and randomisation configuration.

The reproducibility record therefore distinguishes between the response content and the transformation applied to that content.

A generic perturbation record is represented as

$$\mathcal{C}_{\phi,D} : X_{\text{clean}} \rightarrow X_{\text{corr}}, \quad (880)$$

where ϕ denotes the perturbation configuration and D denotes any donor material used by the corresponding transformation.

The public documentation will therefore specify which perturbation families are deterministic, which use donor material, and which rely on controlled randomisation.

14.10 Historical Mechanistic Artifacts

The retained IPIP-NEO-300 geometric analyses are part of the reproducibility record even though they were generated before the final P83.25 architecture.

These artifacts will be explicitly labelled according to their historical development-stage model identity.

In particular, the representations associated with

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 5 \times 6 \quad (881)$$

will be preserved as development-stage mechanistic artifacts and will not be relabelled as P83.25 E14 representations.

This distinction allows the research history to remain reproducible while preventing historical analytical outputs from being confused with final production measurements.

14.11 Machine-Readable Results

Aggregate results will be archived in machine-readable formats together with the corresponding analysis schema.

Where applicable, the package will preserve tables containing:

{RMSE, MAE, r_{Pearson} , ρ_{Spearman} , family-level results, severity-level results, paired comparisons, output-range di
(882)

For final TEST analyses, the corresponding machine-readable summaries will be stored alongside the frozen evaluation manifests and integrity records.

The purpose is to allow independent researchers to reproduce both the aggregate numerical results and their disaggregated error analyses without reconstructing values manually from figures.

14.12 Environment Reproducibility

The reproducibility package will document the computational environment, including Python version, core library versions, machine-learning framework version, numerical dependencies, hardware-relevant configuration where material, precision mode, and execution settings required for reproducibility.

Environment specifications will be versioned together with the corresponding source code.

The target is therefore not merely code-level reproducibility but executable environment reproducibility:

source code + configuration + environment + data provenance $\rightarrow$ reproducible analysis

(883)

14.13 Versioning and Integrity Verification

The public preprint and associated research package will identify a versioned release. Subsequent revisions will retain version history rather than silently replacing the initial record.

Frozen artifacts will be associated with explicit cryptographic digests so that a later release cannot silently substitute a modified checkpoint or evaluation file while retaining the same nominal model identifier.

For an artifact A , integrity verification is represented as

$$H(A)_{\text{released}} = H(A)_{\text{verified}}, \quad (884)$$

where H denotes the SHA-256 digest.

A hash match establishes byte-level artifact identity. It does not establish scientific validity, statistical validity, or independent reproduction.

14.14 Independent Reproduction

The ultimate reproducibility criterion is independent execution of the documented procedure by a separate researcher or research group.

Accordingly,

artifact identity $\neq$ independent scientific reproduction

(885)

The open-science package is therefore intended to make independent reproduction practicable by exposing the model definitions, training configuration, evaluation protocol, frozen artifacts, analysis code, and machine-readable results required to reconstruct the reported findings.

14.15 Future Reproducibility Across Additional Instruments

Future extensions to additional psychometric instrument families will follow the same reproducibility principle but will receive independent versioned records.

In particular, future HEXACO-100 and other instrument-specific studies will require their own:

{instrument definition, Gold construction, perturbation protocol, architecture search, training record, frozen checkpoints, ...} (886)

No future cross-instrument result will be represented as a continuation of an existing checkpoint unless that relationship is explicitly defined and documented.

The transition to additional instrument families is reserved for subsequent research versions because each extension requires additional data, instrument-specific verification, computational resources, and independent validation.

14.16 Open-Science Objective

The open-science objective of the project is to facilitate independent scrutiny, reproduction, extension, and validation rather than to bypass professional psychometric, ethical, regulatory, or clinical standards.

The reproducibility framework therefore follows the principle

transparent provenance $\rightarrow$ independent scrutiny $\rightarrow$ reproduction $\rightarrow$ extension $\rightarrow$ cumulative validation

. (887)

This framework is intended to ensure that future extensions of the Gateway programme can be evaluated against a stable and auditable computational baseline rather than against an undocumented or mutable model state.

15. Conclusion

This study formulated robust psychometric reconstruction under structured response perturbation as a finite-dimensional nonlinear function approximation problem and demonstrated, across two instrument configurations, that a learned Gateway can use information from the complete questionnaire response vector to reconstruct an immutable canonical pre-corruption facet representation more accurately than direct deterministic scoring under the evaluated corrupted-input conditions.

The central empirical result is established on the independent frozen TEST partitions. For both IPIP-NEO-120 and IPIP-NEO-300, the final production Gateway achieved simultaneously lower RMSE and MAE and higher Pearson and Spearman correlations than Traditional scoring under the specified corrupted perturbation protocol.

For IPIP-NEO-120, the final S2 production operating point uses the architecture

$$120 \rightarrow 192 \rightarrow 128 \rightarrow 128 \rightarrow 30, \quad (888)$$

with 69,214 trainable parameters, seed 20260933, and epoch 6.

On the independent corrupted TEST partition, S2 achieved

$$\text{RMSE} = 0.428448503, \quad \text{MAE} = 0.264785287, \quad (889)$$

with

$$r_{\text{Pearson}} = 0.829916061, \quad \rho_{\text{Spearman}} = 0.798439021. \quad (890)$$

Traditional scoring under the same corrupted TEST conditions yielded

$$\text{RMSE} = 0.579328515, \quad \text{MAE} = 0.287756126, \quad (891)$$

with

$$r_{\text{Pearson}} = 0.715059733, \quad \rho_{\text{Spearman}} = 0.712384880. \quad (892)$$

The corresponding S2 reductions were approximately 26.04% in RMSE and 7.98% in MAE, accompanied by increases of +0.114856328 in Pearson correlation and +0.086054141 in Spearman correlation. At the paired row level, S2 was closer to the canonical reference than Traditional scoring for 72.9250% of corrupted TEST observations, with no ties.

For IPIP-NEO-300, the final production architecture is P83.25:

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow \{6, 6, 6, 6\}, \quad (893)$$

with 220,782 trainable parameters and the production checkpoint selected at epoch 14.

On the independent corrupted TEST partition, P83.25 achieved

$$\text{RMSE} = 0.223052573, \quad \text{MAE} = 0.137993711, \quad (894)$$

with

$$r_{\text{Pearson}} = 0.833130759, \quad \rho_{\text{Spearman}} = 0.810581771. \quad (895)$$

Traditional scoring yielded

$$\text{RMSE} = 0.290377488, \quad \text{MAE} = 0.139889697, \quad (896)$$

with

$$r_{\text{Pearson}} = 0.756625041, \quad \rho_{\text{Spearman}} = 0.747784084. \quad (897)$$

The corresponding P83.25 reduction in corrupted TEST RMSE is approximately 23.19%, while MAE decreases by approximately 1.36%. Pearson and Spearman correlations increase by +0.076505718 and +0.062797687, respectively. At the paired row level, P83.25 was closer to the canonical reference than Traditional scoring for 69.9270% of corrupted TEST observations, with no ties.

The agreement of these four principal metrics across both final production systems is important. The result is not driven by a single error criterion: under the specified corrupted-input TEST conditions, the learned Gateways simultaneously reduce absolute reconstruction error, reduce squared-error sensitivity, and increase both linear and rank-based correspondence with the canonical reference.

RMSE ↓ MAE ↓ $r_{\text{Pearson}} \uparrow$ $\rho_{\text{Spearman}} \uparrow$	for both final Gateway configurations on corrupted TEST	(898)
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This constitutes substantially stronger evidence for the reconstruction principle than a result confined to a single metric or a single instrument.

The IPIP-NEO-120 validation programme additionally demonstrated that architectural refinement can be evaluated under predefined guardrails rather than justified retrospectively from final TEST performance. The final S2 operating point differs from V133 by a small increase in corrupted-validation RMSE of 0.000193974 together with a decrease in corrupted-validation MAE of 0.005227696, while Pearson correlation changes by -0.000266882 and Spearman correlation by $+0.000156515$. The independent three-seed confirmation using seeds 20260931, 20260932, and 20260933 satisfied the predefined guardrails in all three evaluated runs and reduced corrupted-validation MAE in all three runs. This provides direct evidence of stability under the specified confirmation protocol while preserving the distinction between validation-stage architecture refinement and independent TEST evaluation.

The IPIP-NEO-300 programme provides complementary evidence concerning architecture discovery. The initial compact structured architecture

$$300 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 5 \times 6 \quad (899)$$

was subjected to a dedicated geometric and mechanistic investigation, including latent displacement, spectral structure, effective dimensionality, diffusion geometry, Jacobian sensitivity, stable rank, persistent homology, CKA, neighbourhood preservation, and output-space deformation. These development-stage observations motivated an architectural hypothesis rather than serving as a hidden optimisation objective. Controlled architecture screening subsequently produced the refined

$$300 \rightarrow 344 \rightarrow 208 \rightarrow 184 \rightarrow 5 \times 6 \quad (900)$$

configuration that became P83.25 after full-TRAIN retraining, validation checkpoint selection, and independent frozen TEST evaluation.

The resulting development sequence is therefore

mechanistic observation $\rightarrow$ architectural hypothesis $\rightarrow$ controlled screening $\rightarrow$ production architecture $\rightarrow$ independent TEST evaluation	(901)
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This provides a reproducible distinction between architectural discovery and final model confirmation. The historical geometric analyses document the scientific process through which the architectural question emerged, while the frozen P83.25 checkpoint defines the authoritative production model.

Taken together, the results strongly support the hypothesis that questionnaire responses contain distributed statistical information extending beyond the fixed item subsets used by conventional facet scoring and that a nonlinear mapping operating on the complete response vector can exploit this information for reconstruction.

The conceptual principle can therefore be stated as

structured response corruption $\rightarrow$ distributed information remains in the full response vector $\rightarrow$ nonlinear reconstruction (902)

The results further suggest that this principle is not intrinsically tied to one particular IPIP-NEO dimensionality or one fixed neural architecture. IPIP-NEO-120 and IPIP-NEO-300 required different instrument-specific Gateway configurations while exhibiting the same qualitative reconstruction advantage on their independent corrupted TEST evaluations.

This provides an encouraging basis for the broader hypothesis that the underlying reconstruction principle may generalise across additional psychometric instrument families. That broader applicability is not yet established, but it is now a directly testable research hypothesis supported by evidence from two instrument configurations with different dimensionalities and architecture-development histories.

The scope of the present conclusion remains deliberately tied to the experimental design. The canonical target is an operational reference defined from the original clean response vector rather than independently observed psychological truth, and the robustness evidence is bounded to the specified semi-synthetic perturbation families, severity conditions, sampled populations, training procedures, and frozen TEST protocols.

Accordingly, the study does not establish universal robustness, universal architectural optimality, independent clinical validity, or automatic cross-instrument transfer. These questions constitute the next stage of the research programme.

Future work will therefore test the reconstruction principle with larger and more diverse participant populations, prospectively collected human perturbations, independently characterized populations, additional psychometric instrument families, external behavioural and psychometric criteria, and eventually clinically oriented validation where scientifically and ethically appropriate.

In particular, extensions to HEXACO-100 and other questionnaire families have been intentionally reserved for a future research version because of the additional time, data, computational, and validation resources required to conduct each extension with the same level of experimental control.

The strongest conclusion supported by the present study is therefore that a learned nonlinear Gateway can exploit distributed information in the complete questionnaire response vector to reconstruct a predefined canonical psychometric representation substantially more accurately than direct deterministic scoring under the evaluated structured corruption conditions, with this effect demonstrated independently for both IPIP-NEO-120 S2 and IPIP-NEO-300 P83.25.

The resulting framework provides a concrete computational basis for a broader research programme investigating whether robust questionnaire reconstruction can become a general methodological capability across psychometric instruments, human response conditions, and ultimately externally validated applications.

Data Availability

The manuscript uses IPIP-derived instrument materials together with derived response representations and internally generated semi-synthetic perturbation data. Exact data-access, redistribution, and licensing terms will be documented in the final release according to the provenance and licensing conditions of the underlying resources.

Code Availability

Source code for the final reproducibility release will be made publicly available in the associated GitHub repository.

Model Availability

Frozen model checkpoints and integrity manifests will be released alongside the reproducibility package, subject to the final artifact distribution policy.

Author Contributions

Oscar Alejandro Laos Medina: conceptualization, methodology, software, dataset engineering, machine learning, validation, statistical analysis, visualization, and manuscript preparation.

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Competing Interests

The author declares no competing interests in the current manuscript version.

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I believe that mental health deserves the same seriousness, scientific discipline, and humanity afforded to every other area of medicine and psychological science. I hope that the broader research programme to which this work belongs may one day contribute, even in a small way, to research and tools that help people receive better understanding, assessment, and support.

This acknowledgement is also a personal tribute to Chester Bennington. His work continues to stand as an example of resilience and overcoming adversity that holds deep meaning for me. So, Chester, you will always be a hero to me.

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TheseusSoft S.A.C. was the organizational environment from which the broader research direction underlying this work emerged. The experience of building the company, developing its

technical vision, and working collectively on its ongoing projects provided the context in which the central idea explored in this manuscript was conceived and subsequently developed.

The author therefore recognizes that this research did not emerge in isolation, but within a shared entrepreneurial and technical effort made possible by the continued work of the TheseusSoft S.A.C. team. Their contributions to the development of the company helped create the environment, continuity, and practical foundation from which this research became possible.

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A. Extended Model Specification

The final supplement will provide the complete instrument mappings, reverse-key audit, full training schedule, complete family- and severity-level tables, full facet-level results, geometric-analysis outputs, propagation matrices, regression tables, and machine-readable artifact references.

B. Artifact Registry

Table 41: Planned public research artifacts.

Artifact	Purpose
IPIP-120 V133 checkpoint	Frozen control architecture for the S2 comparison
IPIP-120 S2 checkpoint	Frozen final reconstruction model for the IPIP-NEO-120 study
IPIP-300 current Gateway checkpoint	Frozen validation benchmark model for the current IPIP-NEO-300 program
IPIP-300 S2 checkpoint	Reserved artifact for the future independently confirmed S2 study
HEXACO-100 S2 checkpoint	Reserved artifact for the future independently confirmed S2 study
Model manifest	Architecture, objective, version, parameter count, provenance
SHA-256 registry	Integrity verification
Reproduction code	Independent rerun of preprocessing, inference, and evaluation
Figure-generation code	Reproduction of publication figures
Machine-readable tables	Exact global, family, severity, facet, and geometry results
Supplementary materials	Detailed methods and extended analyses